

Amendment No. 3 (04/ 2026)

to

**AIS-150: Requirements for approval of vehicles of categories M2, M3, N and T
with regard to braking**

1.0 Clause 5.2.1.32

Substitute following text for existing text for clause 5.2.1.32 and note thereunder;

All vehicles of the following categories shall be equipped with a vehicle stability function:

(a) M2, M3, N2:

(b) N3 having no more than 3 axles; and

(c) N3 with 4 axles, with a maximum mass not exceeding 25 t and a maximum wheel diameter code not exceeding 19.5.

(d) N3 with 4 axles, with two power-driven axles in the rear axle group and only non-driven axles in the other axle group(s), including off-road vehicles regardless of Note.

The vehicle stability function shall include roll-over control and directional control and meet the technical requirements of Annex V to this Standard.

Note: Off-road vehicles, special purpose vehicles (e.g. mobile plant using non-standard vehicle chassis - mobile cranes, hydro-static driven vehicles in which the hydraulic drive system is also used for braking and auxiliary functions, N2 vehicles which have all of the following features: a gross vehicle weight between 3.5 and 7.5 tonnes, a nonstandard low-frame chassis, more than 2 axles and hydraulic transmission), Type I as defined in AIS 052 (Rev.1) and articulated vehicles of categories M2 and M3, N2 tractors for semitrailer with a Gross Combination Weight (GCW) between 3.5 and 7.5 tonnes and Airport Passenger Buses (Tarmac Buses), shall be excluded from this requirement.

2.0 Clause 5.2.2.23

Substitute following text for existing text

All vehicles of categories T3 and T4 (see Note) having no more than 3 axles and equipped with air suspension shall be equipped with a vehicle stability function. This shall include at least rollover control and meet the technical requirements of Annex V.

NOTE— Trailers for exceptional load transport and trailers with areas for standing passengers shall be excluded from this requirement.

3.0 Clause C-1.8.1.1

Substitute following text for existing text

Interurban motor coaches (Type II Buses as defined in AIS 052, Rev. 01) and long-distance touring motor coaches (Type III Buses, Type IV Sleeper Coaches as defined in AIS 052, Rev. 01) of category M3.

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THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA

P. B. NO. 832, PUNE 411 004

ON BEHALF OF

AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER

CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY

MINISTRY OF ROAD TRANSPORT & HIGHWAYS

GOVERNMENT OF INDIA

9th April 2026

AMENDMENT NO. 2 (07/2023)

to

AIS-150 Requirements for approval of vehicles of categories M2, M3, N and T
with regard to braking

1.0 Clause 5.2.1.32, page 31/287

Substitute following text for existing text of note under clause 5.2.1.32

NOTE – Mandatory requirements for fitting vehicle stability function will be in accordance with decision of statutory authority. Off-road vehicles, special purpose vehicles (e.g. mobile plant using non-standard vehicle chassis - mobile cranes, hydro-static driven vehicles in which the hydraulic drive system is also used for braking and auxiliary functions, N2 vehicles which have all of the following features: a gross vehicle weight between 3.5 and 7.5 tonnes, a non-standard low-frame chassis, more than 2 axles and hydraulic transmission), Type I as defined in AIS 052 (Rev.1) and articulated vehicles of categories M2 and M3, N2 tractors for semi-trailer with a Gross Combination Weight (GCW) between 3.5 and 7.5 tonnes and Airport Passenger Buses (Tarmac Buses), shall be excluded from this requirement.

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SET-UP BY

MINISTRY OF ROAD TRANSPORT & HIGHWAYS

(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)

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13th July 2023

**Corrigendum No. 01 to AIS-150 (03/2022) to
AIS-150: Requirements for approval of Vehicles of categories M2, M3, N And T
with regard to braking**

1.0 Clause 5.2.1.32, sub-clause b & c

Delete footnote '12'

2.0 Clause 5.2.1.32,

Substitute following text for existing text of note under clause 5.2.1.32

NOTE – Mandatory requirements for fitting vehicle stability function will be in accordance with decision of statutory authority. Off-road vehicles, special purpose vehicles (e.g. mobile plant using non-standard vehicle chassis - mobile cranes, hydro-static driven vehicles in which the hydraulic drive system is also used for braking and auxiliary functions, N2 vehicles which have all of the following features: a gross vehicle mass between 3.5 and 7.5 tonnes, a non-standard low-frame chassis, more than 2 axles and hydraulic transmission), Type I as defined in AIS 052 (Rev.1) and articulated vehicles of categories M2 and M3, N2 tractors for semi-trailer with a Gross Vehicle Mass (GVM) between 3.5 and 7.5 tonnes, shall be excluded from this requirement.

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ON BEHALF OF
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UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

30th March 2022

AUTOMOTIVE INDUSTRY STANDARD

Requirements for Approval of Vehicles of Categories M2, M3, N and T with Regard to Braking

PRINTED BY
THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA
P.B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

May 2018

Status chart of the standard to be used by the purchaser for updating the record

Sr. No.	Corrigenda.	Amendment	Revision	Date	Remark	Misc.
General remarks :						

INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CTSC). After approval, the Automotive Research Association of India, (ARAI), Pune, being the Secretariat of the AIS Committee, has published this standard. For better dissemination of this information ARAI may publish this standard on their web site.

This Standard cover requirement for M2, M3 and N category of vehicles. For N1 category manufacturer has an option to test their vehicles as per AIS-151 or this standard. An informative Annex X has been incorporated in this document indicating corresponding clauses of this document and UN R 13 (Revision 8 Amendment 4).

This standard is aligned with UN R 13 (Revision 8 Amendment 4). However, keeping in view Indian context this standard is having following differences as compared to UN R:

- a) Trailers are designated as 'T' against 'O'
- b) Coefficient of adhesion of road surface to be at least 0.8 whereas in ECE requirement is given as subjective.
- c) Type II A test is applicable only for few categories of vehicles
- d) Additional India specific requirements are added as below:
 - 1) Drive away chassis test procedure;
 - 2) Criteria for extension of approval to be used for selection of vehicle for testing and extension of approvals added;
 - 3) Test procedure Annex added to avoid interpretation issue; and
 - 4) Tolerance on test speed added to ± 2 percent;
- e) Requirements of 'adhesion utilization' for N1 & N2 category vehicles have been kept in abeyance till the time ABS is made mandatory for these vehicles.
- f) For clarity the testing procedure is given in Annex B which also contains the requirements described in other Annexes. Hence there is a possibility of duplication in case of duplication; whatever is prescribed in Annex B overrides the others.

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible preparation of this standard are given in Annex-Y and Annex-Z respectively.

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REQUIREMENTS FOR APPROVAL OF VEHICLES OF CATEGORIES M2, M3, N AND T WITH REGARD TO BRAKING

1.0 SCOPE

- 1.1 This standard applies to the braking of power-driven vehicles individually and to trailers individually of categories M2, M3, N and T as defined in AIS 053 / IS 14272:2011 Automotive Vehicles –Types – Terminology (first revision).
- 1.2 This standard does not cover:
- a) Vehicles with a maximum design speed not exceeding 25 km/h,
 - b) Trailers which may not be coupled to power-driven vehicles with a design speed exceeding 25 km/h, and
 - c) Vehicles fitted for invalid drivers.
- 1.3 Subject to the applicable provisions of this standard method of measuring reaction (response) times in brakes other than compressed-air brakes are not covered by this standard.

NOTE – AIS 151 is prescribing alternate performance requirements for vehicle categories N1. Vehicle manufacturer may opt for compliance to AIS 151 instead of AIS 150.

2.0 REFERENCES

The following standards contain provisions, which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

Standard	Title
UN R 13 (Revision 8, Amendment 4)	Uniform provisions concerning the approval of vehicles of categories M, N and O with regard to braking

3.0 DEFINITIONS

For the purpose of this standard, the following definitions shall apply.

- 3.1 **Approval of a Vehicle** – The approval of a vehicle type with regard to braking.
- 3.2 **Vehicle Type** – A category of vehicles which do not differ in such essential respects as

a) In the case of power-driven vehicle,

- 1) The vehicle category (see 1.1);
- 2) The maximum mass (see 3.17);
- 3) The distribution of mass among the axles;
- 4) The maximum design speed;
- 5) A different type of braking equipment, with more particular reference to the presence or otherwise of equipment for braking a trailer, or any presence of an electric regenerative braking system;
- 6) The number and arrangement of the axles;
- 7) The engine type;
- 8) The number and ratios of gears;
- 9) The final drive ratios; and
- 10) The tyre dimensions;

b) In the case of trailers,

- 1) The vehicle category (see 1.1);
- 2) The maximum mass (see 3.17);
- 3) The distribution of mass among the axles;
- 4) A different type of braking equipment;
- 5) The number and arrangement of the axles; and
- 6) The tyre dimensions.

3.3 **Braking System** – The combination of parts whose function is progressively to reduce the speed of a moving vehicle or bring it to a halt, or to keep it stationary, if it is already halted; these functions are specified in 5.1.2. The system consists of the control, the transmission, and the brake proper.

3.4 **Control** – The part actuated directly by the driver (or in the case of some trailers, by an assistant) to furnish to the transmission the energy required for braking or controlling it. This energy may be the muscular energy of the driver, or energy from another source controlled by the driver, or in appropriate cases the kinetic energy of a trailer, or a combination of these various kinds of energy.

3.4.1 **Actuation** – Both application and release of the control.

3.5 **Transmission** – The combination of components comprised between the control and the brake and linking them functionally. The transmission may be mechanical, hydraulic, pneumatic, electric or mixed. Where the braking power is derived from or assisted by a source of energy independent of the driver, the reserve of energy in the system is likewise part of the transmission.

The transmission is divided into two independent functions: the control transmission and the energy transmission. Whenever the term 'transmission' is used alone in this standard, it means both the 'control transmission' and the 'energy transmission'. The control and supply lines between towing vehicles and trailers shall not be considered as parts of the transmission.

- 3.5.1 **Control Transmission** – The combination of the components of the transmission which control the operation of the brakes, including the control function and the necessary reserve(s) of energy.
- 3.5.2 **Energy Transmission** – The combination of the components which supply to the brakes the necessary energy for their function, including the reserve(s) of energy necessary for the operation of the brakes.
- 3.6 **Brake** – The part in which the forces opposing the movement of the vehicle develop. It may be a friction brake (when the forces are generated by friction between two parts of the vehicle moving relatively to one another); an electrical brake (when the forces are generated by electro-magnetic action between two parts of the vehicle moving relatively to but not in contact with one another); a fluid brake (when the forces are generated by the action of a fluid situated between two parts of the vehicle moving relatively to one another); or an engine brake (when the forces are derived from an artificial increase in the braking action, transmitted to the wheels, of the engine).
- 3.7 **Different Types of Braking Systems** – Systems which differ in such essential respects as,
 - a) Components having different characteristics;
 - b) A component made of materials having different characteristics, or a component differing in shape or size; and
 - c) A different assembly of the components.
- 3.8 **Component of a Braking System** – One of the individual parts which, when assembled, constitutes the braking system.
- 3.9 **Continuous Braking** – The braking of a combination of vehicles through an installation having the following characteristics:
 - a) A single control which the driver actuates progressively, by a single movement, from his driving seat;
 - b) The energy used for braking the vehicles constituting the combination is furnished by the same source (which may be the muscular energy of the driver); and
 - c) The braking installation ensures simultaneous or suitably-phased braking of each of the constituent vehicles of the combination, whatever their relative positions.

- 3.10 **Semi-Continuous Braking** – The braking of a combination of vehicles through an installation having the following characteristics:
- a) A single control which the driver actuates progressively, by a single movement, from his driving seat;
 - b) The energy used for braking the vehicles constituting the combination is furnished by two different sources (one of which may be the muscular energy of the driver); and
 - c) The braking installation ensures simultaneous or suitably-phased braking of each of the constituent vehicles of the combination, whatever their relative positions.
- 3.11 **Automatic Braking** – Braking of the trailer or trailers occurring automatically in the event of separation of components of the combination of coupled vehicles, including such separation through the breakage of a coupling, the effectiveness of the braking of the remainder of the combination not being thereby destroyed.
- 3.12 **Inertia (or Overrun) Braking** – Braking by utilizing the forces generated by the trailers moving up on the towing vehicle.
- 3.13 **Progressive and Graduated Braking** – Braking during which, within the normal operating range of the equipment, and during actuation of the brakes (see 3.4.1).
- a) The driver can at any moment increase or decrease the braking force by acting on the control,
 - b) The braking force varies proportionally as the action on the control (monotonic function), and
 - c) The braking force can be easily regulated with sufficient precision.
- 3.14 **Phased Braking** – It is a means which may be used where two or more sources of braking are operated from a common control, whereby one source may be given priority by phasing back the other source(s) so as to make increased control movement necessary before they begin to be brought into operation.
- 3.15 **Endurance Braking System** – An additional braking system having the capability to provide and to maintain a braking effect over a long period of time without a significant reduction in performance. The term endurance braking system covers the complete system including the control device.
- 3.15.1 The endurance braking system may comprise a single device or a combination of several devices. Each device may have its own control.
- 3.15.2 Control Configurations for Endurance Braking Systems
- 3.15.2.1 **Independent Endurance Braking System** – An endurance braking system whose control device is separated from that of the service and other braking systems.

- 3.15.2.2 **Integrated Endurance Braking System** – An endurance braking system whose control device is integrated with that of the service braking system in such a way that both endurance and service braking systems are applied simultaneously or suitably phased by operation of the combined control device.
- 3.15.2.3 **Combined Endurance Braking System** – An integrated endurance braking system, which in addition has a cut-out device, which allows the combined control to apply the service braking system alone.
- 3.16 **Laden Vehicle** – Except where otherwise stated, a vehicle so laden as to attain its maximum mass (see 3.17).
- 3.17 **Maximum Mass** – The maximum mass stated by the vehicle manufacturer to be technically permissible (this mass may be higher than the ‘permissible maximum mass laid down by the statutory authority).
- 3.18 **Distribution of Mass Among the Axles** – The distribution of the effect of the gravity on the mass of the vehicle and/or its contents among the axles.
- 3.19 **Wheel/Axle Load** – The vertical static reaction (force) of the road surface in the contact area on the wheel/ wheels of the axle.
- 3.20 **Maximum Stationary Wheel/Axle Load** – The stationary wheel/axle load achieved under the condition of the laden vehicle.
- 3.21 **Electric Regenerative Braking** – A braking system which, during deceleration, provides for the conversion of vehicle kinetic energy into electrical energy.
 - 3.21.1 **Electric Regenerative Braking Control** – A device which modulates the action of the electric regenerative braking system.
 - 3.21.2 **Electric Regenerative Braking System of Category A** – An electric regenerative braking system which is not part of the service braking system.
 - 3.21.3 **Electric Regenerative Braking System of Category B** – An electric regenerative braking system which is part of the service braking system.
 - 3.21.4 **Electric State of Charge** – The instantaneous ratio of electric quantity of energy stored in the traction battery relative to the maximum quantity of electric energy which could be stored in this battery.
 - 3.21.5 **Traction Battery** – An assembly of accumulators constituting the storage of energy used for powering the traction motor(s) of the vehicle.
- 3.22 **Hydraulic Braking System with Stored Energy** – A braking system where energy is supplied by a hydraulic fluid under pressure, stored in one or more accumulators fed from one or more pressure pumps, each fitted with a means of limiting the pressure to a maximum value. This value shall be specified by the manufacturer.

- 3.23 **Simultaneous Lockup of the Front and Rear Wheels** – It refers to the condition when the time interval between the first occurrence of lockup of the last (second) wheel on the rear axle and the first occurrence of lockup on the last (second) wheel on the front axle is less than 0.1 s.
- 3.24 **Electric Control Line** – The electrical connection between power-driven vehicle and trailer which provides the braking control function to the trailer. It comprises the electrical wiring and connector and includes the parts for data communication and the electrical energy supply for the trailer control transmission.
- 3.25 **Data Communication** – The transfer of digital data under the rules of a protocol.
- 3.26 **Point-to-Point** – A topology of a communication network with only two units. Each unit has an integrated termination resistor for the communication line.
- 3.27 **Coupling Force Control** – A system/function to balance automatically the braking rate of towing vehicle and trailer.
- 3.28 **Nominal Value**
- Definitions for braking reference performance are required to put a value on the transfer function of the braking system, relating output to input for vehicles individually and when used in combination.
- a) **Nominal Value** is defined, for a power-driven vehicle, as the characteristic which can be demonstrated at Type Approval and which relates the braking rate of the vehicle on its own to the level of the braking input variable.
 - b) **Nominal Value** is defined, for a trailer, as the characteristic which can be demonstrated at Type Approval and which relates the braking rate to the coupling head signal.
 - c) **Nominal Demand Value** is defined, for coupling force control, as the characteristic which relates the coupling head signal to the braking rate and which can be demonstrated at Type Approval, within the limits of the compatibility bands of Annex J.
- 3.29 **Automatically Commanded Braking** – A function within a complex electronic control system where actuation of the braking system(s) or brakes of certain axles is made for the purpose of generating vehicle retardation with or without a direct action of the driver, resulting from the automatic evaluation of on-board initiated information.
- 3.30 **Selective Braking** – A function within a complex electronic control system where actuation of individual brakes is made by automatic means in which vehicle retardation is secondary to vehicle behaviour modification.

- 3.31 **Reference Braking Forces** - The braking forces of one axle generated at the circumference of the tyre on a roller brake tester, relative to brake actuator pressure and declared at the time of type approval.
- 3.32 **Braking signal** - The logic signal indicating brake activation as specified in paragraph. 5.2.1.30.
- 3.33 **Emergency braking signal** - The logic signal indicating emergency braking as specified in paragraph 5.2.1.31.
- 3.34. **Vehicle Stability Function** - means an electronic control function for a vehicle which improves the dynamic stability of the vehicle.
 - 3.34.1. A vehicle stability function includes one or both of the following:
 - (a) Directional control;
 - (b) Roll-over control.
 - 3.34.2. Control functions within a vehicle stability function:
 - 3.34.2.1. **"Directional control"** means a function within a vehicle stability function that assists the driver, in the event of under steer and over steer conditions, within the physical limits of the vehicle in maintaining the direction intended by the driver in the case of a power-driven vehicle, and assists in maintaining the direction of the trailer with that of the towing vehicle in the case of a trailer.
 - 3.34.2.2. **"Roll-over control"** means a function within a vehicle stability function that reacts to an impending roll-over in order to stabilise the power-driven vehicle or towing vehicle and trailer combination or the trailer during dynamic manoeuvres within the physical limits of the vehicle.
- 3.35. **Subject trailer** – It is a trailer representative of the trailer type for which type approval is sought.
- 3.36. **Brake factor (BF)** - It is the input to output amplification ratio of the brake.
- 3.37. **Identification Code** – It identifies the brake discs or brake drums covered by the braking system approval according to this standard. It contains at least the manufacturer's trade name or trademark and an identification number / part number.
- 3.38. **Axle group** - Refer definition given in AIS-053 / IS: 14272.
- 3.39. **Character of the vehicle** - A descriptive term for a vehicle – tractor for semi-trailer, truck, bus, semi-trailer, full trailer, centre-axle trailer.
- 3.40. **Brake electric/electronic interface** means the part of a separable electrical/electronic connection between the towing vehicle and the towed vehicle which is dedicated to the braking system.
- 3.41. **Automated connector** means a system through which the electric and pneumatic connection, between the towing vehicle and towed vehicle is made automatically without direct intervention of a human operator.

4.0 APPLICATION FOR APPROVAL

- 4.1 The application for approval of a vehicle type with regard to braking shall be submitted by the vehicle manufacturer or by his duly accredited representative.
- 4.2 It shall be accompanied by the details given in Annex A. In case the specifications are already submitted, these need not be submitted again.
- 4.3 A vehicle, representative of the vehicle type to be approved, shall be submitted to the testing agency for conducting the approval tests. See Annex K for guidelines for selecting representative vehicle.

5.0 SPECIFICATIONS

5.1 General

5.1.1 Braking System

- 5.1.1.1 The braking system shall be so designed, constructed and fitted as to enable the vehicle in normal use, despite the vibration to which it may be subjected, to comply with the provisions of this standard.
- 5.1.1.2 In particular, the braking system shall be so designed, constructed and fitted as to be able to resist the corroding and ageing phenomena to which it is exposed.
- 5.1.1.3 Brake linings shall not contain asbestos.
- 5.1.1.4 The effectiveness of the braking systems, including the electric control line, shall not be adversely affected by magnetic or electrical fields. This shall be demonstrated by compliance with AIS 004 (Part 3).
- 5.1.1.5 A failure detection signal may interrupt momentarily (< 10 ms) the demand signal in the control transmission, provided that the braking performance is thereby not reduced.
- 5.1.2 Functions of the Braking System – The braking system defined in 3.3 shall fulfil the following functions.
 - 5.1.2.1 **Service braking system** – The service braking system shall make it possible to control the movement of the vehicle and to halt it safely, speedily and effectively, whatever its speed and load, on any up or down gradient. It shall be possible to graduate this braking action. The driver shall be able to achieve this braking action from his driving seat without removing his hands from the steering control.
 - 5.1.2.2 **Secondary braking system** – The secondary braking system shall make it possible to halt the vehicle within a reasonable distance in the event of failure of the service braking system. It shall be possible to graduate this braking action. The driver shall be able to obtain this braking action from his driving seat while keeping at least one hand on the steering control. For the purposes of these provisions it is assumed that not more than one failure of the service braking system can occur at one time.

5.1.2.3 **Parking braking system** – The parking braking system shall make it possible to hold the vehicle stationary on an up or down gradient even in the absence of the driver, the working parts being then held in the locked position by a purely mechanical device. The driver shall be able to achieve this braking action from his driving seat, subject, in the case of a trailer, to the provisions of 5.2.2.10. The trailer air brake and the parking braking system of the towing vehicle may be operated simultaneously provided that the driver is able to check, at any time, that the parking brake performance of the vehicle combination, obtained by the purely mechanical action of the parking braking system, is sufficient.

5.1.3 Connections for Compressed Air Braking Systems, Between Power-Driven Vehicles and Trailers

5.1.3.1 The connections of the compressed-air braking systems between power-driven vehicles and trailers shall be provided according to 5.1.3.1.1, 5.1.3.1.2 or 5.1.3.1.3.

5.1.3.1.1 One pneumatic supply line and one pneumatic control line.

5.1.3.1.2 One pneumatic supply line, one pneumatic control line and one electric control line.

5.1.3.1.3 One pneumatic supply line and one electric control line.

NOTE – Until uniform procedures are framed, which ensure compatibility and safety, connections between power-driven vehicles and trailers conforming to 5.1.3.1.3 shall not be permitted.

5.1.3.2 The electric control line of the power-driven vehicle shall provide information as to whether the requirements of 5.2.1.18.2 can be satisfied by the electric control line, without assistance from the pneumatic control line. It shall also provide information as to whether it is equipped according to 5.1.3.1.2 with two control lines or according to 5.1.3.1.3 with only an electric control line.

5.1.3.3 A power-driven vehicle equipped according to 5.1.3.1.3 shall recognize that the coupling of a trailer equipped according to 5.1.3.1.1 is not compatible. When such vehicles are electrically connected via the electric control line of the towing vehicle, the driver shall be warned by the red optical warning signal specified in 5.2.1.29.1.1 and when the system is energized, the brakes on the towing vehicle shall be automatically applied. This brake application shall provide at least the prescribed parking braking performance required by C-2.3.1.

5.1.3.4 In the case of a power-driven vehicle equipped with two control lines as defined in 5.1.3.1.2, when electrically connected to a trailer which is also equipped with two control lines, the following provisions shall be fulfilled:

- a) Both signals shall be present at the coupling head and the trailer shall use the electric control signal unless this signal is deemed to have failed. In this case the trailer shall automatically switch to the pneumatic control line;
- b) Each vehicle shall conform to the relevant provisions of Annex J for both electric and pneumatic control lines; and
- c) When the electric control signal has exceeded the equivalent of 100 kPa

(1 bar) for more than 1 s, the trailer shall verify that a pneumatic signal is present; should no pneumatic signal be present, the driver shall be warned from the trailer by the separate yellow warning signal specified in 5.2.1.29.2.

- 5.1.3.5 A trailer may be equipped as defined in 5.1.3.1.3, provided that it can only be operated in conjunction with a power-driven vehicle with an electric control line which satisfies the requirements of 5.2.1.18.2. In any other case, the trailer, when electrically connected, shall automatically apply the brakes or remain braked. The driver shall be warned by the separate yellow warning signal specified in 5.2.1.29.2.
- 5.1.3.6. (a) The electric control line shall conform to ISO 11992-1 and 11992-2:2003 including its amendment 1:2007 and be a point-to-point type using:
- (i) The seven pin connector according to ISO 7638-1 or 7638-2:2003 or,
 - (ii) In the case of systems where the connection of the electric control line is automated, the automated connector shall, as a minimum, provide the same number of pins as the above mentioned ISO 7638 connector and meet the requirements specified in Annex W.
- (b) The data contacts of the ISO 7638 connector shall be used to transfer information exclusively for braking (including ABS) and running gear (steering, tyres and suspension) functions as specified in ISO 11992-2:2003 including its Amd.1:2007. The braking functions have priority and shall be maintained in the normal and failed modes. The transmission of running gear information shall not delay braking functions.
- (c) The power supply, provided by the ISO 7638 connector, shall be used exclusively for braking and running gear functions and that required for the transfer of trailer related information not transmitted via the electric control line. However, in all cases the provisions of 5.2.2.18. shall apply. The power supply for all other functions shall use other measures.
- 5.1.3.6.1 The support of messages defined within ISO 11992-2:2003 including Amd.1:2007 is specified within Annex Q to this Standard for the towing vehicle and trailer as appropriate.
- 5.1.3.6.2 The functional compatibility of towing and towed vehicles equipped with electric control lines as defined above shall be assessed at the time of type approval by checking that the relevant provisions of ISO 11992:2003, including ISO 11992-2:2003 and its Amd.1:2007 parts 1 and 2 are fulfilled. Annex R provides an example of tests that may be used to perform this assessment.
- 5.1.3.6.3 When a power-driven vehicle is equipped with an electric control line and electrically connected to a trailer equipped with an electric control line, a continuous failure (> 40 ms) within the electric control line shall be detected in the power-driven vehicle and shall be signalled to the driver by the yellow warning signal specified in 5.2.1.29.1.2, when such vehicles are connected via the electric control line.
- 5.1.3.7 If the operation of the parking braking system on the power-driven vehicle

also operates a braking system on the trailer, as permitted by 5.1.2.3, then the following additional requirements shall be met:

- a) When the power-driven vehicle is equipped according to 5.1.3.1.1, the actuation of the parking brake system of the power-driven vehicle shall actuate a braking system on the trailer via the pneumatic control line.
- b) When the power-driven vehicle is equipped according to 5.1.3.1.2, the actuation of the parking brake system on the power-driven vehicle shall actuate a braking system on the trailer as prescribed in 5.1.3.7 (a). In addition, the actuation of the parking brake system may also actuate a braking system on the trailer via the electric control line.
- c) When the power-driven vehicle is equipped according to 5.1.3.1.3 or, if it satisfies the requirements of 5.2.1.18.2 without assistance from the pneumatic control line, 5.1.3.1.2, the actuation of the parking braking system on the power-driven vehicle shall actuate a braking system on the trailer via the electric control line. When the electrical energy for the braking equipment of the power-driven vehicle is switched off, the braking of the trailer shall be effected by evacuation of the supply line (in addition, the pneumatic control line may remain pressurized); the supply line may only remain evacuated until the electrical energy for the braking equipment of the power-driven vehicle is restored and simultaneously the braking of the trailer via the electric control line is restored.

5.1.3.8. Shut-off devices which are not automatically actuated shall not be permitted.

5.1.3.9. In the case of tractor and semi-trailer combinations, the flexible hoses and cables shall be a part of the power-driven vehicle. In all other cases, the flexible hoses and cables shall be a part of the trailer. In the case of an automated connector, this requirement regarding the allocation of flexible hoses and cables is not applicable.

5.1.4 Provisions for the Periodic Technical Inspection of Braking Systems

5.1.4.1 It shall be possible to assess the wear condition of the components of the service brake that are subject to wear, for example friction linings and drums/discs (in the case of drums or discs, wear assessment may not necessarily be carried out at the time of periodic technical inspection). The method by which this may be realized is defined in 5.2.1.11.2 and 5.2.2.8.2.

5.1.4.2 For the purpose of determining the in-use braking forces of each axle of the vehicle, with a compressed-air braking system, air pressure test connections are required,

- a) In each independent circuit of the braking system, at the closest readily accessible position to the brake cylinder which is the least favourably placed as far as the response time described in Annex E is concerned.
- b) In a braking system which incorporates a pressure modulation device as referred to in J-7.2, located in the pressure line upstream and downstream of this device at the closest accessible position. If this device is pneumatically controlled an additional test connection is required to simulate the laden condition. Where no such device is fitted, a single

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pressure test connection, equivalent to the downstream connector mentioned above, shall be provided. These test connections shall be so located as to be easily accessible from the ground or within the vehicle.

c) At the closest readily accessible position to the least favourably placed energy storage device within the meaning of F-2.4.

d) In each independent circuit of the braking system so it is possible to check the input and output pressure of the complete transmission line.

e) The pressure test connections shall comply with 4 of ISO 3583.

5.1.4.3 The accessibility of required pressure test connections shall not be obstructed by modifications and assembly of accessories or the vehicle body.

5.1.4.4 It shall be possible to generate maximum braking forces under static conditions on a rolling road or roller brake tester.

5.1.4.5 Data for braking systems

5.1.4.5.1 The data of the compressed-air braking system for the functional and efficiency test shall be specified at the vehicle in a visible position in indelible form, or made freely available in another way (for example handbook, electronic data recorder).

5.1.4.5.2 For vehicles equipped with compressed-air braking systems at least the following data are required:

Pneumatic characteristic data:		
Compressor/unloader valve ⁽¹⁾	Maximum cut-out pressure = (kPa)	Minimum cut-in pressure = (kPa)
Four-circuit protection valve	Static closing pressure = (kPa)	
Trailer control valve or relay ⁽⁴⁾ emergency valve, as appropriate	Corresponding delivery pressure for a control pressure of 150 kPa = (kPa)	
Minimum design pressure in the service braking system for calculation ⁽¹⁾⁽²⁾	Axle(s)	
Brake cylinder type ⁽³⁾ Service / Parking	/	/
Maximum stroke ⁽³⁾ s _{max} = (mm)		
Lever length ⁽³⁾ = (mm)		

NOTES

1 Not applicable for trailers.

- 2 When different from minimum cut-in pressure.
- 3 Only applicable for trailers.
- 4 Not applicable for vehicles with electronic control of braking systems.

5.1.4.6 Reference braking forces

- 5.1.4.6.1 Reference braking forces shall be defined for vehicles with compressed air operated brakes using a roller brake tester.
- 5.1.4.6.2 Reference braking forces are to be determined for a brake actuator pressure range from 100 kPa (1 bar) to the pressure generated under Type O conditions for each axle. The applicant for type approval shall nominate reference-braking forces for a brake activator pressure range from 100 kPa (1 bar). These data shall be made available, by the vehicle manufacturer, according to 5.1.4.5.1.

- 5.1.4.6.3 The reference braking forces shall be declared such that the vehicle is capable of generating a braking rate equivalent to that defined in Annex C for the relevant vehicle (50 percent in the case of vehicles of category M2, M3, N2, N3, T3 and T₄ except semi-trailers, 45 percent in the case of semi-trailers), whenever the measured roller braking force, for each axle irrespective of load, is not less than the reference braking force for a given brake actuator pressure within the declared operating pressure range (see Note).

NOTE – For the purpose of periodic technical inspection, the minimum limit braking rate values defined for the whole vehicle may need adjustment to reflect national or international in-service requirements-

- 5.1.4.7 It shall be possible to verify, in a simple way, the correct operational status of those complex electronic systems which have control over braking. If special information is needed, this shall be made freely available.

- 5.1.4.7.1 Where the operational status is indicated to the driver by warning signals, as specified in this Standard, it shall be possible at a periodic technical inspection to confirm the correct operational status by visual observation of the warning signals following a power-on.

- 5.1.4.7.2 At the time of type approval, the means implemented to protect against simple unauthorized modification of the operation to the verification means chosen by the manufacturer (for example, warning signal) shall be confidentially outlined.

Alternatively, this protection requirement is fulfilled when a secondary means of checking the correct operational status is available.

- 5.1.5 The requirements of Annex S shall be applied to the safety aspects of all complex electronic vehicle control systems which provide or form part of the control transmission of the braking function included those which utilize the braking system(s) for automatically commanded braking or selective braking.

However, systems or functions, which use the braking system as the means of achieving a higher level objective, are subject to Annex S only in so far as they have a direct effect on the braking system. If such systems are provided, they shall not be deactivated during type approval testing of the braking

system.

NOTE – These are general requirements and no verification is needed for this clause for compliance to this standard.

5.2 Characteristics of Braking Systems

5.2.1 Vehicles of Categories M2, M3 and N

5.2.1.1 The set of braking systems with which a vehicle is equipped shall satisfy the requirements laid down for service, secondary and parking braking systems.

5.2.1.2 The systems providing service, secondary and parking braking may have common components so long as they fulfil the following conditions.

5.2.1.2.1 There shall be at least two controls, independent of each other and readily accessible to the driver from his normal driving position.

For all categories of vehicles, except M2 and M3, every brake control (excluding an endurance braking system control) shall be designed such that it returns to the fully off position when released. This requirement shall not apply to a parking brake control (or that part of a combined control) when it is mechanically locked in an applied position.

5.2.1.2.2 The control of the service braking system shall be independent of the control of the parking braking system.

5.2.1.2.3 If the service braking system and the secondary braking system have the same control, the effectiveness of the linkage between that control and the different components of the transmission systems shall not be liable to diminish after a certain period of use.

5.2.1.2.4 If the service braking system and the secondary braking system have the same control, the parking braking system shall be so designed that it can be actuated when the vehicle is in motion. This requirement shall not apply if the vehicle's service braking system can be actuated, even partially, by means of an auxiliary control.

5.2.1.2.5 Without prejudice to the requirements of 5.1.2.3, the service braking system and the parking braking system may use common components in their transmission(s), provided that in the event of a failure in any part of the transmission(s) the requirements for secondary braking are still ensured.

5.2.1.2.6 In the event of breakage of any component other than the brakes (see 3.6) or the components referred to in 5.2.1.2.8, or of any other failure of the service braking system (malfunction, partial or total exhaustion of an energy reserve), the secondary braking system or that part of the service braking system which is not affected by the failure, shall be able to bring the vehicle to a halt in the conditions prescribed for secondary braking.

5.2.1.2.7 In particular, where the secondary braking system and the service braking system have a common control and a common transmission.

5.2.1.2.7 If service braking is ensured by the action of the driver's muscular energy

assisted by one or more energy reserves, secondary braking shall, in the event of failure of that assistance, be capable of being ensured by the driver's muscular energy assisted by the energy reserves, if any, which are unaffected by the failure, the force applied to the control not exceeding the prescribed maxima.

- 5.2.1.2.7 If the service braking force and transmission depend exclusively on the use, controlled by the driver, of an energy reserve, there shall be at least two completely independent energy reserves, each provided with its own transmission likewise independent; each of them may act on the brakes of only two or more wheels so selected as to be capable of ensuring by themselves the prescribed degree of secondary braking without endangering the stability of the vehicle during braking; in addition, each of the aforesaid energy reserves shall be equipped with a warning device as defined in 5.2.1.13. In each service braking circuit in at least one of the air reservoirs a device for draining and exhausting is required in an adequate and easily accessible position.
- 5.2.1.2.7 If the service braking force and transmission depend exclusively on the use of an energy reserve, one energy reserve for the transmission is deemed to be sufficient, provided that the prescribed secondary braking is ensured by the action of the driver's muscular energy acting on the service brake control and the requirements of 5.2.1.6 are met.
- 5.2.1.2.8 Certain parts, such as the pedal and its bearing, the master cylinder and its piston or pistons (hydraulic systems), the control valve (hydraulic and/or pneumatic systems), the linkage between the pedal and the master cylinder or the control valve, the brake cylinders and their pistons (hydraulic and/or pneumatic systems), and the lever-and-cam assemblies of brakes, shall not be regarded as liable to breakage if they are amply dimensioned, are readily accessible for maintenance, and exhibit safety features at least equal to those prescribed for other essential components (such as the steering linkage) of the vehicle. Any such part as aforesaid whose failure would make it impossible to brake the vehicle with a degree of effectiveness at least equal to that prescribed for secondary braking shall be made of metal or of a material with equivalent characteristics and shall not undergo notable distortion in normal operation of the braking systems.
- 5.2.1.3 Where there are separate controls for the service braking system and the secondary braking system, simultaneous actuation of the two controls shall not render both the service braking system and the secondary braking system inoperative, either when both braking systems are in good working order or when one of them is faulty.
- 5.2.1.4 The service braking system shall, whether or not it is combined with the secondary braking system, be such that in the event of failure in a part of its transmission a sufficient number of wheels are still braked by actuation of the service brake control; these wheels shall be so selected that the residual performance of the service braking system satisfies the requirements laid down in C-2.4.
- 5.2.1.4.1 However, the foregoing provisions shall not apply to tractor vehicles for semi-trailers when the transmission of the semi-trailer's service braking system is independent of that of the tractor vehicle's service braking system.

- 5.2.1.4.2 The failure of a part of a hydraulic transmission system shall be signalled to the driver by a device comprising a red warning signal, as specified in 5.2.1.29.1.1. Alternatively, the lighting up of this device when the fluid in the reservoir is below a certain level specified by the manufacturer shall be permitted.
- 5.2.1.5 Where use is made of energy other than the muscular energy of the driver, there need not be more than one source of such energy (hydraulic pump, air compressor, etc.), but the means by which the device constituting that source is driven shall be as safe as practicable.
 - 5.2.1.5.1 In the event of failure in any part of the transmission of a braking system, the feed to the part not affected by the failure shall continue to be ensured if required for the purpose of halting the vehicle with the degree of effectiveness prescribed for residual and/or secondary braking. This condition shall be met by means of devices which can be easily actuated when the vehicle is stationary, or by automatic means.
 - 5.2.1.5.2 Furthermore, storage devices located down-circuit of this device shall be such that in the case of a failure in the energy supply after four full-stroke actuations of the service brake control, under the conditions prescribed in F-1.2, it is still possible to halt the vehicle at the fifth application, with the degree of effectiveness prescribed for secondary braking.
 - 5.2.1.5.3 However, for hydraulic braking systems with stored energy, these provisions can be considered to be met provided that the requirements of F-3 and F-4 for vacuum assisted hydraulic brakes and F-5 and F-6 for Hydraulic brakes with stored energy devices to be satisfied.
- 5.2.1.6 The requirements of 5.2.1.2, 5.2.1.4 and 5.2.1.5 shall be met without the use of any automatic device of a kind such that its ineffectiveness might pass unnoticed through the fact that parts normally in a position of rest come into action only in the event of failure in the braking system.
- 5.2.1.7 The service braking system shall act on all wheels of the vehicle and shall distribute its action appropriately among the axles.
 - 5.2.1.7.1 In the case of vehicles with more than two axles, in order to avoid wheel-locking or glazing of the brake linings, the brake force on certain axles may be reduced to zero automatically when carrying a much reduced load, provided that the vehicle meets all the performance requirements prescribed in Annex C.
 - 5.2.1.7.2 In the case of N1 category vehicles with electric regenerative braking systems of category B, the braking input from other sources of braking, may be suitably phased to allow the electric regenerative braking system alone to be applied, provided that both the following conditions are met:
 - a) Intrinsic variations in the torque output of the electrical regenerative braking system (for example; as a result of changes in the electric state of charge in the traction batteries) are automatically compensated by appropriate variation in the phasing relationship as long as the requirements (See Note) of one of C-1.3.2, or M-5.3 (including the case

with the electric motor engaged) are satisfied.

- b) Wherever necessary, to ensure that braking rate (See Note) remains related to the driver's braking demand, having regard to the available tyre/road adhesion, braking shall automatically be caused to act on all wheels of the vehicle.

Note: The Testing Agency, which is to grant approval, shall have the right to check the service braking system by additional vehicle test procedures.

- 5.2.1.8 The action of the service braking system shall be distributed between the wheels of one and the same axle symmetrically in relation to the longitudinal median plane of the vehicle. Compensation and functions, such as anti-lock, which may cause deviations from this symmetrical distribution, shall be declared.
- 5.2.1.8.1 Compensation by the electric control transmission for deterioration or defect within the braking system shall be indicated to the driver by means of the yellow warning signal specified in 5.2.1.29.1.2. This requirement shall apply for all conditions of loading when compensation exceeds the following limits:
 - a) A difference in transverse braking pressures on any axle:
 - 1) Of 25 percent of the higher value for vehicle decelerations $\geq 2\text{ m/s}^2$; and
 - 2) A value corresponding to 25 percent at 2 m/s^2 for decelerations below this rate.
 - b) An individual compensating value on any axle:
 - 1) > 50 percent of the nominal value for vehicle decelerations $\geq 2\text{ m/s}^2$; and
 - 2) A value corresponding to 50 percent of the nominal value at $\geq 2\text{ m/s}^2$ for decelerations below this rate.
- 5.2.1.8.2 Compensation as defined above is permitted only when the initial brake application is made at vehicle speeds greater than 10 km/h.
- 5.2.1.9 Malfunctions of the electric control transmission shall not apply the brakes contrary to the driver's intentions.
- 5.2.1.10 The service, secondary and parking braking systems shall act on braking surfaces connected to the wheels through components of adequate strength.

Where braking torque for a particular axle or axles is provided by both a friction braking system and an electrical regenerative braking system of Category B, disconnection of the latter source is permitted, providing that the friction braking source remains permanently connected and able to provide the compensation referred to in 5.2.1.7.2 (a).

However, in the case of short disconnection transients, incomplete compensation is accepted, but within 1s, this compensation shall have attained at least 75 percent of its final value.

Nevertheless, in all cases the permanently connected friction braking source shall ensure that both the service and secondary braking systems

continue to operate with the prescribed degree of effectiveness.

Disconnection of the braking surfaces of the parking braking system shall be permitted only on condition that the disconnection is controlled exclusively by the driver from his driving seat, by a system incapable of being brought into action by a leak.

5.2.1.11 Wear of the brakes shall be capable of being easily taken up by means of a system of manual or automatic adjustment. In addition, the control and the components of the transmission and of the brakes shall possess a reserve of travel and, if necessary, suitable means of compensation such that, when the brakes become heated, or the brake linings have reached a certain degree of wear, effective braking is ensured without immediate adjustment being necessary.

5.2.1.11.1 Wear adjustment shall be automatic for the service brakes. However, the fitting of automatic brake adjustment devices is optional for off road vehicles of categories N2 and N3. Brakes equipped with automatic brake adjustment devices shall, after heating followed by cooling, be capable of free running as defined in C-1.5.4 following the Type I test also defined in Annex C.

5.2.1.11. Checking the Wear of the Service Brake Friction Components

5.2.1.11.2.1 It shall be possible to easily check this wear on service brake linings from the outside or underside of the vehicle utilizing only the tools or equipment normally supplied with the vehicle, for instance by the provision of appropriate inspection holes or by some other means. This may be achieved by utilizing simple standard workshop tools or common inspection equipment for vehicles. Alternatively, acoustic or optical devices warning the driver at his driving position when lining replacement is necessary are acceptable. The yellow warning signal specified in 5.2.1.29.1.2 may be used as the optical warning signal.

5.2.1.11.2.2 Assessment of the wear condition of the friction surfaces of brake discs or drums may only be performed by direct measurement of the actual component or examination of any brake disc or drum wear indicators, which may necessitate some level of disassembly. Therefore, at the time of type approval, the vehicle manufacturer shall define the following:

- a) Method by which wear of the friction surfaces of drums and discs may be assessed, including the level of disassembly required and tools and process required to achieve this; and
- b) Information defining the maximum acceptable wear limit at the point at which replacement becomes necessary.

This information shall be made freely available; for example vehicle handbook or electronic data record.

5.2.1.12 In hydraulic-transmission braking systems, the filling ports of the fluid reservoirs shall be readily accessible; in addition, the receptacles containing the reserve fluid shall be so designed and constructed that the level of the reserve fluid can be easily checked without the receptacles having to be opened. If this latter condition is not fulfilled, the red warning signal specified in 5.2.1.29.1.1 shall draw the driver's attention to any fall in the level of reserve fluid liable to cause a failure of the braking

system. The type of fluid to be used in the hydraulic transmission braking systems shall be identified by the symbol in accordance with Fig. 1 or 2 of ISO 9128:2006. The symbol shall be affixed in a visible position in indelible form within 100 mm of the filling ports of the fluid reservoirs; additional information may be provided by the manufacturer.

5.2.1.13 Warning Device

5.2.1.13.1 Any vehicle fitted with a service brake actuated from an energy reservoir shall, where the prescribed secondary braking performance cannot be obtained by means of this braking system without the use of the stored energy, be provided with a warning device, in addition to a pressure gauge, where fitted, giving an optical or acoustic signal when the stored energy, in any part of the system, falls to a value at which without re-charging of the reservoir and irrespective of the load conditions of the vehicle, it is possible to apply the service brake control a fifth time after four full-stroke actuations and obtain the prescribed secondary braking performance (without faults in the service brake transmission and with the brakes adjusted as closely as possible). This warning device shall be directly and permanently connected to the circuit. When the engine is running under normal operating conditions and there are no faults in the braking system, as is the case in approval tests for this type, the warning device shall give no signal except during the time required for charging the energy reservoir(s) after start-up of the engine. The red warning signal specified in 5.2.1.29.1.1 shall be used as the optical warning signal.

5.2.1.13.1.1 However, in the case of vehicles which are only considered to comply with the requirements of 5.2.1.5.1 by virtue of meeting the requirements of F-5.2.2, the warning device shall consist of an acoustic signal in addition to an optical signal. These devices need not operate simultaneously, provided that each of them meets the above requirements and the acoustic signal is not actuated before the optical signal. The red warning signal specified in 5.2.1.29.1.1 shall be used as the optical warning signal.

5.2.1.13.1.2 This acoustic device may be rendered inoperative while the handbrake is applied and/or, at the choice of the manufacturer, in the case of automatic transmission the selector is in the 'Park' position.

5.2.1.14 Without prejudice to the requirements of 5.1.2.3, where an auxiliary source of energy is essential to the functioning of a braking system, the reserve of energy shall be such as to ensure that, if the engine stops or in the event of a failure of the means by which the energy source is driven, the braking performance remains adequate to bring the vehicle to a halt in the prescribed conditions. In addition, if the muscular effort applied by the driver to the parking braking system is reinforced by a servo device, the actuation of the parking braking system shall be ensured in the event of a failure of the servo device, if necessary by using a reserve of energy independent of that normally supplying the servo device. This reserve of energy may be that intended for the service braking system.

5.2.1.15 In the case of a power-driven vehicle to which the coupling of a trailer equipped with a brake controlled by the driver of the towing vehicle is

authorized, the service braking system of the towing vehicle shall be equipped with a device so designed that in the event of failure of the trailer's braking system, or in the event of an interruption in the air supply pipe (or of such other type of connection as may be adopted) between the towing vehicle and its trailer, it shall still be possible to brake the towing vehicle with the effectiveness prescribed for secondary braking; it is accordingly prescribed, in particular, that this device shall be situated on the towing vehicle.

- 5.2.1.16 The pneumatic/hydraulic auxiliary equipment shall be supplied with energy in such a way that during its operation the prescribed deceleration values can be reached and that even in the event of damage to the source of energy the operation of the auxiliary equipment cannot cause the reserves of energy feeding the braking systems to fall below the level indicated in 5.2.1.13.
- 5.2.1.17 If the trailer is of category T3 or T4, the service braking system shall be of the continuous or semi-continuous type.
- 5.2.1.18 In the case of a vehicle authorized to tow a trailer of category T3 or T4, its braking systems shall satisfy the following conditions.
 - 5.2.1.18.1 When the towing vehicle's secondary braking system comes into action, there shall also be a graduated braking action in the trailer.
 - 5.2.1.18.2 In the event of failure of the towing vehicles service braking system, where that system consists of at least two independent parts, the part or parts not affected by the failure shall be capable of partially or fully actuating the brakes of the trailer. It shall be possible to graduate this braking action. If this operation is achieved by a valve which is normally at rest, then such a valve may only be incorporated if its correct functioning can easily be checked by the driver, either from within the cab or from outside the vehicle, without the use of tools.
 - 5.2.1.18.3 In the event of a failure (for example breakage or leak) in one of the pneumatic connecting lines, interruption or defect in the electric control line, it shall nevertheless be possible for the driver, fully or partially, to actuate the brakes of the trailer by means either of the service braking control or of the secondary braking control or of the parking braking control, unless the failure automatically causes the trailer to be braked with the performance prescribed in C-3.3.
 - 5.2.1.18.4 The automatic braking in 5.2.1.18.3 shall be considered to be met when the following conditions are fulfilled.
 - 5.2.1.18.4.1 When the designated brake control of those controls mentioned in 5.2.1.18.3 is fully actuated, the pressure in the supply line shall fall to 150 kPa (1.5 bar) within the following 2 s; in addition, when the brake control is released, the supply line shall be re-pressurized.
 - 5.2.1.18.4.2 When the supply line is evacuated at the rate of at least 100 kPa (1 bar) per second the automatic braking of the trailer shall start to operate before the pressure in the supply line falls to 200 kPa (2 bar).

- 5.2.1.18.5 In the event of a failure in one of the control lines connecting two vehicles equipped according to 5.1.3.1.2, the control line not affected by the failure shall automatically ensure the braking performance prescribed for the trailer in C-3.1.
- 5.2.1.19 In the case of a power-driven vehicle equipped to draw a trailer with an electrical braking system, according to N-1.1, the following requirements shall be met:
- a) The power supply (generator and battery) of the power-driven vehicle shall have a sufficient capacity to provide the current for an electrical braking system. With the engine running at the idling speed recommended by the manufacturer and all electrical devices supplied by the manufacturer as standard equipment of the vehicle switched on, the voltage in the electrical lines shall at maximum current consumption of the electrical braking system (15 A) not fall below the value of 9.6 V measured at the connection. The electrical lines shall not be capable of short circuiting even when overloaded;
 - b) In the event of a failure in the towing vehicle's service braking system, where that system consists of at least two independent parts, the part or parts not affected by the failure should be capable of partially or fully actuating the brakes of the trailer;
 - c) The use of the stop-lamp switch and circuit for actuating the electrical braking system is permissible only if the actuating line is connected in parallel with the stop-lamp and the existing stop-lamp switch and circuit are capable of taking the extra load.
- 5.2.1.20 In the case of a pneumatic service braking system comprising two or more independent sections, any leakage between those sections at or downstream of the control shall be continuously vented to atmosphere.
- 5.2.1.21 In the case of a power-driven vehicle authorized to tow a trailer of categories T3 or T4, the service braking system of the trailer may only be operated in conjunction with the service, secondary or parking braking system of the towing vehicle. However, automatic application of the trailer brakes alone is permitted where the operation of the trailer brakes is initiated automatically by the towing vehicle for the sole purpose of vehicle stabilization.
- 5.2.1.22 Power-driven vehicles of categories N1, M2, M3, N2 and N3 shall be in accordance to Annex J, if equipped with anti-lock systems shall conform to the ~~of~~ category 1 in accordance with Annex M.
- 5.2.1.23 Power-driven vehicles of category N1 equipped with temporary-use spare wheels/tyres shall satisfy the technical requirements of AIS-110, as amended from time to time.
- 5.2.1.24. Power driven vehicles authorized to tow a trailer equipped with an anti-lock system shall also be equipped with either one or both of the following, for the electric control transmission:
- (a) A special electrical connector conforming to ISO 7638:2003;

(See Note)

- (b) An automated connector meeting the requirements specified in Annex W."

NOTE – ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.

5.2.1.25 Additional requirements for vehicles of categories M2, N1 and category N2 < 5 t equipped with an electric regenerative braking system.

5.2.1.25.1 Vehicles fitted with an electric regenerative braking system of category A.

5.2.1.25.1.1 The electric regenerative braking shall only be actuated by the accelerator control and/or the gear selector neutral position for vehicles of category N1.

5.2.1.25.1.2 In addition, for vehicles of categories M2 and N2 (< 5 t), the electric regenerative braking control can be a separate switch or lever.

5.2.1.25.1.3 The requirements of 5.2.1.25.3 and 5.2.1.25.4 also apply to Category A regenerative braking systems.

5.2.1.25.2 Vehicles fitted with an electric regenerative braking system of category B.

5.2.1.25.2.1 It shall not be possible to disconnect, partially or totally, one part of the service braking system other than by automatic means. This should not be construed as a departure from the requirements of 5.2.1.10.

5.2.1.25.2.2 The service braking system shall have only one control device.

5.2.1.25.2.3 For vehicles fitted with an electric regenerative braking system of both categories, all the relevant prescriptions shall apply except 5.2.1.25.1.1.

In this case, the electric regenerative braking may be actuated by the accelerator control and/or the gear selector neutral position for vehicles of category N1.

Additionally, the action on the service braking control shall not reduce the above braking effect generated by the release of accelerator control.

5.2.1.25.2.4 The service braking system shall not be adversely affected by the disengagement of the motor(s) or by the gear ratio used.

5.2.1.25.2.5 If the operation of the electric component of braking is ensured by a relation established between the information coming from the control of the service brake and the braking force at the respective wheels, a failure of this relation leading to the modification of the braking distribution among the axles (Annex J or Annex M, whichever is applicable) shall be signalled to the driver by an optical warning signal at the latest at the moment when the control is actuated and this signal shall remain lit as long as this defect exists and that the vehicle control switch (key) is in the 'ON' position.

- 5.2.1.25.3 The operation of the electric regenerative braking shall not be adversely affected by magnetic or electric fields.
- 5.2.1.25.4 For vehicles equipped with an anti-lock device, the anti-lock device shall control the electric regenerative braking system.
- 5.2.1.26 **Special additional requirements for the electric transmission of the parking braking system.**
 - 5.2.1.26.1 In the case of a failure within the electric transmission, any unintended actuation of the parking braking system shall be prevented.
 - 5.2.1.26.2. The following requirements shall be fulfilled in the event of an electrical failure as specified:
 - 5.2.1.26.2.1 Vehicles of Categories M2, M3, N2 and N3

In the case of a break in the wiring within the electric control transmission external to the electronic control unit(s) and excluding the energy supply, or a failure in the control, it shall remain possible to apply the parking braking system from the driver's seat and thereby be capable of holding the laden vehicle stationary on an 8 percent up or down gradient. Alternatively, in this case, an automatic actuation of the parking brake is allowed when the vehicle is stationary, provided that the above performance is achieved and, once applied, the parking brake remains engaged independently of the status of the ignition (start) switch. In this alternative, the parking brake shall be automatically released as soon as the driver starts to set the vehicle in motion again. It shall also be possible to release the parking braking system, if necessary by the use of tools and/or an auxiliary device carried/fitted on the vehicle.
 - 5.2.1.26.2.2. Vehicles of category N1:

In the case of an electrical failure in the control or a break in the wiring within the electric control transmission between the control and the ECU directly connected with it, excluding the energy supply, it shall remain possible to apply the parking braking system from the driver's seat and thereby be capable of holding the laden vehicle stationary on an 8 per cent up or down gradient. Alternatively, in this case, an automatic actuation of the parking brake is allowed when the vehicle is stationary, provided that the above performance is achieved and, once applied, the parking brake remains engaged independently of the status of the ignition (start) switch. In this alternative, the parking brake shall be automatically released as soon as the driver starts to set the vehicle in motion again. The engine/manual transmission or the automatic transmission (park position) may be used to achieve or assist in achieving the above performance.
 - 5.2.1.26.2.3 A break in the wiring within the electric transmission, or an electric failure in the control of the parking braking system shall be signalled to the driver by the yellow warning signal specified in 5.2.1.29.1.2. When caused by a break in the wiring within the electric control transmission of the parking braking system, this yellow warning signal shall be signalled as soon as the break occurs. In addition, such an electric failure in the control or break in the wiring external to the electronic control unit(s) and

excluding the energy supply shall be signalled to the driver by flashing the red warning signal specified in 5.2.1.29.1.1 as long as the ignition (start) switch is in the 'ON' (run) position including a period of not less than 10 s thereafter and the control is in the 'ON' (activated) position.

However, if the parking braking system detects correct clamping of the parking brake, the flashing of the red warning signal may be suppressed and the non-flashing red signal shall be used to indicate parking brake applied.

Where actuation of the parking brake is normally indicated by a separate red warning signal, satisfying all the requirements of 5.2.1.29.3, this signal shall be used to satisfy the above requirement for a red signal.

- 5.2.1.26.3 Auxiliary equipment may be supplied with energy from the electric transmission of the parking braking system provided that the supply of energy is sufficient to allow the actuation of the parking braking system in addition to the vehicle electrical load under non-fault conditions. In addition, where the energy reserve is also used by the service braking system, the requirements of 5.2.1.27.7 shall apply.
- 5.2.1.26.4 After the ignition/start switch which controls the electrical energy for the braking equipment has been switched off and/or the key removed, it shall remain possible to apply the parking braking system, whereas releasing shall be prevented.
- 5.2.1.27 **Special additional requirements for service braking systems with electric control transmission.**
- 5.2.1.27.1 With the parking brake released, the service braking system shall be able to generate a static total braking force at least equivalent to that required by the prescribed Type 0 test, even when the ignition/start switch has been switched off and/or the key has been removed. In the case of power-driven vehicles authorized to tow trailers of category T3 or T4, such vehicles shall provide a full control signal for the service braking system of the trailer. It should be understood that sufficient energy is available in the energy transmission of the service braking system.
- 5.2.1.27.2 In the case of a single temporary failure (< 40 ms) within the electric control transmission, excluding its energy supply (for example non-transmitted signal or data error) there shall be no distinguishable effect on the service braking performance.
- 5.2.1.27.3 A failure within the electric control transmission (See Note) not including its energy reserve that affects the function and performance of systems addressed in this standard shall be indicated to the driver by the red or yellow warning signal specified in 5.2.1.29.1.1 and 5.2.1.29.1.2, respectively, as appropriate. When the prescribed service braking performance can no longer be achieved (red warning signal), failures resulting from a loss of electrical continuity (for example breakage, disconnection) shall be signalled to the driver as soon as they occur, and the prescribed residual braking performance shall be fulfilled by operating the service braking control in accordance with C-2.4 of Annex C. These requirements shall not be construed as a departure from the requirements concerning secondary braking.

Note: Until uniform test procedures have been agreed, the manufacturer shall provide the Testing Agency with an analysis of potential failures within the control transmission and their effects. This information shall be subject to discussion and agreement between the Testing Agency and the vehicle manufacturer.

- 5.2.1.27.4 A power-driven vehicle, electrically connected to a trailer via an electric control line, shall provide a clear warning to the driver whenever the trailer provides the failure information that the stored energy in any part of the service braking system on the trailer falls below the warning level, as specified in 5.2.2.16. A similar warning shall also be provided when a continuous failure (> 40 ms) within the electric control transmission of the trailer, excluding its energy reserve, precludes achievement of the prescribed service braking performance of the trailer, as specified in 5.2.2.15.2.1. The red warning signal specified in 5.2.1.29.2.1 shall be used for this purpose.
- 5.2.1.27.5 In the event of a failure of the energy source of the electric control transmission, starting from the nominal value of the energy level, the full control range of the service braking system shall be guaranteed after twenty consecutive full stroke actuations of the service braking control. During the test, the braking control shall be fully applied for 20 s and released for 5 s on each actuation. It should be understood that during the above test sufficient energy is available in the energy transmission to ensure full actuation of the service braking system. This requirement shall not be construed as a departure from the requirements of Annex F.
- 5.2.1.27.6 When the battery voltage falls below a value nominated by the manufacturer at which the prescribed service braking performance can no longer be guaranteed and / or which precludes at least two independent service braking circuits from each achieving the prescribed secondary or residual braking performance, the red warning signal specified in 5.2.1.29.1.1 shall be activated. After the warning signal has been activated, it shall be possible to apply the service braking control and obtain at least the residual performance prescribed in C-2.4. It should be understood that sufficient energy is available in the energy transmission of the service braking system. This requirement shall not be construed as a departure from the requirement concerning secondary braking.
- 5.2.1.27.7 If auxiliary equipment is supplied with energy from the same reserve as the electric control transmission, it shall be ensured that, with the engine running at a speed not greater than 80 percent of the maximum power speed, the supply of energy is sufficient to fulfil the prescribed deceleration values by either provision of an energy supply which is able to prevent discharge of this reserve when all auxiliary equipment is functioning or by automatically switching off pre-selected parts of the auxiliary equipment at a voltage above the critical level referred to in 5.2.1.27.6 such that further discharge of this reserve is prevented. Compliance may be demonstrated by calculation or by a practical test. For vehicles authorized to tow a trailer of category T3 or T4 the energy consumption of the trailer shall be taken into account by a load of 400 W. This clause does not apply to vehicles where the prescribed deceleration

values can be reached without the use of electrical energy.

- 5.2.1.27.8 If the auxiliary equipment is supplied with energy from the electric control transmission, the following requirements shall be fulfilled:
- a) In the event of a failure in the energy source, whilst the vehicle is in motion, the energy in the reservoir shall be sufficient to actuate the brakes when the control is applied.
 - b) In the event of a failure in the energy source, whilst the vehicle is stationary and the parking braking system applied, the energy in the reservoir shall be sufficient to actuate the lights even when the brakes are applied.
- 5.2.1.27.9 In the case of a failure in the electric control transmission of the service braking system of a towing vehicle equipped with an electric control line according to 5.1.3.1.2 or 5.1.3.1.3, the full actuation of the brakes of the trailer shall remain ensured.
- 5.2.1.27.10 In the case of a failure in the electric control transmission of a trailer, electrically connected via an electric control line only, according to 5.1.3.1.3, braking of the trailer shall be ensured according to 5.2.1.18.4.1. This shall be the case whenever the trailer provides the ‘supply line braking request’ signal via the data communication part of the electric control line or in the event of the continuous absence of this data communication. This paragraph shall not apply to power-driven vehicles which cannot be operated with trailers connected via an electric control line only, as described in 5.1.3.5.
- 5.2.1.28 **Special requirements for coupling force control**
- 5.2.1.28.1 Coupling force control shall only be permitted in the towing vehicle.
- 5.2.1.28.2 The action of the coupling force control shall be to reduce the difference between the dynamic braking rates of towing and towed vehicles. The operation of the coupling force control shall be checked at the time of type approval. The method by which this check is carried out shall be agreed between the vehicle manufacturer and the Test Agency with the method of assessment and results being appended to the type approval report.
- 5.2.1.28.2.1 The coupling force control may control the braking rate T_M/P_M and/or the brake demand value(s) for the trailer. In the case of a towing vehicle equipped with two control lines according to 5.1.3.1.2, both signals shall be subject to similar control adjustments.
- 5.2.1.28.2.2 The coupling force control shall not prevent the maximum possible braking pressure(s) from being applied.
- 5.2.1.28.3 The vehicle shall fulfil the laden compatibility requirements of Annex J, but to achieve the objectives of 5.2.1.28.2 the vehicle may deviate from these requirements when the coupling force control is in operation.
- 5.2.1.28.4 A coupling force control failure shall be detected and indicated to the driver by a yellow warning signal such as that specified in 5.2.1.29.1.2. In the event of a failure the relevant requirements of Annex J shall be

fulfilled.

- 5.2.1.28.5 Compensation by the coupling force control system shall be indicated by means of the yellow warning signal (see Fig. 1 and Fig. 2) specified in 5.2.1.29.1.2 if this compensation exceeds 150 kPa (1.5 bar) away from the nominal demand value defined in 3.27.3 up to a limit, in p_m , of 650 kPa (6.5 bar) (or the equivalent digital demand). Above the level of 650 kPa (6.5 bar) the warning shall be given, if the compensation causes the operating point to lie outside the laden compatibility band as specified in Annex J for the motor vehicle.
- 5.2.1.28.6 A coupling force control system shall control only the coupling forces generated by the service braking system of the motor vehicle and the trailer. Coupling forces resulting from the performance of endurance braking systems shall not be compensated by the service braking system of either the motor vehicle or trailer. It is considered that endurance braking systems are not part of the service braking systems.
- 5.2.1.29 **Brake Failure and Defect Warning Signal**
 The general requirements for optical warning signals whose function is to indicate to the driver certain specified failures (or defects) within the braking equipment of the power-driven vehicle or, where appropriate, its trailer, are set out in the following sub-paragraphs. Other than as described in 5.2.1.29.6, these signals shall be used exclusively for the purposes prescribed by this standard.
 - 5.2.1.29.1 Power-driven vehicles shall be capable of providing optical brake failure and defect warning signals, as follows:
 - 5.2.1.29.1.1 A red warning signal, indicating failures defined elsewhere in this standard within the vehicle braking equipment which preclude achievement of the prescribed service braking performance and / or which preclude the functioning of at least one of two independent service braking circuits.
 - 5.2.1.29.1.2 Where applicable, a yellow warning signal indicating an electrically detected defect within the vehicle braking equipment, which is not indicated by the red warning signal described in 5.2.1.29.1.1.

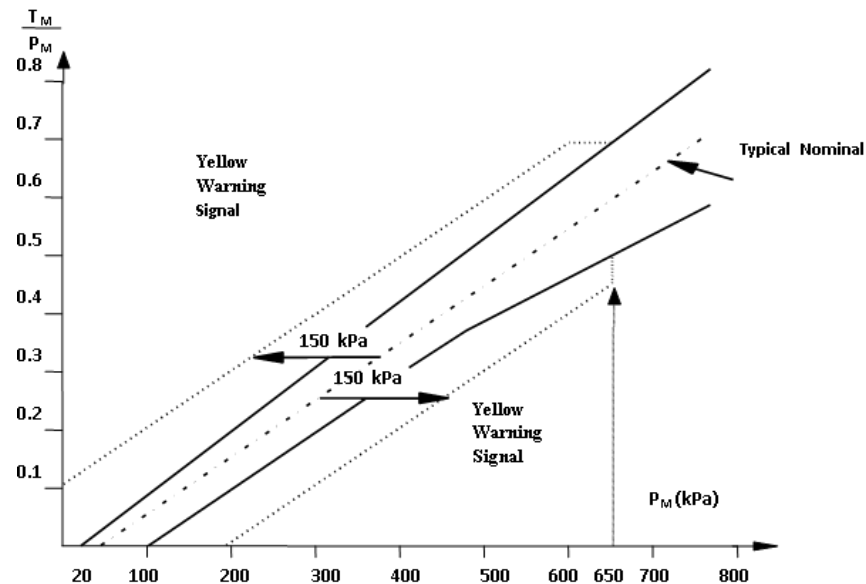


Fig. 1 Towing Vehicles for Trailers (Except Semi-Trailers)

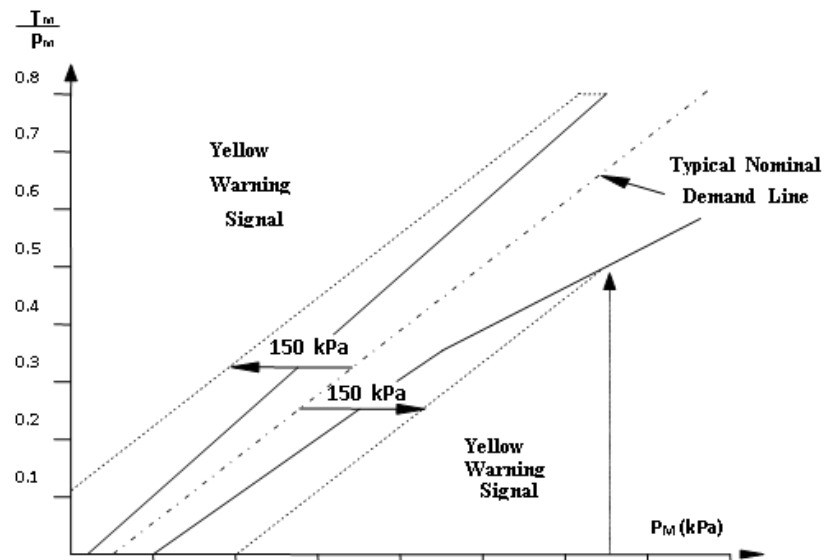


Fig. 2 Tractive Units for Semi-Trailers

- 5.2.1.29.2 Power-driven vehicles equipped with an electric control line and/or authorized to tow a trailer equipped with an electric control transmission shall be capable of providing a separate yellow warning signal to indicate a defect within the electric control transmission of the braking equipment of the trailer. The signal shall be activated from the trailer via pin 5 of the electric connector conforming to ISO 7638:2003 and in all cases the signal transmitted by the trailer shall be displayed without significant delay or modification by the towing vehicle. This warning signal shall not light up when coupled to a trailer without an electric control line and/or electric control transmission and / or anti-lock braking system or when no trailer is coupled. This function shall be automatic.

NOTE – ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.

- 5.2.1.29.2.1 In the case of a power-driven vehicle equipped with an electric control line, when electrically connected to a trailer with an electric control line, the red warning signal specified in 5.2.1.29.1.1 shall also be used to indicate certain specified failures within the braking equipment of the trailer, whenever the trailer provides corresponding failure information via the data communication part of the electric control line. This indication shall be in addition to the yellow warning signal specified in 5.2.1.29.2. Alternatively, instead of utilizing the red warning signal specified in 5.2.1.29.1.1 and the accompanying yellow warning signal above, a separate red warning signal may be provided in the towing vehicle to indicate such a failure within the braking equipment of the trailer.
- 5.2.1.29.3 The warning signals shall be visible, even by daylight; the satisfactory condition of the signals shall be easily verifiable by the driver from the driver's seat; the failure of a component of the warning devices shall not entail any loss of the braking system's performance.
- 5.2.1.29.4 Except where stated otherwise,
- a) A specified failure or defect shall be signalled to the driver by the above-mentioned warning signal(s) not later than on actuation of the relevant braking control;
 - b) The warning signal(s) shall remain displayed as long as the failure/defect persists and the ignition (start) switch is in the 'ON' (run) position; and
 - c) The warning signal shall be constant (not flashing).
- 5.2.1.29.5 The warning signal(s) mentioned above shall light up when the electrical equipment of the vehicle (and the braking system) is energized. With the vehicle stationary, the braking system shall verify that none of the specified failures or defects are present before extinguishing the signals. Specified failures or defects which should activate the warning signals mentioned above, but which are not detected under static conditions, shall be stored upon detection and be displayed at start-up and at all times when the ignition (start) switch is in the 'ON' (run) position, as long as the failure or defect persists.
- 5.2.1.29.6 Non-specified failures (or defects), or other information concerning the brakes and/or running gear of the power-driven vehicle, may be indicated by the yellow signal specified in 5.2.1.29.1.2, provided that all the following conditions are fulfilled:
- a) The vehicle is stationary;
 - b) After the braking equipment is first energized and the signal has indicated that, following the procedures detailed in 5.2.1.29.5, no specified failures (or defects) have been identified; and
 - c) Non-specified faults or other information shall be indicated only by

the flashing of the warning signal. However, the warning signal shall be extinguished by the time when the vehicle first exceeds 10 km/h.

5.2.1.30. Generation of a braking signal to illuminate stop lamps

5.2.1.30.1. Activation of the service braking system by the driver shall generate a signal that will be used to illuminate the stop lamps.

5.2.1.30.2. Requirements for vehicles that utilize electronic signalling to control initial application of the service braking system, and equipped with endurance braking and/or regenerative braking system of category A:

Deceleration by the endurance braking and/or regenerative braking system

$\leq 1.3 \text{ m/s}^2$	$> 1.3 \text{ m/s}^2$
May generate the signal	Shall generate the signal

5.2.1.30.3. In the case of vehicles equipped with a braking system of a specification different to that defined in 5.2.1.30.2., the operation of the endurance braking system and/or regenerative braking system of category A may generate the signal irrespective of the deceleration produced.

5.2.1.30.4. The signal shall not be generated when retardation is produced by the natural braking effect of the engine alone.

5.2.1.30.5. Activation of the service braking system by "automatically commanded braking" shall generate the signal mentioned above. However, when the retardation generated is less than 0.7 m/s^2 , the signal may be suppressed (See Note).

Note: At the time of type approval, compliance with this requirement shall be confirmed by the vehicle manufacturer.

5.2.1.30.6. Activation of part of the service braking system by "selective braking" shall not generate the signal mentioned above (See Note).

Note: During a "selective braking" event, the function may change to "automatically commanded braking".

5.2.1.30.7. In the case of vehicles equipped with an electric control line the signal shall be generated by the motor vehicle when a message "illuminate stop lamps" is received via the electric control line from the trailer.

5.2.1.31. When a vehicle is equipped with the means to indicate emergency braking, activation and de-activation of the emergency braking signal shall only be generated by the application of the service braking system when the following conditions are fulfilled (See Note below 5.2.1.30.5).

- 5.2.1.31.1. The signal shall not be activated when the vehicle deceleration is below the values defined in the following table but it may be generated at any deceleration at or above those values, the actual value being defined by the vehicle manufacturer:

	The signal shall not be activated below
N1	6 m/s ²
M2, M3, N2 and N3	4 m/s ²

The signal shall be de-activated for all vehicles at the latest when the deceleration has fallen below 2.5 m/s².

- 5.2.1.31.2. The following conditions may also be used:

- (a) The signal may be generated from a prediction of the vehicle deceleration resulting from the braking demand respecting the activation and de-activation thresholds defined in 5.2.1.31.1;

Or

- (b) The signal may be activated when the service braking system is applied at a speed above 50 km/h and when the antilock system is fully cycling (as defined in M-2). The signal shall be de-activated when the antilock system is no longer fully cycling.

- 5.2.1.32. All vehicles of the following categories may be equipped with a vehicle stability function (See Note):

- (a) M₂, M₃, N₂;
- (b) N₃¹² having no more than 3 axles;
- (c) N₃¹² with 4 axles, with a maximum mass not exceeding 25 t and a maximum wheel diameter code not exceeding 19.5.

The vehicle stability function shall include roll-over control and directional control and meet the technical requirements of Annex V to this Standard.

NOTE – Mandatory requirements for fitting vehicle stability function will be in accordance with decision of statutory authority.

Off-road vehicles, special purpose vehicles (e.g. mobile plant using non-standard vehicle chassis - mobile cranes, hydro-static driven vehicles in which the hydraulic drive system is also used for braking and auxiliary functions, N2 vehicles which have all of the following features: a gross vehicle mass between 3.5 and 7.5 tonnes, a non-standard low-frame chassis, more than 2 axles and hydraulic transmission), Class I, Class A and articulated vehicles of categories M2 and M3, N2 tractors for semi-trailer with a Gross Vehicle Mass (GVM) between 3.5 and 7.5 tonnes, shall be excluded from this requirement.

- 5.2.1.33. Category N1 vehicles having no more than 3 axles may be equipped with a vehicle stability function. If fitted, it shall include roll-over control and

directional control and meet the technical requirements of Annex V to this Standard.

5.2.2 Vehicles of Category T

- 5.2.2.1 Trailers of category T1 need not be equipped with a service braking system; however, if a trailer of this category is equipped with a service braking system, it shall satisfy the same requirements as a trailer of category T2.
- 5.2.2.2 Trailers of category T2 shall be equipped with a service braking system either of the continuous or semi-continuous or of the inertia (overrun) type. The latter type shall be authorized only for centre axle trailers. However, electrical braking systems conforming to the requirements of Annex N shall be permitted.
- 5.2.2.3 Trailers of categories T3 and T4 shall be equipped with a service braking system of the continuous or semi-continuous type.
- 5.2.2.4 The service braking system,
- a) Shall act on all the wheels of the vehicle;
 - b) Shall distribute its action appropriately among the axles; and
 - c) Shall contain in at least one of the air reservoirs a device for draining and exhausting in an adequate and easily accessible position.
- 5.2.2.5 The action of the service braking system shall be distributed between the wheels of one and the same axle symmetrically in relation to the longitudinal median plane of the vehicle. Compensation and functions, such as anti-lock, which may cause deviations from this symmetrical distribution, shall be declared.
- 5.2.2.5.1 Compensation by the electric control transmission for deterioration or defect within the braking system shall be indicated to the driver by means of the separate yellow optical warning signal specified in 5.2.1.29.2. This requirement shall apply for all conditions of loading when compensation exceeds the following limits:
- a) A difference in transverse braking pressures on any axle:
 - 1) Of 25 percent of the higher value for vehicle decelerations $\geq 2\text{m/s}^2$; and
 - 2) A value corresponding to 25 percent at 2m/s^2 for decelerations below this rate.
 - b) An individual compensating value on any axle:
 - 1) >50 percent of the nominal value for vehicle decelerations $\geq 2\text{m/s}^2$; and
 - 2) A value corresponding to 50 percent of the nominal value at 2m/s^2 for decelerations below this rate.
- 5.2.2.5.2 Compensation as defined above is permitted only when the initial brake application is made at vehicle speeds greater than 10 km/h.

- 5.2.2.6 Malfunctions of the electric control transmission shall not apply the brakes contrary to the driver's intentions.
- 5.2.2.7 The braking surfaces required to attain the prescribed degree of effectiveness shall be in constant connection with the wheels, either rigidly or through components not liable to failure.
- 5.2.2.8 Wear of the brakes shall be capable of being easily taken up by means of a system of manual or automatic adjustment. In addition, the control and the components of the transmission and of the brakes shall possess a reserve of travel and, if necessary, suitable means of compensation such that, when the brakes become heated, or the brake linings have reached a certain degree of wear, effective braking is ensured without immediate adjustment being necessary.
- 5.2.2.8.1 Wear adjustment shall be automatic for the service brakes. However, the fitting of automatic adjustment devices is optional for vehicles of categories T1 and T2. Brakes equipped with automatic brake adjustment devices shall, after heating followed by cooling, be capable of free running as defined in C-1.7.3 following the Type I or Type III test also defined in Annex C as appropriate.
- 5.2.2.8.1.1 In the case of trailers of category T4 the performance requirements of 5.2.2.8.1 shall be deemed to be satisfied by fulfilling the requirements of C-1.7.3.
- 5.2.2.8.1.2 In the case of trailers of categories T2 and T3 the performance requirements of 5.2.2.8.1 shall be deemed to be satisfied by fulfilling the requirements of C-1.7.3 (See Note).
- Note: Until uniform technical provisions have been agreed that correctly assess the function of the automatic brake adjustment device, the free running requirement shall be deemed to be fulfilled when free running is observed during all brake tests prescribed for the relevant trailer.
- 5.2.2.8.2 Checking the wear of the service brake friction components.
- 5.2.2.8.2.1 It shall be possible to easily check this wear on service brake linings from the outside or underside of the vehicle, without the removal of the wheels, utilizing only the tools or equipment normally supplied with the vehicle, for instance, by the provision of appropriate inspection holes or by some other means.
- Alternatively, a trailer mounted display providing information when lining replacement is necessary or a sensing device per wheel (twin wheels are considered as a single wheel), which will warn the driver at his driving position when lining replacement is necessary, is acceptable. In the case of an optical warning, the yellow warning signal specified in 5.2.1.29.2. may be used provided that the signal complies with the requirements of 5.2.1.29.6.
- 5.2.2.8.2.2 Assessment of the wear condition of the friction surfaces of brake discs or drums may only be performed by direct measurement of the actual components or examination of any brake disc or drum wear indicators, which may necessitate some level of disassembly. Therefore, at the time

of type approval, the vehicle manufacturer shall define the following:

- a) Method by which wear of the friction surfaces of drums and discs may be assessed, including the level of disassembly required and tools and process required to achieve this; and
- b) Information defining the maximum acceptable wear limit at the point at which replacement becomes necessary.

This information shall be made freely available, for example vehicle handbook or electronic data record.

5.2.2.9 The braking systems shall be such that the trailer is stopped automatically if the coupling separates while the trailer is in motion.

5.2.2.10 On every trailer which is required to be equipped with a service braking system, parking braking shall be assured even when the trailer is separated from the towing vehicle. The parking braking device shall be capable of being actuated by a person standing on the ground; however, in the case of a trailer used for the carriage of passengers, this brake shall be capable of being actuated from inside the trailer.

5.2.2.11 If the trailer is fitted with a device enabling compressed-air actuation of the braking system other than the parking braking system to be cut out, the first-mentioned system shall be so designed and constructed that it is positively restored to the position of rest not later than on resumption of the supply of compressed air to the trailer.

5.2.2.12 Trailers of categories T3 and T4 shall satisfy the conditions specified in 5.2.1.18.4.2. An easily accessible pressure test connection is required downstream of the coupling head of the control line.

In the case of trailers equipped with an electric control line and electrically connected to a towing vehicle with an electric control line the automatic braking action specified in 5.2.1.18.4.2 may be suppressed as long as the pressure in the compressed air reservoirs of the trailer is sufficient to ensure the braking performance specified in C-3.3.

5.2.2.13 Trailers of categories T3 shall be equipped with anti-lock systems in accordance with the requirements of Annex M. Trailers of category T4 shall be equipped with an anti-lock system in accordance with the category A requirements of Annex M.

5.2.2.14 Where the auxiliary equipment is supplied with energy from the service braking system, the service braking system shall be protected to ensure that the sum of the braking forces exerted at the periphery of the wheels shall be at least 80 percent of the value prescribed for the relevant trailer as defined in C-3.1.2.1. This requirement shall be fulfilled under both of the following operating conditions:

- a) During operation of the auxiliary equipment; and
- b) In the event of breakage or leakage from the auxiliary equipment, unless such breakage or leakage affects the control signal referred to in J-6, in which case the performance requirements of that clause shall

apply.

- 5.2.2.14.1 The above provisions are deemed to be fulfilled when the pressure in the service brake storage device(s) is maintained at a pressure of at least 80 percent of the control line demand pressure or equivalent digital demand as defined in C-3.1.2.2.
- 5.2.2.15 Special additional requirements for service braking systems with electric control transmission.
 - 5.2.2.15.1 In the case of a single temporary failure (< 40 ms) within the electric control transmission, excluding its energy supply (for example, non-transmitted signal or data error) there shall be no distinguishable effect on the service braking performance.
 - 5.2.2.15.2 In the case of a failure within the electric control transmission (See Note) (for example, breakage, disconnection), a braking performance of at least 30 percent of the prescribed performance for the service braking system of the relevant trailer shall be maintained. For trailers, electrically connected via an electric control line only, according to 5.1.3.1.3, and fulfilling 5.2.1.18.4.2 with the performance prescribed in C-3.3, it is sufficient that the provisions of 5.2.1.27.10 are invoked, when a braking performance of at least 30 percent of the prescribed performance for the service braking system of the trailer can no longer be ensured, by either providing the supply line braking request signal via the data communication part of the electric control line or by the continuous absence of this data communication.

Note: Until uniform test procedures have been agreed, the manufacturer shall provide the Testing Agency with an analysis of potential failures within the control transmission, and their effects. This information shall be subject to discussion and agreement between the Testing Agency and the vehicle manufacturer.
 - 5.2.2.15.2.1 A failure within the electric control transmission of the trailer, that affects the function and performance of systems addressed in this standard, and failures of energy supply available from the ISO 7638:2003 connector shall be indicated to the driver by the separate warning signal specified in 5.2.1.29.2 via pin 5 of the electrical connector conforming to ISO 7638:2003 (See Note). In addition, trailers equipped with an electric control line, when electrically connected to a towing vehicle with an electric control line, shall provide the failure information for activation of the red warning signal specified in 5.2.1.29.2.1 via the data communication part of the electric control line, when the prescribed service braking performance of the trailer can no longer be ensured.

Note: The ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.
- 5.2.2.16 When the stored energy in any part of the service braking system of a trailer equipped with an electric control line and electrically connected to a towing vehicle with an electronic control line falls to the value determined in accordance with 5.2.2.16.1, a warning shall be provided to the driver of the towing vehicle. The warning shall be provided by activation of the red signal specified in 5.2.1.29.2.1 and the trailer shall

provide the failure information via the data communication part of the electric control line. The separate yellow warning signal specified in 5.2.1.29.2 shall also be activated via pin 5 of the electrical connector conforming to ISO 7638:2003 (See Note below 5.2.2.15.2.1) to indicate to the driver that the low-energy situation is on the trailer.

5.2.2.16.1 The low energy value referred to in 5.2.2.16 shall be that at which, without re-charging of the energy reservoir and irrespective of the load condition of the trailer, it is not possible to apply the service braking control a fifth time after four full-stroke actuations and obtain at least 50 percent of the prescribed performance of the service braking system of the relevant trailer.

5.2.2.17. Trailers equipped with an electric control line and T3 and T4 category trailers equipped with an anti-lock system, shall be fitted with either one or both of the following, for the electric control transmission:

(a) A special electrical connector for the braking system and/or anti-lock system, conforming to ISO 7638:2003; (See Note 1,2)

(b) An automated connector meeting the requirements specified in Annex W.

Failure warning signals required from the trailer by this Standard shall be activated via the above connectors. The requirement to be applied to trailers with respect to the transmission of failure warning signals shall be those, as appropriate, which are prescribed for motor vehicles in paragraphs 5.2.1.29.4., 5.2.1.29.5. and 5.2.1.29.6. of this Standard.

Trailers equipped with an ISO 7638:2003 connector as defined above shall be marked in indelible form to indicate the functionality of the braking system when the ISO 7638:2003 connector is connected and disconnected. (See Note 3).

The marking is to be positioned so that it is visible when connecting the pneumatic and electrical interface connections.

NOTE -

- 1) The ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.
- 2) The conductor cross sections specified in ISO 7638:2003 for the trailer may be reduced if the trailer is installed with its own independent fuse. The rating of the fuse shall be such that the current rating of the conductors is not exceeded. This derogation shall not apply to trailers equipped to tow another trailer.
- 3) In the case of a trailer equipped with both an ISO 7638 connector and automated connector, the marking shall show that the ISO 7638 connector should not be connected when an automated connector is in use.

5.2.2.17.1 Trailers that utilize selective braking as a means to enhance vehicle

stability shall in the event of a failure within the electric control transmission of the stability system indicate the failure by the separate yellow warning signal specified in 5.2.1.29.2 above via pin 5 of the ISO 7638 (Part 1 or 2) connector-

- 5.2.2.17.2 It is permitted to connect the braking system to a power supply in addition to that available from the ISO 7638:2003 connector above. However, when an additional power supply is available the following provisions will apply:
- (a) In all cases the ISO 7638:2003 power supply is the primary power source for the braking system, irrespective of any additional power supply that is connected. The additional supply is intended to provide a backup should a failure of the ISO 7638:2003 power supply occur;
 - (b) It shall not have an adverse effect on the operation of the braking system under normal and failed modes;
 - (c) In the event of a failure of the ISO 7638:2003 power supply the energy consumed by the braking system shall not result in the maximum available power from the additional supply being exceeded;
 - (d) The trailer shall not have any marking or label to indicate that the trailer is equipped with an additional power supply;
 - (e) A failure warning device is not permitted on the trailer for the purposes of providing a warning in the event of a failure within the trailer braking system when the braking system is powered from the additional supply;
 - (f) When an additional power supply is available it shall be possible to verify the operation of the braking system from this power source;
 - (g) Should a failure exist within the electrical supply of energy from the ISO 7638:2003 connector the requirements of paragraphs 5.2.2.15.2.1 and M-4.1. with respect to failure warning shall apply irrespective of the operation of the braking system from the additional power supply.
- 5.2.2.18 Whenever power supplied by the ISO 7638:2003 connector is used for the functions defined in 5.1.3.6, the braking system shall have priority and be protected from an overload external to the braking system. This protection shall be a function of the braking system.
- 5.2.2.19 In the case of a failure in one of the control lines connecting two vehicles equipped according to 5.1.3.1.2 the trailer shall use the control line not affected by the failure to ensure, automatically, the braking performance prescribed for the trailer in C-3.1.
- 5.2.2.20 When the supply voltage to the trailer falls below a value nominated by the manufacturer at which the prescribed service braking performance can no longer be guaranteed, the separate yellow warning signal specified in 5.2.1.29.2 shall be activated via pin 5 of the ISO 7638:2003 (See Note) connector. In addition, trailers equipped with an electrical control line, when electrically connected to a towing vehicle with an electric control

line, shall provide the failure information for actuation of the red warning signal specified in 5.2.1.29.2.1 via the data communication part of the electric control line.

Note: The ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.

- 5.2.2.21. In addition to the requirements of paragraphs 5.2.1.18.4.2. and 5.2.1.21. above, the brakes of the trailer may also be applied automatically when this is initiated by the trailer braking system itself following the evaluation of onboard generated information.

5.2.2.22. **Activation of the service braking system.**

- 5.2.2.22.1. In the case of trailers equipped with an electric control line the message "illuminate stop lamps" shall be transmitted by the trailer via the electric control line when the trailer braking system is activated during "automatically commanded braking" initiated by the trailer. However, when the retardation generated is less than 0.7 m/s^2 , the signal may be suppressed (See Note).

Note: At the time of type approval, compliance with this requirement shall be confirmed by the vehicle manufacturer.

- 5.2.2.22.2. In the case of trailers equipped with an electric control line the message "illuminate stop lamps" shall not be transmitted by the trailer via the electrical control line during "selective braking" initiated by the trailer (See Note).

Note: During a "selective braking" event, the function may change to "automatically commanded braking".

- 5.2.2.23. Subject to the provisions of paragraph 12.3. of this Standard, all vehicles of categories T3 and T4 (See Note) having no more than 3 axles and equipped with air suspension shall be equipped with a vehicle stability function. This shall include at least roll-over control and meet the technical requirements of Annex V.

Note: Trailers for exceptional load transport and trailers with areas for standing passengers shall be excluded from this requirement.

6.0 TESTS AND TEST PROCEDURE

Braking tests which the vehicles submitted for approval are required to undergo, and the braking performance required are described in Annex B, C, D, G, H, K, L, P, Q, T U, V and W.

7.0 MODIFICATION OF VEHICLE TYPE OR BRAKING SYSTEM AND EXTENSION OF APPROVAL

- 7.1 Every modification of the vehicle type or of its braking equipment with regard to the characteristics in Annex A shall be notified to the certifying agency. Certifying agency may then either,

- a) Consider that the modifications made are unlikely to have an appreciable adverse effect and that in any case the vehicle still meets the requirements; or
- b) Require a further testing. For considering whether testing is required or not, guidelines given in Annex K shall be followed. In this case, checks for those parameters which are affected by the modifications only need to be carried out.

7.2 After successful compliance to requirements, the certificate of compliance shall be validated for the modified version.

ANNEX A

(See Clauses 4.2 and 7.1)

**INFORMATION ON TECHNICAL SPECIFICATIONS
TO BE SUBMITTED BY THE MANUFACTURER**

Sl. No.	Specification
1.0	Details of Vehicle Manufacturer
1.1	Manufacturer's name and address
1.2	Telephone No.
1.3	Fax No.
1.4	E-Mail address
1.5	Contact person
1.6	Name of the model and variants
1.7	Plant(s) of manufacturer
2.0	Vehicle Type
2.1	Type of vehicle
2.1.1	Control (Forward/Semi-forward/Normal/Others)
2.1.2	Drive (4 × 2 or 4 × 4 or 6 x 4 or Others)
2.1.3	Cab type (Fully built cab/Sleeper cab/Cowl/Front end structure/With wind shield/Without wind shield
2.1.4	Load body (Fitted/Not fitted)
2.2	Category of vehicle
3.0	Engine
3.1	Type
3.2	Manufacturer's name
3.3	Working principle (4/2 Stroke)
3.4	Model name (if any)
3.5	Type of fuel used
3.6	Swept volume, cc
3.7	Compression ratio (Specify tolerance)

Sl. No.	Specification
3.8	Engine performance
3.9	Maximum net power of the engine (kW) (Specify tolerance)
3.10	Engine RPM at maximum power (Specify tolerance)
4.0	Vehicle Performance
4.1	Maximum designed speed, km/h
5.0	Weights
5.1	Vehicle kerb weight, kg
5.1.1	Front axle, kg
5.1.2	Rear axle, kg
5.1.3	Total, kg
5.2	Gross vehicle weight, kg
5.2.1	Maximum permissible weight
5.2.2	Front axle
5.2.3	Rear axle
5.3	Reference mass, kg
5.4	Seating capacity
5.4.1	Maximum (including driver)
5.4.2	Crew (including driver) NOTE — In case of incompletely built vehicles, this shall be manufacturer's recommendations for completely built vehicles.
6.0	Tyres
6.1	Number and arrangement of wheels
6.1.1	Front
6.1.2	Rear
6.2	Tyre type (Radial/Cross ply), Size/Ply rating
6.2.1	Front
6.2.2	Rear

Sl. No.	Specification
6.3	Rolling radius, mm
6.3.1	Static
6.3.2	Dynamic (if data is available)
6.4	Inflation pressure (Laden)
6.4.1	Front
6.4.2	Rear
7.0	Transmission
7.1	Type (Manual/Automatic/Semi-Auto) NOTE — If automatic, give all pertinent data.
7.2	Gear box
7.2.1	Type
7.2.2	Model name (if any) :
7.2.3	Gear shifting control system
7.2.4	No. of gears
7.3	Gear ratio a) High : b) Low :
7.4	Final drive (Crown wheel)
7.4.1	Type
7.4.2	Reduction ratio
7.4.4	Final drive ratio

7.5 Gear ratio

	G/B Ratio	Overall Ratio
1 st		
2 nd		
3 rd		
4 th		
5 th		
6 th		
And so on		
Over drive		
Reverse		

8.0	Number of Axles and Arrangement Steered Non-Steered a) Driven b) Non-driven
9.0	Vehicle Dimensions
9.1	Wheel base, mm
9.2	Wheel track, mm
9.2.1	Front
9.2.2	Rear
9.3	Body overhang, mm
9.3.1	Front end
9.3.2	Rear end
9.4	Frame overhang (in case of vehicle without complete body), mm
9.4.1	Front end
9.4.2	Rear end

10.0	Service Brake
10.1	Name of producer
10.2	Type (Mechanical/Hydraulic/Air/Air assisted/Others) 90
10.3	Location and No. of braking wheels
10.4	Schematic layout indicating a) Method of split of brake system b) Location of valves, reservoirs, etc
11.0	Lining Or Pad
11.1	Area (cm ²) total per braked wheel
11.1.1	Front wheel
11.1.2	Rear wheel
11.1.3	Others
11.2	Make and material designation
11.2.1	Front wheel
11.2.2	Rear wheel
12.0	Brake Drum or Disc
12.1	Effective dia, mm
12.1.1	Front wheel
12.1.2	Rear wheel
12.1.3	Others
12.2	Material (if the braking surface is not ferrous)
12.2.1	Front wheel
12.2.2	Rear wheel
12.2.3	Others
13.0	Master Cylinder or Brake
13.1	Name of producer
13.2	Type
13.3	Inner dia of master cylinder, mm
13.4	Type of supply tank

14.0	Wheel Cylinder
14.1	Dia, mm
14.1.1	Front
14.1.2	Rear
14.1.3	Others
14.2	Type (Single/Double acting)
14.2.1	Front
14.2.2	Rear
14.2.3	Others
15.0	Booster
15.1	Name of producer
15.2	Type
15.3	Boost ratio
16.0	Vacuum or Air Assistance
16.1	Pressure, kg/cm ² / Vacuum level
16.1.1	Nominal pressure, P ₂
16.1.2	Cut in pressure
16.1.3	Cut out pressure, P _{Max}
16.2	Type : Vacuum pump/Air compressor
16.2.1	Free air delivery, l/min
16.3	Type of pressure regulator
16.4	No. of reservoirs
16.5	Reservoir capacity, l
17.0	Brake Hose
17.1	Make and identification
17.2	Length(s) of hoses
17.3	Nominal bore dia

17.4	End fitting type
18.0	Failure Warning Device for Braking
18.1	Type (Visual/Audible/Others)
18.2	Operation pressure, kg/cm ²
18.3	Type of safety device
19.0	Parking Brake
19.1	Type (Mechanical/Spring Brake/Graduated}
19.2	Acting on transmission/wheel
19.3	Acting on transmission/wheel
19.4	Lining/pad
19.4.1	Name of producer
19.4.2	Area, cm ²
19.4.3	Material
19.4.4	Dia of Brake drum/disc, mm
19.4.5	Control (Hand/Foot operated)
20.0	Secondary Brake
20.1	Type
20.2	Description
21.0	Additional Retarding Devices
21.1	Type
21.2	Description
22.0	Brake Fluid
22.1	Name of producer
22.2	Trade name
22.3	Specification /Grade as per IS
22.4	Brake fluid level indicator (Yes/No)
23.0	Brake Adjustment

23.1	Type (Manual/Auto) Front and Rear			
23.2	Lever length			
24.0	Provision of Liner Wear Indicator			
24.1	Type (Window/Acoustic/Optical) Front and Rear			
25.0	Pneumatic Brake Actuator			
25.1	Type (as per IS)			
26.0	Load Distribution			
			Laden kg	Unladen kg
				Unladen Front/Rear Ratio
		Front Axle		
		Rear Axle 1		
		Rear Axle 2		
		Total		

27.0	Proportioning/Appportioning/Load Sensing/‘G’ Valve
27.1	Name of Manufacturer
27.2	Identification
27.3	Characteristics graph in unladen +2 and laden condition
28.0	Anti-Lock Braking System Provided (Yes/No/Optional)
28.1	If yes, details of ABS
28.2	Make
28.3	Category of ABS
28.4	Nos. of directly controlled wheel(s)
28.5	Brief description of failure warning tell–tale
28.6	Wheel speed sensors
28.6.1	No. of sensors
28.6.2	Make of sensors
28.6.3	Type of sensors
28.7	Modulator

28.7.1	Nos. of modulators
28.7.2	Make of modulators
28.7.3	Identification No. / Part No. of modulator
28.7.4	Brief description and features
28.8	Controller
28.8.1	Nos. of controller
28.8.2	Make of controller
28.8.3	Identification No. / Part No. of controller
28.8.4	Brief description and features
28.9	Height of center of gravity (mm)
28.9.1	Unladen condition
28.9.2	Laden condition
28.10	Slack adjuster
28.10.1	Front (Automatic / Manual)
28.10.1.1	Make of slack adjuster
28.10.1.2	Model and Type
28.10.2	Rear (Automatic / Manual)
28.10.2.1	Make of slack adjuster
28.10.2.2	Model and type
29.0	ADDITIONAL INFORMATION
29.1	Motor Vehicle stability function information document
29.1.1	System
29.1.2	System variants
29.1.3	System options
29.1.3.1	Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control
29.1.4	System configurations (where appropriate)
29.1.5	System identification including software level identifier

29.2	Applications:
29.2.1	List of motor vehicles by description and configuration that are covered by the information document
29.2.2	Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:
29.2.2.1	Lift axles
29.2.2.2	Steering axles
29.2.2.3	Anti-lock braking configurations
29.2.3	Scope of application with respect to suspension:
29.2.3.1	Air
29.2.3.2	Mechanical
29.2.3.3	Rubber
29.2.3.4	Mixed
29.2.3.4	Anti-roll bar
29.2.4	Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:
29.2.4.1	Wheelbase, track, centre of gravity height
29.2.4.2	Wheel type (single or twin) and tyre type (e.g. structure, category of use, size)
29.2.4.3	Gearbox type (e.g. manual, automated manual, semi-automatic, automatic)
29.2.4.4	Drive train options (e.g. retarder)
29.2.4.5	Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected)
29.2.4.1	Management of the engine or any other source(s) of motive power e.g. Torque Management, control of fuel supply etc.
29.3	Component description:
29.3.1	Sensor(s) External to the Controller
29.3.1.1	Function
29.3.1.2	Limitations on the location of the sensors
29.3.1.3	Identification (e.g. part numbers)

29.3.2	Controller(s)
29.3.2.1	General description and function
29.3.2.2	Functionality of internal sensors (if applicable)
29.3.2.3	Hardware identification (e.g. part numbers)
29.3.3.4	Software identification
29.3.3.5	Limitations on the location of the controller(s)
29.3.3.6	Additional features
29.3.3	Modulators
29.3.3.1	General description and function
29.3.3.2	Hardware identification (e.g. part numbers)
29.3.3.3	Software identification (if applicable)
29.3.3.4	Limitations
29.3.4	Electrical Equipment
29.3.4.1	Circuit diagrams
29.3.4.2	Powering methods
29.3.5	Pneumatic circuits
29.3.6	Safety aspects of the electronic system in accordance with Annex S of AIS 150
29.3.6.1	List of complex functions covered:
29.3.6.2	Manufacturer's documents contain information showing the interaction of complex braking functions with other vehicle systems and/or how the system directly controls output variables.
29.3.6.3	Manufacturer's documents explain the functionality and safety concept of the system and describe how the operational status may be checked.
29.3.6.4	A list of input and output variables, including their working range, is provided.
29.3.6.5	Documentation includes an inventory of components, outlines the function of each unit, its interconnection/interaction with other systems and defines signal flow priorities.

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29.3.6.6	Each unit of the system is clearly identified in a manner which defines clearly the hardware and software version as appropriate.
29.3.6.7	Safety concept statement verifies that the system will not prejudice the safe operation of the braking system under non-fault conditions.
29.3.6.8	System architecture, design methods and tools identified.
29.3.6.9	Information regarding design provisions in case of failure is provided including any error messages, warning signals, partial performance conditions, back-up modes and/or removal of high-level functions.
29.3.6.10	Additional material and analysis (FMEA, FTA) of fault conditions made available for inspection and maintained by the manufacturer.
29.3.6.11	Document reference and date of inspection:
29.3.6.12	Vehicle complies with all other performance requirements of AIS 150 and meets manufacturer's specifications under non-fault conditions.
29.3.6.13	Under fault conditions, vehicle response corresponds to that described in the manufacturer's documents / failure analysis and safety concept is verified.
	(see annex 4 of this report for test data)
29.3.7	Electro-magnetic compatibility
29.3.7.1	Documentation demonstrating compliance with AIS-004(Part 3).

ANNEX B
(Foreword and Clause 6)
TEST PROCEDURES

B-1.0 TECHNICAL SPECIFICATIONS OF VEHICLE

- B-1.1 Technical specifications of vehicle as relevant to brake system shall be declared by the details given in Annex A.
- B-1.2 If this information has been declared for type approval of the vehicle it is not necessary to submit the information again.

B-2.0 MODIFICATIONS/CHANGES

- B-2.1 Every functional modification pertaining to technical specifications of brake system of vehicle declared in accordance with B-1.1 shall be intimated to the testing agency.
- B-2.1.1 Testing agency may then consider validating the test report as per guidelines provided in Annex K.

B-3.0 CLASSIFICATIONS OF VEHICLES

- B-3.1 For the purpose of this standard, the definitions given in AIS 053/ IS 14272:2011 shall be referred.

B-3.2 Applicability of the Tests

Based on the applicable Annex the various tests shall be conducted on the vehicle-submitted for test.

- B-3.2.1 The initial speed shall be the speed at the moment when the driver begins to actuate the control of the braking system; the initial speed shall not be less than 98 percent of the prescribed speed for the test in question.
- B-3.2.2 The test shall be carried out at the initial speeds prescribed for each type of the test or 90 percent of maximum design speed of the vehicle, whichever is lower.
- B-3.2.3 During the tests, the force applied to the control of the braking system in order to obtain the prescribed performance shall not exceed the maximum force laid down for the test vehicle's category. Values for force measured at speeds below 5 km/h shall not be considered.
- B-3.2.4 The road surface shall be dry with a road adhesion coefficient of at least 0.8.

B-4.0 VISUAL CHECKS

- B-4.1 Check and verify following:

- a) Two independent controls are readily accessible to driver from the driver

seat while wearing safety belt (see 5.2.1.2.1).

- b) It is possible to actuate the service brake control even with both hands on steering control even while wearing safety belts (see 5.1.2.1).
- c) Service, secondary, and parking brake surfaces are permanently connected to the wheels except transmission parking brake.
- d) It is possible to actuate the secondary brake control while keeping at least one hand on steering control and even while wearing safety belts (see 5.1.2.2).
- e) It shall be possible to lock the parking brake in the applied by a purely mechanical device (see 5.1.2.3).
- f) All brakes shall be provided with automatic adjustment for wear. Provision for assessment of condition of liner shall be present. Alternatively, acoustical or optical devices, warning the driver about lining replacement may be fitted (see 5.2.1.11., 5.2.1.13.1.1) (Optional for N2, N3 off road and M1 and N1 rear application).

B-5.0 INSTRUMENTATION

B-5.1 Accuracies and least count of the instruments shall be as below or better:

Sl No.	Parameter	Accuracy	Least Count
i)	Time	± 0.01 s	0.01 s
ii)	a) Pneumatic line pressure	± 20 kPa (0.2 bar)	10 kPa (0.1 bar)
	b) Hydraulic line pressure	± 200 kPa (2 bar)	100 kPa (1 bar)
iii)	Control force	± 1 kg	0.5 kg
iv)	Speed	± 1 percent of the measured value	0.1 km/h
v)	Distance	± 1 percent of the measured value	0.1 m
vi)	Deceleration	± 0.01 percent of the measured value	0.01 m/s^2
vii)	Temperature	± 2 °C	1°C
viii)	Wind velocity	± 0.5 percent of the measured value	0.5 m/s
ix)	Towing load		

	a) Up to 1.5 t	10 kg	10 kg
	b) Above 1.5 t	20 kg	20 kg
x)	Weights		
	a) Up to 1.0 t	10 kg	10 kg
	b) Above 1.0 t to 12.0 t	50 kg	10 kg
	c) Above 12.0 t	100 kg	10 kg

NOTES

1. It is recommended to use only pressure transducers.
2. The range given for towing load and Weigh Bridge are the values to be measured.

B-5.2 Ensure that the instruments/equipment are calibrated as per the recommendations of manufacturer of instruments/equipment.

B-5.3 Any other instruments can be used after confirmation of satisfactory calibration.

B-6.0 TEST TRACK

B-6.1 Longitudinal and lateral slopes of the track shall not exceed 1 percent and 3 percent respectively. The track shall have sufficient length and width for achieving the test speed and carrying out the test safely.

B-6.2 If the tests are carried out at the test track of the vehicle manufacturer, the conditions specified for the test track are deemed to be complied with if the manufacturer issues a certificate to that effect.

B-6.3 For parking brake test, gradient of the track shall be constant within ± 0.5 percent.

B-7.0 VEHICLE PREPARATION

B-7.1 Before testing the vehicle, ensure that the vehicle is run in as per manufacturer's recommendation and that the brakes are properly bedded in for brake liner/pad contact.

B-7.2 Tyres which have covered at least 500 km and not more than maximum 10 percent of their expected life shall be fitted on the vehicle. These conditions are deemed to be complied with if the manufacturer gives a written undertaking to that effect.

Tyre shall be inflated to the pressure specified by vehicle manufacturer for respective load condition of the vehicle when the tyres are cold.

- B-7.3 Actual weight during testing shall not differ from specified laden or unladen weight by:

a)	Unladen	:	10 percent, Max
b)	Laden	:	5 percent, Max

- B-7.4 The distribution of weight among axles shall be as specified by the vehicle manufacturer. If sum of recommended maximum axle weights exceeds the gross vehicle weight, the actual weight on the axle shall be in the same proportion of the ratios of the gross vehicle weight to the sum of the maximum recommended axle weights. Actual load condition shall be recorded in the report.

- B-7.5 Weight of the testing personnel and instrumentation shall be considered as part of the vehicle load and shall be 180 kg in the case of unladen test.

- B-7.6 In the case of tractive units for semi-trailers the required laden condition shall be achieved by loading the tractive unit and not by using a semi-trailer.

- B-7.7 Where the vehicle is not having standard production body or is having more than one version of standard production body, the following cab/body allowance shall apply:

Sl No.	Allowance in Weight		
	GVW Range (kg)	Load Body Weight (kg)	Cab Weight(kg)
(1)	(2)	(3)	(4)
i)	Up to 2 500	120	300
ii)	Above 2 500 but below 5 000	200	400
iii)	Above 5 000 but below 7 500	400	500
iv)	Above 7 500 but below 12 000	700	600
v)	Above 12 000	1 000	700

- B-7.8 In the case of incompletely built vehicle (such as drive away chassis) suitable arrangements for safety of testing personnel during testing shall be provided. The weight of such provisions shall be part of the load prescribed in B-7.7.

B-8.0 SEQUENCE OF TESTING

- B-8.1 All the static tests, the result of, which are required in dynamic test, are to be done before the dynamic tests.
- B-8.2 There is no preferred sequence during static tests.
- B-8.3 If more than one type of dynamic tests are to be carried out, sequence of tests shall be Type 0 test, Type I and / or Type II / II-A test. There is no preferred sequence among the various Type 0 tests.
- B-8.4 Parking brake test(s) shall be conducted before Type I tests.

B-9.0 REQUIREMENTS DURING TESTING

- B-9.1 Brake shall not be applied prior to the test within 800 m immediately before the point at which test commences.
- B-9.2 The ambient temperature shall be between 15 to 45°C.
- B-9.3 The test shall be valid if vehicle does not go out of the lane width one and half times the width of the vehicle. Wheel locking towards the end of deceleration in last three meter shall be permitted.
- B-9.4 In the case of diagonal split secondary brake system the force required at the time of steering wheel to hold a straight course may be measured and, if felt excessive, shall be recorded in the test report.
- B-9.5 Suitable methods for measurement of temperature on the disc or outside of the drum may be used.

NOTE – Measurement of lining/pad temperature is not a method for assessing the temperature on the disc or outside of the drum.

B-9.6 Final Result of the Test

Number of readings to be taken during the test shall be as follows:

- a) For all Type 0 tests, maximum six numbers of tests shall be carried out; the test need not be carried out further once it meets the performance requirement during these tests.
- b) All dynamic tests other than Type 0 test shall be conducted only once and the result obtained during the test shall be the final result.
- c) All static tests other than capacity of the energy source shall be conducted only once and the result obtained during the test shall be the final result.
- d) Parking brake gradient holding test shall be conducted only once. The test shall be conducted preferably on the specified gradient. If such a gradient is not available, the nearest gradient available shall be used. The Gross Vehicle Weight (GVW) can be simulated by the following formula.

$$G = 100 \times \tan [\sin^{-1}((\sin \theta \times W_T)/W_R)]$$

where

G = Gradient of vehicle in percentage;

θ = Slope angle of the test track used for the test;

W_T = Test weight of the vehicle; and

W_R = Maximum recommended GVW of the vehicle.

ANNEX C

(Clauses 5.1.3.3, 5.1.4.6.3, 5.2.1.4, 5.2.1.7.1, 5.2.1.7.2, 5.2.1.11.1, 5.2.1.18.3, 5.2.1.18.5, 5.2.2.8.1.1, 5.2.2.8.1.2, 5.2.2.12, 5.2.2.14, 5.2.2.14.1, 5.2.2.14.2, 5.2.2.15.2, 5.2.2.19 and 6)

C-0.0 BRAKING TESTS AND PERFORMANCE OF BRAKING SYSTEMS**C-1.0 BRAKING TESTS****C-1.1 General**

C-1.1.1 The performance prescribed for braking systems is based on the stopping distance and/or the mean fully developed deceleration. The performance of a braking system shall be determined by measuring the stopping distance in relation to the initial speed of the vehicle and/or by measuring the mean fully developed deceleration during the test.

C-1.1.2 The stopping distance shall be the distance covered by the vehicle from the moment when the driver begins to actuate the control of the braking system until the moment when the vehicle stops; the initial speed shall be the speed at the moment when the driver begins to actuate the control of the braking system; the initial speed shall be within tolerance prescribed in B 3.2.1.

The mean fully developed deceleration (d_m) shall be calculated as the deceleration averaged with respect to distance over the interval v_b to v_e , according to the following formula:

$$d_m = \frac{v_b^2 - v_e^2}{25.92 (s_e - s_b)} \quad [\text{m/s}^2]$$

where

v_b = vehicle speed at 0.8 v_o , in km/h;

v_e = vehicle speed at 0.1 v_o , in km/h;

v_o = initial vehicle speed, in km/h;

s_b = distance travelled between v_o and v_b , in m; and

s_e = distance travelled between v_o and v_e , in m.

The speed and the distance shall be determined using instrumentation prescribed in B-5. The mean fully developed deceleration may be determined by other methods than the measurement of speed and distance; in this case, the accuracy of the mean fully developed deceleration shall be within ± 3 percent.

C-1.2 For the approval of any vehicle, the braking performance shall be measured during road tests conducted in the following conditions.

C-1.2.1 The vehicle's condition as regards mass shall be as prescribed for each type of test and be specified in the test report.

C-1.2.2 The test shall be carried out at the speeds prescribed for each type of test; if

the maximum design speed of a vehicle is lower than the speed prescribed for a test, the test shall be performed at the vehicle's maximum speed.

- C-1.2.3 During the tests, the force applied to the control of the braking system in order to obtain the prescribed performance shall not exceed the maximum force laid down for the test vehicle's category.
- C-1.2.4 The road surface shall be dry with a coefficient of adhesion as specified in B-3.2.4
- C-1.2.5 The tests shall be performed when there is no wind liable to affect the results.
- C-1.2.6 At the start of the tests, the tyres shall be cold and at the pressure prescribed as per B-7.2.
- C-1.2.7 The prescribed performance shall be obtained without locking of the wheels, without deviation of the vehicle from its course and abnormal vibration of the vehicle.
- C-1.2.8 For vehicles powered completely or partially by an electric motor (or motors), permanently connected to the wheels, all tests shall be carried out with the motor(s) connected.
- C-1.2.9 For vehicles as described in C-1.2.8, fitted with an electric regenerative braking system of category A, behaviour tests defined in C-1.4.3.1 shall be carried out on a track with a low adhesion coefficient (as defined in M-5.2.2). However, the maximum test speed shall not exceed the maximum test speed specified in M-5.3.1. for a low adhesion surface and the relevant vehicle category.
- C-1.2.9.1 Moreover, for vehicles fitted with an electric regenerative braking system of category A, transient conditions as gear changes or accelerator control release shall not affect the behaviour of the vehicle in condition of tests described in C-1.2.9.
- C-1.2.10 During the tests specified in C-1.2.9 and C-1.2.9.1, wheel locking is not allowed. However, steering correction is permitted if the angular rotation of the steering control is within 120° during the initial 2 s and not more than 240° in total.
- C-1.2.11 For a vehicle with electrically actuated service brakes powered from traction batteries (or an auxiliary battery) which receive(s) energy only from an independent external charging system, these batteries shall, during braking performance testing, be at an average of not more than 5 percent above that state of charge at which the brake failure warning prescribed in 5.2.1.27.6 is required to be given.

If this warning is given, the batteries may be recharged during the tests, to keep them in the required state of charge range.

C-1.3 Behaviour of the Vehicle During Braking

- C-1.3.1 In braking tests, and in particular in those at high speed, the general behaviour of the vehicle during braking shall be checked.

- C-1.3.2 Behaviour of the Vehicle During Braking on A Road on which Adhesion is

Reduced

The behaviour of vehicles of categories M2, M3, N1, N2, N3, T3 and T4 on a road on which adhesion is reduced, shall meet the relevant requirements of Annex J and/or Annex M.

- C-1.3.2.1 In the case of a braking system according to 5.2.1.7.2, where the braking for a particular axle (or axles) is comprised of more than one source of braking torque, and any individual source can be varied with respect to the other(s), the vehicle shall satisfy the requirements of Annex J, or alternatively, Annex M under all relationships permitted by its control strategy (See Note).

Note: The manufacturer shall provide the Testing Agency with the family of braking curves permitted by the automatic control strategy employed. These curves may be verified by the Testing Agency.

C-1.4 **Type 0 Test (Ordinary Performance Test with Brakes Cold)**

C-1.4.1 **General**

- C-1.4.1.1 The brakes shall be cold; a brake is deemed to be cold when the temperature measured on the disc or on the outside of the drum is below 100°C.

C-1.4.1.2 **The test shall be conducted in the following conditions:**

- C-1.4.1.2.1 The vehicle shall be laden, the distribution of its mass among the axles being that stated by the manufacturer; where provision is made for several arrangements of the load on the axles the distribution of the maximum mass among the axles shall be such that the load on each axle is proportional to the maximum permissible load for each axle. In the case of tractors for semi-trailers, the load may be re-positioned approximately half-way between the kingpin position resulting from the above loading conditions and the centreline of the rear axle(s).

- C-1.4.1.2.2 Every test shall be repeated on the unladen vehicle. In the case of a power-driven vehicle there may be, in addition to the driver, a second person on the front seat who is responsible for noting the results of the test.

In the case of a tractor for a semi-trailer, the unladen tests will be conducted with the vehicle in its solo condition, including a mass representing the fifth wheel. It will also include a mass representing a spare wheel, if this is included in the standard specification of the vehicle.

In the case of a incompletely built vehicle (such as Drive Away Chassis) a supplementary load shall be added to simulate the mass of the body not exceeding the Vehicle kerb weight, declared by the manufacturer in Annex A. In absence of this data values specified as per B 7.7, shall be used.

In the case of a vehicle equipped with an electric regenerative braking system, the requirements depend on the category of this system:

- a) Category A – Any separate electric regenerative braking control which is provided, shall not be used during the Type 0 tests.
- b) Category B – The contribution of the electric regenerative braking system to the braking force generated shall not exceed that minimum level guaranteed by the system design.

This requirement is deemed to be satisfied if the batteries are at one of the following state of charge (See Note) conditions where state of charge is determined by the method set out in C-5.

Note: By agreement with the Testing Agency, state of charge assessment will not be required for vehicles, which have an on-board energy source for charging the traction batteries and the means for regulating their state of charge.

At the maximum charge level as recommended by the manufacturer in the vehicle specification, or at a level not less than 95 percent of the full charge level, where the manufacturer has made no specific recommendation, or at the maximum level which results from automatic charge control on the vehicle, or when the tests are conducted without a regenerative braking component regardless of the state of charge of the batteries.

C-1.4.1.2.3 The limits prescribed for minimum performance, both for tests with the vehicle unladen and for tests with the vehicle laden, shall be those laid down hereunder for each category of vehicles; the vehicle shall satisfy both the prescribed stopping distance and/or the prescribed mean fully developed deceleration for the relevant vehicle category, but it may not be necessary to actually measure both parameters.

C-1.4.1.2.4 The road shall be level

C-1.4.2 **Type 0 Test with Engine Disconnected**

The test shall be carried out at the speed prescribed for the category to which the vehicle belongs, the Figures prescribed in this connection being subject to not less than 98 percent of the prescribed speed for the test in question. The minimum performance prescribed for each category shall be attained.

C-1.4.3 Type 0 Test with Engine Connected

C-1.4.3.1 Tests shall also be carried out at various speeds, the lowest being equal to 30 percent of the maximum speed of the vehicle and the highest being equal to 80 percent of that speed. In the case of vehicles equipped with a speed limiter, this limiter speed shall be taken as the maximum speed of the vehicle. The maximum practical performance figures shall be measured and the behaviour of the vehicle shall be recorded in the test report. Tractors for semi-trailers, artificially loaded to simulate the effects of a laden semi-trailer shall not be tested beyond 80 km/h.

C-1.4.3.2 Further tests shall be carried out with the engine connected, from the speed prescribed for the category to which the vehicle belongs. The minimum performance prescribed for each category shall be attained. Tractive units for semi-trailers, artificially loaded to simulate the effects of a laden semi-trailer shall not be tested beyond 80 km/h.

C-1.4.4 Type 0 Test for Vehicles of Category T, Equipped with Compressed Air Brakes

C-1.4.4.1 The braking performance of the trailer may be calculated either from the braking rate of the towing vehicle plus the trailer and the measured thrust on the coupling or, in certain cases, from the braking rate of the towing vehicle plus the trailer with only the trailer being braked. The engine of the towing vehicle shall be disconnected during the braking test.

In the case where only the trailer is braked to take account of the extra mass being retarded, the performance will be taken to be the mean fully developed deceleration.

C-1.4.4.2 With the exception of cases according to C-1.4.4.3 and C-1.4.4.4, it is necessary for the determination of the braking rate of the trailer to measure the braking rate of the towing vehicle plus the trailer and the thrust on the coupling. The towing vehicle has to meet the requirements laid down in Annex J with regard to the relationship between the ratio T_M/P_M and the pressure p_m . The braking rate of the trailer is calculated according to the following formula:

$$Z_R = Z_{R+M} + \frac{D}{P_R}$$

where

Z_R = Braking rate of the trailer;

Z_{R+M} = Braking rate of the towing vehicle plus the trailer;

D = Thrust on the coupling; (tractive force: $+D$),
(compressive force: $-D$); and

P_R = Total normal static reaction between road surface and wheels of trailer (see Annex J).

- C-1.4.4.3 If a trailer has a continuous or semi-continuous braking system where the pressure in the brake actuators does not change during braking despite the dynamic axle load shifting and in the case of semi-trailers the trailer alone may be braked. The braking rate of the trailer is calculated according to the following formula:

$$z_R = (z_{R+M} - R) \cdot \frac{P_M + P_R}{P_R} + R$$

where

R = Rolling resistance value = 0.01; and

P_M = Total normal static reaction between road surface and wheels of towing vehicles for trailers (see Annex J).

- C-1.4.4.4 Alternatively, the evaluation of the braking rate of the trailer may be done by braking the trailer alone. In this case the pressure used shall be the same as that measured in the brake actuators during the braking of the combination.

C-1.5 Type I Test (Fade Test)

C-1.5.1 With Repeated Braking

- C-1.5.1.1 The service braking systems of all power-driven vehicles shall be tested by successively applying and releasing the brakes a number of times, the vehicle being laden, in the conditions shown below:

Category of Vehicles	Conditions			
	v ₁ (km/h)	v ₂ (km/h)	Δt (s)	n
M2	80 percent v _{max} ≤ 100	1/2 v ₁	55	15
N1	80 percent v _{max} ≤ 120	1/2 v ₁	55	15
M3, N2, N3	80 percent v _{max} ≤ 60	1/2 v ₁	60	20

Where

v₁ = Initial speed, at beginning of braking;

v₂ = Speed at end of braking;

v_{max} = Maximum speed of vehicle;

n = Number of brake applications; and

Δt = Duration of a braking cycle: time elapsing between the initiation of one brake application and the initiation of the next.

- C-1.5.1.2 If the characteristics of the vehicle make it impossible to abide by the

duration prescribed for Δt , the duration may be increased; in any event, in addition to the time necessary for braking and accelerating the vehicle, a period of 10 s shall be allowed in each cycle for stabilizing the speed v_1 .

- C-1.5.1.3 In these tests, the force applied to the control shall be so adjusted as to attain the mean fully developed deceleration of 3 m/s^2 at the first brake application; this force shall remain constant throughout the succeeding brake applications.
- C-1.5.1.4 During brake applications, the highest gear ratio (excluding overdrive, etc.) shall be continuously engaged.
- C-1.5.1.5 For regaining speed after braking, the gearbox shall be used in such a way as to attain the speed v_1 in the shortest possible time (maximum acceleration allowed by the engine and gearbox).
- C-1.5.1.6 For vehicles not having sufficient autonomy to carry out the cycles of heating of the brakes, the tests shall be carried out by achieving the prescribed speed before the first braking application and thereafter by using the maximum acceleration available to regain speed and then braking successively at the speed reached at the end of each time cycle duration as specified, for the appropriate vehicle category in C-1.5.1.1.
- C-1.5.1.7 In the case of vehicles equipped with automatic brake adjustment devices the adjustment of the brakes shall, prior to the Type I test above, be set according to the following procedures as appropriate.
- C-1.5.1.7.1 In the case of vehicles equipped with air operated brakes the adjustment of the brakes shall be such as to enable the automatic brake adjustment device to function. For this purpose the actuator stroke shall be adjusted to

$s_0 \geq 1.1 A_{s_{\text{re-adjust}}}$ (the upper limit shall not exceed a value recommended by the manufacturer)

where

$s_{\text{re-adjust}}$ = Re-adjustment stroke according to the specification of the manufacturer of the automatic brake adjustment device, that is the stroke, where it starts to readjust the running clearance of the brake with an actuator pressure of 15 percent of the brake system operating pressure but not less than 100 kPa (1 bar).

Where, by agreement with the Testing Agency, it is impractical to measure the actuator stroke, the initial setting shall be agreed with the Testing Agency.

From the above condition the brake shall be operated with an actuator pressure of 30 percent of the brake system operating pressure but not less than 200 kPa (2 bar), 50 times in succession. This shall be followed by a single brake application with an actuator pressure of $\geq 650 \text{ kPa}$ (6.5 bar).

- C-1.5.1.7.2 In the case of vehicles equipped with hydraulically operated disc brakes no setting requirements are deemed necessary.

C-1.5.1.7.3 In the case of vehicles equipped with hydraulically operated drum brakes the adjustment of the brakes shall be as specified by the manufacturer.

C-1.5.1.8 For vehicles equipped with an electric regenerative braking system of category B, the condition of the vehicle batteries at the start of the test, shall be such that the braking force contribution provided by the electric regenerative braking system does not exceed the minimum guaranteed by the system design.

This requirement is deemed to be satisfied if the batteries are at one of the state of charge conditions listed in C-1.4.1.2.2.

C-1.5.2 With Continuous Braking

C-1.5.2.1 The service brakes of trailers of categories T2 and T3 (when the T3 trailer has not passed alternatively the Type-III test according to C-1.7.) shall be tested in such a manner that, the vehicle being laden, the energy input to the brakes is equivalent to that recorded in the same period of time with a laden vehicle driven at a steady speed of 40 km/h on a 7 percent down-gradient for a distance of 1.7 km.

C-1.5.2.2 The test may be carried out on a level road, the trailer being drawn by a towing vehicle; during the test, the force applied to the control shall be adjusted so as to keep the resistance of the trailer constant (7 percent of the maximum total stationary axle load of the trailer). If the power available for hauling is insufficient, the test may be conducted at a lower speed but over a greater distance as shown below:

Speed (km/h)	Distance (m)
40	1 700
30	1 950
20	2 500
15	3 100

C-1.5.2.3 In the case of trailers equipped with automatic brake adjustment devices the adjustment of the brakes shall, prior to the Type I test prescribed above, be set according to the procedure as laid down in C-1.7.1.1.

C-1.5.3 Hot Performance

C-1.5.3.1 At the end of the Type I test (test described in C-1.5.1 or C-1.5.2) the hot performance of the service braking system shall be measured in the same conditions (and in particular at a constant control force no greater than the mean force actually used) as for the Type 0 test with the engine disconnected (the temperature conditions may be different).

C-1.5.3.1.1 For power-driven vehicles the hot performance shall not be less than 80 percent of that prescribed for the category in question, nor less than 60 percent of the figure recorded in the Type 0 test with the engine disconnected. In calculating the residual performance requirements, the

percentages are to be applied to the second term of polynomial expression of stopping distance given in C-2.1.1.

- C-1.5.3.1.2 For vehicles fitted with an electric regenerative braking system of category A, during brake applications, the highest gear shall be continuously engaged and the separate electric regenerative braking control, if any, shall not be used.
- C-1.5.3.1.3 In the case of vehicles equipped with an electric regenerative braking system of category B, having carried out the heating cycles according to C-1.5.1.6, the hot performance test shall be carried out at the maximum speed which can be reached by the vehicle at the end of the brake heating cycles, unless the speed specified in C-1.4.2 can be reached.

For comparison, the Type 0 test with cold brakes shall be repeated from this same speed and with a similar electric regenerative braking contribution, as set by an appropriate state of battery charge, as was available during the hot performance test.

Reconditioning of the linings shall be permitted before the test is made to compare this second Type 0 cold performance with that achieved in the hot test, against the criteria of C-1.5.3.1.1 and C-1.5.3.2.

The tests may be conducted without a regenerative braking component. In this case, the requirement on the state of charge of the batteries is not applicable.

- C-1.5.3.1.4 However, in the case of trailers, the hot brake force at the periphery of the wheels when tested at 40 km/h shall not be less than 36 percent of the maximum stationary wheel load, nor less than 60 percent of the figure recorded in the Type 0 test at the same speed.
- C-1.5.3.2 In the case of a power-driven vehicle which satisfies the 60 percent requirement specified in C-1.5.3.1.1, but which cannot comply with the 80 percent requirement of C-1.5.3.1.1, a further hot performance test may be carried out using a control force not exceeding that specified in C-2 for the relevant vehicle category. The results of both tests shall be entered in the report.

C-1.5.4 Free Running Test

In the case of motor vehicles equipped with automatic brake adjustment devices, the brakes after completing the tests defined in C-1.5.3 shall be allowed to cool to a temperature representative of a cold brake (that is $\leq 100^\circ\text{C}$) and it shall be verified that the vehicle is capable of free running by fulfilling one of the following conditions:

- a) Wheels are running freely (that is may be rotated by hand), and
- b) It is ascertained that when the vehicle is driven at a constant speed of $v = 60 \text{ km/h}$ with the brakes released the asymptotic temperatures shall not exceed a drum/disc temperature increase of 80°C , then the residual

brake moments are regarded as acceptable.

C-1.6 **Type II Test (Downhill Behaviour Test)**

C-1.6.1 Laden power-driven vehicles shall be tested in such a manner that the energy input is equivalent to that recorded in the same period of time with a laden vehicle driven at an average speed of 30 km/h on a 6 percent down-gradient for a distance of 6 km, with the appropriate gear engaged and the endurance braking (See Note) system such as retarders and exhaust brake systems, if the vehicle is equipped with one, being used. The gear engaged shall be such that the speed of the engine (min^{-1}) does not exceed the maximum value prescribed by the manufacturer.

NOTE – Endurance brake means, a supplementary brake which is capable of operating for a long duration, without or minimum wear on service brake friction material and drum. Example engine brake, exhaust brake, hydraulic or electromagnetic retarders. These brakes are optional to enhance the safety and reduce friction.

C-1.6.2 For vehicles in which the energy is absorbed by the braking action of the engine alone, a tolerance of ± 5 km/h on the average speed shall be permitted, and the gear enabling the speed to be stabilized at the value closest to 30 km/h on the 6 percent down-gradient shall be engaged. If the performance of the braking action of the engine alone is determined by a measurement of deceleration, it shall be sufficient if the mean deceleration measured is at least 0.5 m/s^2 .

C-1.6.3 At the end of the test, the hot performance of the service braking system shall be measured in the same conditions as for the Type 0 test with the engine disconnected (the temperature conditions may be different). This hot performance shall give a stopping distance not exceeding the following values and a mean fully developed deceleration not less than the following values, using a control force not exceeding 70 daN (kg):

a)	category M3	$0.15v + (1.33 v^2/130)$ (the second term corresponds to a mean fully developed deceleration $d_m = 3.75 \text{ m/s}^2$); and
b)	category N3	$0.15v + (1.33 v^2/115)$ (the second term corresponds to a mean fully developed deceleration $d_m = 3.3 \text{ m/s}^2$).

C-1.6.4 Vehicles cited in C-1.8.1.1, C-1.8.1.2 and C-1.8.1.3 shall satisfy the Type IIA test described in C-1.8 instead of the Type II test.

C-1.7 **Type III Test (Fade Test for Vehicles of Category T4 or alternatively of category T3)**

C-1.7.1 **Track Test**

C-1.7.1.1 The adjustment of the brakes shall, prior to the Type III test below, be set according to the following procedures as appropriate:

- C-1.7.1.1.1 In the case of trailers equipped with air operated brakes the adjustment of the brakes shall be such as to enable the automatic brake adjustment device to function. For this purpose the actuator stroke shall be adjusted to $s_0 \geq 1.1 \cong s_{re-adjust}$ (the upper limit shall not exceed a value recommended by the manufacturer)

where

$s_{re-adjust}$ is the re-adjustment stroke according to the specification of the manufacturer of the automatic brake adjustment device, that is the stroke, where it starts to re-adjust the running clearance of the brake with an actuator pressure of 100 kPa (1 bar).

Where, by agreement with the Certifying Agency, it is impractical to measure the actuator stroke, the initial setting shall be agreed with the Certifying Agency.

From the above condition the brake shall be operated with an actuator pressure of 200 kPa (2 bar), 50 times in succession. This shall be followed by a single brake application with an actuator pressure of ≥ 650 kPa (6.5 bar).

- C-1.7.1.1.2 In the case of trailers equipped with hydraulically operated disc brakes no setting requirements are deemed necessary.
- C-1.7.1.1.3 In the case of trailers equipped with hydraulically operated drum brakes the adjustment of the brakes shall be as specified by the manufacturer.
- C-1.7.1.2 For the road test the conditions shall be as follows:

Number of brake applications	20
Duration of a braking cycle	60 s
Initial speed at the beginning of braking	60 km/h
Braking applications	In these tests, the force applied to the control shall be so adjusted as to attain the mean fully developed deceleration of 3 m/s^2 in respect to the trailer mass P_R at the first brake application; this force shall remain constant throughout the succeeding brake applications

The braking rate of a trailer is calculated according to the formula given in C-1.4.4.3.

The speed at the end of braking:

$$z_R = (z_{R+M} - R) \cdot \frac{(P_M + P_R)}{P_R} + R$$

$$v_2 = v_1 \cdot \sqrt{\frac{P_M + P_1 + P_2/4}{P_M + P_1 + P_2}}$$

Where

- z_R = Braking rate of the trailer;
- z_{R+M} = Braking rate of the vehicle combination (motor vehicle and trailer);
- R = Rolling resistance value = 0.01;
- P_M = Total normal static reaction between the road surface and the wheels of towing vehicle for trailer (kg);
- P_R = Total normal static reaction between the road surface and the wheels of trailer (kg);
- P_1 = Part of the mass of the trailer borne by the unbraked axle(s) (kg);
- P_2 = Part of the mass of the trailer borne by the braked axle(s) (kg);
- v_1 = Initial speed (km/h);
- v_2 = Final speed (km/h).

C-1.7.2 Hot Performance

At the end of the test according to C-1.7.1, the hot performance of the service braking system shall be measured under the same conditions as for the Type 0 test with, however, different temperature conditions and starting from an initial speed of 60 km/h. The hot brake-force at the periphery of the wheels shall then not be less than 40 percent of the maximum stationary wheel load, and not less than 60 percent of the figure recorded in the Type 0 test at the same speed.

C-1.7.3 Free Running Test

After completing the tests defined in C-1.7.2, the brakes shall be allowed to cool to a temperature representative of a cold brake (that is ≤ 100 °C) and it shall be verified that the trailer is capable of free running by fulfilling one of the following conditions:

- a) Wheels are running freely (that is, may be rotated by hand); and
- b) It is ascertained that when the trailer is driven at a constant speed of $v = 60$ km/h with the brakes released the asymptotic temperatures shall not exceed a drum/disc temperature increase of 80 °C, then the residual brake moments are regarded as acceptable.

C-1.8 Type IIA Test (Endurance Braking Performance)

C-1.8.1 Vehicles of the following categories shall be subject to the Type-IIA test:

C-1.8.1.1 Interurban motor coaches and long distance touring motor coaches of category M3.

- C-1.8.1.2 Vehicles of category N3 which are authorized to tow a trailer of category T4.

If the maximum mass exceeds 26 t, the test mass is limited to 26 t or, in the case where the unladen mass exceeds 26 t, this mass is to be taken into account by calculation.

- C-1.8.1.3 Certain vehicles subject to ADR (Annex D)

NOTE – Mandatory requirements for Type II A Test will be in accordance with decision of statutory authority.

- C-1.8.2 Test Conditions and Performance Requirements

- C-1.8.2.1 The performance of the endurance braking system shall be tested at the maximum mass of the vehicle or of the vehicle combination.

- C-1.8.2.2 Laden vehicles shall be tested in such a manner that the energy input is equivalent to that recorded in the same period of time with a laden vehicle driven at an average speed of 30 km/h on a 7 percent down-gradient for a distance of 6 km. During the test, the service, secondary and parking braking systems shall not be engaged. The gear engaged shall be such that the speed of the engine does not exceed the maximum value prescribed by the manufacturer. An integrated endurance braking system may be used, provided that it is suitably phased such that the service braking system is not applied; this may be verified by checking that its brakes remain cold, as defined in C-1.4.1.1.

- C-1.8.2.3 For vehicles in which the energy is absorbed by the braking action of the engine alone, a tolerance of ± 5 km/h on the average speed shall be permitted, and the gear enabling the speed to be stabilized at a value closest to 30 km/h on a 7 percent down-gradient shall be engaged. If the performance of the braking action of the engine alone is determined by measuring the deceleration, it shall be sufficient if the mean deceleration measured is at least 0.6 m/s^2 .

C-2 PERFORMANCE OF BRAKING SYSTEMS OF VEHICLES OF CATEGORIES M2, M3 AND N

- C-2.1 **Service Braking System**

- C-2.1.1 The service brakes of vehicles of categories M2, M3 and N shall be tested under the conditions shown in the following table:

Category	M2	M3	N1	N2	N3
Type of test	0-I	0-I-II or IIA	0-I	0-I	0-I-II
v	60	60	80	60	60

Type 0 test with engine disconnected	$s \leq$	$0.15v + \frac{v^2}{130}$				
	$d_m \geq$	5.0				
Type 0 test with engine connected	$v = 0.80 v_{\max}$ but not exceeding	100	90	120	100	90
	$s \leq$	$0.15v + \frac{v^2}{103.5}$				
	$d_m \geq$	4.0				
	$F \leq$	70				

where

- v = Prescribed test speed, in km/h;
 s = Stopping distance, in m;
 d_m = Mean fully developed deceleration, in m/s^2 ;
 F = Force applied to foot control, in daN (kg); and
 $v_{\max}$ = Maximum speed of the vehicle, in km/h.

C-2.1.2 In the case of a power-driven vehicle authorized to tow an unbraked trailer, the minimum performance prescribed for the corresponding power-driven vehicle category (for the Type 0 test with engine disconnected) shall be attained with the unbraked trailer coupled to the power-driven vehicle and with the unbraked trailer laden to the maximum mass declared by the power-driven vehicle manufacturer.

The combination performance shall be verified by calculations referring to the maximum braking performance actually achieved by the power-driven vehicle alone (laden) during the Type 0 test with the engine disconnected, using the following formula (no practical tests with a coupled unbraked trailer are required):

$$d_{M+R} = d_M \cdot \frac{P_M}{P_M + P_R}$$

Where:

- d_{M+R} = Calculated mean fully developed deceleration of the power-driven vehicle when coupled to an unbraked trailer, in m/s^2 ;
 d_M = Maximum mean fully developed deceleration of the power-driven vehicle alone achieved during the Type 0 test with engine disconnected, in m/s^2 ;

- P_M = Mass of the power-driven vehicle (laden);
- P_R = Maximum mass of an unbraked trailer which may be coupled, as declared by the power-driven vehicle manufacturer.

C-2.2 Secondary Braking System

- C-2.2.1 The secondary braking system, even if the control which actuates it is also used for other braking functions, shall give a stopping distance not exceeding the following values and a mean fully developed deceleration not less than the following values:
- a) Category M2, M30.15 $v + (2v^2/130)$ (the second term corresponds to a mean fully developed deceleration $d_m = 2.5 \text{ m/s}^2$)
 - b) Category N 0.15 $v + (2v^2/115)$ (the second term corresponds to a mean fully developed deceleration $d_m = 2.2 \text{ m/s}^2$)
- C-2.2.2 If the control is hand control, the prescribed performance shall be obtained by applying to the control a force not exceeding 60 daN (kg) in the case of other vehicles, and the control shall be so placed that it can be easily and quickly grasped by the driver.
- C-2.2.3 If it is a foot control, the prescribed performance shall be obtained by applying to the control a force not exceeding 70 daN (kg) in the case of other vehicles, and the control shall be so placed that it can be easily and quickly actuated by the driver.
- C-2.2.4 The performance of the secondary braking system shall be checked by the Type 0 test with engine disconnected from the following initial speeds:
- a) M2: 60 km/h M3: 60 km/h
 - b) N1: 70 km/h N2: 50 km/h N3: 40 km/h
- C-2.2.5 The secondary braking effectiveness test shall be conducted by simulating the actual failure conditions in the service braking system.
- C-2.2.6 For vehicles employing electric regenerative braking systems, the braking performance shall additionally be checked under the two following failure conditions.
- C-2.2.6.1 For a total failure of the electric component of the service braking output.
 - C-2.2.6.2 In the case where the failure condition causes the electric component to deliver its maximum braking force.

C-2.3 Parking Braking System

- C-2.3.1 The parking braking system shall, even if it is combined with one of the other braking systems, be capable of holding the laden vehicle stationary on an 18 percent up or down-gradient.
- C-2.3.2 On vehicles to which the coupling of a trailer is authorized, the parking braking system of the towing vehicle shall be capable of holding the combination of vehicles stationary on a 12 percent up or down-gradient.
- C-2.3.3 If the control is manual, the force applied to it shall not exceed 60 daN

(kg).

C-2.3.4 If it is a foot control, the force exerted on the control shall not exceed 70 daN (kg).

C-2.3.5 A parking braking system which has to be actuated several times before it attains the prescribed performance is admissible.

C-2.3.6 To check compliance with the requirement specified in 5.2.1.2.4, a Type 0 test shall be carried out with the engine disconnected at an initial test speed of 30 km/h. The mean fully developed deceleration on application of the control of the parking brake system and the deceleration immediately before the vehicle stops shall not be less than 1.5 m/s². The test shall be carried out with the laden vehicle.

The force exerted on the braking control device shall not exceed the specified values.

C-2.4 Residual Braking After Transmission Failure

C-2.4.1 The residual performance of the service braking system, in the event of failure in a part of its transmission, shall give a stopping distance not exceeding the following values and a mean fully developed deceleration not less than the following values, using a control force not exceeding 70 daN (kg), when checked by the Type 0 test with the engine disconnected from the following initial speeds for the relevant vehicle category:

Stopping Distance (m) and Mean Fully Developed Deceleration (d _m) [m/s ²]					
Vehicle Category	v (km/h)	LADEN		UNLADEN	
		Stopping distance (m)	d _m (m/s ²)	Stopping distance (m)	d _m (m/s ²)
M2	60	0.15v + (100/30).(v ² /130)	1.5	0.15v + (100/25).(v ² /130)	1.3
M3	60	0.15v + (100/30).(v ² /130)	1.5	0.15v + (100/30).(v ² /130)	1.5
N1	70	0.15v + (100/30).(v ² /115)	1.3	0.15v + (100/25).(v ² /115)	1.1
N2	50	0.15v + (100/30).(v ² /115)	1.3	0.15v + (100/25).(v ² /115)	1.1
N3	40	0.15v + (100/30).(v ² /115)	1.3	0.15v + (100/30).(v ² /115)	1.3

C-2.4.2 The residual braking effectiveness test shall be conducted by simulating the actual failure conditions in the service braking system.

C-3.0 PERFORMANCE OF BRAKING SYSTEMS OF VEHICLES OF CATEGORY T

C-3.1 Service Braking System

C-3.1.1 Provision Relating to Tests of Vehicles of Category T1

Where the provision of a service braking system is mandatory, the performance of the system shall meet the requirements laid down for

vehicles of categories T2 and T3.

C-3.1.2 Provisions Relating to Tests of Vehicles of Categories T2 and T3

C-3.1.2.1 If the service braking system is of the continuous or semi-continuous type, the sum of the forces exerted on the periphery of the braked wheels shall be at least x percent of the maximum stationary wheel load, x having the following values:

x [%]

Full trailer, laden and unladen:50

Semi-trailer, laden and unladen:45

Centre-axle trailer, laden and unladen:50

C-3.1.2.2 If the trailer is fitted with a compressed-air braking system, the pressure in the supply line shall not exceed 700 kPa (7 bar) during the brake test and the signal value in the control line shall not exceed the following values, depending on the installation:

a) 650 kPa (6.5 bar) in the pneumatic control line; and

b) A digital demand value corresponding to 650 kPa (6.5 bar) as defined in ISO 11992:2003 including ISO 11992-2:2003 and its Amd 1:2007 in the electric control line.

The test speed is 60 km/h. A supplementary test at 40 km/h shall be carried out with the laden trailer for comparison with the Type I test result.

C-3.1.2.3 If the braking system is of the inertia type, it shall comply with the requirements of Annex L.

C-3.1.2.4 In addition, the vehicles shall undergo the Type I test or alternatively a Type III test in the case of T3 Trailer.

C-3.1.2.5 In the Type I or the Type III test of a semi-trailer, the mass braked by the latter's axle(s) shall correspond to the maximum axle load(s) (not including the king pin load).

C-3.1.3 Provisions Relating to Tests of Vehicles of Category T4

C-3.1.3.1 If the service braking system is of the continuous or semi-continuous type, the sum of the forces exerted on the periphery of the braked wheels shall be at least x percent of the maximum stationary wheel load, x having the following values:

x [%]

Full trailer, laden and unladen:50

Semi-trailer, laden and unladen:45

Centre-axle trailer, laden and unladen:50

C-3.1.3.2 If the trailer is fitted with a compressed air braking system, the pressure in the control line shall not exceed 650 kPa (6.5 bar) and the pressure in the

supply line shall not exceed 700 kPa (7 bar) during the brake test. The test speed is 60 km/h.

C-3.1.3.3 In addition, the vehicles shall undergo the Type III test.

C-3.1.3.4 In the Type III test of a semi-trailer, the mass braked by the latter's axle(s) shall correspond to the maximum axle load(s).

C-3.2 Parking Braking System

C-3.2.1 The parking braking system with which the trailer is equipped shall be capable of holding the laden trailer stationary, when separated from the towing vehicle, on an 18 percent up or down-gradient. The force applied to the control device shall not exceed 60 daN (kg).

C-3.3 Automatic Braking System

C-3.3.1 The automatic braking performance in the event of a failure, as described in 5.2.1.18.3, when testing the laden vehicle from a speed of 40 km/h, shall not be less than 13.5 percent of the maximum stationary wheel load. Wheel-locking at performance levels above 13.5 percent is permitted.

C-4.0 RESPONSE TIME

C-4.1 Where a vehicle is equipped with a service braking system which is totally or partially dependent on a source of energy other than the muscular effort of the driver, the following requirements shall be satisfied.

C-4.1.1 In an emergency manoeuvre, the time elapsing between the moment when the control device begins to be actuated and the moment when the braking force on the least favourably placed axle reaches the level corresponding to the prescribed performance shall not exceed 0.6 s.

C-4.1.2 In the case of vehicles fitted with compressed-air braking systems, the requirements of C-4.1.1 are considered to be satisfied if the vehicle complies with the provisions of Annex E.

C-4.1.3 In the case of vehicles fitted with hydraulic braking systems, the requirements of C-4.1.1 are considered to be satisfied if in an emergency manoeuvre, the deceleration of the vehicle or the pressure at the least favourable brake cylinder, reaches a level corresponding to the prescribed performance within 0.6 s.

C-5.0 PROCEDURE FOR MONITORING THE STATE OF BATTERY CHARGE

This procedure is applicable to vehicle batteries used for traction and regenerative braking. The procedure requires the use of a bi-directional dc watt-hour meter.

C-5.1 Procedure

C-5.1.1 If the batteries are new or have been subject to extended storage, they shall be cycled as recommended by the manufacturer. A minimum 8 h soak period at ambient temperature shall be allowed after completion of cycling.

C-5.1.2 A full charge shall be established using the manufacturer's recommended

charging procedure.

- C-5.1.3 When the braking tests of C-1.2.11, C-1.4.1.2.2, C-1.5.1.6, and C-1.5.3.1.3 are conducted the watt-hours consumed by the traction motors and supplied by the regenerative braking system shall be recorded as a running total which shall then be used to determine the state of charge existing at the beginning or end of a particular test.
- C-5.1.4 To replicate a level of state of charge in the batteries for comparative tests, such as those of C-1.5.3.1.3, the batteries shall be either recharged to that level or charged to above that level and discharged into a fixed load at approximately constant power until the required state of charge is reached. Alternatively, for vehicles with battery powered electric traction only, the state of charge may be adjusted by running the vehicle. Tests conducted with a battery partially charged at their start shall be commenced as soon as possible after the desired state of charge has been reached.

ANNEX D

(Clauses 6 and C-1.8.1.3)

ADDITIONAL PROVISIONS APPLICABLE TO CERTAIN VEHICLES MEANT FOR CARRIAGE OF DANGEROUS AND HAZARDOUS GOODS

D-1.0 SCOPE

This Annex applies to certain vehicles meant for the carriage of dangerous and hazardous goods vehicles with respect to specific requirements concerning anti-lock and endurance braking performance.

D-2.0 REQUIREMENTS

D-2.1 General Provisions

Power-driven vehicles and trailers intended for use as transport units for dangerous goods shall fulfil all relevant technical requirements of this standard. In addition, the following technical provisions shall apply as appropriate.

D-2.2 Anti-Lock Braking System of Trailers

Trailers of category T4 shall be equipped with category 1 anti-lock systems as defined in Annex M.

D-2.3 Endurance Braking System

D-2.3.1 Power-driven vehicles having a maximum mass exceeding 16 t, or authorized to tow a trailer of category T4 shall be fitted with an endurance braking system according to 3.15 which complies with the following requirements:

- a) The endurance braking control configurations shall be from a type described in 3.15.2.1 to 3.15.2.3;
- b) In the case of an electrical failure of the anti-lock system, integrated or combined endurance braking systems shall be switched off automatically;
- c) The effectiveness of the endurance braking system shall be controlled by the anti-lock braking system such that the axle(s) braked by the endurance braking system cannot be locked by that system at speeds above 15 km/h. However, this requirement shall not apply to that part of the braking system constituted by the natural engine braking;
- d) The endurance braking system shall comprise several stages of effectiveness, including a low stage appropriate for the unladen condition. Where the endurance braking system of a power-driven vehicle is constituted by its engine, the different gear ratios shall be considered to provide the different stages of effectiveness; and
- e) The performance of the endurance braking system shall be such that it fulfils the requirements of C-1.8 (Type IIA test), with a laden vehicle mass comprising the laden mass of the motor vehicle and its authorized maximum towed mass but not exceeding a total of 44 t.

D-2.3.2 If a trailer is equipped with an endurance braking system it shall fulfil the requirements of D-2.3.1 (a to d) as appropriate.

D-2.4. Braking requirements for EX/III vehicles of categories T1 and T2

D-2.4.1. Notwithstanding the provisions of paragraph 5.2.2.9. of this Standard, EX/III vehicles, irrespective of their mass, shall be equipped with a braking system which automatically brakes the trailer to a stop if the coupling device separates while the trailer is in motion.

ANNEX E

(Clauses 5.1.4.2 and C-4.1.2)

**METHOD OF MEASURING THE RESPONSE TIME ON VEHICLES
EQUIPPED WITH COMPRESSED AIR BRAKING SYSTEMS**

E-1.0 GENERAL

- E-1.1 The response times of the service braking system shall be determined on the stationary vehicle, the pressure being measured at the intake to the cylinder of the least favourably placed brake. In the case of vehicles fitted with combined compressed air/hydraulic braking systems, the pressure

may be measured at the opening of the least favourably placed pneumatic unit. For vehicles equipped with load sensing valves, these devices shall be set in the 'laden' position.

- E-1.2 During the test, the stroke of the brake cylinders of the various axles shall be that required for brakes adjusted as closely as possible.
- E-1.3 The response times determined in accordance with the provisions of this Annex shall be rounded to the nearest tenth of a second. If the figure representing the hundredth is five or more, the response time shall be rounded up to the next higher tenth.

E-2.0 POWER DRIVEN VEHICLES

- E-2.1 At the beginning of each test, the pressure in the energy storage device shall be equal to the pressure at which the governor restores the feed to the system. In systems not equipped with a governor (for example, pressure-limited compressors) the pressure in the energy storage device at the beginning of each test shall be 90 percent of the pressure specified by the manufacturer and defined in F-1.2.2.1, used for the tests prescribed in this Annex.
- E-2.2 The response times as a function of the actuating time (t_r) shall be obtained by a succession of full actuations, beginning with the shortest possible actuating time and increasing to a time of about 0.4 s. The measured values shall be plotted on a graph.
- E-2.3 The response time to be taken into consideration for the purpose of the test is that corresponding to an actuating time of 0.2 s. This response time can be obtained from the graph by interpolation.
- E-2.4 For an actuating time of 0.2 s, the time elapsing from the initiation of the braking system control actuation to the moment when the pressure in the brake cylinder reaches 75 percent of its asymptotic value shall not exceed 0.6 s.
- E-2.5 In the case of power-driven vehicles having a pneumatic control line for trailers, in addition to the requirements of E 1.1, the response time shall be measured at the extremity of a pipe 2.5 m long with an internal diameter of 13 mm which shall be joined to the coupling head of the control line of the service braking system. During this test, a volume of $385 \text{ cm}^3 \pm 5 \text{ cm}^3$ (which is deemed to be equivalent to the volume of a pipe 2.5 m long with an internal diameter of 13 mm and under a pressure of 650 kPa) shall be connected to the coupling head of the supply line.

Tractors for semi-trailers shall be equipped with flexible pipes for making the connection to semi-trailers. The coupling heads will, therefore, be at the extremity of those flexible pipes. The length and internal diameter of the pipes shall be recorded in test report.

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In the case of an automated connector the measurement including the use of a 2.5 m pipe and $385 \text{ cm}^3 \pm 5 \text{ cm}^3$ of volume as described above shall be made, considering the connector interface as the coupling heads."

- E-2.6 The time elapsing from the initiation of brake pedal actuation to the moment when
- The pressure measured at the coupling head of the pneumatic control line; and
 - The digital demand value in the electric control line measured according to ISO 11992:2003 including ISO 11992-2:2003 and its Amd.1:2007 reaches x percent of its asymptotic, respectively final value shall not exceed the times shown in the table below:

x (percent)	t (s)
10	0.2
75	0.4

- E-2.7 In the case of power-driven vehicles authorized to tow trailers of category T3 or T4 fitted with compressed-air braking systems, in addition to the above mentioned requirements, the prescriptions in 5.2.1.18.4.1 shall be verified by conducting the following test:
- By measuring the pressure at the extremity of a pipe 2.5 m long with an internal diameter of 13 mm which shall be joined to the coupling head of the supply line,
 - By simulating a failure of the control line at the coupling head, and
 - By actuating the service braking control device in 0.2 s, as described in F-2.3.

E-3 TRAILERS

- E-3.1 The trailer's response times shall be measured without the power-driven vehicle. To replace the power-driven vehicle, it is necessary to provide a simulator to which the coupling heads of the supply line, the pneumatic control line and/or the connector of the electric control line are connected.
- E-3.2 The pressure in the supply line shall be 650 kPa (6.5 bar).
- E-3.3 The simulator for pneumatic control lines shall have the following characteristics:
- It shall have a reservoir with a capacity of 30 liters which shall be charged to a pressure of 650 kPa (6.5 bar) before each test and which

shall not be recharged during each test. At the outlet of the braking control device, the simulator shall incorporate an orifice with a diameter of from 4.0 to 4.3 mm inclusive. The volume of the pipe measured from the orifice up to and including the coupling head shall be $385 \pm 5 \text{ cm}^3$ [which is deemed to be equivalent to the volume of a pipe 2.5 m long with an internal diameter of 13 mm and under a pressure of 650 kPa (6.5 bar)]. The control line pressures referred to in F-3.3.3 shall be measured immediately downstream of the orifice.

- b) The braking system control shall be so designed that its performance in use is not affected by the tester.
- c) The simulator shall be set, e.g. through the choice of orifice in accordance with E 3.3.a in such a way that, if a reservoir of $385 \text{ cm}^3 \pm 5 \text{ cm}^3$ is joined to it, the time taken for the pressure to increase from 65 to 490 kPa (10 and 75 per cent respectively of the nominal pressure of 650 kPa) shall be $0.2 \text{ second} \pm 0.01 \text{ second}$. If a reservoir of $1,155 \text{ cm}^3 \pm 15 \text{ cm}^3$ is substituted for the above-mentioned reservoir, the time taken for the pressure to increase from 65 to 490 kPa without further adjustment shall be $0.38 \text{ second} \pm 0.02 \text{ second}$. Between these two pressure values, the pressure shall increase in an approximately linear way. These reservoirs shall be connected to the coupling head without using flexible pipes. The connection between the reservoirs and the coupling head shall have an internal diameter of not less than 10 mm. The setting shall be carried out using a coupling head arrangement that is representative of the type fitted to the trailer for which type approval is sought.
- d) Figures 3 to 5 (see E-4) give an example of the correct configuration of the simulator for setting and use.

E-3.4

The simulator for checking the response to signals transmitted via the electric control line shall have the following characteristics:

- a) The simulator shall produce a digital demand signal in the electric control line according to ISO 11992-2:2003 and its Amd.1:2007 and shall provide the appropriate information to the trailer via pins 6 and 7 of the ISO 7638:2003 (Part 1 or 2) connector. For the purpose of response time measurement the simulator may at the manufacturer's request transmit to the trailer information that no pneumatic control line is present and that the electric control line demand signal is generated from two independent circuits (see 6.4.2.2.24 and 6.4.2.2.25 of ISO 11992-2:2003 and its Amd.1:2007).
- b) The braking system control shall be so designed that its performance in use is not affected by the tester.
- c) For the purpose of response time measurement the signal produced by the electric simulator shall be equivalent to a linear pneumatic pressure increase from 0 to 650 kPa in 0.2 ± 0.01 s.
- d) The diagrams in E-4 give an example of the correct configuration of the simulator for setting and use.

E-3.5 Performance Requirements

E-3.5.1 For trailers with a pneumatic control line the time elapsing between the moment when the pressure produced in the control line by the simulator reaches 650 kPa and the moment when the pressure in the brake actuator of the trailer reaches 75 percent of its asymptotic value shall not exceed 0.4 s.

E-3.5.1.1 Trailers equipped with a pneumatic control line and having electric control transmission shall be checked with the electrical power supplied to the trailer via ISO 7638:2003 connector (5 or 7 pin).

E-3.5.2 For trailers with an electric control line the time elapsing between the moment when the signal produced by the simulator exceeds the equivalent of 65 kPa (0.65 bar) and the moment when the pressure in the brake actuator of the trailer reaches 75 percent of its asymptotic value shall not exceed 0.4 s.

E-3.5.3 In the case of trailers equipped with a pneumatic and an electric control line, the response time measurement for each control line shall be determined independently according to the relevant procedure defined above.

E-4.0 EXAMPLE OF A SIMULATOR

E-4.1 Setting the Simulator (see Fig. 3)

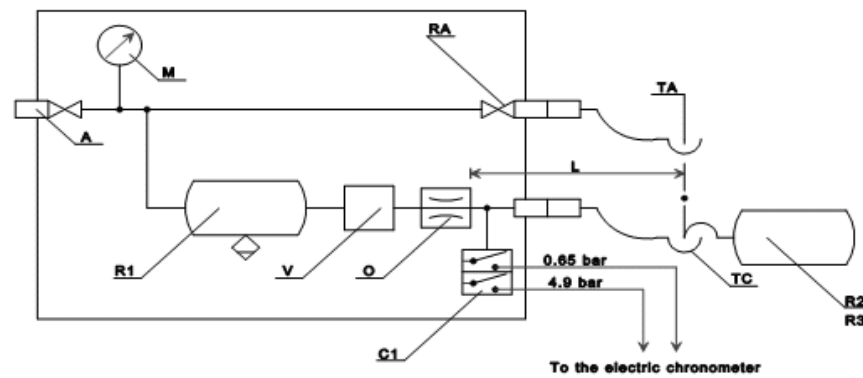


FIG. 3 SETTING THE SIMULATOR

E-4.2 Testing the Trailer (see Fig. 4)

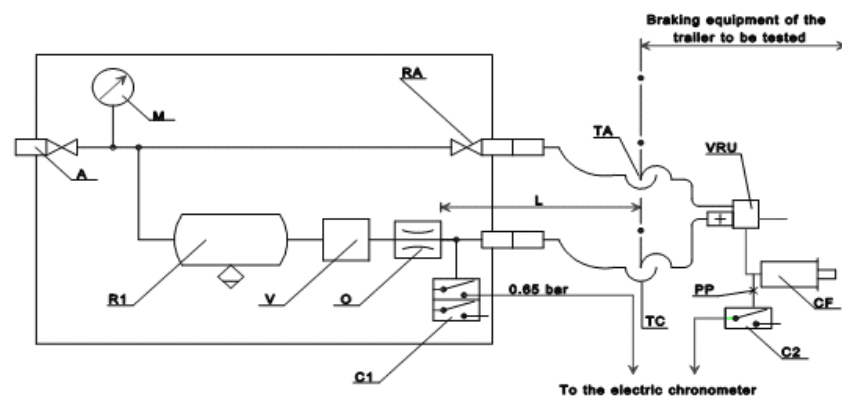


FIG. 4 TESTING THE TRAILER

E-4.3 Example of a Simulator for Electric Control Lines (see Fig. 5)

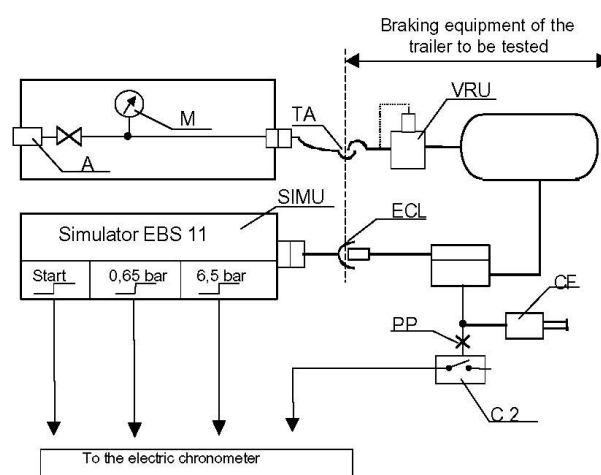


FIG. 5 EXAMPLE OF A SIMULATOR FOR ELECTRICAL CONTROL LINES

ECL	=	Electric control line corresponding to ISO 7638 (Part 1 or 2);
SIMU	=	Simulator of byte 3, 4 of EBS 11 according to ISO 11992 with output signals at start, 65 kPa (0.65 bar) and 650 kPa (6.5 bar);
A	=	Supply connection with shut-off valve;
C1	=	Pressure switch in the simulator, set at 65 kPa (0.65 bar) and at 490 kPa (4.9 bar);
C2	=	Pressure switch to be connected to the brake actuator of the trailer, to operate at 75 percent of the asymptotic pressure in the brake actuator CF;
CF	=	Brake cylinder;
L	=	Line from orifice O up to and including its coupling head TC, having an inner volume of $385 \pm 5 \text{ cm}^3$ under a pressure of 650 kPa (6.5 bar);
M	=	Pressure gauge;
O	=	Orifice with a diameter of not less than 4 mm and not more than 4.3 mm;
PP	=	Pressure test connection;
R1	=	30-litre air reservoir with drain valve;
R2	=	Calibrating reservoir, including its coupling head TC, to be $385 \pm 5 \text{ cm}^3$;
R3	=	Calibrating reservoir, including its coupling head TC, to be $1155 \pm 15 \text{ cm}^3$;
RA	=	Shut-off valve;
TA	=	Coupling head, supply line;
TC	=	Coupling head, control line;
V	=	Braking system control device; and
VRU	=	Emergency relay valve

ANNEX F

(Clauses 5.1.4.2, 5.2.1.5.2, 5.2.1.5.3, 5.2.1.13.1.1, 5.2.1.27.5, 5.2.1.27.6,
E-2.1, E-2.7 and E-3.3)

PROVISIONS RELATING TO ENERGY SOURCES AND ENERGY STORAGE DEVICES (ENERGY ACCUMULATORS)

SECTION A COMPRESSED AIR BRAKING SYSTEMS

F-1.0 CAPACITY OF ENERGY STORAGE DEVICES (ENERGY RESERVOIRS)

F-1.1 General

F-1.1.1 Vehicles on which the operation of the braking system requires the use of compressed-air shall be equipped with energy storage devices (energy accumulators) of a capacity meeting the requirements of F-1.2 and F-1.3.

F-1.1.2 It shall be possible to easily identify the reservoirs of the different circuits.

F-1.1.3 However, the energy storage devices shall not be required to be of a prescribed capacity if the braking system is such that in the absence of any energy reserve it is possible to achieve a braking performance at least equal to that prescribed for the secondary braking system.

F-1.1.4 In verifying compliance with the requirements of F-1.2 and F-1.3, the brakes shall be adjusted as closely as possible.

F-1.2 Power-Driven Vehicles

F-1.2.1 The energy storage devices (energy reservoirs) of power-driven vehicles shall be such that after eight full-stroke actuations of the service braking system control the pressure remaining in the energy storage device(s) shall be not less than the pressure required to obtain the specified secondary braking performance.

F-1.2.2 Testing shall be performed in conformity with the following requirements

F-1.2.2.1 The initial energy level in the energy storage device(s) shall be that specified by the manufacturer. It shall be such as to enable the prescribed performance of the service braking system to be achieved.

NOTE - The initial energy level shall be stated in the approval document.

F-1.2.2.2 The energy storage device(s) shall not be fed; in addition, any energy storage device(s) for auxiliary equipment shall be isolated.

F-1.2.2.3 In the case of power-driven vehicle to which the coupling of a trailer is authorized and with a pneumatic control line, the supply line shall be stopped and a compressed-air reservoir of 0.5 l capacity shall be connected directly to the coupling head of the pneumatic control line. Before each braking operation, the pressure in this compressed-air reservoir shall be completely eliminated. After the test referred to in F-1.2.1 the energy level supplied to the pneumatic control line shall not fall below a level equivalent to one-half the Figure obtained at the first brake application.

F-1.3 Trailers

F-1.3.1 The energy storage devices (energy reservoirs) with which trailers are equipped shall be such that, after eight full-stroke actuations of the towing vehicle's service braking system, the energy level supplied to the operating members using the energy does not fall below a level equivalent to one-half of the Figure obtained at the first brake application and without actuating either the automatic or the parking braking system of the trailer.

F-1.3.2 Testing shall be performed in conformity with the following requirements:

- a) The pressure in the energy storage devices at the beginning of each test shall be 850 kPa (8.5 bar);
- b) The supply line shall be stopped; in addition, any energy storage device(s) for auxiliary equipment shall be isolated;
- c) The energy storage devices shall not be replenished during the test;
- d) At each brake application, the pressure in the pneumatic control line shall be 750 kPa (7.5 bar); and
- e) At each brake application, the digital demand value in the electric control line shall be corresponding to a pressure of 750 kPa (7.5 bar).

F-2.0 CAPACITY OF ENERGY SOURCES

F-2.1 General

The compressors shall meet the requirements set forth in the following paragraphs.

F-2.2 Definitions

F-2.2.1 'p₁' is the pressure corresponding to 65 percent of the pressure 'p₂' defined in F-2.2.2.

F-2.2.2 'p₂' is the value specified by the manufacturer and referred to in F-1.2.2.1.

F-2.2.3 't₁' is the time required for the relative pressure to rise from 0 to p₁, and 'T₂' is the time required for the relative pressure to rise from 0 to p₂.

F-2.3 Conditions of Measurement

F-2.3.1 In all cases, the speed of the compressor shall be that obtained when the engine is running at the speed corresponding to its maximum power or at the speed allowed by the governor.

F-2.3.2 During the tests to determine the time t₁ and the time t₂, the energy storage device(s) for auxiliary equipment shall be isolated.

F-2.3.3 If it is intended to attach a trailer to a power-driven vehicle, the trailer shall be represented by an energy storage device whose maximum relative pressure p (expressed in kPa / 100) is that which can be supplied through the towing vehicle's supply circuit and whose volume V, expressed in

litres, is given by the formula $p \times V = 20 R$ (R being the permissible maximum mass, in tonnes, on the axles of the trailer).

F-2.4 Interpretation of Results

F-2.4.1 The time t_1 recorded for the least favoured energy storage device shall not exceed:

F-2.4.1.1 Three minutes in the case of vehicles to which the coupling of a trailer is not authorized; or

F-2.4.1.2 Six minutes in the case of vehicles to which the coupling of a trailer is authorized.

F-2.4.2 The time t_2 recorded for the least favoured energy storage device shall not exceed:

F-2.4.2.1 Six minutes in the case of vehicles to which the coupling of a trailer is not authorized; or

F-2.4.2.2 Nine minutes in the case of vehicles to which the coupling of a trailer is authorized.

F-2.5 Additional Test

F-2.5.1 If the power-driven vehicle is equipped with one or more energy storage devices for auxiliary equipment having a total capacity exceeding 20 percent of the total capacity of the braking energy storage devices, an additional test shall be performed during which no irregularity shall occur in the operation of the valves controlling the filling of the energy storage device(s) for auxiliary equipment.

F-2.5.2 It shall be verified during the aforesaid test that the time t_3 necessary to raise the pressure from 0 to p_2 in the least favoured braking energy storage device is less than.

F-2.5.2.1 Eight minutes in the case of vehicles to which the coupling of a trailer is not authorized; or

F-2.5.2.2 Eleven minutes in the case of vehicles to which the coupling of a trailer is authorized.

F-2.5.3 The test shall be performed in the conditions prescribed in F-2.3.1 and F-2.3.3.

F-2.6 Towing Vehicles

F-2.6.1 Power driven vehicles to which the coupling of a trailer is authorized shall also comply with the above requirements for vehicles not so authorized. In that case, the tests in F-2.4.1 and F-2.4.2 (and F-2.5.2) shall be conducted without the energy storage device mentioned in F-2.3.3.

SECTION B VACUUM BRAKING SYSTEMS

F-3.0 CAPACITY OF ENERGY STORAGE DEVICES (ENERGY RESERVOIRS)

F-3.1 General

F-3.1.1 Vehicles on which operation of the braking system requires the use of a vacuum shall be equipped with energy storage devices (energy reservoirs) of a capacity meeting the requirements of F-3.2 and F-3.3.

F-3.1.2 However, the energy storage devices shall not be required to be of a prescribed capacity if the braking system is such that in the absence of any energy reserve it is possible to achieve a braking performance at least equal to that prescribed for the secondary braking system.

F-3.1.3 In verifying compliance with the requirements of F-3.2 and F-3.3, the brakes shall be adjusted as closely as possible.

F-3.2 Power Driven Vehicles

F-3.2.1 The energy storage devices (energy reservoirs) of power-driven vehicles shall be such that it is still possible to achieve the performance prescribed for the secondary braking system:

- a) After eight full stroke actuations of the service braking system control where the energy source is a vacuum pump; and
- b) After four full stroke actuations of the service brake control where the energy source is the engine.

F-3.2.2 Testing shall be performed in conformity with the following requirements:

- a) The initial energy level in the energy storage device(s) shall be that specified by the manufacturer. It shall be such as to enable the prescribed performance of the service braking system to be achieved and shall correspond to a vacuum not exceeding 90 percent of the maximum vacuum furnished by the energy source.

NOTE – The initial energy level shall be stated in the approval document.

- a) The energy storage device(s) shall not be fed; in addition any energy storage device(s) for auxiliary equipment shall be isolated.
- b) In the case of a power-driven vehicle authorized to tow a trailer, the supply line shall be stopped and an energy storage device of 0.5 litre capacity shall be connected to the control line. After the test referred to in F-3.2.1 the vacuum level provided at the control line shall not have fallen below a level equivalent to one-half of the figure obtained at the first brake application.

F-3.3 Trailers (Categories T1 and T2 Only)

F-3.3.1 The energy storage devices (energy reservoirs) with which trailers are equipped shall be such that the vacuum level provided at the user points shall not have fallen below a level equivalent to one-half of the value obtained at the first brake application after a test comprising four full-stroke actuations of the trailer's service braking system.

F-3.3.2 Testing shall be performed in conformity with the following requirements:

- a) The initial energy level in the energy storage device(s) shall be that specified by the manufacturer. It shall be such as to enable the prescribed performance of the service braking system to be achieved.

NOTE – The initial energy level shall be stated in the approval document.

- a) The energy storage device(s) shall not be fed; in addition, any energy storage device(s) for auxiliary equipment shall be isolated.

F-4.0 CAPACITY OF ENERGY SOURCES

F-4.1 General

Starting from the ambient atmospheric pressure, the energy source shall be capable of achieving in the energy storage device(s), in 3 min, the initial level specified in F-3.2.2 (a). In the case of a power-driven vehicle to which the coupling of a trailer is authorized, the time taken to achieve that level in the conditions specified in F-4.2 shall not exceed 6 min.

F-4.2 Conditions of Measurement

F-4.2.1 The speed of the vacuum source shall be,

- a) Where the vacuum source is the vehicle engine, the engine speed obtained with the vehicle stationary, the neutral gear engaged and the engine idling;
- b) Where the vacuum source is a pump, the speed obtained with the engine running at 65 percent of the speed corresponding to its maximum power output; and
- c) Where the vacuum source is a pump and the engine is equipped with a governor, the speed obtained with the engine running at 65 percent of the maximum speed allowed by the governor.

F-4.2.2 Where it is intended to couple to the power-driven vehicle a trailer whose service braking system is vacuum-operated, the trailer shall be represented by an energy storage device having a capacity V in litres determined by the formula $V = 15 R$, where R is the maximum permissible mass, in tonnes, on the axles of the trailer.

SECTION C

HYDRAULIC BRAKING SYSTEMS WITH STORED ENERGY

- F-5.0 CAPACITY OF ENERGY STORAGE DEVICES (ENERGY RESERVOIRS)**
- F-5.1 General**
- F-5.1.1 Vehicles on which operation of the braking system requires the use of stored energy provided by hydraulic fluid under pressure shall be equipped with energy storage devices (energy reservoirs) of a capacity meeting the requirements of F-5.2.
- F-5.1.2 However, the energy storage devices shall not be required to be of a prescribed capacity, if the braking system is such that in the absence of any energy reserve it is possible with the service braking system control to achieve a braking performance at least equal to that prescribed for the secondary braking system.
- F-5.1.3 In verifying compliance with the requirements of F-5.2.1, F-5.2.2 and F-6.1, the brakes shall be adjusted as closely as possible and for F-5.2.1, the rate of full-stroke actuations shall be such as to provide an interval of at least 60 s between each actuation.
- F-5.2 Power-Driven Vehicles**
- F-5.2.1 Power-driven vehicles equipped with a hydraulic braking system with stored energy shall meet the following requirements:
- a) After eight full-stroke actuations of the service braking system control, it shall still be possible to achieve, on the ninth application, the performance prescribed for the secondary braking system; and
 - b) Testing shall be performed in conformity with the following requirements:
 - 1) Testing shall commence at a pressure that may be specified by the manufacturer but is not higher than the cut-in pressure; and
 - 2) The energy storage device(s) shall not be fed; in addition, any energy storage device(s) for auxiliary equipment shall be isolated.
- F-5.2.2 Power-driven vehicles equipped with a hydraulic braking system with stored energy which cannot meet the requirements of 5.2.1.5.1 shall be deemed to satisfy if the following requirements are met:
- a) After any single transmission failure it shall still be possible after eight full-stroke actuations of the service braking system control, to achieve, at the ninth application, at least the performance prescribed for the secondary braking system or, where secondary performance requiring the use of stored energy is achieved by a separate control, it shall still be possible after eight full-stroke actuations to achieve, at the ninth application, the residual performance prescribed in 5.2.1.4.

b) Testing shall be performed in conformity with the following requirements:

- 1) With the energy source stationary or operating at a speed corresponding to the engine idling speed, any transmission failure may be induced. Before inducing such a failure, the energy storage device(s) shall be at a pressure that may be specified by the manufacturer but not exceeding the cut-in pressure; and
- 2) The auxiliary equipment and its energy storage devices, if any, shall be isolated.

F-6.0 CAPACITY OF HYDRAULIC FLUID ENERGY SOURCES

F-6.1 The energy sources shall meet the requirements set out in the following paragraphs:

F-6.1.1 Definitions

F-6.1.1.1 ‘**p₁**’ represents the maximum system operational pressure (cut-out pressure) in the energy storage device(s) specified by the manufacturer.

F-6.1.1.2 ‘**p₂**’ represents the pressure after four full-stroke actuations with the service braking system control, starting at p₁, without having fed the energy storage device(s).

F-6.1.1.3 ‘**t**’ represents the time required for the pressure to rise from p₂ to p₁ in the energy storage device(s) without application of the service braking system control.

F-6.1.2 Conditions of Measurement

F-6.1.2.1 During the test to determine the time t, the feed rate of the energy source shall be that obtained when the engine is running at the speed corresponding to its maximum power or at the speed allowed by the over-speed governor.

F-6.1.2.2 During the test to determine the time t, energy storage device(s) for auxiliary equipment shall not be isolated other than automatically.

F-6.1.3 Interpretation of Results

F-6.1.3.1 In the case of all vehicles except those of categories M3, N2 and N3, the time t shall not exceed 20 s.

F-6.1.3.2 In the case of vehicles of categories M3, N2 and N3, the time t shall not exceed 30 s.

F-7.0 CHARACTERISTICS OF WARNING DEVICES

With the engine stationary and commencing at a pressure that may be specified by the manufacturer but does not exceed the cut-in pressure, the warning device shall not operate following two full-stroke actuations of the service braking system control.

ANNEX G

(Clause 6)

**PROVISIONS RELATING TO SPECIFIC CONDITIONS
FOR SPRING BRAKING SYSTEMS**

G-1 DEFINITIONS

- G-1.1 Spring Braking Systems – Braking systems in which the energy required for braking is supplied by one or more springs acting as an energy storage device (energy accumulator).
- G-1.1.1 The energy necessary to compress the spring in order to release the brake is supplied and controlled by the ‘control’ actuated by the driver (see 3.4).
- G-1.2 Spring Compression Chamber – The chamber where the pressure variation that induces the compression of the spring is actually produced.
- G-1.3 Pressure – If the compression of the springs is obtained by means of a vacuum device, pressure shall mean negative pressure everywhere in this Annex.

G-2 GENERAL

- G-2.1 A spring braking system shall not be used as a service braking system. However, in the event of a failure in a part of the transmission of the service braking system, a spring braking system may be used to achieve the residual performance prescribed in 5.2.1.4 provided that the driver can graduate this action. In the case of power-driven vehicles, with the exception of tractors for semi-trailers meeting the requirements specified in 5.2.1.4.1, the spring braking system shall not be the sole source of residual braking. Vacuum spring braking systems shall not be used for trailers.
- G-2.2 A small variation in any of the pressure limits which may occur in the spring compression chamber feed circuit shall not cause a significant variation in the braking force.
- G-2.3 The following requirements shall apply to power-driven vehicles equipped with spring brakes:
- a) The feed circuit to the spring compression chamber shall either include an own energy reserve or shall be fed from at least two independent energy reserves. The trailer supply line may be branched from this feed line under the condition that a pressure drop in the trailer supply line shall not be able to apply the spring brake actuators.
 - b) Auxiliary equipment may only draw its energy from the feed line for the spring brake actuators under the condition that its operation, even in the event of damage to the energy source, cannot cause the energy reserve for the spring brake actuators to fall below a level from which one release of the spring brake actuators is possible.

- c) In any case, during re-charging of the braking system from zero pressure, the spring brakes shall remain fully applied irrespective of the position of the control device until the pressure in the service braking system is sufficient to ensure at least the prescribed secondary braking performance of the laden vehicle, using the service braking system control.
- d) Once applied, the spring brakes shall not release unless there is sufficient pressure in the service braking system to at least provide the prescribed residual braking performance of the laden vehicle by application of the service braking control.

G-2.4 In power-driven vehicles, the system shall be so designed that it is possible to apply and release the brakes at least three times if the initial pressure in the spring compression chamber is equal to the maximum design pressure. In the case of trailers, it shall be possible to release the brakes at least three times after the trailer has been uncoupled, the pressure in the supply line being 750 kPa (7.5 bar) before the uncoupling. However, prior to the check the emergency brake shall be released. These conditions shall be satisfied when the brakes are adjusted as closely as possible. In addition, it shall be possible to apply and release the parking braking system as specified in 5.2.2.10 when the trailer is coupled to the towing vehicle.

G-2.5 For power-driven vehicles, the pressure in the spring compression chamber beyond which the springs begin to actuate the brakes, the latter being adjusted as closely as possible, shall not be greater than 80 percent of the minimum level of the normal available pressure.

In the case of trailers, the pressure in the spring compression chamber beyond which the springs begin to actuate the brakes shall not be greater than that obtained after four full-stroke actuations of the service braking system in accordance with F-1.3. The initial pressure is fixed at 700 kPa.

G-2.6 When the pressure in the line feeding energy to the spring compression chamber - excluding lines of an auxiliary release device using a fluid under pressure falls to the level at which the brake parts begin to move, an optical or audible warning device shall be actuated. Provided this requirement is met, the warning device may comprise the red warning signal specified in 5.2.1.29.1.1. This provision does not apply to trailers.

G-2.7 If a power-driven vehicle authorized to tow a trailer with a continuous or semi-continuous braking system is fitted with a spring braking system, automatic application of the said system shall cause application of the trailer's brakes.

G-2.8 Trailers which utilise the service braking system energy reserves to fulfil the requirements for the automatic brake as defined in C-3.3 shall also fulfil one of the following requirements when the trailer is uncoupled from the towing vehicle and the trailer park brake control is in the released position (spring brakes not applied):

- (a) When the energy reserves of the service braking system reduce to a pressure no lower than 280 kPa the pressure in the spring brake

compression chamber shall reduce to 0 kPa to fully apply the spring brakes. This requirement shall be verified with a constant service braking system energy reserve pressure of 280 kPa;

- (b) A reduction in the pressure within the service braking system energy reserve results in a corresponding reduction in the pressure in the spring compression chamber.

G-3.0 AUXILIARY RELEASE SYSTEM

- G-3.1 A spring braking system shall be so designed that, in the event of a failure in that system, it is still possible to release the brakes. This may be achieved by the use of an auxiliary release device (pneumatic, mechanical, etc.).

Auxiliary release devices using an energy reserve for releasing shall draw their energy from an energy reserve which is independent from the energy reserve normally used for the spring braking system. The pneumatic or hydraulic fluid in such an auxiliary release device may act on the same piston surface in the spring compression chamber which is used for the normal spring braking system under the condition that the auxiliary release device uses a separate line. The junction of this line with the normal line connecting the control device with the spring brake actuators shall be at each spring brake actuator immediately before the port to the spring compression chamber, if not integrated in the body of the actuator. This junction shall include a device which prevents an influence of one line on the other. The requirements of 5.2.1.6 also apply to this device.

- G-3.1.1 For the purposes of the requirement of G-3.1, components of the braking system transmission shall not be regarded as subject to failure, if under the terms of 5.2.1.2.7 they are not regarded as liable to breakage, provided that they are made of metal or of a material having similar characteristics and do not undergo significant distortion in normal braking.
- G-3.2 If the operation of the auxiliary device referred to in G-3.1 requires the use of a tool or spanner, the tool or spanner shall be kept on the vehicle.
- G-3.3 Where an auxiliary release system utilizes stored energy to release the spring brakes the following additional requirements shall apply:
 - G-3.3.1 Where the control of the auxiliary spring brake release system is the same as that used for the secondary/parking brake, the requirements defined in G-2.3 shall apply in all cases.
 - G-3.3.2 Where the control for the auxiliary spring brake release system is separate to the secondary/parking brake control, the requirements defined in G-2.3 shall apply to both control systems. However, the requirements of G-2.3 (d) shall not apply to the auxiliary spring brake release system. In addition the auxiliary release control shall be located so that it is protected against application by the driver from the normal driving position.
- G-3.4 If compressed air is used in the auxiliary release system, the system should be activated by a separate control, not connected to the spring brake control.

ANNEX H
(Clause 6)

**PROVISIONS RELATING TO PARKING BRAKING SYSTEMS EQUIPPED
WITH A MECHANICAL BRAKE-CYLINDER LOCKING DEVICE (LOCK
ACTUATORS)**

H-1.0 DEFINITION

- H-1.1 Mechanical Brake-Cylinder Locking Device – A device which ensures braking operation of the parking braking system by mechanically locking the brake piston rod. Mechanical locking is effected by exhausting the compressed fluid held in the locking chamber; it is so designed that unlocking can be effected by restoring pressure in the locking chamber.

H-2.0 SPECIAL REQUIREMENTS

- H-2.1 When the pressure in the locking chamber approaches the level at which mechanical locking occurs, an optical or audible warning device shall come into action. Provided this requirement is met, the warning device may comprise the red warning signal specified in 5.2.1.29.1.1. This provision shall not apply to trailers.

In the case of trailers, the pressure corresponding to mechanical locking shall not exceed 400 kPa (4 bar). It shall be possible to achieve parking braking performance after any single failure of the trailer service braking system. In addition, it shall be possible to release the brakes at least three times after the trailer has been uncoupled, the pressure in the supply line being 650 kPa (6.5 bar) before the uncoupling. These conditions shall be satisfied when the brakes are adjusted as closely as possible. It shall also be possible to apply and release the parking braking system as specified in 5.2.2.10 when the trailer is coupled to the towing vehicle.

- H-2.2 In cylinders equipped with a mechanical locking device, movement of the brake piston shall be ensured by energy from either of two independent energy storage devices.
- H-2.3 It shall not be possible to release the locked brake cylinder unless it is certain that after such release the brake can be applied again.
- H-2.4 In the event of a failure of the energy source feeding the locking chamber, an auxiliary release device (for example; mechanical, or pneumatic which may use the air contained in one of the vehicle's tyres) shall be available.
- H-2.5 The control shall be such that, when actuated, it performs the following operations in sequence: it applies the brakes so as to provide the degree of efficiency required for parking braking, locks the brakes in that position and then cancels out the brake application force.

ANNEX J

(Clauses 3.15.2.2, 3.28, 5.2.1.25.2.5, 5.2.1.28.3, 5.2.1.28.4, 5.2.1.28.5, 5.2.2.14, C-1.3.2, C-1.3.2.1 and C-1.4.4.2)

DISTRIBUTION OF BRAKING AMONG THE AXLES OF VEHICLES AND REQUIREMENTS FOR COMPATIBILITY BETWEEN TOWING VEHICLES AND TRAILERS

J- 0.0 Requirement of this Annex and clauses where this Annex has been referred to are not mandatory.

NOTE – Mandatory requirements for complying with this Annex will be in accordance with decision of statutory authority.

J-1.0 GENERAL REQUIREMENTS

J-1.1 Vehicles of categories M2, M3, N, T2, T3 and T4 shall meet all the requirements of this annex. If a special device is used, this shall operate automatically (See Note). However, vehicles in the above categories which are equipped with an antilock braking system and fulfil the relevant requirements of Annex M, shall also fulfil all the relevant requirements of this annex with the following exceptions: (a) Compliance with the adhesion utilization requirements associated with diagrams 1A, 1B or 1C, as appropriate, is not required. (b) In the case of towing vehicles and trailers, equipped with a compressed air braking system, compliance with the unladen compatibility requirements associated with diagrams 2, 3 or 4, as appropriate, is not required. However, for all load conditions, a braking rate shall be developed between a pressure of 20 kPa and 100 kPa or the equivalent digital demand value at the coupling head of the control line(s).

Note: In the case of trailers with electronically controlled brake force distribution, the requirements of this annex shall only apply when the trailer is electrically connected to the towing vehicle by the ISO 7638:2003 connector

J-1.1.1 Where a vehicle is installed with an endurance braking system the retarding force produced by that system shall not be taken into consideration when determining the vehicle performance with respect to the provisions of this Annex.

J-1.2 The requirements relating to the diagrams specified in J-3.1.5, J-3.1.6, J-4.1, J-5.1 and J-5.2, are valid both for power-driven vehicles with a pneumatic control line according to 5.1.3.1.1 and for vehicles with an electric control line according to 5.1.3.1.3. In both cases, the reference value (abscissa of the diagrams) will be the value of the transmitted pressure in the control line: for vehicles equipped according to 5.1.3.1.1, this will be the actual pneumatic pressure in the control line (p_m); for vehicles equipped according to 5.1.3.1.3, this will be the pressure corresponding to the transmitted digital demand value in the electric control line, according to ISO 11992:2003 and its Amd. 1:2007. Vehicles equipped according to 5.1.3.1.2 (with both pneumatic and electric control lines) shall satisfy the requirements of the diagrams related to both control lines. However, identical braking characteristic curves related to both control lines are not required.

J-1.3 Validation of the Development of Braking Force

J-1.3.1 At the time of type approval it shall be checked that the development of a braking on an axle of each independent axle group shall be within the

following pressure ranges:

- (a) Laden vehicles: At least one axle shall commence to develop a braking force when the pressure at the coupling head is within the pressure range 20 to 100 kPa or equivalent digital demand value.

At least one axle of every other axle group shall commence to develop a braking force when the coupling head is at a pressure < 120 kPa or equivalent digital demand value.

- (b) Unladen vehicles: At least one axle shall commence to develop a braking force when the pressure at the coupling head is within the pressure range 20 to 100 kPa or equivalent digital demand value.

J-1.3.1.1 With the wheel(s) of the axle(s) raised off the ground and free to rotate, apply an increasing brake demand and measure the coupling head pressure corresponding to when the wheel(s) may no longer be rotated by hand. This condition is defined as the development of the braking force.

J-1.4 In the case of vehicles of Category T with pneumatic braking systems, when the alternative type approval procedure defined in Annex U is utilized, the relevant calculations required in this Annex shall be made using the performance characteristics obtained from the relevant Annex T verification reports and the centre of gravity height determined by the method defined in U-9.

J-2.0 SYMBOLS

I	=	Axle index (i = 1, front axle; i = 2, second axle; etc)
P _i	=	Normal reaction of road surface on axle i under static conditions
N _i	=	Normal reaction of road surface on axle i under braking
T _i	=	Force exerted by the brakes on axle i under normal braking conditions on the road
f _i	=	T _i /N _i , adhesion utilized by axle i (see Note 1)
J	=	Deceleration of vehicle
g	=	Acceleration due to gravity: g = 10 m/s ²
z	=	Braking rate of vehicle = J/g (see Note 2)
P	=	Mass of vehicle
h	=	Height above ground of centre of gravity specified by the manufacturer and agreed by the test agency conducting the approval test
E	=	Wheelbase
k	=	Theoretical coefficient of adhesion between tyre and road
K _c	=	Correction factor: semi-trailer laden

K_v	=	Correction factor: semi-trailer unladen
T_M	=	Sum of braking forces at the periphery of all wheels of towing vehicles for trailers
P_M	=	Total normal static reaction of road surface on wheels of towing vehicles for trailers (see Note 3)
p_m	=	Pressure at coupling head of control line
T_R	=	Sum of braking forces at periphery of all wheels of trailer
P_R	=	Total normal static reaction of road surface on all wheels of trailer (see Note 4)
$P_{R_{max}}$	=	Value of P_R at maximum mass of trailer
E_R	=	Distance between king-pin and centre of axle or axles of semi-trailer
h_R	=	Height above ground of centre of gravity of semi-trailer specified by the manufacturer and agreed by the test agency conducting the approval test

NOTES

- 1 Adhesion utilization curves of a vehicle means curve showing, for specified load conditions, the adhesion utilized by each axle i plotted against the braking rate of the vehicle.
- 2 For semi-trailers, z is the braking force divided by static load on the semi-trailer axle(s).
- 3 As referred to in C-1.4.4.3.
- 4 As referred to in C-1.4.4.3.

J-3.0 REQUIREMENTS FOR POWER-DRIVEN VEHICLES

J-3.1 Two-Axle Vehicles

J-3.1.1 For all categories of vehicles for k values between 0.2 and 0.8 (see Note)

$$z \geq 0.10 + 0.85 (k - 0.20)$$

NOTE – The provisions of J-3.1.1 or J-5.1.1 do not effect the requirements of Annex C relating to the braking performance. However, if, in tests made under the provisions of J-3.1.1 or J-5.1.1, braking performances are obtained which are higher than those prescribed in Annex C, the provisions relating to the adhesion utilization curves shall be applied within the areas of Figs. 6, 7 and 8 defined by the straight lines $k=0.8$ and $z=0.8$.

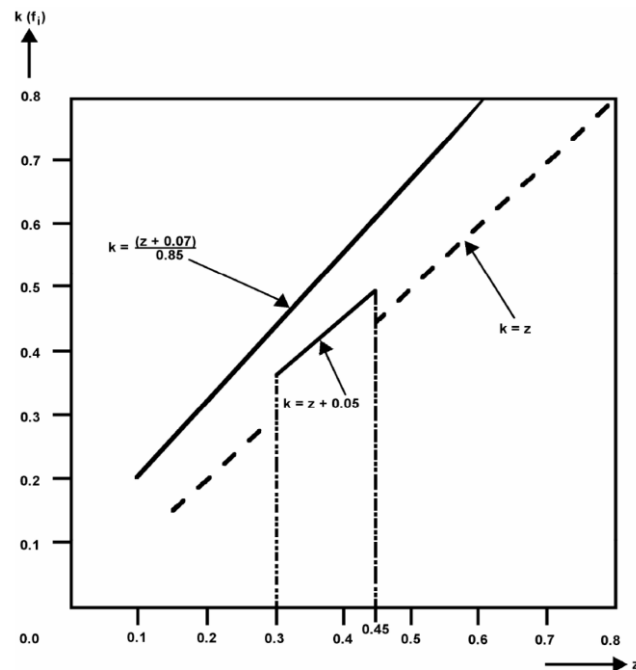
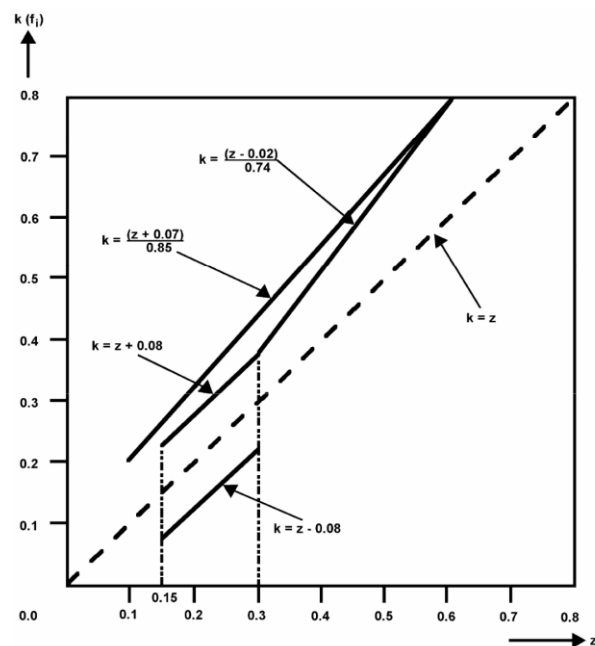


FIG. 6 ADHESION UTILIZATION CURVES FOR CERTAIN VEHICLES OF CATEGORY N1)

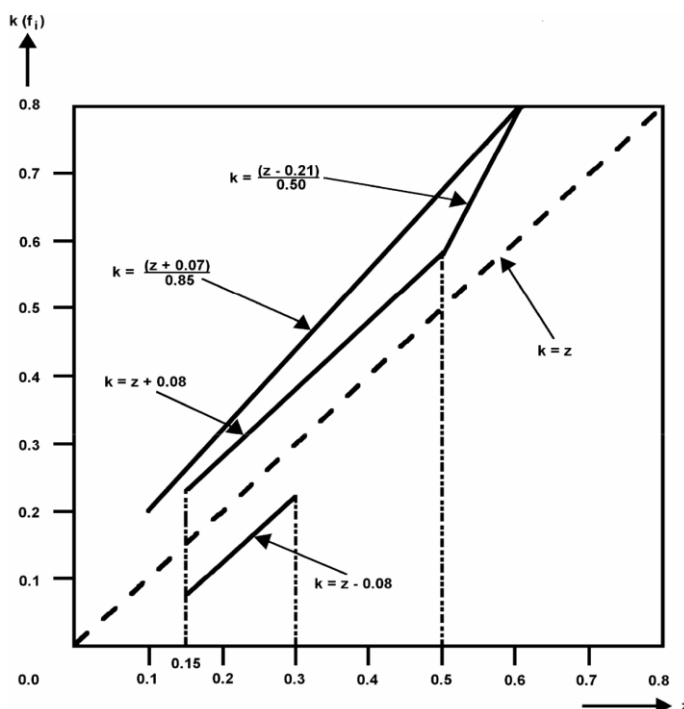
(SEE J-3.1.2.1)



NOTE – The lower limit $k = z - 0.08$ is not applicable for the adhesion utilisation of the rear axle.

FIG. 7 ADHESION UTILIZATION CURVES FOR VEHICLES OTHER THAN N1 AND FULL TRAILERS

(SEE J-3.1.2.3 AND J-5.1.1.2)



NOTE – The lower limit $k = z - 0.08$ is not applicable for the adhesion utilization of the rear axle.

FIG. 8 ADHESION UTILIZATION CURVES FOR VEHICLES OF CATEGORY N1 (WITH CERTAIN EXCEPTIONS)
(SEE J-3.1.2.2)

- J-3.1.2 For all states of load of the vehicle, the adhesion utilization curve of the rear axle shall not be situated above that for the front axle.
- J-3.1.2.1 For all braking rates between 0.15 and 0.80 in the case of vehicles of category N1 with a laden/unladen rear axle loading ratio not exceeding 1.5 or having a maximum mass of less than 2 tonnes, in the range of z values between 0.3 and 0.45, an inversion of the adhesion utilization curves is permitted provided that the adhesion utilization curve of the rear axle does not exceed by more than 0.05, the line defined by the formula $k = z$ (line of ideal adhesion utilization in Fig. 8).
- J-3.1.2.2 For all braking rates between 0.15 and 0.50 in the case of other vehicles of category N1 (see Note).

This condition is considered satisfied if, for braking rates between 0.15 and 0.30, the adhesion utilization curves for each axle are situated between two lines parallel to the line of ideal adhesion utilization given by the equation $k = z \pm 0.08$ as shown in Fig. 8 where the adhesion utilization curve for the rear axle may cross the line $k = z - 0.08$; and complies for a braking rate between 0.30 and 0.50, with the relation $z \geq k - 0.08$; and between 0.50 and 0.61 with the relation $z \geq 0.5k + 0.21$.

- J-3.1.2.3 For all braking rates between 0.15 and 0.30 in the case of vehicles of other categories.

This condition is also considered satisfied if, for braking rates between 0.15 and 0.30, the adhesion utilization curves for each axle are situated between two lines parallel to the line of ideal adhesion utilization given by the equation $k = z \pm 0.08$ as shown in Fig. 7 and the adhesion utilization curve for the rear axle for braking rates $z \geq 0.3$ complies with the relation

$$z \geq 0.3 + 0.74 (k - 0.38).$$

- J-3.1.3 In the case of a power-driven vehicle authorized to tow trailers of category T3 or T4 fitted with compressed-air braking systems.

- J-3.1.3.1 When tested with the energy source stopped, the supply line blocked off, a reservoir of 0.5 l capacity connected to the pneumatic control line, and the system at cut-in and cut-out pressures, the pressure at full application of the braking control shall be between 650 and 850 kPa at the coupling heads of the supply line and the pneumatic control line, irrespective of the load condition of the vehicle.

- J-3.1.3.2 For vehicles equipped with an electric control line; a full application of the control of the service braking system shall provide a digital demand value corresponding to a pressure between 650 and 850 kPa (see ISO 11992:2003 including ISO 11992-2:2003 and its Amd.1:2007).

- J-3.1.3.3 These values shall be demonstrably present in the power-driven vehicle when uncoupled from the trailer. The compatibility bands in the diagrams specified in J-3.1.5, J-3.1.6, J-4.1, J-5.1 and J-5.2, should not be extended beyond 750 kPa (7.5 bar) and/or the corresponding digital demand value (see ISO 11992: 2003 including ISO 11992-2:2003 and its Amd.1:2007).

- J-3.1.3.4 It shall be ensured that at the coupling head of the supply line a pressure of at least 700 kPa (7 bar) is available when the system is at cut-in pressure. This pressure shall be demonstrated without applying the service brakes.

J-3.1.4 Verification of the Requirements of J-3.1.1 and J-3.1.2

- J-3.1.4.1 In order to verify the requirements of J-3.1.1 and J-3.1.2, the manufacturer shall provide the adhesion utilization curves for the front and rear axles calculated by the formulae:

$$f_1 = \frac{T_1}{N_1} = \frac{T_1}{P_1 + z \cdot \frac{h}{E} \cdot P \cdot g}$$

$$f_2 = \frac{T_2}{N_2} = \frac{T_2}{P_2 - z \cdot \frac{h}{E} \cdot P \cdot g}$$

The curves shall be plotted for both the following load conditions:

- J-3.1.4.1.1 Unladen, in running order with the driver on board; in the case of a vehicle

presented as a bare chassis-cab, a supplementary load may be added to simulate the mass of the body, not exceeding the minimum mass declared by the manufacturer in Annex A.

- J-3.1.4.1.2 Laden; where provision is made for several possibilities of load distribution, the one whereby the front axle is the most heavily laden shall be the one considered.

- J-3.1.4.2 If it is not possible, for vehicles with (permanent) all-wheel drive, to carry out the mathematical verification pursuant to J-3.1.4.1, the manufacturer may instead verify by means of a wheel lock sequence test that, for all braking rates between 0.15 and 0.8, lockup of the front wheels occurs either simultaneously with or before the lockup of the rear wheels.

- J-3.1.4.3 Procedure to Verify the Requirements of J-3.1.4.2

- J-3.1.4.3.1 The wheel lock sequence test shall be conducted on road surfaces with a coefficient of adhesion of not more than 0.3 and of about 0.8 (dry road) from the initial test speeds specified in J-3.1.4.3.2.

- J-3.1.4.3.2 Test speeds

60 km/h, but not exceeding $0.8 v_{\max}$ for decelerations on low coefficient of friction road surfaces; 80 km/h, but not exceeding $v_{\max}$ for decelerations on high coefficient of friction road surfaces.

- J-3.1.4.3.3 The pedal force applied may exceed the permissible actuation forces pursuant to C-2.1.1.

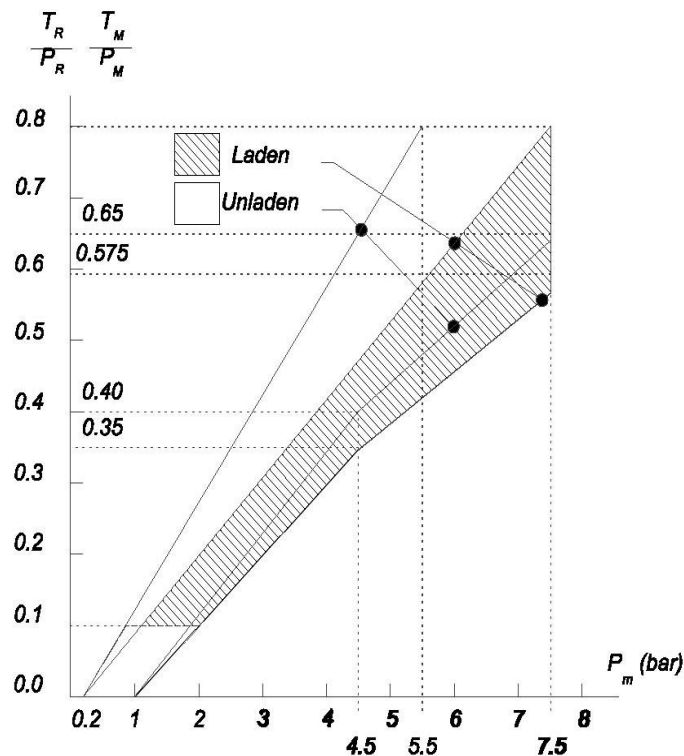
- J-3.1.4.3.4 Pedal force is applied and increased such that the second wheel on the vehicle will reach lockup between 0.5 and 1 s after initiating the brake application, until lockup of both wheels on one axle occurs (additional wheels may also lock during the test, for example; in the case of simultaneous lockup).

- J-3.1.4.4 The tests prescribed in J-3.1.4.2 shall be carried out twice on each road surface. If the result of one test fails, a third, hence decisive test shall be carried out.

- J-3.1.4.5 For vehicles fitted with an electric regenerative braking system of Category B, where the electric regenerative braking capacity is influenced by the electric state of charge, the curves shall be plotted by taking account of the electric regenerative braking component under the minimum and maximum conditions of delivered braking force. This requirement is not applicable if the vehicle is equipped with an anti-lock device which controls the wheels connected to the electric regenerative braking and shall be replaced by the requirements of Annex M.

- J-3.1.5 **Towing Vehicles Other Than Tractors for Semi-Trailers**

- J-3.1.5.1 In the case of a power-driven vehicle authorized to tow trailers of category T3 or T4 fitted with a compressed air braking system, the permissible relationship between the braking rate T_M/P_M and the pressure p_m shall lie within the areas shown on Fig. 9 for all pressures between 20 and 750 kPa.



NOTE – The relationships required by the diagram shall apply progressively for intermediate states of loading between the laden and the unladen states and shall be achieved by automatic means.

**FIG. 9 TOWING VEHICLES AND TRAILERS
(EXCEPT TRACTORS FOR SEMI-TRAILERS AND SEMI-
TRAILERS)
(SEE J-3.1.5.1)**

J-3.1.6 Tractors for Semi-Trailers

J-3.1.6.1 Tractors with unladen semi-trailer

An unladen combination is understood to be a tractor in running order, with the driver on board, coupled to an unladen semi-trailer. The dynamic load of the semi-trailer on the tractor shall be represented by a static mass P_s mounted at the fifth wheel coupling equal to 15 percent of the maximum mass on the coupling. The braking forces shall continue to be regulated between the state of the tractor with unladen semi-trailer and that of the tractor alone; the braking forces relating to the tractor alone shall be verified.

J-3.1.6.2 Tractors with laden semi-trailer

A laden combination is understood to be a tractor in running order, with the driver on board, coupled to a laden semi-trailer. The dynamic load of the semi-trailer on the tractor shall be represented by a static mass P_s mounted at the fifth wheel coupling equal to:

$$P_s = P_{s0} (1 + 0.45z)$$

Where

P_{so} represents the difference between the maximum laden mass of the tractor and its unladen mass.

For h the following value shall be taken:

$$h = \frac{h_o \cdot P_o + h_s \cdot P_s}{P}$$

where

h_o = Height of the centre of gravity of the tractor ;

h_s = Height of the coupling on which the semi-trailer rests;

P_o = Unladen mass of the tractor alone; and

$$P = P_o + P_s = \frac{P_1 + P_2}{g}$$

J-3.1.6.3 In the case of a vehicle fitted with a compressed air braking system, the permissible relationship between the braking rate T_M / P_M and the pressure p_m shall be within the areas shown on Fig. 10 for all pressures between 20 and 750 kPa.

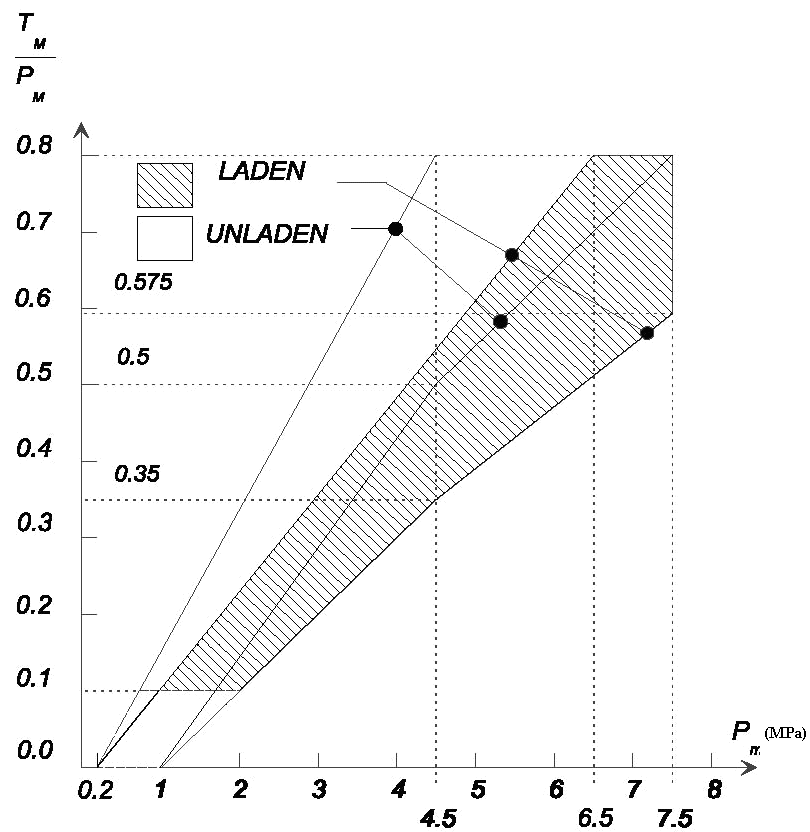
J-3.2 **Vehicles with More than Two Axles**

The requirements of J-3.1 shall apply to vehicles with more than two axles. The requirements of J-3.1.2 with respect to wheel lock sequence shall be considered to be met, if in the case of braking rates between 0.15 and 0.30, the adhesion utilized by at least one of the front axles is greater than that utilized by at least one of the rear axles.

J-4.0 **REQUIREMENTS FOR SEMI-TRAILERS**

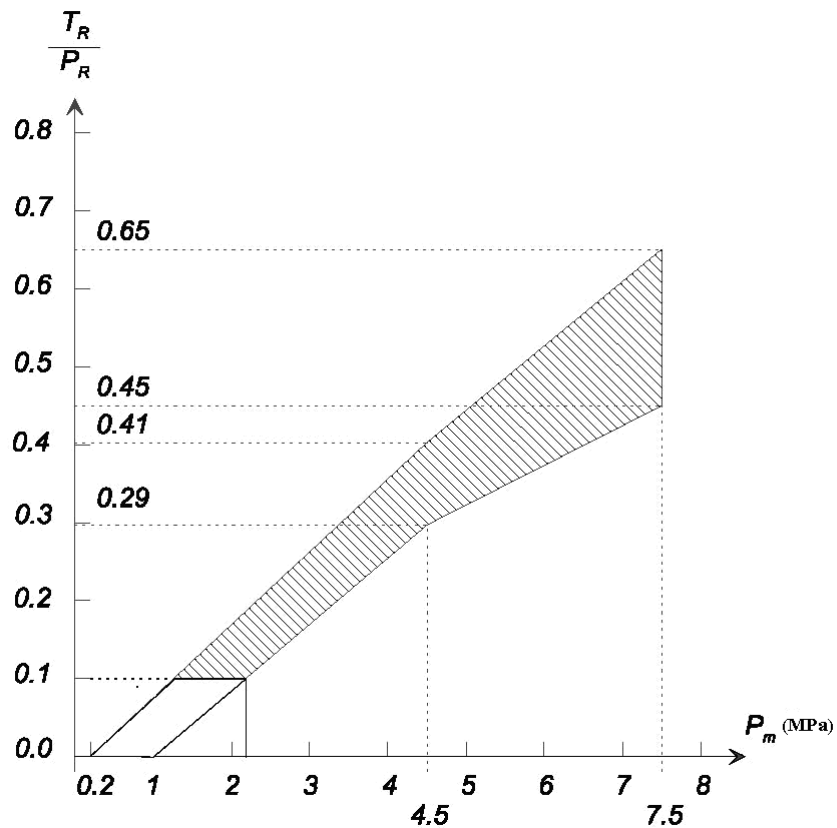
J-4.1 **For Semi-Trailers Fitted with Compressed-Air Braking Systems**

J-4.1.1 The permissible relationship between the braking rate T_R / P_R and the pressure p_m shall lie within two areas derived from Fig. 11 and Fig. 12 for all pressures between 20 and 750 kPa, in both the laden and unladen states of load. This requirement shall be met for all permissible load conditions of the semi-trailer axles.



NOTE - The relationships required by the diagram shall apply progressively for intermediate states of loading between the laden and the unladen states and shall be achieved by automatic means.

FIG. 10 TRACTORS FOR SEMI-TRAILERS
(SEE J-3.1.6)



NOTE - The relation between the braking rate T_R/P_R and the control line pressure for the laden and unladen conditions is determined as follows:

The factors K_c (laden), K_v (unladen) are obtained by reference to Fig. 12. To determine the areas corresponding to the laden and unladen conditions, the values of the ordinates of the upper and lower limits of the hatched area in Fig. 11 are multiplied by the factors K_c and K_v respectively.

FIG. 11

**RELATIONSHIP BETWEEN BRAKING RATE (T_R/P_R) AND
PRESSURE (p_m) FOR SEMI-TRAILERS**

(SEE J-4)

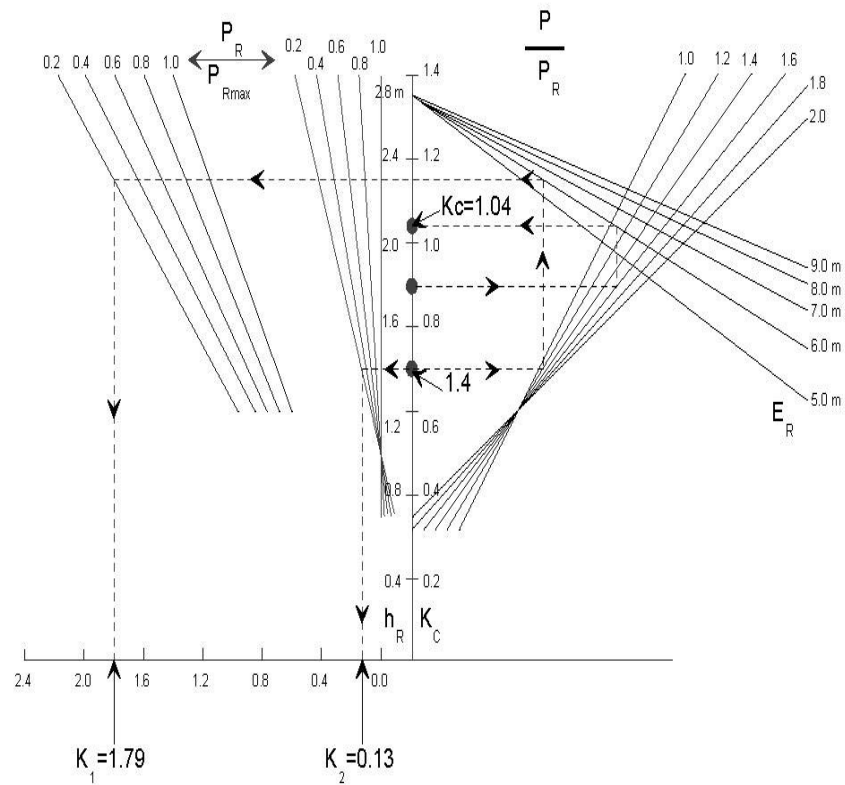


FIG. 12

FACTORS K_C (LADEN) AND K_V (UNLADEN) FOR SEMI TRAILERS

(SEE J-4 AND FIG. 11)

EXPLANATORY NOTES ON THE USE OF FIG. 12

1.0 Formula from which Fig. 12 is derived:

$$K = \left[1.7 - \frac{0.7P_R}{P_{Rmax}} \right] \left[1.35 - \frac{0.96}{E_R} \left(1.0 + (h_R - 1.2) \frac{g \cdot P}{P_R} \right) \right] - \left[1.0 - \frac{P_R}{P_{Rmax}} \right] \left[\frac{h_R - 1.0}{2.5} \right]$$

2.0 Description of method of use with practical example.

2.1. The broken lines shown on Fig. 12 refer to the determination of the factors K_C and K_V for the following vehicle, where:

	Laden	Unladen
P	24 tonnes (240 kN)	4.2 tonnes (42 kN)
P_R	150 kN	30 kN
P_{Rmax}	150 kN	150 kN
h_R	1.8 m	1.4 m
E_R	6.0 m	6.0 m

In the following paragraphs the Figures in parentheses relate only to the vehicle being used for the purpose of illustrating the method of using Fig. 12.

2.2 Calculation of ratios

(a) $\left[\frac{g \cdot P}{P_R} \right]$ laden (= 1.6)

(b) $\left[\frac{g \cdot P}{P_R} \right]$ unladen (= 1.4)

(c) $\left[\frac{P_R}{P_{Rmax}} \right]$ unladen (= 0.2)

2.3 Determination of the correction factor when laden, K_C :

- Start at the appropriate value of h_R ($h_R = 1.8$ m)
- Move horizontally to the appropriate $g \cdot P/P_R$ line ($g \cdot P/P_R = 1.6$)
- Move vertically to the appropriate E_R line ($E_R = 6.0$ m)
- Move horizontally to the K_C scale; K_C is the laden correction factor required ($K_C = 1.04$)

2.4 Determination of the correction factor when unladen, K_V :

2.4.1 Determination of the factor K_2 :

- a) Start at appropriate h_R ($h_R = 1.4$ m)
- b) Move horizontally to the appropriate P_R/P_{Rmax} line in the group of curves nearest to vertical axis ($P_R/P_{Rmax} = 0.2$)
- c) Move vertically to the horizontal axis and read off the value of K_2 ($K_2 = 0.13$ m).

2.4.2 Determination of the factor K_1 :

- a) Start at the appropriate value of h_R ($h_R = 1.4$ m)
- b) Move horizontally to the appropriate g. P/P_R line (g. $P/P_R = 1.4$)
- c) Move vertically to the appropriate E_R line ($E_R = 6.0$ m)
- d) Move horizontally to the appropriate P_R/P_{Rmax} line in the group of curves furthest from the vertical axis ($P_R/P_{Rmax} = 0.2$)
- e) Move vertically to the horizontal axis and read off the value of K_1 ($K_1 = 1.79$).

2.4.3 Determination of the factor K_V :

The unladen correction factor K_V is obtained from the following expression:

$$K_V = K_1 - K_2 \cdot (K_V = 1.66)$$

J-4.1.2 The provision of paragraph J-4.1.1. does not have to be fulfilled, if a semi-trailer with a K_c factor less than 0.95 meets at least the braking performance specified in C-3.1.2.1. or in C-3.1.3.1., as appropriate.

J-5.0 REQUIREMENTS FOR FULL AND CENTRE-AXLE TRAILERS

J-5.1 For Full Trailers Fitted with Compressed-Air Braking Systems

J-5.1.1 For full trailers with two axles the following requirements apply:

J-5.1.1.1 For k values between 0.2 and 0.8 (see Note)

$$z \geq 0.1 + 0.85(k-0.2)$$

NOTE - The provisions of J-3.1.1 or J-5.1.1 do not effect the requirements of Annex C relating to the braking performance. However, if, in tests made under the provisions of J-3.1.1 or J-5.1.1, braking performances are obtained which are higher than those prescribed in Annex C, the provisions relating to the adhesion utilization curves shall be applied within the areas of Figs. 6, 7 and 8 defined by the straight lines $k=0.8$ and $z=0.8$.

- J-5.1.1.2 For all states of load of the vehicle, the adhesion utilization curve of the rear axle shall not be situated above that for the front axle for all braking rates between 0.15 and 0.30. This condition is also considered satisfied if, for braking rates between 0.15 and 0.30, the adhesion utilization curves for each axle are situated between two lines parallel to the line of ideal adhesion utilization given by equations $k = z + 0.08$ and $k = z - 0.08$ as shown in Fig. 7 and the adhesion utilization curve for rear axle for braking rates $z \geq 0.3$ complies with the relation:
- $$z \geq 0.3 + 0.74 (k - 0.38)$$
- J-5.1.1.3 For the verification of the requirements of J-5.1.1.1 and J-5.1.1.2 the procedure should be as that in the provisions of J-3.1.4.
- J-5.1.2 Full trailers with more than two axles shall be subject to the requirements of J-5.1.1 shall apply. The requirements of J-5.1.1 with respect to wheel lock sequence shall be considered to be met if, in the case of braking rates between 0.15 and 0.30, the adhesion utilized by at least one of the front axles is greater than that utilized by at least one of the rear axles.
- J-5.1.3 The permissible relationship between the braking rate T_R/P_R and the pressure p_m shall lie within the designated areas in Fig. 9 for all pressures between 20 and 5 kPa, in both the laden and unladen state.
- J-5.2 For Centre-Axle Trailers Fitted with Compressed-Air Braking Systems
- J-5.2.1 The permissible relationship between the braking rate T_R/P_R and the pressure p_m shall lie within two areas derived from Fig. 9, by multiplying the vertical scale by 0.95. This requirement shall be met at all pressures between 20 and 75 kPa,, in both the laden and unladen states of load.
- J-5.2.2 If the requirements of C-3.1.2.1 may not be satisfied due to lack of adhesion, then the centre-axle trailer shall be fitted with an anti-lock system complying with Annex M.

J-6.0 REQUIREMENTS TO BE MET IN CASE OF FAILURE OF THE BRAKING DISTRIBUTION SYSTEM

When the requirements of this Annex are fulfilled by means of a special device (for example controlled mechanically by the suspension of the vehicle), or if the vehicle is equipped with such a special device, it shall be possible, in the event of the failure of its control, to stop the vehicle under the conditions specified for secondary braking in the case of power-driven vehicles; for those power-driven vehicles authorized to tow a trailer fitted with compressed-air braking systems, it shall be possible to achieve a pressure at the coupling head of the control line within the range specified in J-3.1.3. In the event of failure of the control of the device on trailers, a service braking performance of at least 30 percent of that prescribed for the vehicle in question shall be attained.

J-7.0 MARKINGS

- J-7.1 Vehicles, which meet the requirements of this Annex by means of a device mechanically controlled by the suspension of the vehicle, or if the vehicle is equipped with such a device, shall be marked to show the useful travel of the device between the positions corresponding to vehicle unladen and laden states, respectively, and any further information to enable the setting of the device to be checked.
- J-7.1.1 When a brake load sensing device is controlled via the suspension of the vehicle by any other means, the vehicle shall be marked with information to enable the setting of the device to be checked.
- J-7.2 When the requirements of this Annex are met by means of a device which modulates the air pressure in the brake transmission, the vehicle shall be marked to show the axle loads at the ground, the nominal outlet pressures of the device and an inlet pressure of not less than 80 percent of the maximum design inlet pressure, as declared by the vehicle manufacturer, for the following states of load:
- a) Technically permissible maximum axle load for the axle(s) which control(s) the device;
 - b) Axle load(s) corresponding to the unladen mass of the vehicle in running order as stated in 26 of Annex A.
 - c) The axle load(s) approximating to the vehicle with proposed bodywork in running order where the axle load(s) mentioned in J-7.2 (b) relate(s) to the vehicle chassis with cab;
 - d) The axle load(s) designated by the manufacturer to enable the setting of the device to be checked in service, if this is (these are) different from the loads specified in J-7.2 (a to c).
- J-7.3 Information to enable compliance with the requirements of J-7.1 and J-7.2 shall be recorded.
- J-7.4 The markings referred to in J-7.1 and J-7.2 shall be affixed in a visible position in indelible form. An example of the markings for a mechanically controlled device in a vehicle fitted with compressed-air braking system is shown in Fig. 13.

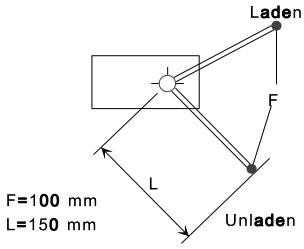
Control data	Vehicle loading	Axle No. 2 load at the ground [daN]	Inlet pressure [kPa]	Nominal outlet pressure [kPa]
	Laden	10 000	600	600
	Unladen	1 500	600	240

FIG. 13
BRAKE LOAD SENSING DEVICE

J-7.5 Electronically controlled brake force distribution systems that may not fulfill the requirements of J-7.1, J-7.2, J-7.3 and J-7.4 shall have a self-checking procedure of the functions which influence brake force distribution. In addition, when the vehicle is stationary, it shall be possible to carry out the checks defined in J-1.3.1 by generating the nominal demand pressure value associated with the commencement of braking for both the laden and the unladen condition.

J-8.0 VEHICLE TESTING

At the time of type approval test agency shall verify conformity with the requirements contained in the present Annex and carry out any further tests considered necessary to this end. The report of any further tests shall be recorded and appended to the type approval report.

(Clauses 4.3, 6 and B-2.1.1)

**CASES IN WHICH TYPE 0, TYPE- I AND/OR TYPE II (OR TYPE IIA) OR
TYPE III TESTS DO NOT HAVE TO BE CARRIED OUT [CRITERIA FOR
EXTENSION OF APPROVAL (CEA)]**

- K-1.1 This Annex gives the factors to be considered while selecting a vehicle for testing to represent a range of variants for establishing compliance of a model for type approval to meet the test requirements.
- K-1.2 This also applies to extension of type approval for changes in technical specifications of an already type approved model as declared by manufacturer as per Annex A for establishing compliance of new model/variant(s) based on already type approved model.
- K-1.3 Extension may be granted based on guidelines given in Table 1 for the changes to the technical specifications of an already type approved vehicle.
- K-2.0** Every modification pertaining to the information declared in accordance with Table 1 shall be intimated by the manufacturer to the certifying agency.
- K-2.1 If the changes in parameters are not related to the provisions, no further action needs to be taken.
- K-2.2 If the changes in parameters are related to the provisions, the certifying agency may then consider whether,
 - a) the model with the changed specifications still complies with provisions, or
 - b) any further verification is required to establish compliance.
- K-2.2.1 For considering whether any further verification is required or not, guidelines given in Table 1 shall be used.
- K-2.2.2 In case of K-2.2 (b), verification for only those parameters, which are affected by the modifications, needs to be carried out.
- K-2.2.3 In case of fulfillment of criteria of K-2.2 (a), or after results of further verification as per K- 2.2 (b) are successful, the type approval of compliance shall be extended for the changes carried out.
- K-2.3 Guidelines for selection / extension as per Table 1 are applicable to ABS as well as non ABS part of the brake system. Additional guidelines are provided in Annex M for ABS part of brake system also.

Table 1 Guide lines for Deciding Test Vehicle / Extension criteria (Clauses K-2.2.1, K-2.3 and 2.5)		
Sl No. (1)	Parameter and Changes (2)	Test to be Conducted (3)
i)	Change in vehicle category:	
	a) Change of category M1 to any other	Only the additional tests prescribed for the changed category
	b) Change of category M2 to N1	Only Type 0 tests need to be done if the following are not satisfied: The initial test speed prescribed for new category is not more than that used testing of the existing category Performance requirements of type approved category are within the prescribed limits of the new category.
	c) Change of category M2 to N2 or N2 to M2	No test. Secondary brake to be done for N2 to M2 subject to actual test results.
	d) Change of category M2 to M3	Type II or II A (if applicable) For other tests applicability dependent on change in GVW, if any (see Sl. No. ii)
	e) Change of category M3 to N3	No test
	f) Change of category N3 to M3	Type IIA (if applicable) to be carried out.
	g) Change of category N3 to N2	Engine connected test applicability dependent on change in Test Speed (see Sl. No. iii)
	h) Change of category N3 to M2	Engine connected test applicability dependent on change in Test Speed (see Sl. No. iii)
	j) Change of category M3 to N2	Engine connected test applicability dependent on change in Test Speed (see Sl. No. iii)

ii)	Change in vehicle mass:	
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	a) Increase in GVW exceeding 10 percent	All dynamic tests in laden condition (Type 0, Type I and Type II or II A (if applicable) and parking brake
	b) Increase in GVW upto 10 percent	If the stopping distance calculated from the previous type approval corrected for new GVW by formula (see K-2.4) is not within limits Type 0 test engine disconnected , If within limits, no test
	c) Any decrease in GVW exceeding 10 percent	Secondary brake test (Type 0) Laden to be done, if laden F/R ratio is increased in excess by 10 percent
	d) Any decrease in GVW upto 10 percent	No test
	e) Any change in unladen weight	All dynamic tests - (Type 0) test in the unladen condition, if F/R ratio in unladen condition is increased in excess by 10 percent,
iii)	Change in vehicle speed:	
	a) Decrease in maximum speed or increase in maximum speed which does not cause the initial speed for Type 0 test to be increased by more than 8 percent of the initial speed used in the testing	No test
	b) Increase in maximum speed which causes the initial speed for Type 0 test to be increased by more than 8 percent of the initial speed used in the testing	Type 0 and Type I and if applicable Type II tests
	c) A change in drive line ratio that increases engine rpm corresponding to gear and the initial speed for 0 type test	No test
	d) A change in drive line ratio that decreases engine rpm corresponding to gear and the initial speed for 0 type test in excess of 8 percent	Engine connected Type 0 test and Type I tests need to be conducted

iv)	Change in braking effect:	
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	a) Changes in engine which increase or do not reduce the engine braking effect. (Changes such as increase in swept volume, increase in compression ratio, 2 strokes to 4 strokes, SI to CI etc. are considered to increase the engine braking effect.)	No test
	b) Changes which reduces the engine braking effect (Changes such as decrease in swept volume, decrease in compression ratio, 4 stroke to 2 stroke, CI to SI, etc, are considered to decrease the engine braking effect)	Engine connected brake Type 0 test to be conducted
	c) Increase in the brake lining area	No test
	d) Decrease in the brake lining area	Type I and If applicable Type II
	e) Changes in foundation brakes which causes a increase in brake torque	{ Type 0 test in unladen condition to be done. If the braking torque is increased by 15 percent then carry out laden test also
	f) (Other than those listed in k.)	
	g) Changes in brakes which causes a decrease in brake torque	Type 0 and Type I tests to be done
	h) Changes in number of braked axles, if decreased	All test in dynamic condition (Type 0, Type I)
	j) Changes in number of unbraked axles	No test
	k) Setting of load sensing valves	All dynamic tests (Type 0, Type I) in unladen condition if F/R ratio in unladen condition is increased in excess by 10 percent
v)	Change in wheelbase	
	a) Decrease in wheel base in excess of 10 percent	All Type 0 dynamic tests
	b) Decrease in wheel base less than 10 percent	No test
	c) Increase in wheel base	Reaction time, if applicable. These tests may be conducted on a shorter wheelbase variant by simulating the piping for the longer wheelbase
vi)	Change in brake system:	

a) Brake lining material	Either all applicable dynamic tests on vehicle or Verification of inertia dynamometer (as per Annex J) if the performance of brake system, for which the type approval is to be extended, established on inertia dynamometer is within ± 15 percent of the performance of the lining already type approved, established on inertia dynamometer, no tests need to be conducted on vehicles
b) Reservoir capacity for Individual Circuits- Air or Vacuum brakes	If increased, charging time to be tested. If decreased, depletion tests to be done
c) Booster ratio, any decrease	Type 0 test except booster disconnection test
d) Supplementary braking system if performance reduced	Type II test to be done
e) Addition of relay valve in the circuit	Response time test
f) Addition of load sensing valve in the circuit	Dynamic tests (Type 0, Type I) in unladen condition and response time test
	If load-sensing valve does not reduce the pressure in laden condition then no test in Laden condition
g) Addition of Supplementary brake	No test
h) Increase in the system pressure 'P2' up to 10 percent	Charging time test.
j) Drum to disc or vice versa	All dynamic tests to be done
k) Hydraulic to air or vice versa	All tests to be done.

vii)	Change in tyre:	
	a) Tyre: Increase in rolling radius in excess of 10 percent	All dynamic test Type 0, Type I and parking brake to be done
	b) Changes affecting adhesion such as change from radial ply to cross ply or high aspect ratio to low aspect ratio	Type 0 and Type I Test to be done All dynamic test except Type II and static parking brake to be done
	c) Change from single tyre rear axle to twin tyre rear axle or vice versa	No test.
viii)	Other:	
	a) Any change of type of transmission (Manual to automatic or automatic to manual)	No Test
	b) Change of manufacturer of brake	Type 0 and Type I tests to be done if the specifications are changed Only to be tested for foundation brakes and changes in spec for other items only
K-2.4	The stopping distance or the mean fully developed deceleration shall be calculated to the condition of new specified mass by following formulae:	
	$S_{ext} = \{ (S_{ta} - 0.1 V_s) \times (M_{ext} / M_{ta}) \} + 0.1 V_s$ $d_{ext} = d_{ta} \times M_{ta} / M_{ext}$	
	Where	
	S_{ta}	= Stopping distance reported in the earlier type approval test;
	S_{ext}	= Stopping distance calculated for the new G.V.W. for which extension is sought;
	M_{ta}	= Mass specified for the earlier type approval test, in kg;
	M_{ext}	= Mass for which extension is sought, in kg;
	d_{ta}	= Mean fully developed deceleration reported in earlier type approval test, in m/s^2 ;
	d_{ext}	= Mean fully developed deceleration (m/s^2) calculated for the new G.V.W. for which extension is sought, in m/s^2 ; and
	V_s	= Test speed applicable to the vehicle under consideration.

K-2.5 Changes other than those listed in Table 1 are considered to be having no adverse effect on the brake performance.

K-3.0 Alternative procedures for Type-I and Type-III tests for trailer brakes

K-3.1. General

K-3.1.1. Type-I or Type-III test may be waived at the time of type approval of the vehicle provided that the braking system components comply with the requirements of K3 and that the resulting predicted braking performance meets the requirements of this Standard for the appropriate vehicle category.

K-3.1.2. Tests carried out in accordance with the methods detailed in this K-3 shall be deemed to meet the above requirements.

K-3.1.2.1. Tests carried out according to K-3.3.5.1., which were positive, are deemed to fulfil the provisions of K-3.3.5.1., as last amended. If use is made of this alternative procedure, the test report shall make reference to the original test report from which the test results are taken for the new updated report. However, new tests have to be carried out to the requirements of the latest amended version of this Standard.

K-3.1.2.2. Tests carried out in accordance with this K-3, together with any supporting data from the vehicle/axle/brake manufacturer, provide sufficient information to meet the requirements of Supplement 2 to the 11 series of amendments can be used for a new report or the extension of an existing test report without the need to carry-out actual tests.

K-3.1.3. Tests carried out in accordance with K-3.3.6. and the results reported in section 2 of K-4 or K-5 to this annex shall be acceptable as a means of proving compliance with the requirements of 5.2.2.8.1.

K-3.1.4. The adjustment of the brake(s) shall, prior to the Type-III test below, be set according to the following procedures as appropriate:

K-3.1.4.1. In the case of air operated trailer brake(s), the adjustment of the brakes shall be such as to enable the automatic brake adjustment device to function. For this purpose the actuator stroke shall be adjusted to:

$s_0 > 1.1 \cdot s_{re-adjust}$ (the upper limit shall not exceed a value recommended by the manufacturer),

Where:

$s_{re-adjust}$ is the re-adjustment stroke according to the specification of the manufacturer of the automatic brake adjustment device, i.e. the stroke, where it starts to re-adjust the running clearance of the brake with an actuator pressure of 100 kPa.

Where, by agreement with the Test Agency, it is impractical to measure the actuator stroke, the initial setting shall be agreed with the Test Agency.

From the above condition the brake shall be operated with an actuator pressure of 200 kPa, 50 times in succession. This shall be followed by a single brake application with an actuator pressure of ≥ 650 kPa.

- K-3.1.4.2. In the case of hydraulically operated trailer disc brakes no setting requirements are deemed necessary.
- K-3.1.4.3. In the case of hydraulically operated trailer drum brakes the adjustment of the brakes shall be as specified by the manufacturer.
- K-3.1.5. In the case of trailers equipped with automatic brake adjustment devices the adjustment of the brakes shall, prior to the Type-I test below, be set according to the procedure as laid down in paragraph 1.4. above.

K-3.2. Symbols and definitions

K-3.2.1. Symbols

P	=	part of the vehicle mass borne by the axle under static conditions
F	=	normal reaction of road surface on the axle under static conditions $= P \cdot g$
F_R	=	total normal static reaction of road surface on all wheels of trailer
F_e	=	test axle load
P_e	=	F_e / g
g	=	acceleration due to gravity: $g = 9.81 \text{ m/s}^2$
C	=	brake input torque
C_O	=	brake input threshold torque, definition see paragraph 2.2.2. below
$C_{0,dec}$	=	declared brake input threshold torque
C_{max}	=	maximum brake input torque
R	=	tyre rolling radius (dynamic)
T	=	brake force at tyre/road interface
T_R	=	total brake force at tyre/road interface of the trailer
M	=	brake torque = $T \cdot R$
z	=	braking rate = T/F or $M/(R \cdot F)$
s	=	actuator stroke (working stroke plus free stroke)
s_p	=	see T-16
Th_A	=	see T-16
l	=	lever length
r	=	internal radius of brake drums or effective radius of brake discs
p	=	brake actuation pressure

Note: Symbols with the suffix "e" relate to the parameters associated with the reference brake test and may be added to other symbols as appropriate.

K-3.2.2. Definitions

K-3.2.2.1. Mass of a disc or drum

K-3.2.2.1.1. The "declared mass" is the mass declared by the manufacturer which is a representative mass for the brake identifier (see K-3.3.7.2.2.).

K-3.2.2.1.2. The "nominal test mass" is the mass which the manufacturer specifies for the disc or drum with which the relevant test is carried out by the Test Agency.

K-3.2.2.1.3. The "actual test mass" is the mass measured by the Test Agency prior to the test.

K-3.2.2.2. "Brake input threshold torque":

K-3.2.2.2.1. The brake input threshold torque "C0" is the input torque necessary to produce a measurable brake torque. This torque may be determined by extrapolation of measurements within a range not exceeding 15 percent braking rate or other equivalent methods (e.g. J- 1.3.1.1.).

K-3.2.2.2.2. The brake input threshold torque "C0,dec" is the brake input threshold torque declared by the manufacturer which is a representative brake input threshold torque for the brake (see K-3.3.7.2.2.1.) and is needed to produce Fig-24.

K-3.2.2.2.3. The brake input threshold torque "C0,e" is determined by the procedure as defined in K-3. 2.2.2.1., measured by the Test Agency at the end of the test.

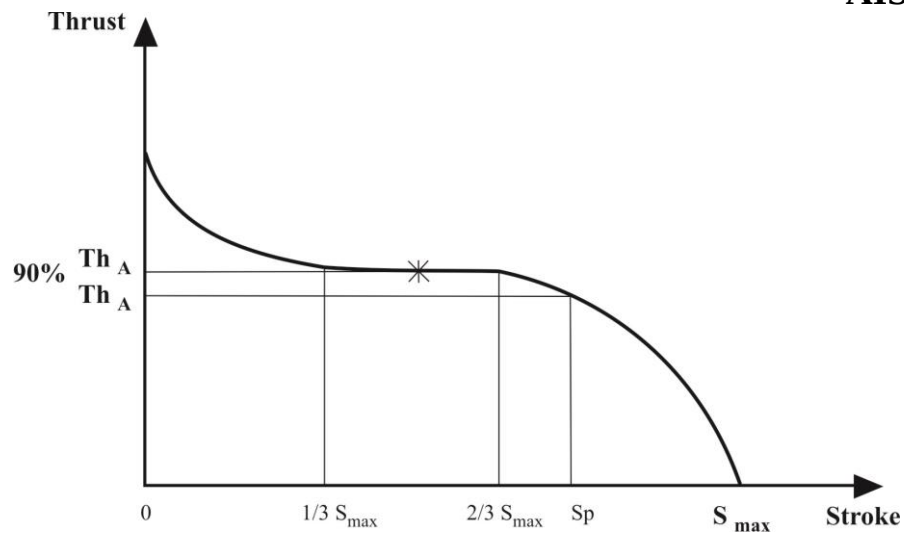
K-3.2.2.3. **"External diameter of a disc":**

K-3.2.2.3.1. The "declared external diameter" is the external diameter of a disc declared by the manufacturer which is a representative external diameter for the disc (**see K-3.3.7.2.2.1.**).

K-3.2.2.3.2. The "nominal external diameter" is the external diameter which the manufacturer specifies for the disc on which the relevant test is carried out by the Test Agency.

K-3.2.2.3.3. The "actual external diameter" is the external diameter measured by the Test Agency prior to the test.

K-3.2.2.4. The "effective length of the cam shaft" is the distance from the centre line of the S-cam to the centre line of the operating lever.



K-3.3. Test methods

K-3.3.1. Track tests

K-3.3.1.1. The brake performance tests should preferably be carried out on a single axle only.

K-3.3.1.2. The results of tests on a combination of axles may be used, provided that each axle contributes equal braking energy input during the drag and hot brake tests.

K-3.3.1.2.1. This is ensured if the following are identical for each axle: braking geometry, lining, wheel mounting, tyres, actuation and pressure distribution in the actuators.

K-3.3.1.2.2. The documented result for a combination of axles will be the average for the number of axles, as though a single axle had been used.

K-3.3.1.3. The axle(s) should preferably be loaded with the maximum static axle load, though this is not essential provided that due allowance is made during the tests for the difference in rolling resistance caused by a different load on the test axle(s).

K-3.3.1.4. Allowance shall be made for the effect of the increased rolling resistance resulting from a combination of vehicles being used to carry out the tests.

K-3.3.1.5. The initial speed of the test shall be that prescribed. The final speed shall be calculated by the following formula:

$$v_2 = v_1 \sqrt{\frac{P_o + P_1}{P_o + P_1 + P_2}}$$

Where:

v_1 = Initial speed (km/h),

v_2 = Final speed (km/h),

P_o = Mass of the towing vehicle (kg) under test conditions,

P_1 = Part of the mass of the trailer borne by the unbraked axle(s) (kg),

P_2 = Part of the mass of the trailer borne by the braked axle(s) (kg).

K-3.3.2. Inertia dynamometer tests

K-3.3.2.1. The test machine shall have a rotary inertia simulating that part of the linear inertia of the vehicle mass acting upon one wheel, necessary for the cold performance and hot performance tests, and capable of being operated at constant speed for the purpose of the test described in paragraphs K-3.3.5.2. and K-3.3.5.3.

K-3.3.2.2. The test shall be carried out with a complete wheel, including the tyre, mounted on the moving part of the brake, as it would be on the vehicle. The inertia mass may be connected to the brake either directly or via the tyres and wheels.

K-3.3.2.3. Air cooling at a velocity and air flow direction simulating actual conditions may be used during the heating runs, the speed of the air flow being

$$v_{\text{air}} = 0.33 v$$

Where:

v = vehicle test speed at initiation of braking.

The temperature of the cooling air shall be the ambient temperature.

K-3.3.2.4. Where the tyre rolling resistance is not automatically compensated for in the test, the torque applied to the brake shall be modified by subtracting a torque equivalent to a rolling resistance coefficient of 0.01.

K-3.3.3. Rolling road dynamometer tests

K-3.3.3.1. The axle should preferably be loaded with the maximum static axle mass though this is not essential provided that due allowance is made during the tests for the difference in rolling resistance caused by a different mass on the test axle.

K-3.3.3.2. Air cooling at a velocity and air flow direction simulating actual conditions may be used during the heating runs, the speed of the air flow being

$$v_{\text{air}} = 0.33 v$$

Where:

v = vehicle test speed at initiation of braking.

The temperature of the cooling air shall be the ambient temperature.

K-3.3.3.3. The braking time shall be 1 second after a maximum build-up time of

0.6 second.

K-3.3.4. Test conditions (General)

K-3.3.4.1. The test brake(s) shall be instrumented so that the following measurements can be taken:

K-3.3.4.1.1. A continuous recording to enable the brake torque or force at the periphery of the tyre to be determined.

K-3.3.4.1.2. A continuous recording of air pressure in the brake actuator.

K-3.3.4.1.3. Vehicle speed during the test.

K-3.3.4.1.4. Initial temperature on the outside of the brake drum or brake disc.

K-3.3.4.1.5. Brake actuator stroke used during Type-0 and Type-I or Type-III tests.

K-3.3.5. Test procedures

K-3.3.5.1. Supplementary cold performance test

The preparation of the brake shall be in accordance with T-4.4.2.

In the case, that the verification of the brake factor BF and brake threshold torque has been carried out according to T-4.4.3, the bedding in procedure for the supplementary cold performance test shall be identical with the procedure used for the verification according to T-4.4.3.

It is permissible to carry out the cold performance tests after the verification for the brake factor BF in accordance with T-4.

It is also permissible to carry out the two fade tests, Type-I and Type- III, one after the other.

Some brake applications according to T-4.4.2.6. may be done between each of the fade tests, and between the verification and the cold performance tests. The quantity of applications is to be declared by the brake manufacturer.

K-3.3.5.1.1. This test is carried out at an initial speed equivalent to 40 km/h in the case of Type-I test and 60 km/h in the case of Type-III test in order to evaluate the hot braking performance at the end of Type-I and Type-III tests. The Type-I and/or Type-III fade test has/have to be done immediately after this cold performance test.

K-3.3.5.1.2. Three brake applications are made at the same pressure (p) and at an initial speed equivalent to 40 km/h (in the case of Type-I test) or 60 km/h (in the case of Type-III test), with an approximately equal initial brake temperature not exceeding 100 °C, measured at the outside surface of the drums or discs. The applications shall be at the brake actuator pressure required to give a brake torque or force equivalent to a braking rate (z) of at least 50 per cent. The brake actuator pressure shall not exceed 650 kPa, and the brake input torque (C) shall not exceed the maximum permissible brake input torque (Cmax). The average of the three results shall be taken as the cold performance.

K-3.3.5.2. **Fade test (Type-I test)**

- K-3.3.5.2.1. This test is carried out at a speed equivalent to 40 km/h with an initial brake temperature not exceeding 100 °C, measured at the outside surface of the drum or brake disc.
- K-3.3.5.2.2. A braking rate is maintained at 7 per cent, including the rolling resistance (see K-3.3.2.4.).
- K-3.3.5.2.3. The test is made during 2 minutes and 33 seconds or during 1.7 km at a vehicle speed of 40 km/h. If the test velocity cannot be achieved, then the duration of the test can be lengthened according to C-1.5.2.2.
- K-3.3.5.2.4. Not later than 60 seconds after the end of the Type-I test, a hot performance test is carried out in accordance with C-1.5.3. at an initial speed equivalent to 40 km/h. The brake actuator pressure shall be that used during the Type-0 test.

K-3.3.5.3. Fade test (Type-III test)

K-3.3.5.3.1. Test methods for repeated braking

K-3.3.5.3.1.1. Track tests (see C-1.7)

K-3.3.5.3.1.2. Inertia dynamometer test

For the bench test as in K-2.3.2. the conditions may be as for the road test according to paragraph C-1.7.1. to this Standard with:

$$v_2 = \frac{v_1}{2}$$

K-3.3.5.3.1.3. Rolling road dynamometer test

For the bench test as in K-3.3.3., the conditions shall be as follows:

Number of brake applications

20 Duration of braking cycle

(braking time 25 s and recovery time 35 s)60 s

Test speed 30 km/h

Braking rate 0.06

Rolling resistance 0.01

- K-3.3.5.3.2. Not later than 60 seconds after the end of the Type-III test a hot performance test is carried out in accordance with paragraph C-1.7.2. The brake actuator pressure shall be that used during the Type-0 test.

K-3.3.6. Performance requirements for automatic brake adjustment devices

- K-3.3.6.1. The following requirements shall apply to an automatic brake adjustment device which is installed on a brake, the performance of which is being verified according to the provisions of this K-3.

On completion of the tests defined in K-3.3.5.2.4. (Type-I test) or K-3.3.5.3.2 (Type-III test) above, the requirements of K-3.3.6.3. shall be verified.

- K-3.3.6.2. The following requirements shall apply to an alternative automatic brake adjustment device installed on a brake for which an Appendix 3

test report already exists.

K-3.3.6.2.1. Brake performance

Following heating of the brake(s) carried out in accordance with the procedures defined in K-3.3.5.2. (Type-I test) or K-3.3.5.3 (Type-III test), as appropriate, one of the following provisions shall apply:

- (a) The hot performance of the service braking system shall be ≥ 80 per cent of the prescribed Type-0 performance; or
- (b) The brake shall be applied with a brake actuator pressure as used during the Type-0 test; at this pressure the total actuator stroke (s_A) shall be measured and shall be $\leq 0.9 s_p$ value of the brake chamber.

s_p = The effective stroke means the stroke at which the output thrust is 90 per cent of the average thrust (Th_A) - K-3.2.

K-3.3.6.2.2. On completion of the tests defined in paragraph K-3.3.6.2.1. above the requirements of paragraph K-3.3.6.3. below shall be verified.

K-3.3.6.3. Free running test

After completing the tests defined in paragraphs K-3.3.6.1. or K-3.3.6.2. above, as applicable, the brake(s) shall be allowed to cool to a temperature representative of a cold brake (i.e. ≤ 100 °C) and it should be verified that the trailer/wheel(s) is capable of free running by fulfilling one of the following conditions:

- (a) Wheels are running freely (i.e. wheels can be rotated by hand);
- (b) It is ascertained that at a constant speed equivalent to $v = 60$ km/h with the brake(s) released the asymptotic temperature shall not exceed a drum/disc temperature increase of 80 °C, then this residual brake moment is regarded as acceptable.

K-3.3.7. Identification

K-3.3.7.1. The axle shall carry in a visible position at least the following identification information grouped together, in any order, in a legible and indelible manner:

- (a) Axle manufacturer and/or make;
- (b) Axle identifier (see paragraph K-3.3.7.2.1.);
- (c) Brake identifier (see paragraph K-3.3.7.2.2.);
- (d) F_e identifier (see paragraph K-3. 3.7.2.3.);
- (e) Base part of test report number (see paragraph K-3.3.9.).

An example is given below:

Axle manufacturer and/or make ABC

ID1-XXXXXX

ID2-YYYYYY

ID3-11200

ID4-ZZZZZZZ

- K-3.3.7.1.1. A non-integrated automatic brake adjustment device shall carry in a visible position at least the following identification information grouped together, in a legible and indelible manner:
- (a) Manufacturer and/or make;
 - (b) Type;
 - (c) Version.
- K-3.3.7.1.2. The make and type of each brake lining shall be visible when the lining/pad is mounted on the brake shoe / back plate in a legible and indelible manner.
- K-3.3.7.2. **Identifiers**
- K-3.3.7.2.1. Axle identifier
- The axle identifier categorizes an axle in terms of its braking force/torque capability as stated by the axle manufacturer.
- The axle identifier shall be an alphanumeric number consisting of the four characters "ID1-" followed by a maximum of 20 characters.
- K-3.3.7.2.2. Brake identifier
- The brake identifier shall be an alphanumeric number consisting of the four characters "ID2-" followed by a maximum of 20 characters.
- A brake with the same identifier is a brake which does not differ with regard to the following criteria:
- (a) Type of brake (e.g. drum (S-cam, wedge, etc.) or disc brake (fixed, floating, single or twin disc, etc.));
 - (b) Base material (e. g. ferrous or non ferrous) with respect to calliper housing, brake carrier, brake disc and brake drum;
 - (c) Dimensions with the suffix "e" according to Figures 2A and 2B of K-6;
 - (d) The basic method used within the brake to generate the braking force;
 - (e) In the case of disc brakes, the friction ring mounting method: fixed or floating;
 - (f) Brake factor B_F ;
 - (g) Different brake characteristics with respect to the requirements of Annex K which are not covered by subparagraph K-3.3.7.2.2.1.
- K-3.3.7.2.2.1. Differences allowed within the same brake identifier
- The same brake identifier may include different brake characteristics with regard to the following criteria:
- (a) Increase in maximum declared brake input torque C_{max} ;
 - (b) Deviation of declared brake disc and brake drum mass m_{dec} : ± 20

per cent;

- (c) Method of attachment of the lining / pad on the brake shoe / back plate;
- (d) In the case of disc brakes, increase of maximum stroke capability of the brake;
- (e) Effective length of the cam shaft;
- (f) Declared threshold torque $C_{0,dec}$;
- (g) ± 5 mm from the declared external diameter of the disc;
- (h) Type of cooling of the disc (ventilated/non-ventilated);
- (i) Hub (with or without integrated hub);
- (j) Disc with integrated drum – with or without parking brake function;
- (k) Geometric relationship between disc friction surfaces and disc mounting;
- (l) Brake lining type;
- (m) Material variations (excluding changes in base material, see paragraph 3.7.2.2. above) for which the manufacturer confirms that such a material variation does not change the performance with respect to the required tests;
- (n) Back plate and shoes.

K-3.3.7.2.3. Fe identifier

The Fe identifier indicates the test axle load. It shall be an alphanumeric number consisting of the four characters "ID3-" followed by the Fe value in daN, without the "daN" unit identifier.

K-3.3.7.2.4. Test report identifier

The test report identifier shall be an alphanumeric number consisting of the four characters "ID4-" followed by the base part of the test report number.

K-3.3.7.3. Automatic brake adjustment device (integrated and non-integrated)

K-3.3.7.3.1. Types of automatic brake adjustment device

The same type of automatic brake adjustment device does not differ with regard to the following criteria:

- (a) Body: base material (e. g. ferrous or non-ferrous, cast iron or forged steel);
- (b) Maximal permitted brake shaft moment;
- (c) Adjustment operating principle, e.g. stroke (travel) dependent, force dependent or electronic/mechanical.

K-3.3.7.3.2. Versions of automatic brake adjustment device, with respect to the adjustment behaviour

Automatic brake adjustment devices within a type that have an impact on the running clearance of the brake are considered to be different versions.

K-3.3.8. **Test criteria**

The testing shall demonstrate compliance with all the requirements laid down in K-3. In the case that a new test report, or a test report extension, is required for a modified axle/brake within the limits specified in paragraph K-3.3.7.2.2.1. above the following criteria are used to determine the necessity for further testing taking into account worst case configurations agreed with the Test Agency.

Abbreviations shown below are used in the subsequent table:

CT (complete test)	Test according to K-3: 3.5.1.: Supplementary cold performance test 3.5.2.: Fade test (Type-I test)* 3.5.3.: Fade test (Type-III test)* Test according to Annex T: 4.: Cold performance characteristics for trailer brakes*
FT (fade test)	Test according to K-3: 3.5.1.: Supplementary cold performance test 3.5.2.: Fade test (Type-I test)* 3.5.3.: Fade test (Type-III test)*
*If applicable.	
Differences according to paragraph 3.7.2.2.1. above	Test criteria
(a) Increase in maximum declared brake input torque $C_{\max}$	Change allowed without additional testing
(b) Deviation of declared brake disc and brake drum mass m_{dec} : ± 20 per cent	CT: The lightest variant shall be tested if the nominal test mass for a new variant deviates less than 5 per cent from a previously tested variant with a higher nominal value then the test of the lighter version may be dispensed with. The actual test mass of the test specimen may vary ± 5 per cent from the nominal test mass.

(c) Method of attachment of the lining / pad on the brake shoe / back plate	The worst case specified by the manufacturer and agreed by the Test Agencies conducting the test
(d) In the case of disc brakes, increase of maximum stroke capability of the brake	Change allowed without additional testing
(e) Effective length of the cam shaft	The worst case is considered to be the lowest cam shaft torsional stiffness and shall be verified by either: (i) FT; or (ii) Change allowed without additional testing if by calculation the influence with respect to stroke and braking force can be shown. In this case the test report shall indicate the following extrapolated values: s_e, C_e, T_e, T_e/F_e.
(f) Declared threshold torque $C_{0,dec}$	It shall be checked that the brake performance remains within the corridors of Fig.24
(g) ± 5 mm from the declared external diameter of the disc	The worst case test is considered the smallest diameter The actual external diameter of the test specimen may vary ± 1 mm from the nominal external diameter specified by the axle manufacturer.
(h) Type of cooling of the disc (ventilated/non-ventilated)	Each type shall be tested
(i) Hub (with or without integrated hub)	Each type shall be tested
(j) Disc with integrated drum – with or without parking brake function	Testing is not required for this feature

(k) Geometric relationship between disc friction surfaces and disc mounting	Testing is not required for this feature
(l) Brake lining type	Each type of brake lining
(m) Material variations (excluding changes in base material, see paragraph 3.7.2.2.) for which the manufacturer confirms that such a material variation does not change the performance with respect to the required tests	Test not required for this condition
(n) Back plate and shoes	Worst case test conditions:* Back plate: minimum thickness Shoe: lightest brake shoe
* No test is required if the manufacturer can demonstrate that a change does not affect the stiffness.	

K-3.3.8.1. If an automatic brake adjustment device deviates from a tested one according to K-3.3.7.3.1. and K-3.3.7.3.2. an additional test according to K-3.3.6.2. is necessary.

K-3.3.9. Test report

K-3.3.9.1. Test report number

The test report number consists of two parts: a base part and a suffix which identifies the issue level of the test report.

The base part, consisting of a maximum of 20 characters, and suffix shall be clearly separated from each other using e.g. a dot or slash.

The base part of the test report number shall only cover brakes with the same brake identifier and the same brake factor (T-4).

K-3.3.9.2. Test code

In addition to the test report number a "test code" consisting of up to eight characters (e.g. ABC123) shall indicate the test results applicable to the identifiers and the test specimen, which is described by the details given in paragraph 3.7. above.

K-3.3.9.3. Test results

K-3.3.9.3.1. The result of tests carried out in accordance with paragraphs K-3.3.5. and K-3. 3.6.1. shall be reported on a form, a model of which is shown in K-4.

K-3.3.9.3.2. In the case of a brake installed with an alternative brake adjustment device the results of tests carried out in accordance with K-3.3.6.2 shall be reported on a form, a model of which is shown in K-5.

K-3.3.9.4. Information document

An information document, provided by the axle or vehicle manufacturer, containing at least the information defined in K-6 shall be part of the Test Report.

The information document shall identify, if applicable, the various variants of the brake/axle equipment with respect to the essential criteria listed in paragraph K-3. 3.7.2.2.1. above.

K-3.4. Verification

K-3.4.1. Verification of components

The brake specification of the vehicle to be type approved shall comply with the requirements laid down in paragraphs K-3.3.7. and K-3.3.8. above.

K-3.4.2. Verification of brake energy absorbed

K-3.4.2.1. The brake forces (T) for each subject brake (for the same control line pressure pm) necessary to produce the drag force specified for both Type-I and Type-III test conditions shall not exceed the values Te as stated in K-4.2.3.1. and K-4.2.3.2., which were taken as a basis for the test of the reference brake.

K-3.4.3. Verification of hot performance

K-3.4.3.1. The brake force (T) for each subject brake for a specified pressure (p) in the actuators and for a control line pressure (pm) used during the Type-0 test of the subject trailer is determined as follows:

K-3.4.3.1.1. The predicted actuator stroke (s) of the subject brake is calculated as follows:

$$s = 1. \frac{S_e}{l_e}$$

This value shall not exceed sp. Where sp has been verified and reported in accordance with the procedure defined in item T-2 to this Standard and may only be applied within the pressure range recorded in paragraph 3.3.1. of the test report defined in T-6.

K-3.4.3.1.2. The average thrust output (ThA) of the actuator fitted to the subject brake at the pressure specified in K-3.4.3.1. is measured.

K-3.4.3.1.3. The brake input torque (C) is then calculated as follows:

$$C = Th_A \cdot l$$

C shall not exceed C_{max}

K-3.4.3.1.4. The predicted brake performance for the subject brake is given by:

$$T = (T_e - 0.01 \cdot F_e) \frac{C - C_o}{C_e - C_{oe}} \cdot \frac{R_e}{R} + 0.01 \cdot F$$

R shall not be less than $0.8 R_e$

K-3.4.3.2. The predicted brake performance for the subject trailer is given by:

$$\frac{T_R}{F_R} = \frac{\Sigma T}{\Sigma F}$$

K-3.4.3.3. The hot performances following the Type-I or Type-III tests shall be determined in accordance with paragraphs K-3.4.3.1.1. to K-3.4.3.1.4. above. The resulting predictions given by paragraph K-3.4.3.2. above shall satisfy the requirements of this Standard for the subject trailer. The value used for:

"The figure recorded in the Type-0 test as prescribed in paragraph C-1.5.3. or C-1.7.2. "shall be the figure recorded in the Type-0 test of the subject trailer.

K-4.0 Model test report form as prescribed in paragraph 3.9. of K-3

Test Report No.

Base part: ID4-

Suffix:

K-4.1. General

K-4.1.1. Axle manufacturer (name and address):

K-4.1.1.1. Make of axle manufacturer:

K-4.1.2. Brake manufacturer (name and address):

K-4.1.2.1. Brake identifier ID2-:

K-4.1.2.2. Automatic brake adjustment device: integrated/non-integrated⁽¹⁾

K-4.1.3. Manufacturer's information document:

⁽¹⁾ Strike out what does not apply.

K-4.2. **Test Record**

The following data has to be recorded for each test:

- K-4.2.1. Test code (K-3.3.9.2.):
- K-4.2.2. Test specimen: (precise identification of the variant tested related to the Manufacturer's information document. See also paragraph K-3.3.9.2.
- K-4.2.2.1. Axle
 - K-4.2.2.1.1. Axle identifier: ID1-
 - K-4.2.2.1.2. Identification of tested axle:
 - K-4.2.2.1.3. Test axle load (Fe identifier): ID3-daN
- K-4.2.2.2. **Brake**
 - K-4.2.2.2.1. Brake identifier: ID2-
 - K-4.2.2.2.2. Identification of tested brake:
 - K-4.2.2.2.3. Maximum stroke capability of the brake⁽²⁾:
 - K-4.2.2.2.4. Effective length of the cam shaft⁽³⁾:
 - K-4.2.2.2.5. Material variation as per paragraph K-3.3.8 (m):
 - K-4.2.2.2.6. Brake drum / disc⁽¹⁾
 - K-4.2.2.2.6.1. Actual test mass of disc / drum⁽¹⁾
 - K-4.2.2.2.6.2. Nominal external diameter of disc ⁽²⁾
 - K-4.2.2.2.6.3. Type of cooling of the disc ventilated/non-ventilated⁽¹⁾
 - K-4.2.2.2.6.4. With or without integrated hub⁽¹⁾
 - K-4.2.2.2.6.5. Disc with integrated drum – with or without parking brake function^{(1),(2)}
 - K-4.2.2.2.6.6. Geometric relationship between disc friction surfaces and disc mounting:
 - K-4.2.2.2.6.7. Base material:
 - K-4.2.2.2.7. Brake lining or pad⁽¹⁾

⁽¹⁾ Strike out what does not apply.

⁽²⁾ Applies to disc brakes only.

⁽³⁾ Applies to drum brakes only.

- K-4.2.2.2.7.1. Manufacturer:
- K-4.2.2.2.7.2. Make:
- K-4.2.2.2.7.3. Type:
- K-4.2.2.2.7.4. Method of attachment of the lining / pad on the brake shoe / back plate:1
- K-4.2.2.2.7.5. Thickness of back plate, weight of shoes or other describing information (Manufacturer's information document)⁽¹⁾:
- K-4.2.2.2.7.6. Base material of brake shoe / back plate⁽¹⁾:
- K-4.2.2.3. Automatic brake adjustment device (not applicable in the case of integrated automatic brake adjustment device)⁽¹⁾:
- K-4.2.2.3.1. Manufacturer (name and address):
- K-4.2.2.3.2. Make:
- K-4.2.2.3.3. Type:
- K-4.2.2.3.4. Version:
- K-4.2.2.4. Wheel(s) (dimensions see Figures 1A and 1B in K-6)
- K-4.2.2.4.1. Reference tyre rolling radius (Re) at test axle load (Fe):
- K-4.2.2.4.2. Data of the fitted wheel during testing:
- | Tyre size | Rim size | X _e (mm) | D _e (mm) | E _e (mm) | G _e (mm) |
|-----------|----------|---------------------|---------------------|---------------------|---------------------|
| | | | | | |
- K-4.2.2.5. Lever length le:
- K-4.2.2.6. Brake actuator
- K-4.2.2.6.1. Manufacturer:
- K-4.2.2.6.2. Make:
- K-4.2.2.6.3. Type:
- K-4.2.2.6.4. (Test) Identification number:

⁽¹⁾ Strike out what does not apply.

- K-4.2.3. Test results (corrected to take account of rolling resistance of 0.01 • Fe)

K-4.2.3.1. In the case of vehicles of categories T2 and T3 where the T3 trailer has been subject to the Type-I test:

Test type:	0	I	
K-3, paragraph:	3.5.1.2.	3.5.2.2./3.	3.5.2.4.
Test speed (km/h)	40	40	40
Brake actuator pressure p (kPa)		-	
Braking time (min)	-	2.55	-
Brake force developed T_e (daN)			
Brake efficiency T_e/F_e			
Actuator stroke s_e (mm)		-	
Brake input torque C_e (Nm)		-	
Brake input threshold torque $C_{0,e}$ (Nm)			

K-4.2.3.2. In the case of vehicles of categories T3 and T4 where the T3 trailer has been subject to the Type-III test:

Test type:	0	III	
K-3, paragraph:	3.5.1.2.	3.5.3.1.	3.5.3.2
Test speed initial (km/h)	60		60
Test speed final (km/h)			
Brake actuator pressure p_e (kPa)		-	
Number of brake applications	-	20	-
Duration of braking cycle (s)	-	60	-
Brake force developed T_e (daN)			
Brake efficiency T_e/F_e			
Actuator stroke s_e (mm)		-	
Brake input torque C_e (Nm)		-	
Brake input threshold torque $C_{0,e}$ (Nm)		-	

- K-4.2.3.3. This item is to be completed only when the brake has been subject to the test procedure defined in T-4 to verify the cold performance characteristics of the brake by means of the brake factor (B_F).
- K-4.2.3.3.1 Brake factor B_F :
- K-4.2.3.3.2 Declared threshold torque $C_{0,dec}$ (Nm)
- K-4.2.3.4. Performance of the automatic brake adjustment device (if applicable)
- K-4.2.3.4.1 Free running according to K-3.3.6.3. : yes / no⁽¹⁾
- K-4.3. **Application range**
The application range specifies the axle/brake variants that are covered in this test report, by showing which variables are covered by the individual test codes.
- K-4.4. This test has been carried out and the results reported in accordance with K-3 and where appropriate T-4 as last amended

At the end of the test defined in K-3.3.6.⁽⁴⁾ the requirements of paragraph 5.2.2.8.1. of Standard No. 13 were deemed to be fulfilled / not fulfilled.⁽¹⁾

Test Agency carrying out the test
Signed: Date:
- K-5 Model test report form for an alternative automatic brake adjustment device as prescribed in paragraph K-3.3.7.3.

Test report No
- K-5.1. **Identification**
- K-5.1.1. Axle:

Make:

Type:

Model:

Test axle load (F_e identifier): ID3- daN

K-4, Test Report No.
- K-5.1.2. Brake:

Make:

⁽¹⁾ Strike out what does not apply.

⁽⁴⁾ Only to be completed when an automatic brake wear adjustment device is installed.

Type:

- Model:
- Brake lining:
- Make/Type:
- K-5.1.3. Actuation:
- Manufacturer:
- Type (cylinder/diaphragm)⁽¹⁾
- Model:
- Lever length (l): mm
- K-5.1.4. Automatic brake adjustment device:
- Manufacturer (name and address):
- Make:
- Type:
- Version:
- K-5.2. Record of test results
- K-5.2.1. Performance of the automatic brake adjustment device
- K-5.2.1.1. Hot performance of service braking systems determined according to the test defined in K-3.3.6.2.1. (a) in (%)
- or
- Actuator stroke s_A determined according to the test defined in K-3.3.6.2.1. (b): in (mm)
- K-5.2.1.2. Free running according to K-3.3.6.3. : yes / no⁽¹⁾
- K-5.3. Name of Test Agency/Type Approval Authority⁽¹⁾conducting the test:
- K-5.4. Date of test:
- K-5.5. This test has been carried out and the results reported in accordance with K-3.3.6.2.
- K-5.6. At the end of test defined in item 5 above the requirements of paragraph 5.2.2.8.1. of Standard No. 13 were deemed to be: Fulfilled / Not fulfilled⁽¹⁾

⁽¹⁾Strike out what does not apply.

- K-5.7. Test Agency carrying out the test
Signed:Date:
- K-6.0 Trailer axle and brake information document with respect to the alternative Type-I and Type-III procedure
- K-6.1. General
 - K-6.1.1. Name and address of axle or vehicle manufacturer:
 - K-6.2. Axle data
 - K-6.2.1. Manufacturer (name and address):
 - K-6.2.2. Type/variant:
 - K-6.2.3. Axle identifier: ID1-
 - K-6.2.4. Test axle load (F_e): daN
 - K-6.2.5. Wheel and brake data according to the following Figures 1A and 1B

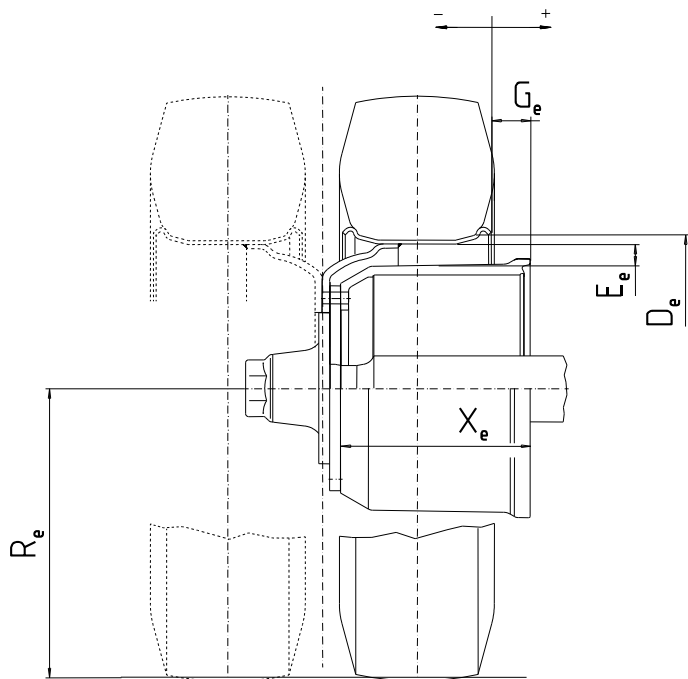


Figure 1A

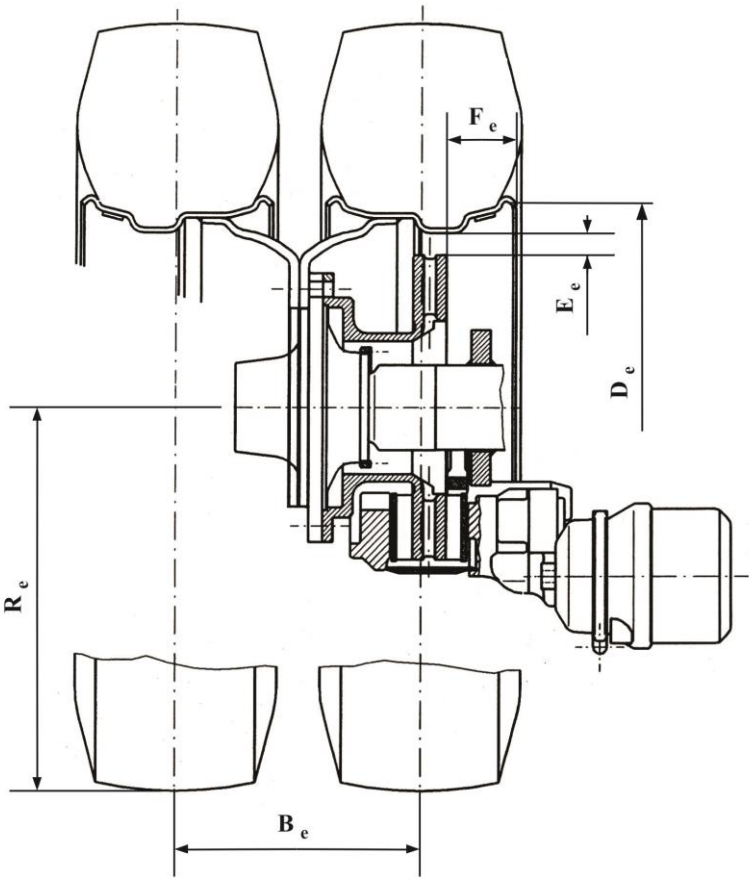


Figure 1B

- K-6.3. Brake
- K-6.3.1. General information
- K-6.3.1.1. Make:
- K-6.3.1.2. Manufacturer (name and address):
- K-6.3.1.3. Type of brake (e.g. drum / disc):
- K-6.3.1.3.1. Variant (e.g. S-cam, single wedge etc.):
- K-6.3.1.4. Brake identifier: ID2-
- K-6.3.1.5. Brake data according to the following Figures 2A and 2B:

Figure 2A

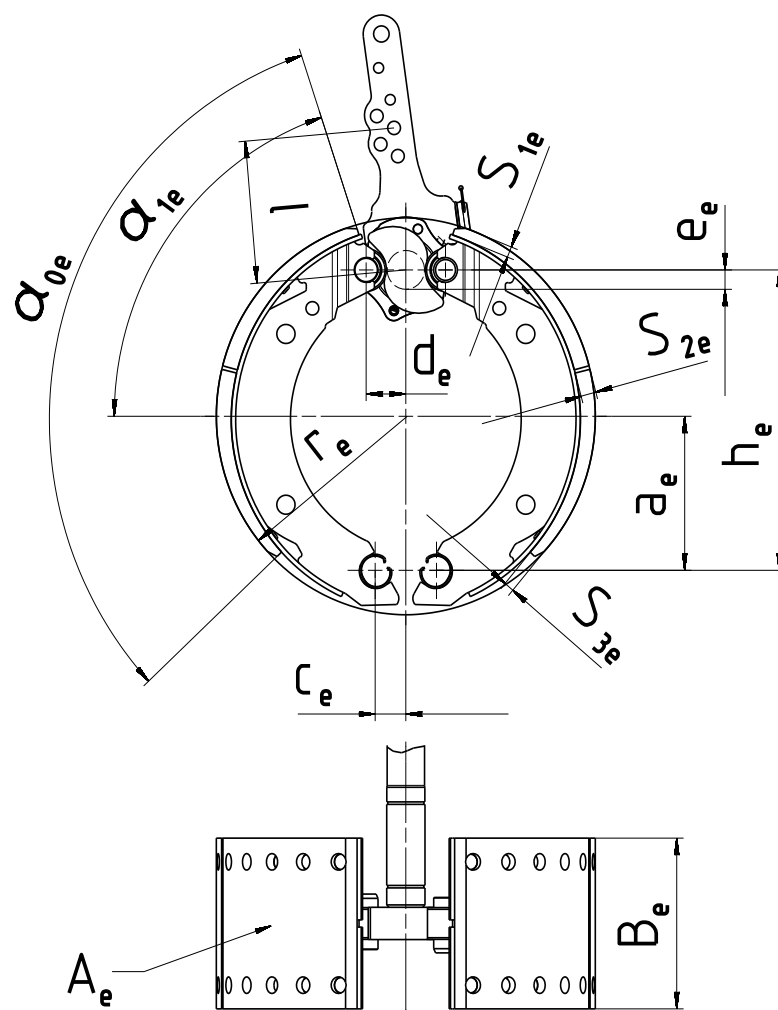
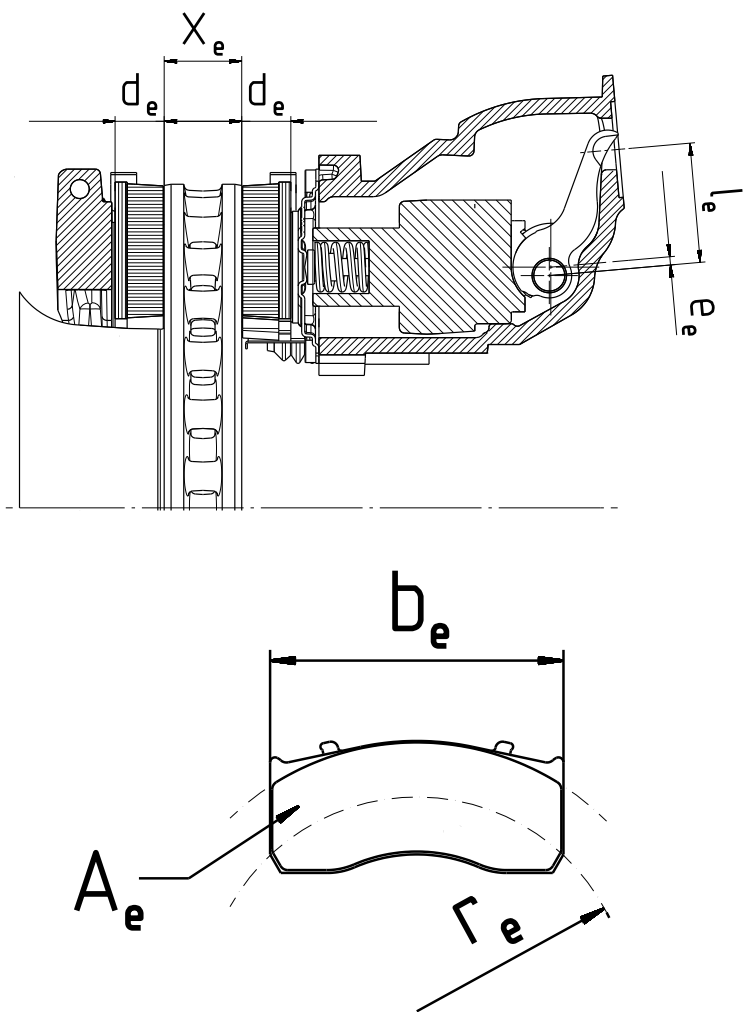


Figure 2B



x_e (mm)	h_e (mm)	c_e (mm)	d_e (mm)	e_e (mm)	α_{0e}	α_{1e}	b_e (mm)	r_e (mm)	A_e (cm ²)	S_{1e} (m m)	S_{2e} (m m)	S_{3e} (mm)
a_e (mm)												

- K-6.3.2. Drum brake data
- K-6.3.2.1. Brake adjustment device (external/integrated):
- K-6.3.2.2. Declared maximum brake input torque C_{max} : Nm
- K-6.3.2.3. Mechanical efficiency: η =
- K-6.3.2.4. Declared brake input threshold torque $C_{0,dec}$: Nm
- K-6.3.2.5. Effective length of the cam shaft: mm
- K-6.3.3. Brake drum
- K-6.3.3.1. Max diameter of friction surface (wear limit): mm
- K-6.3.3.2. Base material:

K-6.3.3.3	Declared mass: kg
K-6.3.3.4.	Nominal mass: kg
K-6.3.4.	Brake lining
K-6.3.4.1.	Manufacturer and address:
K-6.3.4.2.	Make:
K-6.3.4.3.	Type:
K-6.3.4.4.	Identification (type identification on lining):
K-6.3.4.5	Minimum thickness (wear limit): mm
K-6.3.4.6.	Method of attaching friction material to brake shoe:
K-6.3.4.6.1.	Worst case of attachment (in the case of more than one):
K-6.3.5.	Disc brake data
K-6.3.5.1.	Connection type to the axle (axial, radial, integrated, etc.):
K-6.3.5.2.	Brake adjustment device (external/integrated):
K-6.3.5.3	Max. actuation stroke: mm
K-6.3.5.4.	Declared maximum input force Th_{Amax} : daN
K-6.3.5.4.1	$C_{max} = Th_{Amax} \cdot l_e$: Nm
K-6.3.5.5.	Friction radius: r_e = mm
K-6.3.5.6.	Lever length: l_e = mm
K-6.3.5.7.	Input/output ratio (l_e/e_e) : i =
K-6.3.5.8.	Mechanical efficiency η =
K-6.3.5.9.	Declared brake input threshold force $Th_{A0,dec}$: N
K-6.3.5.9.1	$C_{0,dec} = Th_{A0,dec} \cdot l_e$: Nm
K-6.3.5.10.	Minimum rotor thickness (wear limit): mm
K-6.3.6.	Brake disc data:
K-6.3.6.1.	Disc type description:
K-6.3.6.2.	Connection/mounting to the hub:
K-6.3.6.3	Ventilation (yes/no):
K-6.3.6.4.	Declared mass: kg
K-6.3.6.5.	Nominal mass: kg
K-6.3.6.6.	Declared external diameter: mm
K-6.3.6.7.	Minimum external diameter: mm
K-6.3.6.8.	Inner diameter of friction ring: mm
K-6.3.6.9.	Width of ventilation channel (if appl.): mm

- K-6.3.6.10. Base material:
- K-6.3.7. Brake pad data:
 - K-6.3.7.1. Manufacturer and address:
 - K-6.3.7.2. Make:
 - K-6.3.7.3. Type:`
 - K-6.3.7.4. Identification (type identification on pad back plate):
 - K-6.3.7.5. Minimum thickness (wear limit): mm
 - K-6.3.7.6. Method of attaching friction material to pad back plate:
 - K-6.3.7.6.1. Worst case of attachment (in the case of more than one):

ANNEX L

(Clauses 6 and C-3.1.2.3)

CONDITIONS GOVERNING THE TESTING OF VEHICLES EQUIPPED WITH INERTIA (OVERRUN) BRAKING SYSTEMS

L-1 GENERAL PROVISIONS

- L-1.1 The inertia (overrun) braking system of a trailer comprises the control device, the transmission and the wheel brakes, hereinafter called brakes.
- L-1.2 The control device is the aggregate of the components integral with the traction device (coupling head).
- L-1.3 The transmission is the aggregate of the components comprised between the last part of the coupling head and the first part of the brake.
- L-1.4 The brake is the part in which the forces opposing the movement of the vehicle develop. The first part of the brake is either the lever actuating the brake cam or similar components (mechanical-transmission inertia braking system), or the brake cylinder (hydraulic-transmission inertia braking system).
- L-1.5 Braking systems in which accumulated energy (for example; electric, pneumatic or hydraulic energy) is transmitted to the trailer by the towing vehicle and is controlled only by the thrust on the coupling do not constitute inertia braking systems within the meaning of this Standard.
- L-1.6 **Tests**
 - L-1.6.1 Determination of essential components of the brake.
 - L-1.6.2 Determination of essential components of the control device and verification of the latter's conformity with the provisions of this Standard.
 - L-1.6.3 Checking on the vehicle:
 - a) The compatibility of the control device and the brake; and
 - b) The transmission.

L-2 SYMBOLS AND DEFINITIONS

L-2.1 Units Used

- a) Mass: kg;
- b) Force: N;
- c) Acceleration due to gravity: $g = 10 \text{ m/s}^2$
- d) Torques and moments: Nm;
- e) Areas: cm^2 ;
- f) Pressures: kPa; and
- g) Lengths: unit specified in each case.

L-2.2 Symbols Valid for All Types of Brakes (see Fig. 14)

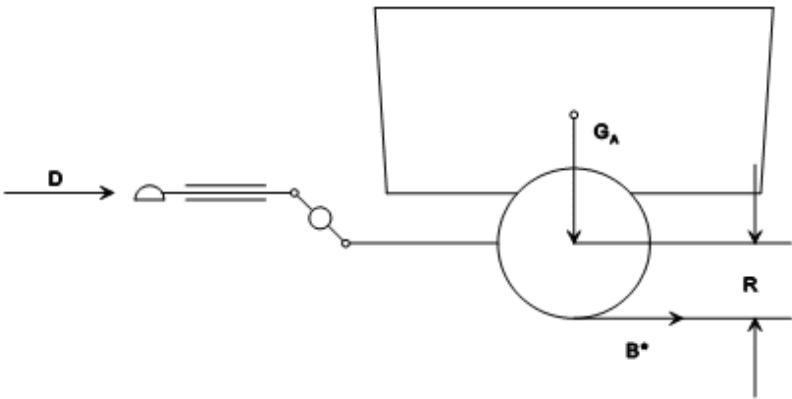


FIG. 14

SYMBOLS VALID FOR ALL TYPES OF BRAKES

L-2.2.1	G_A	Trailer's technically permissible maximum mass as declared by the manufacturer;
L-2.2.2	G'_A	Trailer's maximum mass capable of being braked by the control device, as declared by the manufacturer;
L-2.2.3	G_B	Trailer's maximum mass capable of being braked by joint operation of all of the trailer's brakes
	$G_B = n \cdot G_{B0}$;	
L-2.2.4	G_{B0}	Fraction of trailer's permissible maximum mass capable of being braked by one brake, as declared by the manufacturer;
L-2.2.5	B^*	Required braking force;
L-2.2.6	B	Required braking force taking account of rolling resistance;
L-2.2.7	D^*	Permissible thrust on coupling;
L-2.2.8	D	Thrust on coupling;
L-2.2.9	P'	Control device output force;
L-2.2.10	K	Supplementary force of control device, conventionally designated by the force D corresponding to the point of intersection with the axis of the abscissae of the extrapolated curve expressing P' in terms of D , measured with the device in the mid-travel position (see Figs. 15 and 16);

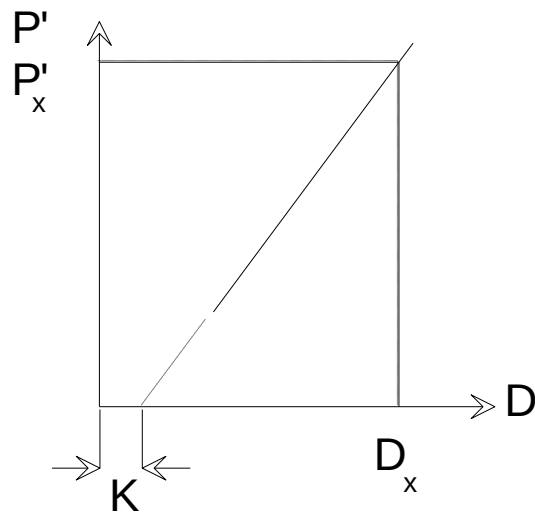


FIG. 15

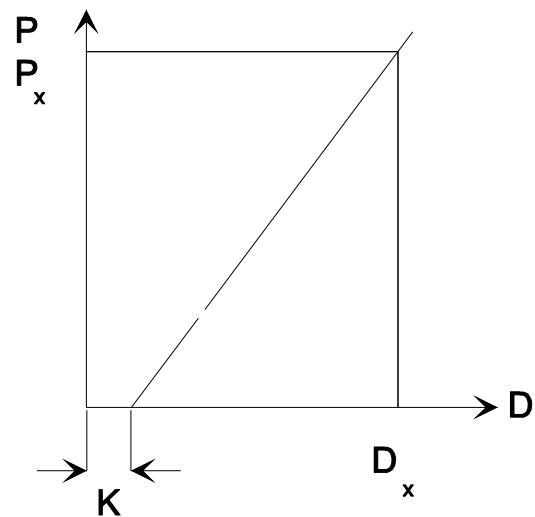
MECHANICAL-TRANSMISSION

FIG. 16

HYDRAULIC TRANSMISSION

- | | | |
|----------|-------------|---|
| L-2.2.11 | K_A | Stress threshold of control device, i.e., the maximum thrust on the coupling head which can be applied for a short time without placing any stress on the output side of the control device. The symbol K_A is conventionally applied to the force measured when the coupling head begins to be pushed home at a speed of 10 to 15 mm/s, the control device transmission being uncoupled; |
| L-2.2.12 | D_1 | Maximum force applied to the coupling head when it is being pushed home at a speed of $s \text{ mm/s} \pm 10 \text{ percent}$, the transmission being uncoupled; |
| L-2.2.13 | D_2 : | The maximum force applied to the coupling head when it is being pulled at a speed of $s \text{ mm/s} \pm 10 \text{ percent}$ out of the position of maximum compression, the transmission being uncoupled; |
| L-2.2.14 | η_{H0} | Efficiency of inertia control device; |

L-2.2.15	η_{H1}	Efficiency of transmission system;
L-2.2.16	η_H	Overall efficiency of control device and transmission $\eta_H = \eta_{H0} \cdot \eta_{H1}$;
L-2.2.17	s	Travel of control in mm;
L-2.2.18	s'	Effective (useful) travel of control in mm, determined as required by L 10.4;
L-2.2.19	s''	Spare travel of master cylinder, measured in mm at coupling head;
L-2.2.19.1	s_{Hz}	Stroke of the master cylinder, in mm according to Fig. 20;
L-2.2.19.2	s''_{Hz}	Spare travel of the master cylinder, in mm at piston rod, according to Fig. 20;
L-2.2.20	s_o	Loss of travel, that is, travel, in mm of the coupling head when the latter is so actuated as to move from 300 mm above to 300 mm below the horizontal, the transmission remaining stationary;
L-2.2.21	$2s_B$	Brake-shoe lift (brake-shoe application travel), in mm, measured on diameter parallel to applying device, the brakes not being adjusted during the test;
L-2.2.22	$2s_B^*$	Minimum brake shoe centre lift (minimum brake shoe application travel) (in mm) for wheel brakes with drum brakes

$$2s_B^* = 2.4 + \frac{4}{1000} \cdot 2r$$

2r being the diameter of the brake drum in mm (see Fig. 17)

For wheel brakes with disc brakes with hydraulic transmission

$$2s_B^* = 1.1 \cdot \frac{10 \cdot V_{60}}{F_{RZ}} + \frac{1}{1000} \cdot 2r_A$$

where

V_{60} = Fluid volume absorbed by one wheel brake at a pressure corresponding to a braking force of $1.2 B^* = 0.6 \cdot G_{B0}$ and a maximum tyre radius.

$2r_A$ = outer diameter of brake disc.

(V_{60} in cm^3 , F_{RZ} in cm^2 and r_A in mm)

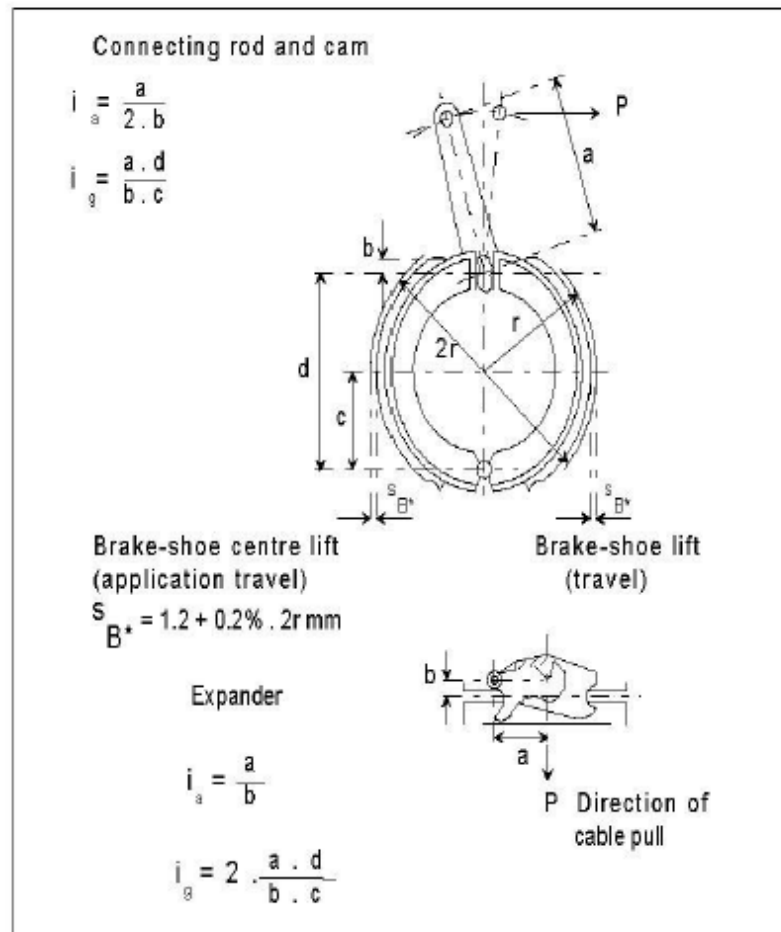


FIG. 17 BRAKE CHECKS

L-2.2.23	M*	Braking torque as specified by the manufacturer in L 9.2 (5). This braking torque shall produce at least the prescribed braking force B*;
L-2.2.23.1	M _T	Test braking torque in the case when no overload protector is fitted (according to L-6.2.1);
L-2.2.24	R	Dynamic tyre rolling radius (m);
L-2.2.25	n	Number of brakes.
L-2.2.26	M _r	Maximum braking torque resulting from the maximum permissible travel s _r or the maximum permissible fluid volume V _r when the trailer moves rearward (including rolling resistance = 0.01 · g · G _{B0});
L-2.2.27	s _r	Maximum permissible travel at the brake control lever when the trailer moves rearward;
L-2.2.28	V _r	Maximum permissible fluid volume absorbed by one braking wheel when the trailer moves rearward;
L-2.2.29	D _A	Application force at input side of the control device, at which the overload protector is activated;
L-2.2.30	M _A	Braking torque at which the overload protector is activated.

L-2.3 Symbols Valid for Mechanical-Transmission Braking Systems (see Fig. 18);

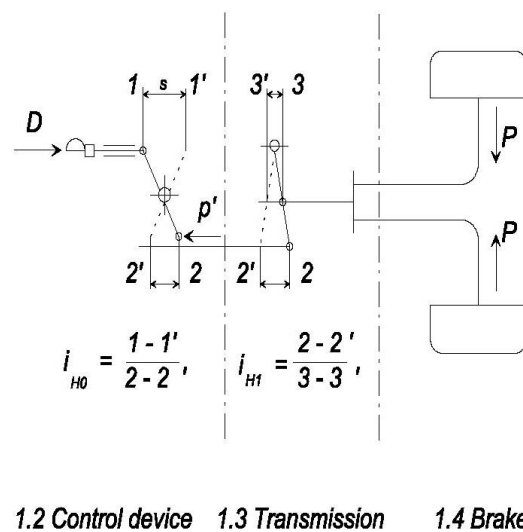


FIG. 18

MECHANICAL-TRANSMISSION BRAKING SYSTEM

- | | | |
|---------|----------|--|
| L-2.3.1 | i_{H0} | Reduction ratio between travel of coupling head and travel of lever at output side of control device; |
| L-2.3.2 | i_{H1} | Reduction ratio between travel of lever at output side of control device and travel of brake lever (gearing down of transmission); |
| L-2.3.3 | i_H | Reduction ratio between travel of coupling head and travel of brake lever |

$$i_H = i_{H0} \cdot i_{H1}$$

- | | | |
|-----------|----------|--|
| L-2.3.4 | i_g | Reduction ratio between travel of brake lever and lift (application travel) at brake-shoe centre (see Fig. 17); |
| L-2.3.5 | P | Force applied to the brake control lever (see Fig. 17); |
| L-2.3.6 | P_o | Brake-retraction force when the trailer moves forward; that is, in graph $M = f(P)$, the value of the force P at the point of intersection of the extrapolation of this function with the abscissa (see Fig. 19); |
| L-2.3.6.1 | P_{or} | Brake-retraction force when the trailer moves rearward (see Fig. 19); |

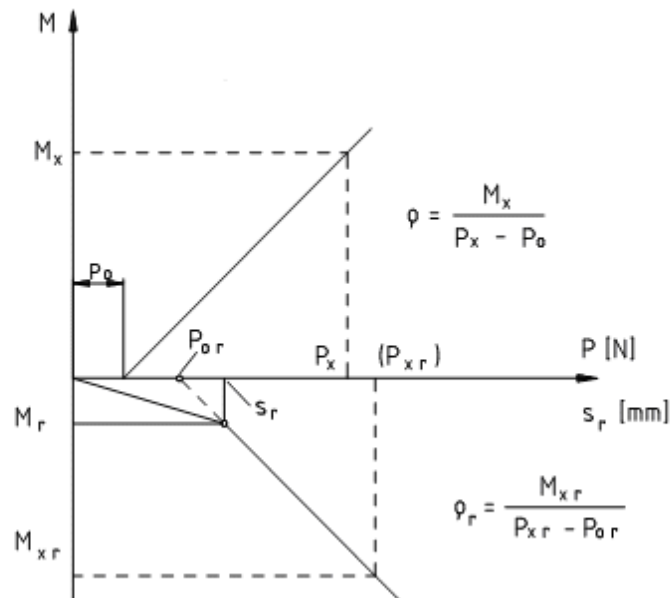


FIG. 19
MECHANICAL BRAKE

- L-2.3.7 P^* Force applied to the brake control lever to produce the braking force B^* ;
- L-2.3.8 P_T Test force according to L-6.2.1;
- L-2.3.9 ρ Characteristic of the brake when the trailer moves forward as defined from:

$$M = \rho (P - P_0)$$

- L-2.3.9.1 ρ_r Characteristic of the brake when the trailer moves rearward as defined from:

$$M_r = \rho_r (P_r - P_{0r})$$

- L-2.3.10. s_{cf} : Rear cable or rod travel at compensator when brakes operate in forward direction ⁽¹⁾
- L-2.3.11. s_{cr} Rear cable or rod travel at compensator when brakes operate in rearward direction ⁽¹⁾
- L-2.3.12. s_{cd} : Differential travel at compensator when only one brake operates in the forward direction and the other in the reverse direction ⁽¹⁾

Where:

$$s_{cd} = s_{cr} - s_{cf} \text{ (see Figure 5A of Appendix 1)}$$

⁽¹⁾ Paragraphs 2.3.10., 2.3.11. and 2.3.12. only apply to the parking brake differential travel calculation method.

L-2.4 Symbols Valid for Hydraulic-Transmission Braking Systems (see Fig. 20)

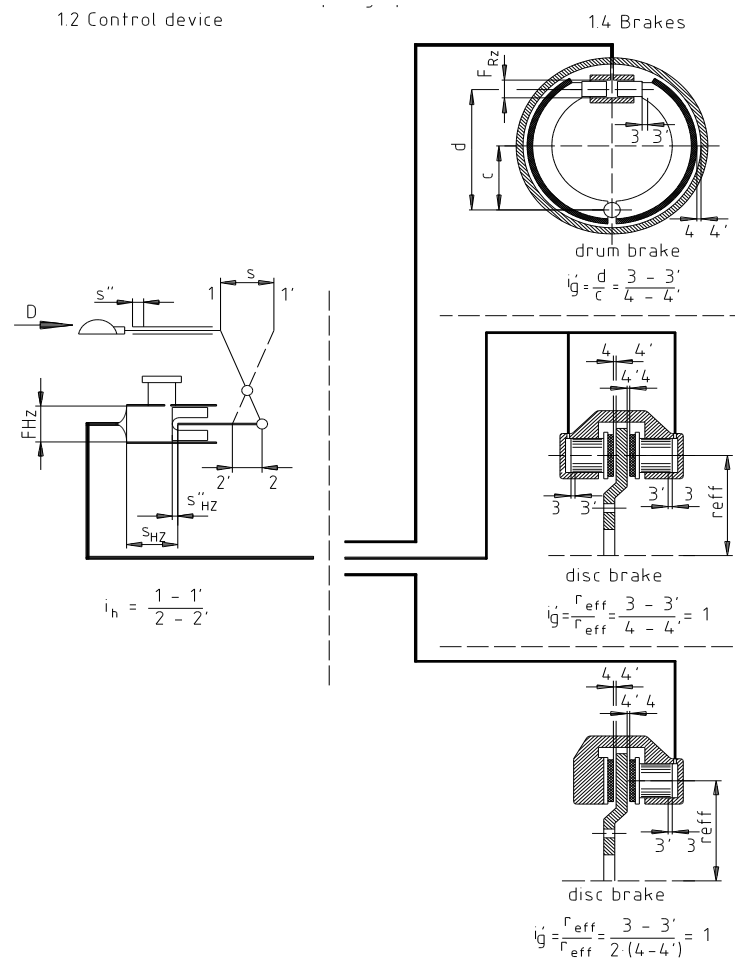


FIG. 20
HYDRAULIC TRANSMISSION BRAKING SYSTEM

- | | | |
|---------|----------|---|
| L-2.4.1 | i_h | Reduction ratio between travel of coupling head and travel of piston in master cylinder; |
| L-2.4.2 | i'_g | Reduction ratio between travel of cylinder thrust point and lift (application travel) of brake-shoe centre; |
| L-2.4.3 | F_{RZ} | Surface area of piston of one wheel cylinder for drum brake(s); for disc brake(s), sum of the surface area of the caliper piston(s) on one side of the disc; |
| L-2.4.4 | F_{HZ} | Surface area of piston in master cylinder; |
| L-2.4.5 | p | Hydraulic pressure in brake cylinder; |
| L-2.4.6 | p_o | Retraction pressure in the brake cylinder when the trailer moves forward; that is, in graph of $M = f(p)$, the value of the pressure p at the point of intersection of the extrapolation of this function with the abscissa (see Fig. 21); |

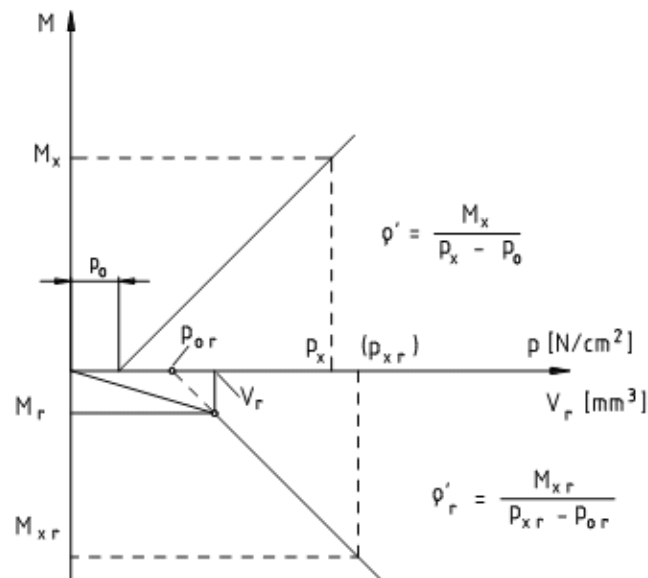


FIG. 21

HYDRAULIC BRAKE

- L-2.4.6.1 p_{or} : Brake retraction pressure when the trailer moves rearward (see Fig. 21);
- L-2.4.7 p^* : Hydraulic pressure in the brake cylinder to produce the braking force B^* ;
- L-2.4.8 p_T : Test pressure according to L-6.2.1
- L-2.4.9 ρ' : Characteristic of the brake when the trailer moves forward as defined from: $M = \rho' (p - p_o)$
- L-2.4.9.1 ρ'_r : Characteristic of the brake when the trailer moves rearward as defined from:

$$M_r = \rho'_r (p_r - p_{or})$$

L-2.5 **Symbols with Respect to the Braking Requirements Relating to Overload Protectors**

- L-2.5.1 D_{op} : Application force at the input side of the control device, at which the overload protector is activated;
- L-2.5.2 M_{op} : Brake torque at which the overload protector is activated (as declared by the manufacturer);
- L-2.5.3 M_{Top} : Minimum test braking torque in the case when an overload protector is fitted (see L-6.2.2.2);
- L-2.5.4 P_{op_Min} : Force applied to the brake at which the overload protector is activated (see L-6.2.2.1);
- L-2.5.5 P_{op_Max} : Maximum force (when the coupling head is pushed fully home) which is applied by the overload protector to the brake (see L-6.2.2.3);

L-2.5.6	p_{op_Min}	Pressure applied to the brake at which the overload protector is activated (see L-6.2.2.1);
L-2.5.7	p_{op_Max}	Maximum hydraulic pressure (when the coupling head is pushed fully home) which is applied by the overload protector to the brake actuator (see L-6.2.2.3);
L-2.5.8	P_{Top}	Minimum test brake force in the case when an overload protector is fitted (see L-6.2.2.2); and
L-2.5.9	p_{Top}	Minimum test brake pressure in the case when an overload protector is fitted (see L-6.2.2.2).

L-3 GENERAL REQUIREMENTS

- L-3.1 The transmission of force from the coupling head to the trailer's brakes shall be effected either by rod linkage or by one or more fluids. However, a sheathed cable (Bowden cable) may provide part of the transmission; this part shall be as short as possible. The control rods and cables shall not contact the trailer frame or other surfaces that may affect the application or release of the brake.
- L-3.2 All bolts at joints shall be adequately protected. In addition, these joints shall be either self-lubricating or readily accessible for lubrication.
- L-3.3 Inertia braking devices shall be so arranged that in the case when the coupling head travels to its fullest extent, no part of the transmission seizes, undergoes permanent distortion, or breaks. This shall be checked by uncoupling the end of the transmission from the brake control levers.
- L-3.4 The inertia braking system shall allow the trailer to be reversed with the towing vehicle without imposing a sustained drag force exceeding $0.08 g_{GA}$. Devices used for this purpose shall act automatically and disengage automatically when the trailer moves forward.
- L-3.5 Any special device incorporated for the purpose of L-3.4 shall be such that the parking performance when facing up a gradient shall not be adversely affected.
- L-3.6 Inertia braking systems may incorporate overload protectors. They shall not be activated at a force of less than $D_{op} = 1.2 \cdot D^*$ (when fitted at the control device) or at a force of less than $P_{op} = 1.2 \cdot P^*$ or at a pressure of less than $p_{op} = 1.2 \cdot p^*$ (when fitted at the wheel brake) where the force P^* or the pressure p^* corresponds to a braking force of $B^* = 0,5 \cdot g \cdot G_{Bo}$.

L-4 REQUIREMENTS FOR CONTROL DEVICES

- L-4.1 The sliding members of the control device shall be long enough to enable the full travel to be used even when the trailer is coupled.
- L-4.2 The sliding members shall be protected by a bellows or some equivalent device. They shall either be lubricated or be constructed of self-lubricating materials. The surfaces in frictional contact shall be made of a material such that there is neither electrochemical torque nor any mechanical incompatibility liable to cause the sliding members to seize.
- L-4.3 The stress threshold (K_A) of the control device shall be not less than

0.02 g G'A and not more than 0.04 g G'A.

L-4.4 The maximum insertion force D₁ may not exceed 0.10 g G'A in trailers with rigid drawbars and 0.067 g G'A in multi-axled trailers with pivoted drawbars.

L-4.5 The maximum tractive force D₂ shall be not less than 0.1 g G'A and not more than 0.5 g G'A.

L-5 TESTS AND MEASUREMENTS TO BE CARRIED OUT ON THE CONTROL DEVICES

L-5.1 Control devices submitted to the certifying agency conducting the tests shall be checked for conformity with the requirements of L-3 and L-4.

L-5.2 The following shall be measured in respect of all types of brakes:

- a) Travel s and effective travel s';
- b) Supplementary force K;
- c) Force threshold K_A;
- d) Insertion force D₁;
- e) Tractive force D₂.

L-5.3 In the case of mechanical-transmission inertia braking systems, the following shall be determined:

- a) The reduction ratio i_{H0} measured at the mid-travel position of the control;
- b) The control-device output force P' as a function of the thrust D on the drawbar. The supplementary force K and the efficiency are derived from the representative curve obtained from these measurements (see Fig. 15)

$$\eta_{H0} = \frac{1}{i_{H0}} \cdot \frac{P'}{D - K}$$

L-5.4 In the case of hydraulic-transmission inertia braking systems, the following shall be determined:

- a) The reduction ratio i_h measured at the mid-travel position of the control;
- b) The master cylinder output pressure p as a function of the thrust D on the drawbar and of the surface area F_{HZ} of the master-cylinder piston, as specified by the manufacturer. The supplementary force K and the efficiency are derived from the representative curve obtained from these measurements (see Fig. 16);

$$\eta_H = \frac{1}{i_h} \cdot \frac{p \cdot F_{HZ}}{D - K}$$

- c) The spare travel of the master cylinder s , as referred to in L-2.2.19;
- d) Surface area F_{HZ} of the piston in the master cylinder;
- e) Stroke s_{HZ} of the master cylinder, (in mm); and
- f) Spare travel s''_{HZ} of the master cylinder, (in mm);

L-5.5 In the case of inertia braking system on multi-axled trailers with pivoted drawbar, the loss of travel s_0 referred to in L10.4.1 should be measured.

L-6.0 REQUIREMENTS FOR BRAKES

L-6.1 In addition to the brakes to be checked, the manufacturer shall submit to the certifying agency conducting the tests, drawings of the brakes showing the type, dimensions and material of the essential components and the make and type of the linings. In the case of hydraulic brakes, these drawings shall show the surface area F_{RZ} of the brake cylinders. The manufacturer shall also specify the braking torque M^* and the mass G_{Bo} defined in L-2.2.4.

L-6.2 Testing Conditions

L-6.2.1 In the case when an overload protector is neither fitted nor intended to be fitted within the inertia (overrun) braking system, the wheel brake shall be tested with the following test forces or pressures:

$$P_T = 1.8 P^* \text{ or } p_T = 1.8 p^* \text{ and } M_T = 1.8 M^* \text{ as appropriate.}$$

L-6.2.2 In the case when an overload protector is fitted or intended to be fitted within the inertia (overrun) braking system, the wheel brake shall be tested with the following test forces or pressures.

L-6.2.2.1 The minimum design values for overload protector shall be specified by the manufacturer and shall not be less than

$$P_{op} = 1.2 P^* \text{ or } p_{op} = 1.2 p^*$$

L-6.2.2.2 The ranges of minimum test force P_{Top} or minimum test pressure p_{Top} and the minimum test torque M_{Top} are:

$$P_{Top} = 1.1 \text{ to } 1.2 P^* \text{ or } p_{Top} = 1.1 \text{ to } 1.2 p^*$$

And

$$M_{Top} = 1.1 \text{ to } 1.2 M^*$$

L-6.2.2.3 The maximum values (P_{op_max} or p_{op_max}) for the overload protector shall be specified by the manufacturer and shall not be more than P_T or p_T respectively.

L-7.0 TESTS AND MEASUREMENTS TO BE CARRIED OUT ON THE BRAKES

L-7.1 Brakes and components submitted to the certifying agency conducting the tests shall be tested for conformity with the requirements of L-6.

L-7.2 The following should be determined:

- a) The minimum brake-shoe lift (minimum brake-shoe application travel), $2s_{B*}$; and
- b) The brake-shoe center lift (brake-shoe application travel) $2s_B$ (which shall be greater than $2s_{B*}$);

L-7.3 In the case of mechanical brakes, the following shall be determined:

- a) Reduction ratio i_g (see Fig. 17);
- b) Force P^* for braking torque M^* ; and
- c) Torque M^* as a function of the force P^* applied to the control lever in mechanical-transmission systems. The rotational speed of the braking surfaces shall correspond to an initial vehicle speed of 60 km/h when the trailer moves forward and 6 km/h when the trailer moves rearward. The following shall be derived from the curve obtained from these measurements (see Fig. 19):
 - 1) The brake-retraction force P_o and the characteristic value ρ when the trailer moves forward;
 - 2) The brake-retraction force P_{or} and the characteristic value ρ_r when the trailer moves rearward;
 - 3) Maximum braking torque M_r up to the maximum permissible travel s_r when the trailer moves rearward (see Fig. 19); and
 - 4) Maximum permissible travel at the brake control lever when the trailer moves rearward (see Fig. 19).

L-7.4 In the case of hydraulic brakes, the following should be determined:

- a) Reduction ratio i_g' (see Fig. 20);
- b) Pressure p^* for braking torque M^* ;
- c) Torque M^* as a function of the pressure p^* applied to the brake cylinder in hydraulic transmission systems. The rotational speed of the braking surfaces shall correspond to an initial vehicle speed of 60 km/h when the trailer moves forward and 6 km/h when the trailer moves rearward. The following shall be derived from the curve obtained from these measurements (see Fig. 21):
 - 1) The retraction pressure p_o and the characteristic ρ' when the trailer moves forward;
 - 2) The retraction pressure p_{or} and the characteristic ρ'_r when the trailer moves rearward;

- 3) Maximum braking torque M_r up to the maximum permissible fluid volume V_r when the trailer moves rearward (see Fig. 21); and
- 4) Maximum permissible fluid volume V_r absorbed by one braking wheel when the trailer moves rearward (see Fig. 21).

d) Surface area F_{RZ} of the piston in the brake cylinder.

L-7.5. Alternative procedure for the Type-I test

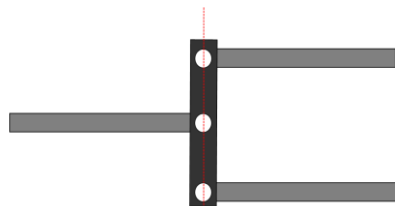
L-7.5.1. The Type-I test according to C-1.5. does not have to be carried out on a vehicle submitted for type approval, if the braking system components are tested on an inertia test bench to meet the prescriptions of C-1.5.2. and C-1.5.3.

L-7.5.2. The alternative procedure for the Type-I test shall be carried out in accordance with the provisions laid down in K-3. 3.5.2. (in analogy also applicable for disc brakes).

L-8.0 Simulated gradient parking brake force differential

L-8.1. Calculation method

L-8.1.1. The pivot points in the compensator shall lie in a straight line with the park brake at the rest position.



All compensator pivots to be in-line

Alternative arrangements can be used, if they provide equal tension in both rear cables, even when there are differences in travel between the rear cables.

L-8.1.2. Drawing details are to be provided to demonstrate that the compensator articulation is sufficient to ensure equal cable tension is applied to each of the rear cables. The compensator needs to have sufficient distance across the width to facilitate the differential travels left to right. The jaws of the yokes also need to be deep enough relative to their width to make sure that they do not prevent articulation when the compensator is at an angle.

Differential travel at compensator (s_{cd}) shall be derived from:

$$s_{cd} \geq 1.2 \bullet (S_{cr} - S_c')$$

Where:

S_c'	=	S'/i_H (travel at compensator - forward operation) and $S_c' = 2 \cdot S_B/i_g$
S_{cr}	=	S_r/i_H (travel at compensator - rearward operation)

L-9.0 TEST REPORTS

Applications for the approval of trailers equipped with inertia braking systems shall be accompanied by the test reports relating to the control device and the brakes and the test report on the compatibility of the inertia type control device, the transmission device and the brakes of the trailer, these reports including at least the particulars prescribed in L 9.1, L 9.2 and L 9.3.

L-9.1 Test Report on Inertia-Braking System Control Device

- 1.0 Manufacture:
- 2.0 Make:
- 3.0 Type :
- 4.0 Characteristics of trailers for which control device intended by manufacturer:
 - 4.1. mass G'_A (kg):
 - 4.2. permissible static vertical force at towing-device head (N):
 - 4.3. trailer with rigid drawbar/multi-axled trailer with pivoted drawbar
- 5.0 Brief description (List of attached plans and dimensioned drawings):
- 6.0 Diagram showing principle of control :
- 7.0 Travel s (mm)
- 8.0 Reduction ratio of control device:
 - 8.1. With mechanical transmission device
 $i_{H0} = \text{from } \dots\dots \text{ to } \dots\dots$
 - 8.2. With hydraulic transmission device
 $i_h = \text{from } \dots\dots\dots \text{ to } \dots\dots$
 F_{HZ} (cm²):
Travel of master cylinder s_{Hz} (mm):
Tpore travel of master cylinder s''_{Hz} (mm):
- 9.0 Test results:
 - 9.1 Efficiency

- with mechanical transmission device, $\eta_H = \dots\dots\dots$
- with hydraulic transmission device, $\eta_H = \dots\dots\dots$
- 9.2 Supplementary force, K (N):
- 9.3 Maximum compressive force, D_1 (N)
- 9.4 Maximum tractive force, D_2 (N)
- 9.5 **Force** threshold, K_A (N)
- 9.6 Loss of travel and spare travel:
Where the position of the drawing device has an effect, s_o (mm)
with a hydraulic-transmission device, $s'' = s''_{Hz} \cdot i_h$ (mm)
- 9.7 Effective (useful) travel of control, s' (mm)
- 9.8 An overload protector according to L-3.6 is provided / not provided
- 9.8.1 If the overload protector is fitted before the transmission lever of the control device
- 9.8.1.1 Threshold force of the overload protector,
 D_{op} (N):
- 9.8.1.2 Where the overload protector is mechanical
max. force which the inertia control device can develop
 $P'_{max}/i_{Ho} = P_{op_max} = (N)$
- 9.8.1.3 Where the overload protector is hydraulic the pressure which the inertia control device can develop
 $p'_{max}/i_h = p_{op_max} = (N/cm^2)$
- 9.8.2 If the overload protector is fitted after the transmission lever of the control device
- 9.8.2.1 Threshold force on the overload protector where the overload protector is mechanical
 $D_{op} \cdot i_{Ho} = (N)$
where the overload protector is hydraulic
 $D_{op} \cdot i_h = (N)$
- 9.8.2.2 Where the overload protector is mechanical max force which the inertia control device can develop
 $P'_{max} = P_{op_max} = (N)$
- 9.8.2.3 Where the overload protector is hydraulic the pressure which the inertia control device can develop

$$p'_{\max} = p_{\text{op_max}} \text{ (N/cm}^2\text{)}$$

10.0 The control device described above complies/does not comply with the requirements of L-3, L- 4 and L-5.

Date

Signature

11.0 This test has been carried out and the results reported in accordance with relevant provisions of Annex L

Certifying agency carrying out the test

Signed:

Date:

L- 9.2 TEST REPORT ON THE BRAKE

1.0 Manufacturer:

2.0 Make:

3.0 Type:

4.0 Permissible maximum mass per wheel

$$G_{B0} = \text{(kg)}$$

5.0 Braking torque M^* (as specified by the manufacturer according to L-2.2.23) = (Nm)

6.0 Dynamic tyre rolling radius

$$R_{\min} = \text{(m)};$$

$$R_{\max} = \text{(m)}$$

7.0 Brief description (List of plans and dimensioned drawings)

8.0 Diagram showing principle of brake

9.	Test result:			
	Mechanical brake			Hydraulic brake
9.1.	Reduction ratio $i_g =$		9.1.A.	Reduction ratio $i'_g =$
9.2.	Lift (application travel) $s_B = \text{mm}$		9.2.A.	Lift (application travel) $s_B = \text{m}$

9.3.	Prescribed lift (prescribed application travel) $s_{B^*} = \text{mm}$		9.3.A.	Prescribed lift (prescribed application travel) $s_{B^*} = \text{mm}$
9.4.	Retraction force $P_o = \text{N}$		9.4.A.	Retraction pressure $p_o = \text{N/cm}^2$
9.5.	Coefficient (characteristic) $\kappa = \text{m}$		9.5.A.	Coefficient (characteristic) $\kappa' = \text{m}$
9.6	An overload protector according to L-3.6 is/is not provided		9.6.A.	An overload protector according to L-3.6 is/is not provided
9.6.1.	Braking torque activating the overload protector $M_{op} = (\text{Nm})$		9.6.1.A.	Braking torque activating the overload protector $M_{op} = (\text{Nm})$
9.7.	Force for M^* $P^* = (\text{N})$		9.7.A.	Pressure for M^* $P^* = (\text{N/cm}^2)$
			9.8.A.	Surface area of wheel cylinder $F_{RZ} = (\text{cm}^2)$
			9.9.A.	(for disk brakes) Fluid volume absorption $V_{60} = (\text{cm}^3)$

9.10 Service brake performance when the trailer moves rearwards (see Fig. 6 and Fig.7)

9.10.1 Maximum Fig. 6 braking torque $M_r = (\text{Nm})$

9.10.1A Maximum Fig. 7 braking torque $M_r = (\text{Nm})$

9.10.2 Maximum permissible travel $s_r = (\text{mm})$

9.10.2.A Maximum permissible fluid volume absorbed $V_r = (\text{cm}^3)$

9.11 Further brake characteristics when the trailer moves rearwards (see Fig. 6 and Fig.7)

9.11.1. Brake-retraction force $P_{or} =$

9.11.1.A Brake-retraction pressure $p_{or} = (\text{N/cm}^2)$

- 9.11.2. Brake characteristic $\rho_r =$ (m)
- 9.11.2.A Brake characteristic $\rho'_r =$ (m)
- 9.12. Tests according to L-7.5 (if applicable) (corrected to take account of the rolling resistance corresponding to $0.01 \cdot g \cdot G_{Bo}$)
- 9.12.1. Brake test Type 0
 Test speed = (km/h)
 Braking ratio = (%)
 Control force = (N)
- 9.12.2 Brake test Type I
 Test speed = (km/h)
 Sustained braking ratio = (%)
 Braking time = (min)
 Hot performance = (%)
 (Expressed as a % of the above Type 0 test result in item 9.12.1.)
 Control force = (N)
10. The above brake does / does not conform to the requirements of L-3 and L-6 of the testing conditions for vehicles fitted with inertia braking systems. The brake may / may not be used for an inertia braking system without an overload protector.
 Date:
 Signature:
11. This test has been carried out and the results reported in accordance with relevant provisions of L-12.
 Certifying agency carrying out the test
 Date:
 Signature:
12. Approval Authority
 Date:
 Signature:

L 9.3 Test Report on the Compatibility of the Inertia Brake Control Device, the Transmission and the Brakes on the Trailer

- 1.0** Control device
 described in the attached test report (see L9.1)
 Reduction ratio selected:

$$i_{Ho}^{(1)} = \dots\dots^{(2)} \text{ or } i_h^{(1)} = \dots\dots^{(2)}$$

(shall be within the limits specified in 8.1. or 8.2. of L 9.1)

2.0

Brakes

described in the attached test report (see L 9.2)

3.0

Transmission devices on the trailer

3.1

Brief description with diagram showing principle

3.2

Reduction ratio and efficiency of the mechanical-transmission device on the trailer

$$i_{H1}^{(1)} = \dots\dots^{(2)}$$

$$\eta_{H1}^{(1)} = \dots\dots$$

4.0

Trailer :

4.1

Manufacturer :

4.2

Make :

4.3

Type :

4.4

Type of drawbar connection: trailer with rigid drawbar/multi-axled trailer with pivoted drawbar ⁽¹⁾

4.5

Number of brakes n =

4.6

Technically permissible maximum mass $G_A = (\text{kg})$

4.7

Dynamic tyre rolling radius $R = \text{m}$

4.8

Permissible thrust on coupling

$$D^* = 0.100 \text{ g } G_A^{(1)} = (\text{N})$$

or

$$D^* = 0.067 \text{ g } G_A^{(1)} = (\text{N})$$

4.9

Required braking force $B^* = 0.50 \text{ g } G_A = (\text{N})$

4.10

Brake force $B = 0.49 \text{ g } G_A = \text{N}$

5.0

Compatibility - Test results

5.1

Force threshold $100 \text{ A } K_A / (\text{g } G_A)$ (shall be between 2 and 4)

5.2

Maximum compressive force $100 \text{ A } D_1 / (\text{g } G_A) =$

(shall not exceed 10 for trailers with rigid drawbar, or 6.7 for multi-axled trailers with pivoted drawbar)

5.3

Maximum tractive force $100 \text{ A } D_2 / (\text{g } G_A) =$ (shall be between 10 and 50)

5.4

Technically permissible maximum mass for inertia control device

$$G'_A = (\text{kg}) \text{ (shall not be less than } G_A)$$

⁽¹⁾ Strike out what does not apply.

⁽²⁾ State lengths used to determine i_{Ho} or i_h .

5.5

Technically permissible maximum mass for all of trailer's brakes

- $G_B = n \cdot G_{Bo} = (\text{kg})$ (shall not be less than G_A)
- 5.6 Braking torque of the brakes
 $n \cdot A \cdot M_{\max} / (B \cdot A \cdot R) =$
 (shall be equal or greater than 1.0)
- 5.6.1 An overload protector within the meaning of paragraph 3.6. of this Annex is / is not fitted on the inertia control device / on the brakes
- 5.6.1.1 where the overload protector is mechanical on the inertia control device
 $n \cdot P^* / (i_{H1} \cdot \eta_{H1} \cdot P'_{\max}) =$
 (shall be equal or greater than 1.2)
- 5.6.1.2 where the overload protector is hydraulic on the inertia control device
 $p^* / p'_{\max} =$
 (shall be equal or greater than 1.2)
- 5.6.1.3 if the overload protector is on the inertia control device:
 Threshold force $D_{op} / D^* =$
 (shall be equal or greater than 1.2)
- 5.6.1.4 if the overload protector is fitted on the brake:
 Threshold torque $n \cdot M_{op} / (B \cdot R) =$
 (shall be equal or greater than 1.2)
- 5.7 Inertia braking system with mechanical transmission device
- 5.7.1 $i_H = i_{H0} \cdot i_{H1} =$
- 5.7.2 $\eta_H = \eta_{H0} \cdot \eta_{H1} =$
- 5.7.3 $\left[\frac{B \cdot R}{\rho} + n \cdot P_o \right] \cdot \frac{1}{(D^* - K) \cdot \eta_H} = \dots$
 (shall not be greater than i_H)
- 5.7.4 $\frac{s'}{s_B^* \cdot i_g} = \dots$
 (shall not be less than i_H)
- 5.7.5 Ratio $s' / i_H =$
 when the trailer moves rearward (shall not exceed s_r)
- 5.7.6 Braking torque when the trailer moves rearward including rolling resistance
 $n \cdot M_r = (\text{Nm})$
 (shall not exceed $0.08 \cdot g \cdot G_A \cdot R$)
- 5.8 Inertia braking system with hydraulic transmission device ⁽¹⁾
- 5.8.1 $i_h / F_{HZ} =$
- 5.8.2 $\left[\frac{B \cdot R}{n \cdot \rho'} + P_o \right] \cdot \frac{1}{(D^* - K) \cdot \eta_H} = \dots$

(shall not be greater than i_h/F_{HZ})

$$5.8.3 \quad \frac{s'}{2 s_B \cdot n \cdot F_{RZ} \cdot i_g} = \dots\dots\dots$$

(Shall not be less than i_h/F_{HZ})

(Shall not be greater than travel of master cylinder actuator as specified in L-9.1.8.2.)

5.8.4 s/i_h = (shall not be greater than travel of master cylinder actuator as specified in 8.2. of L9.1)

5.8.5 Ratio s'/F_{HZ} =
when the trailer moves rearward (shall not exceed V_r)

5.8.6 Braking torque when the trailer moves rearward including rolling resistance

$$n \cdot M_r = (\text{Nm})$$

(shall not exceed $0.08 \cdot g \cdot G_A \cdot R$)

6.0 Differential travel at park brake compensator

6.1.1. Maximum permissible compensator travel (forward)

$$s_{cf} = (\text{mm})$$

6.1.2. Maximum permissible compensator travel (rearward)

$$s_{cr} = (\text{mm})$$

6.1.3 Maximum permissible differential compensator travel

$$s_{cd} = (\text{mm})$$

7.0 The inertia braking system described above complies/does not comply with the requirements of L-3 to L 10.

Date

Signature

8.0 This test has been carried out and the results reported in accordance with relevant provisions of Annex L. Certifying agency carrying out the test

Date:

Signature:

L10 COMPATIBILITY BETWEEN THE CONTROL DEVICE AND THE BRAKES OF A VEHICLE

L10.1 A check shall be made on the vehicle to verify in the light of the characteristics of the control device (L9.1), the characteristics of the brakes (L9.2), and the trailer characteristics referred to in L-9.3 (4), whether the trailer's inertia braking system meets the prescribed requirements.

L10.2 General Checks for All Types of Brakes

L10.2.1 Any parts of the transmission not checked at the same time as the control device or the brakes shall be checked on the vehicle. The results of the

check shall be entered in L9.3 (for example; i_{H1} and η_{H1}).

L10.2.2 Mass

L 10.2.2.1 The maximum mass G_A of the trailer shall not exceed the maximum mass G'_A for which the control device is authorized.

L10.2.2.2 The maximum mass G_A of the trailer shall not exceed the maximum mass G_B which can be braked by joint operation of all of the trailer's brakes.

L10.2.3 Forces

L10.2.3.1 The stress threshold K_A shall not be below $0.02 g G_A$ and not above $0.04 g G_A$.

L10.2.3.2 The maximum insertion force D_1 may not exceed $0.100 g G_A$ in trailers with rigid drawbar and $0.067 g G_A$ in multi-axled trailers with pivoted drawbar.

L10.2.3.3 The maximum tractive force D_2 shall be between $0.1 g G_A$ and $0.5 g G_A$.

L10.3 Check of Braking Efficiency

L10.3.1 The sum of the braking forces exerted on the circumference of the trailer wheels shall not be less than $B^* = 0.50 g G_A$, including a rolling resistance of $0.01 g G_A$: this corresponds to a braking force B of $0.49 g G_A$. In this case, the maximum permissible thrust on the coupling shall be,

$D^* = 0.067 g G_A$ in the case of multi-axled trailers with pivoted drawbar; and

$D^* = 0.10 g G_A$ in the case of trailers with rigid drawbar.

To check whether these conditions are complied with the following inequalities shall be applied:

L 10.3.1.1 In mechanical-transmission inertia braking systems:

$$\left[\frac{B.R}{\rho} + n P_o \right] \frac{1}{(D^* - K) \cdot \eta_H} \leq i_H$$

L 10.3.1.2 In hydraulic-transmission inertia braking systems:

$$\left[\frac{B.R}{n \cdot \rho'} + p_o \right] \frac{1}{(D^* - K) \cdot \eta_H} \leq \frac{i_h}{F_{HZ}}$$

L 10.4 Check of Control Travel

L10.4.1 In control devices for multi-axled trailers with pivoted drawbars where the brake rod linkage depends on the position of the towing device, the control travel s shall be longer than the effective (useful) control travel s' , the difference being at least equivalent to the loss of travel s_o . The travel loss of s_o shall not exceed 10 percent of the effective travel s' .

L10.4.2 The effective (useful) travel of control s' shall be determined for single and multi-axle trailers as follows:

a) If the brake rod linkage is affected by the angular position of the towing device, then,

$$s' = s - s_0$$

b) If there is no loss of travel then,

$$s' = s$$

c) In hydraulic braking systems,

$$s' = s - s''.$$

L10.4.3 The following inequalities shall be applied to check whether control travel is adequate;

a) In mechanical-transmission inertia braking systems:

$$i_H \leq \frac{s'}{s_{B^*} \cdot i_g}$$

b) In hydraulic-transmission inertia braking systems:

$$\frac{i_h}{F_{HZ}} \leq \frac{s'}{2 s_{B^*} \cdot n F_{RZ} \cdot i'_g}$$

L10.5 Additional Checks

L10.5.1 In mechanical transmission inertia braking systems a check shall be made to verify that the rod linkage by which the forces are transmitted from the control device to the brakes is correctly fitted.

L10.5.2 In hydraulic-transmission inertia braking systems a check shall be made to verify that the travel of the master cylinder is not less than s/i_h . A lower level shall not be permitted.

L10.5.3 The general behaviour of the vehicle when braking shall be the subject of a road test carried out at different road speeds with different levels of brake effort and rates of application. Self-excited, undamped oscillations shall not be permitted.

L11.0 GENERAL COMMENTS

The above requirements apply to the most usual embodiments of mechanical-transmission or hydraulic-transmission inertia braking systems where, in particular, all of the trailer's wheels are equipped with the same type of brake and the same type of tyre. For checking less usual embodiments, the above requirements shall be adapted to the circumstances of the particular case.

ANNEX M

(Clauses 3.15.2.2, 5.2.1.7.2, 5.2.1.22, 5.2.1.25.2.5, 5.2.2.13, C-1.2.9, C-1.3.2, C-1.3.2.1, D.2.2, J-3.1.4, J-5.2.2 and K-2.3)

REQUIREMENTS FOR VEHICLES EQUIPPED WITH

ANTI-LOCK BRAKING SYSTEMS

M-1.0 SCOPE

M-1.1 This Annex defines the required braking performance for road vehicles fitted with anti-lock systems.

M-1.2 The anti-lock systems known at present comprise a sensor or sensors, a controller or controllers and a modulator or modulators. Any device of a different design which may be introduced in the future, or where an anti-lock braking function is integrated into another system, shall be deemed to be an anti-lock braking system within the meaning of this Annex and Annex J, if it provides performance equal to that prescribed by this Annex.

M-2.0 DEFINITIONS

M-2.1 **Anti-Lock System** – It is a part of a service braking system which automatically controls the degree of slip, in the direction of rotation of the wheel(s), on one or more wheels of the vehicle during braking.

M-2.2 **Sensor** – A component designed to identify and transmit to the controller the conditions of rotation of the wheel(s) or the dynamic conditions of the vehicle.

M-2.3 **Controller** – A component designed to evaluate the data transmitted by the sensor(s) and to transmit a signal to the modulator.

M-2.4 **Modulator** – A component designed to vary the braking force(s) in accordance with the signal received from the controller.

M-2.5 **Directly Controlled Wheel** – A wheel whose braking force is modulated according to data provided at least by its own sensor (see Note).

NOTE - Anti-lock systems with select-high control are deemed to include both directly and indirectly controlled wheels; in systems with select-low control, all sensed wheels are deemed to be directly controlled wheels.

M-2.6 **Indirectly Controlled Wheel** – A wheel whose braking force is modulated according to data provided by the sensor(s) of other wheel(s) (see Note below M-2.5).

M-2.7 **Full Cycling** – The anti-lock system is repeatedly modulating the brake force to prevent the directly controlled wheels from locking. Brake applications where modulation only occurs once during the stop shall not be considered to meet this definition.

In the case of trailers with pneumatic braking systems, full cycling of the anti-lock braking system is only assured when the pressure available at any brake actuator of a directly controlled wheel is more than 100 kPa (1 bar) above the maximum cycling pressure throughout a given test. The supply pressure available may not be increased above 800 kPa (8 bar).

M-3 TYPES OF ANTI-LOCK SYSTEMS

M-3.1 A power-driven vehicle is deemed to be equipped with an anti-lock system within the meaning of J-1, if one of the following systems is fitted:

- a) Category 1 anti-lock system – A vehicle equipped with a Category 1 anti-lock system shall meet all the relevant requirements of this Annex.
- b) Category 2 anti-lock system – A vehicle equipped with a Category 2 anti-lock system shall meet all the relevant requirements of this Annex, except those of M-5.3.5.
- c) Category 3 anti-lock system – A vehicle equipped with a Category 3 anti-lock system shall meet all the relevant requirements of this Annex, except those of M-5.3.4 and M-5.3.5. On such vehicles, any individual axle (or bogie) which does not include at least one directly controlled wheel shall fulfil the conditions of adhesion utilization and the wheel-locking sequence of Annex J, with regard to the braking rate and the load respectively. These requirements may be checked on high- and low-adhesion road surfaces (about 0.8 and 0.3 maximum) by modulating the service braking control force.

M-3.2 A trailer is deemed to be equipped with an anti-lock system within the meaning of J-1 when at least two wheels on opposite sides of the vehicle are directly controlled and all remaining wheels are either directly or indirectly controlled by the anti-lock system. In the case of full trailers, at least two wheels on one front axle and two wheels on one rear axle are directly controlled with each of these axles having at least one independent modulator and all remaining wheels are either directly or indirectly controlled. In addition, the anti-lock equipped trailer shall meet one of the following conditions:

M-3.2.1 **Category A Anti-lock System**

A trailer equipped with a Category A anti-lock system shall meet all the relevant requirements of this Annex.

M-3.2.2 **Category B Anti-lock System**

A trailer equipped with a Category B anti-lock system shall meet all the relevant requirements of this Annex, except M-6.3.2.

M-4 GENERAL REQUIREMENTS

M-4.1 Any electrical failure or sensor anomaly that affects the system with respect to the functional and performance requirements in this standard, including those in the supply of electricity, the external wiring to the controller(s), the controller(s) (see Note) and the modulator(s) shall be signalled to the driver by a specific optical warning signal. The yellow warning signal specified in 5.2.1.29.1.2 shall be used for this purpose.

NOTE – The manufacturer shall provide the test agency with documentation relating to the controller(s) which follows the format set out in Annex S.

M-4.1.1 Sensor anomalies, which may not be detected under static conditions, shall be detected not later than when the vehicle speed exceeds 10 km/h (see Note). However, to prevent erroneous fault indication when a sensor is not generating a vehicle speed output, due to non-rotation of a wheel, verification may be delayed but detected not later than when the

vehicle speed exceeds 15 km/h.

NOTE - The warning signal may light up again while the vehicle is stationary, provided that it is extinguished before the vehicle speed reaches 10 km/h or 15 km/h, as appropriate, when no defect is present.

M-4.1.2 When the anti-lock braking system is energized with the vehicle stationary, electrically controlled pneumatic modulator valve(s) shall cycle at least once.

M-4.2 Power-driven vehicles equipped with an anti-lock system and authorized to tow a trailer equipped with such a system, with the exception of vehicles of Categories M1 and N1, shall be fitted with a separate optical warning signal for the anti-lock system of the trailer, meeting the requirements of M-4.1. The separate yellow warning signals specified in 5.2.1.29.1 shall be used for this purpose, activated via pin 5 of the electrical connector conforming to ISO 7638:

NOTE— The ISO 7638:2003 connector may be used for 5 pin or 7 pin applications, as appropriate.

M-4.3 In the event of a failure as defined in M-4.1. above, the following requirements shall apply:

Motor vehicles: The residual braking performance shall be that prescribed for the vehicle in question in the event of a failure of part of the transmission of the service braking system as defined in 5.2.1.4. This requirement shall not be construed as a departure from the requirements concerning secondary braking.

Trailers: The residual braking performance shall be that defined in 5.2.2.15.2.

M-4.4 The operation of the anti-lock system shall not be adversely affected by magnetic or electrical fields. This shall be demonstrated by compliance with AIS-004(Part 3):2009.

M-4.5 A manual device may not be provided to disconnect or change the control mode (see Note) of the anti-lock system, except on off-road power-driven vehicles of categories N2 and N3 as defined in IS 14272 (Part 1) where a device is fitted to N2 or N3 category vehicles, the following conditions shall be met.

M-4.5.1 The power-driven vehicle with the anti-lock system disconnected or the control mode changed by the device referred to in M-4.5 shall satisfy all the relevant requirements in Annex J.

M-4.5.2 An optical warning signal shall inform the driver that the anti-lock system has been disconnected or the control mode changed; the yellow anti-lock failure warning signal specified in 5.2.1.29.1.2 may be used for this purpose. The warning signal may be constant or flashing.

M-4.5.3 The anti-lock system shall automatically be reconnected/returned to on-road mode when the ignition (start) device is again set to the 'ON' (run) position.

M-4.5.4 The vehicle user's handbook provided by the manufacturer should warn the driver of the consequences of manual disconnection or mode change

of the anti-lock system.

- M-4.5.5 The device referred to in M-4.5 may, in conjunction with the towing vehicle, disconnect/change the control mode of the anti-lock system of the trailer. A separate device for the trailer alone is not permitted.

NOTE – It is understood that devices changing the control mode of the anti-lock system are not subject to M-4.5 if in the changed control mode condition all requirements for the category of anti-lock systems, with which the vehicle is equipped, are fulfilled. However, in this case, M-4.5.2, M-4.5.3 and M-4.5.4 shall be met.

- M-4.6 Vehicles equipped with an integrated endurance braking system shall also be equipped with an anti-lock braking system acting at least on the service brakes of the endurance braking system's controlled axle and on the endurance braking system itself, and shall fulfill the relevant requirements of this Annex.

M-5 SPECIAL PROVISIONS CONCERNING POWER-DRIVEN VEHICLES

M-5.1 Energy Consumption

Power-driven vehicles equipped with anti-lock systems shall maintain their performance when the service braking control device is fully applied for long periods. Compliance with this requirement shall be verified by means of the following tests:

M-5.1.1 Test Procedure

- M-5.1.1.1 The initial energy level in the energy storage device(s) shall be that specified by the manufacturer. This level shall be at least such as to ensure the efficiency prescribed for service braking when the vehicle is laden. The energy storage device(s) for pneumatic auxiliary equipment shall be isolated.

- M-5.1.1.2 From an initial speed of not less than 50 km/h, on a surface with a coefficient of adhesion of 0.3 or less, (see Note) the brakes of the laden vehicle shall be fully applied for a time t , during which time the energy consumed by the indirectly controlled wheels shall be taken into consideration and all directly controlled wheels shall remain under control of the anti-lock system.

NOTE – Until such test surfaces become generally available, tyres at the limit of wear, and higher values up to 0.4 may be used at the discretion of the test agency. The actual value obtained and the type of tyres and surface shall be recorded.

- M-5.1.1.3 The vehicle's engine shall then be stopped or the supply to the energy transmission storage device(s) cut off.

- M-5.1.1.4 The service braking control device shall then be fully actuated four times in succession with the vehicle stationary.

- M-5.1.1.5 When the control device is applied for the fifth time, it shall be possible to brake the vehicle with at least the performance prescribed for secondary braking of the laden vehicle.

M-5.1.1.6 During the tests, in the case of a power-driven vehicle authorized to draw a trailer equipped with a compressed-air braking system, the supply line shall be stopped and an energy storage device of 0.5 l capacity shall be connected to the pneumatic control line (if fitted in accordance with F-1.2.2.3). When the brakes are applied for the fifth time, as provided in M-5.1.1.5, the energy level supplied to the pneumatic control line shall not be below half the level obtained at a full application starting with the initial energy level.

M-5.1.2 Additional Requirements

M-5.1.2.1 The coefficient of adhesion of the road surface shall be measured with the vehicle under test, by the method described in M-8.1.1.

M-5.1.2.2 The braking test shall be conducted with the engine disconnected and idling, and with the vehicle laden.

M-5.1.2.3 The braking time t shall be determined by the formula:

$$t = v_{\max} \div 7 \text{ (but not less than 15 s)}$$

where, t is expressed in seconds and $v_{\max}$ represents the maximum design speed of the vehicle expressed, in km/h, with an upper limit of 160 km/h.

M-5.1.2.4 If the time t may not be completed in a single braking phase, further phases may be used, up to a maximum of four in all.

M-5.1.2.5 If the test is conducted in several phases, no fresh energy shall be supplied between the phases of the test.

From the second phase, the energy consumption corresponding to the initial brake application may be taken into account, by subtracting one full brake application from the four full applications prescribed in

M-5.1.1.4 (and M-5.1.1.5, M-5.1.1.6 and M-5.1.2.6) for each of the second, third and fourth phases used in the test prescribed in M-5.1.1 as applicable.

M-5.1.2.6 The performance prescribed in M-5.1.1.5 shall be deemed to be satisfied, if at the end of the fourth application, with the vehicle stationary, the energy level in the storage device(s) is at or above that required for secondary braking with the laden vehicle.

M-5.2 Utilization of Adhesion

M-5.2.1 The utilization of adhesion by the anti-lock system takes into account the actual increase in braking distance beyond the theoretical minimum. The anti-lock system shall be deemed to be satisfactory when the condition $\varepsilon \geq 0.75$ is satisfied, where ε represents the adhesion utilized, as defined in M-8.1.2.

M-5.2.2 The adhesion utilization ε shall be measured on road surfaces with a coefficient of adhesion of 0.3 or less, (see Note) and of about 0.8 (dry road), with an initial speed of 50 km/h. To eliminate the effects of differential brake temperatures it is recommended that z_{AL} be determined prior to the determination of k .

NOTE – Until such test surfaces become generally available, tyres at the limit of wear, and higher values up to 0.4 may be used at the discretion

of the test agency. The actual value obtained and the type of tyres and surface shall be recorded.

M-5.2.3 The test procedure to determine the coefficient of adhesion (k) and the formulae for calculation of the adhesion utilization (ϵ) shall be those laid down in M-8.

M-5.2.4 The utilization of adhesion by the anti-lock system shall be checked on complete vehicles equipped with anti-lock systems of categories 1 or 2. In the case of vehicles equipped with category 3 anti-lock systems, only the axle(s) with at least one directly controlled wheel shall satisfy this requirement.

M-5.2.5 The condition $\epsilon \geq 0.75$ shall be checked with the vehicle both laden and unladen (see Note 1).

The laden test on the high adhesion surface may be omitted if the prescribed force on the control device does not achieve full cycling of the anti-lock system.

For the unladen test, the control force may be increased up to 100 daN if no cycling is achieved with its full force value (see Note 2). If 100 daN is insufficient to make the system cycle, then this test may be omitted. For air braking systems the air pressure may not be increased above the cut-out pressure for the purpose of this test.

NOTES

- 1 Until such test surfaces become generally available, tyres at the limit of wear, and higher values up to 0.4 may be used at the discretion of the test agency. The actual value obtained and the type of tyres and surface shall be recorded.
- 2 'Full force' means the maximum force laid down in Annex C for the category of vehicle; a higher force may be used if required to activate the anti-lock system.

M-5.3 Additional Checks

The following additional checks shall be carried out, engine disconnected, with the vehicle laden and unladen.

M-5.3.1 The wheels directly controlled by an anti-lock system shall not lock when the full force (see Note 1) is suddenly applied on the control device, on the road surfaces specified in M-5.2.2, at an initial speed of 40 km/h and at a high initial speed as indicated on the table below (see Notes 2 and 3) :

	Vehicle Category	Maximum Test Speed
High-adhesion surface	N1, M2, M3	$0.8 v_{\max} \leq 120 \text{ km/h}$
	N2, N3 laden	$0.8 v_{\max} \leq 80 \text{ km/h}$
Low-adhesion	N1	$0.8 v_{\max} \leq 120 \text{ km/h}$

surface	M2, M3, N2 except tractors for semi-trailers	$0.8 v_{\max} \leq 80 \text{ km/h}$
	N3 and N2 tractors for semi-trailers	$0.8 v_{\max} \leq 70 \text{ km/h}$

NOTES

- 1 'Full force' means the maximum force laid down in Annex C for the category of vehicle; a higher force may be used if required to activate the anti-lock system.
- 2 The purpose of the test is to check that the wheels do not lock and that the vehicle remains stable; it is not necessary, therefore, to make complete stops and bring the vehicle to a halt on the low adhesion surface.

M-5.3.2 When an axle passes from a high-adhesion surface (k_H) to a low-adhesion surface (k_L) where $k_H \geq 0.5$ and $k_H/k_L \geq 2$, (see Note 1) with the full force (see Note 2) applied on the control device, the directly controlled wheels shall not lock. The running speed and the instant of applying the brakes shall be so calculated that, with the anti-lock system fully cycling on the high-adhesion surface, the passage from one surface to the other is made at high and at low speed, under the conditions laid down in M-5.3.1 (see Note 3).

NOTES

- 1 k_H is the high-adhesion surface coefficient, k_L is the low-adhesion surface coefficient. k_H and k_L are measured as laid down in M-8.
- 2 'Full force' means the maximum force laid down in Annex C for the category of vehicle; a higher force may be used if required to activate the anti-lock system.
- 3 The purpose of these tests is to check that the wheels do not lock and that the vehicle remains stable; it is not necessary, therefore, to make complete stops and bring the vehicle to a halt on the low-adhesion surface.

M-5.3.3 When a vehicle passes from a low-adhesion surface (k_L) to a high-adhesion surface (k_H) where $k_H \geq 0.5$ and $k_H/k_L \geq 2$, (see Note 1 below M-5.3.2) with the full force (see Note 2 below M-5.3.2) applied on the control device, the deceleration of the vehicle shall rise to the appropriate high value within a reasonable time and the vehicle shall not deviate from its initial course. The running speed and the instant of applying the brakes shall be so calculated that, with the anti-lock system fully cycling on the low-adhesion surface, the passage from one surface to the other occurs at approximately 50 km/h.

M-5.3.4 In the case of vehicles equipped with anti-lock systems of Categories 1 or 2, when the right and left wheels of the vehicle are situated on surfaces with differing coefficients of adhesion (k_H and k_L), where $k_H \geq 0.5$ and $k_H/k_L \geq 2$, (see Note 1 below M-5.3.2) the directly controlled wheels shall not lock when the full force (see Note 2 below M-5.3.2) is suddenly applied on the control device at a speed of 50 km/h.

- M-5.3.5 Furthermore, laden vehicles equipped with anti-lock systems of Category 1 shall, under the conditions of M-5.3.4 satisfy the prescribed braking rate in M-9.
- M-5.3.6 However, in the tests provided in M-5.3.1, M-5.3.2, M-5.3.3, M-5.3.4 and M-5.3.5, brief periods of wheel-locking shall be allowed. Furthermore, wheel-locking is permitted when the vehicle speed is less than 15 km/h; likewise, locking of indirectly controlled wheels is permitted at any speed, but stability and steerability shall not be affected.
- M-5.3.7 During the tests provided in M-5.3.4 and M-5.3.5, steering correction is permitted, if the angular rotation of the steering control is within 120° during the initial 2 s, and not more than 240° in all. Furthermore, at the beginning of these tests the longitudinal median plane of the vehicle shall pass over the boundary between the high- and low-adhesion surfaces and during these tests no part of the (outer) tyres shall cross this boundary (see Note).

NOTE – Until a uniform test procedure is established, the tests required by this clause may have to be repeated for vehicles equipped with electrical regenerative braking systems, in order to determine the effect of different braking distribution values provided by automatic functions

M-6.0 SPECIAL PROVISIONS CONCERNING TRAILERS

M-6.1 Energy Consumption

Trailers equipped with anti-lock systems shall be so designed that, even after the service braking control device has been fully applied for some time, the vehicle retains sufficient energy to bring it to a halt within a reasonable distance.

- M-6.1.1 Compliance with the above requirement shall be checked by the procedure specified below, with the vehicle unladen, on a straight and level road with a surface having a good coefficient of adhesion and with the brakes adjusted as closely as possible and with the proportioning/load-sensing valve (if fitted) held in the laden position throughout the test.

NOTE – If the coefficient of adhesion of the test track is too high, preventing the anti-lock braking system from full cycling, then the test may be carried out on a surface with a lower coefficient of adhesion.

- M-6.1.2 In the case of compressed-air braking systems, the initial energy level in the energy transmission storage device(s) shall be equivalent to a pressure of 800 kPa (8 bar) at the coupling head of the trailer's supply line.
- M-6.1.3 With an initial vehicle speed of at least 30 km/h the brakes shall be fully applied for a time $t = 15$ s, during which all wheels shall remain under control of the anti-lock system. During this test, the supply to the energy transmission storage device(s) shall be cut off.

If the time $t = 15$ s may not be completed in a single braking phase, further phases may be used. During these phases no fresh energy shall be supplied to the energy transmission storage device(s) and, as from the

second phase, the additional energy consumption for filling the actuators is to be taken into account, for example; by the following test procedure.

The pressure in the reservoir(s) when starting the first phase is to be that stated in M-6.1.2. At the beginning of the following phase(s) the pressure in the reservoir(s) after application of the brakes shall be not less than the pressure in the reservoir(s) at the end of the preceding phase.

At the subsequent phase(s), the only time to be taken into account is from the point at which the pressure in the reservoir(s) is equal to that at the end of the preceding phase.

- M-6.1.4 At the end of the braking, with the vehicle stationary, the service braking control device shall be fully actuated four times. During the fifth application, the pressure in the operating circuit shall be sufficient to provide a total braking force at the periphery of the wheels equal to not less than 22.5 percent of the maximum stationary wheel load and without causing an automatic application of any braking system not being under the control of the anti-lock system.

M-6.2 Utilization of Adhesion

- M-6.2.1 Braking systems equipped with an anti-lock system shall be deemed acceptable when the condition $\varepsilon \geq 0.75$ is satisfied, where ε represents the adhesion utilized, as defined in M-8.2. This condition shall be verified with the vehicle unladen, on a straight and level road with a surface having a good coefficient of adhesion (see Notes).

NOTES

- 1 If the coefficient of adhesion of the test track is too high, preventing the anti-lock braking system from full cycling, then the test may be carried out on a surface with a lower coefficient of adhesion.
- 2 In the case of trailers equipped with a brake load sensing device, the pressure setting of the device may be increased to ensure full cycling.

- M-6.2.2 To eliminate the effects of differential brake temperatures, it is recommended to determine z_{RAL} prior to the determination of k_R .

M-6.3 Additional Checks

- M-6.3.1 At speeds exceeding 15 km/h, the wheels directly controlled by an anti-lock system shall not lock when the full force (see Note) is suddenly applied on the control device of the towing vehicle. This shall be checked, under the conditions prescribed in M-6.2, at initial speeds of 40 km/h and 80 km/h.

NOTE – ‘Full force’ means the maximum force laid down in Annex C for the category of vehicle; a higher force may be used if required to activate the anti-lock system.

- M-6.3.2 The provisions of this paragraph shall only apply to trailers equipped with an anti-lock system of Category A. When the right and left wheels are situated on surfaces which produce differing maximum braking rates (z_{RALH} and z_{RALL}), where

$$\frac{Z_{RALH}}{\varepsilon_H} \geq 0.5 \text{ and } \frac{Z_{RALH}}{Z_{RALL}} \geq 2$$

The directly controlled wheels shall not lock when the full force (see Note below M-6.3.1) is suddenly applied on the control device of the towing vehicle at a speed of 50 km/h. The ratio Z_{RALH}/Z_{RALL} may be ascertained by the procedure in M-8.2 or by calculating the ratio Z_{RALH}/Z_{RALL} . Under this condition the unladen vehicle shall satisfy the prescribed braking rate in M-9 (see Note).

NOTE – In the case of trailers equipped with a brake load sensing device, the pressure setting of the device may be increased to ensure full cycling.

- M-6.3.3 At vehicle speeds ≥ 15 km/h the directly controlled wheels are permitted to lock for brief periods, but at speeds < 15 km/h any locking is permissible. Indirectly controlled wheels are permitted to lock at any speed but in all cases stability shall not be affected.

M-7.0 SYMBOLS AND DEFINITIONS

Symbol	Notes
E	Wheelbase
E_R	Distance between king-pin and centre of axle or axles of semi-trailer (or distance between drawbar coupling and centre of axle or axles of centre-axle trailer)
ε	Adhesion utilized of the vehicle: quotient of the maximum braking rate with the anti-lock system operative (z_{al}) and the coefficient of adhesion (k)
ε_i	ε -value measured on axle i (in the case of a power-driven vehicle with a category 3 anti-lock system)
ε_H	ε -value on the high-adhesion surface
ε_L	ε -value on the low-adhesion surface
F	Force [N]
F_{bR}	Braking force of the trailer with the anti-lock system inoperative
F_{bRMax}	Maximum value of f_{br}
F_{bRMaxi}	Value of f_{brMax} with only axle i of the trailer braked
F_{bRAL}	Braking force of the trailer with the anti-lock system operative
F_{Cnd}	Total normal reaction of road surface on the unbraked and non-driven axles of the vehicle combination under static conditions
F_{Cd}	Total normal reaction of road surface on the unbraked and driven axles of the vehicle combination under static conditions

F_{dyn}	Normal reaction of road surface under dynamic conditions with the anti-lock system operative
F_{idyn}	F_{dyn} on axle i in case of power-driven vehicles or full trailers
F_i	Normal reaction of road surface on axle i under static conditions
F_M	Total normal static reaction of road surface on all wheels of power-driven (towing) vehicle
F_{Mnd} (see note)	Total normal static reaction of road surface on the unbraked and non-driven axles of the power-driven vehicle
F_{Md} (see note)	Total normal static reaction of road surface on the unbraked and driven axles of the power-driven vehicle
F_R	Total normal static reaction of road surface on all wheels of trailer
F_{Rdyn}	Total normal dynamic reaction of road surface on the axle(s) of semi-trailer or centre-axle trailer
F_{WM} (see note)	$0.01 f_{\text{Mnd}} + 0.015 f_{\text{Md}}$
g	Acceleration due to gravity (9.81 m/s^2)
h	Height of centre of gravity specified by the manufacturer and agreed by the test agency conducting the approval test
h_D	Height of drawbar (hinge point on trailer)
h_K	Height of fifth wheel coupling (king pin)
h_R	Height of centre of gravity of the trailer
k	Coefficient of adhesion between tyre and road
k_f	K-factor of one front axle
k_H	K-value determined on the high-adhesion surface
k_i	K-value determined on axle i for a vehicle with a category 3 anti-lock system
k_L	K-value determined on the low-adhesion surface
k_{lock}	Value of adhesion for 100 percent slip
k_M	K-factor of the power-driven vehicle
k_{peak}	Maximum value of the curve 'adhesion versus slip'
k_r	K-factor of one rear axle
k_R	K-factor of the trailer
P	Mass of individual vehicle [kg]
R	Ratio of k_{peak} to k_{lock}
t	Time interval [s]

t_m	Mean value of t
t_{min}	Minimum value of t
z	Braking rate
z_{AL}	Braking rate z of the vehicle with the anti-lock system operative
z_C	Braking rate z of the vehicle combination, with the trailer only braked and the anti-lock system inoperative
z_{CAL}	Braking rate z of the vehicle combination, with the trailer only braked and the anti-lock system operative
z_{Cmax}	Maximum value of z_c
z_{Cmaxi}	Maximum value of z_c with only axle i of the trailer braked
z_m	Mean braking rate
z_{max}	Maximum value of z
z_{MALS}	z_{al} of the power-driven vehicle on a ‘split surface’
z_R	Braking rate z of the trailer with the anti-lock system inoperative
z_{RAL}	z_{al} of the trailer obtained by braking all the axles, the towing vehicle unbraked and its engine disengaged
z_{RALH}	z_{ral} on the surface with the high coefficient of adhesion
z_{RALL}	z_{ral} on the surface with the low coefficient of adhesion
z_{RALS}	z_{ral} on the split-surface
z_{RH}	z_r on the surface with the high coefficient of adhesion
z_{RL}	z_r on the surface with the low coefficient of adhesion
z_{RHmax}	Maximum value of z_{rh}
z_{RLmax}	Maximum value of z_{rl}
z_{Rmax}	Maximum value of z_r

NOTE – f_{Mnd} and f_{Md} in case of two-axle power-driven vehicles: these symbols may be simplified to corresponding F_i -symbols.

M- 8 UTILIZATION OF ADHESION

M-8.1 Method of Measurement for Power-Driven Vehicles

M-8.1.1 Determination of the Coefficient of Adhesion (k)

M-8.1.1.1 The coefficient of adhesion (k) shall be determined as the quotient of the maximum braking forces without locking the wheels and the corresponding dynamic load on the axle being braked.

M-8.1.1.2 The brakes shall be applied on only one axle of the vehicle under test, at an initial speed of 50 km/h. The braking forces shall be distributed between the wheels of the axle to reach maximum performance. The

anti-lock system shall be disconnected, or inoperative, between 40 km/h and 20 km/h.

- M-8.1.1.3 A number of tests at increments of line pressure shall be carried out to determine the maximum braking rate of the vehicle ($z_{\max}$). During each test, a constant input force shall be maintained and the braking rate will be determined by reference to the time taken (t) for the speed to reduce from 40 km/h to 20 km/h using the formula:

$$z = \frac{0.566}{t}$$

$z_{\max}$ is the maximum value of z ; t is in seconds.

- M-8.1.1.3.1 Wheel lock may occur below 20 km/h.
- M-8.1.1.3.2 Starting from the minimum measured value of t , called $t_{\min}$, then select three values of t comprised within $t_{\min}$ and $1.05 t_{\min}$ and calculate their arithmetical mean value t_m , then calculate

$$z_m = \frac{0.566}{t_m}$$

If it is demonstrated that for practical reasons the three values defined above may not be obtained, then the minimum time $t_{\min}$ may be utilized. However, the requirements of M-8.1.3 shall still apply.

- M-8.1.1.4 The braking forces shall be calculated from the measured braking rate and the rolling resistance of the unbraked axle(s) which is equal to 0.015 and 0.010 of the static axle load for a driven axle and a non-driven axle, respectively.
- M-8.1.1.5 The dynamic load on the axle shall be that given by the formulae in Annex J.
- M-8.1.1.6 The value of k shall be rounded to three decimal places.
- M-8.1.1.7 Then, the test will be repeated for the other axle(s) as defined in M-8.1.1.1 to M-8.1.1.6 (for exemptions see M-8.1.4 and M-8.1.5).
- M-8.1.1.8 For example, in the case of a two-axle rear-wheel drive vehicle, with the front axle (1) being braked, the coefficient of adhesion (k) is given by:

$$k_f = \frac{z_m \cdot P \cdot g - 0.015 \cdot F_2}{F_1 + \frac{h}{E} \cdot z_m \cdot P \cdot g}$$

- M-8.1.1.9 One coefficient will be determined for the front axle k_f and one for the rear axle k_r .
- M-8.1.2 Determination of the Adhesion Utilized (ε)
- M-8.1.2.1 The adhesion utilized (ε) is defined as the quotient of the maximum braking rate with the anti-lock system operative (z_{AL}) and the coefficient of adhesion (k_M) that is
- M-8.1.2.2 From an initial vehicle speed of 55 km/h, the maximum braking rate (z_{AL}) shall be measured with full cycling of the anti-lock braking system and based on the average value of three tests, as in M-8.1.1.3, using the time taken for the speed to reduce from 45 km/h to 15 km/h, according to the following formula:

$$z_{AL} = \frac{0.849}{t_m}$$

$$\varepsilon = \frac{z_{AL}}{k_M}$$

- M-8.1.2.3 The coefficient of adhesion k_M shall be determined by weighting with the dynamic axle loads.

$$k_M = \frac{k_f \cdot F_{f\ dyn} + k_r \cdot F_{r\ dyn}}{P \cdot g}$$

where

$$F_{f\ dyn} = F_f + \frac{h}{E} \cdot z_{AL} \cdot P \cdot g$$

$$F_{r\ dyn} = F_r - \frac{h}{E} \cdot z_{AL} \cdot P \cdot g$$

- M-8.1.2.4 The value of ε shall be rounded to two decimal places.
- M-8.1.2.5 In the case of a vehicle equipped with an anti-lock system of Categories 1 or 2, the value of z_{AL} will be based on the whole vehicle, with the anti-lock system operative, and the adhesion utilized (ε) is given by the same formula quoted in M-8.1.2.1.
- M-8.1.2.6 In the case of a vehicle equipped with an anti-lock system of Category 3, the value of z_{AL} will be measured on each axle which has at least one directly controlled wheel. For example, for a two-axle rear-wheel drive vehicle with an anti-lock system acting only on the rear axle (2), the adhesion utilized (ε) is given by:

$$\varepsilon_2 = \frac{z_{AL} \cdot P \cdot g - 0.010 \cdot F_1}{k_2(F_2 - \frac{h}{E} \cdot z_{AL} \cdot P \cdot g)}$$

This calculation shall be made for each axle having at least one directly controlled wheel.

- M-8.1.3 If $\varepsilon > 1.00$ the measurements of coefficients of adhesion shall be repeated. A tolerance of 10 percent is accepted.
- M-8.1.4 For power-driven vehicles equipped with three axles, only the axle not associated with a close-coupled bogie will be used to establish a k value for the vehicle (see Note).

NOTE – Until a uniform test procedure is agreed, vehicles with more than three axles and special vehicles will be subject to consultation with the test agency.

- M-8.1.5 For vehicles of categories N2 and N3 with a wheelbase less than 3.80 m and with $h/E \geq 0.25$ the determination of the coefficient of adhesion for the rear axle will be omitted.

- M-8.1.5.1 In that case the adhesion utilized (ε) is defined as the quotient of the maximum braking rate with the anti-lock system operative (z_{AL}) and the coefficient of adhesion (k_f), that is

$$\varepsilon = \frac{z_{AL}}{k_f}$$

M-8.2 Method of Measurement for Trailers

M-8.2.1 General

- M-8.2.1.1 The coefficient of adhesion (k) shall be determined as the quotient of the maximum braking forces without locking the wheels and the corresponding dynamic load on the axle being braked.

- M-8.2.1.2 The brakes shall be applied on only one axle of the trailer under test, at an initial speed of 50 km/h. The braking forces shall be distributed between the wheels of the axle to reach maximum performance. The anti-lock system shall be disconnected or inoperative, between 40 km/h and 20 km/h.

- M-8.2.1.3 A number of tests at increments of line pressure shall be carried out to determine the maximum braking rate of the vehicle combination (z_{CMax}) with the trailer only braked. During each test, a constant input force shall be maintained and the braking rate will be determined by reference to the time taken (t) for the speed to reduce from 40 km/h to 20 km/h using the formula:

$$z_c = 0.566 / t$$

- M-8.2.1.3.1 Wheel lock may occur below 20 km/h.

- M-8.2.1.3.2 Starting from the minimum measured value of t , called t_{Min} , then select three values of t comprised within t_{Min} and $1.05 t_{Min}$ and calculate their arithmetical mean value t_m , then calculate

$$z_{C Max} = 0.566 / t_m$$

If it is demonstrated that for practical reasons the three values defined above may not be obtained, then the minimum time t_{min} may be utilized.

- M-8.2.1.4 The adhesion utilized (ε) shall be calculated by means of the formula:

$$\varepsilon = \frac{z_{RAL}}{k_R}$$

The k value has to be determined according to M-8.2.2.3 for full trailers or M-8.2.3.1 for semi-trailers respectively.

- M-8.2.1.5 If $\varepsilon > 1.00$ the measurements of coefficients of adhesion shall be repeated. A tolerance of 10 percent is accepted.

- M-8.2.1.6 The maximum braking rate (z_{RAL}) shall be measured with full cycling of the anti-lock braking system and the towing vehicle unbraked, based on the average value of three tests, as in M-8.2.1.3.

M-8.2.2 Full Trailers

- M-8.2.2.1 The measurement of k (with the anti-lock system being disconnected, or inoperative, between 40 km/h and 20 km/h) will be performed for the front and the rear axles.

For one front axle i :

$$F_{bRmaxi} = z_{Cmaxi}(F_M + F_R) - 0.01F_{Cnd} - 0.015F_{Cd}$$

$$F_{idyn} = F_i + \frac{z_{Cmax}(F_M \cdot h_D + g \cdot P \cdot h_R) - F_{WM} \cdot h_D}{E}$$

$$k_f = \frac{F_{bRmaxi}}{F_{idyn}}$$

For one rear axle i :

$$F_{bRmaxi} = z_{Cmaxi} \cdot (F_M + F_R) - 0.01F_{Cnd} - 0.015F_{Cd}$$

$$F_{idyn} = F_i - \frac{z_{Cmax}(F_M \cdot h_D + g \cdot P \cdot h_R) - F_{WM} \cdot h_D}{E}$$

$$k_r = \frac{F_{bRmaxi}}{F_{idyn}}$$

- M-8.2.2.2 The values of k_f and k_r will be rounded to three decimal places.
- M-8.2.2.3 The coefficient of adhesion k_R shall be determined proportionally according to the dynamic axle loads.

$$k_R = \frac{k_f \cdot F_{fdyn} + k_r \cdot F_{rdyn}}{P \cdot g}$$

- M-8.2.2.4 Measurement of z_{RAL} (with the anti-lock system operative)

$$Z_{RAL} = \frac{Z_{CAL} \cdot (F_M + F_R) - 0.01F_{Cnd} - 0.015F_{Cd}}{F_R}$$

z_{RAL} is to be determined on a surface with a high coefficient of adhesion and, for vehicles with a category A anti-lock system, also on a surface with a low coefficient of adhesion.

M-8.2.3 **Semi-Trailers and Centre-Axle Trailers**

- M-8.2.3.1 The measurement of k (with the anti-lock system being disconnected or inoperative, between 40 km/h and 20 km/h) shall be carried out with wheels fitted only on one axle, the wheels of the other axle(s) are removed.

$$F_{bRmax} = z_{Cmax} \cdot (F_M + F_R) - F_{WM}$$

- M-8.2.3.2 The measurement of z_{RAL} (with the anti-lock system operative) shall be carried out with all wheels fitted.

$$F_{Rdyn} = F_R - \frac{F_{bRmax} \cdot h_K + z_C \cdot g \cdot P \cdot (h_R - h_K)}{E_R}$$

$$k = \frac{F_{bRmax}}{F_{Rdyn}}$$

$$F_{bRAL} = Z_{CAL} \cdot (F_M + F_R) - F_{WM}$$

$$F_{Rdyn} = F_R - \frac{F_{bRAL} \cdot h_K + z_c \cdot g \cdot P \cdot (h_R - h_K)}{E_R}$$

$$Z_{RAL} = \frac{F_{bRAL}}{F_{Rdyn}}$$

z_{RAL} is to be determined on a surface with a high coefficient of adhesion and, for vehicles with a Category A anti-lock system, also on a surface with a low coefficient of adhesion.

M-9.0 PERFORMANCE ON DIFFERING ADHESION SURFACES

M-9.1 Power-Driven Vehicles

M-9.1.1 The prescribed braking rate referred to in M-5.3.5 may be calculated by reference to the measured coefficient of adhesion of the two surfaces on which this test is carried out. These two surfaces shall satisfy the conditions prescribed in M-6.3.4.

M-9.1.2 The coefficient of adhesion (k_H and k_L) of the high- and low- adhesion surfaces, respectively, shall be determined in accordance with the provisions in M-8.1.1.

M-9.1.3 The braking rate (z_{MALS}) for laden power-driven vehicles shall be:

$$z_{MALS} \geq 0.75 \frac{4k_L + k_H}{5} \text{ and } z_{MALS} \geq k_L$$

M-9.2 Trailers

M-9.2.1 The braking rate referred to in M-6.3.2 may be calculated by reference to the measured braking rates z_{RALH} and z_{RALL} on the two surfaces on which the tests are carried out with the anti-lock system operative. These two surfaces shall satisfy the conditions prescribed in M-6.3.2.

M-9.2.2 The braking rate z_{RALS} shall be

$$z_{RALS} \geq \frac{0.75}{\varepsilon_H} \cdot \frac{4z_{RALL} + z_{RALH}}{5}$$

and

$$z_{RALS} > \frac{z_{RALL}}{\varepsilon_H}$$

If $\varepsilon_H > 0.95$, use $\varepsilon_H = 0.95$

M-10 METHOD OF SELECTION OF THE LOW-ADHESION SURFACES

M-10.1 Details of the coefficient of adhesion of the surface selected, as defined in M-5.1.1.2, shall be given to the test agency.

M-10.1.1 These data shall include a curve of the coefficient of adhesion versus slip (from 0 to 100 percent slip) for a speed of approximately 40 km/h (See Note).

Note: Until a uniform test procedure is established for the determination of the adhesion curve for vehicles with a maximum mass exceeding 3.5 tonnes, the curve established for passenger cars may be used. In this case, for such vehicles, the ratio k_{peak} to k_{lock} shall be established using a value of k_{peak} as defined in M-8. With the consent of the Test Agency, the coefficient of adhesion described in this paragraph may be determined by another method provided that the equivalence of the values of k_{peak} and k_{lock} are demonstrated.

M-10.1.1.1 The maximum value of the curve will represent k_{peak} and the value at 100 percent slip will represent k_{lock} .

M-10.1.1.2 The ratio R shall be determined as the quotient of the k_{peak} and k_{lock} .

$$R = \frac{k_{peak}}{k_{lock}}$$

M-10.1.1.3 The value of R shall be rounded to one decimal place.

M-10.1.1.4 The surface to be used shall have a ratio R between 1.0 and 2.0. (See Note)

Note: Until such test surfaces become generally available, a ratio R up to 2.5 is acceptable, subject to discussion with the Test Agency.

M-10.2 Prior to the tests, the test agency shall ensure that the selected surface meets the specified requirements and shall be informed of the following:

- a) Test method to determine R,
- b) Type of vehicle (power-driven vehicle, trailer, ...); and
- c) Axle load and tyres (different loads and different tyres have to be tested and the results shown to the test agency which will decide if they are representative for the vehicle to be approved).

M-10.2.1 The value of R shall be mentioned in the test report.

The calibration of the surface has to be carried out at least once a year with a representative vehicle to verify the stability of R.

ANNEX N

(Clauses 5.2.1.19 and 5.2.2.2)

**TEST CONDITIONS FOR TRAILERS WITH ELECTRICAL
BRAKING SYSTEMS****N-1.0 GENERAL**

- N-1.1 For the purposes of the following provisions electrical braking systems are service braking systems consisting of a control device, an electromechanical transmission device, and friction brakes. The electrical control device regulating the voltage for the trailer shall be situated on the trailer.
- N-1.2 The electrical energy required for the electrical braking system is supplied to the trailer by the towing vehicle.
- N-1.3 Electrical braking systems shall be actuated by operating the service braking system of the towing vehicle.
- N-1.4 The nominal voltage rating shall be 12 V.
- N-1.5 The maximum current consumption shall not exceed 15 A.
- N-1.6 The electrical connection of the electrical braking system to the towing vehicle shall be effected by means of a special plug and socket connection corresponding to the plug of which shall not be compatible with the sockets of the lighting equipment of the vehicle. The plug together with the cable shall be situated on the trailer.

N-2.0 CONDITIONS CONCERNING THE TRAILER

- N-2.1 If there is a battery on the trailer fed by the power supply unit of the towing vehicle, it shall be separated from its supply line during service braking of the trailer.
- N-2.2 With trailers whose unladen mass is less than 75 percent of their maximum mass, the braking force shall be automatically regulated as a function of the loading condition of the trailer.
- N-2.3 Electrical braking systems shall be such that even when the voltage in the connection lines is reduced to a value of 7 V, a braking effect of 20 percent of the (sum of the) maximum stationary axle load(s) is maintained.
- N-2.4 Control devices for regulating the braking force, which react to the inclination in the direction of travel (pendulum, spring-mass-system, liquid-inertia-switch) shall, if the trailer has more than one axle and a vertically adjustable towing device, be attached to the chassis. In the case of single-axle trailers and trailers with close-coupled axles where the axle spread is less than 1 metre, these control devices shall be equipped with a mechanism indicating its horizontal position (for example, spirit level) and shall be manually adjustable to allow the mechanism to be set in the horizontal plane in line with the direction of travel of the vehicle.

- N-2.5 The relay for actuating the braking current in accordance with 5.2.1.19 (b) which is connected to the actuating line shall be situated on the trailer.
- N-2.6 A dummy socket shall be provided for the plug.
- N-2.7 A tell-tale shall be provided at the control device, lighting up at any brake application and indicating the proper functioning of the trailer electrical braking system.
- N-3.0 PERFORMANCE**
- N-3.1 Electrical braking systems shall respond at a deceleration of the tractor/trailer combination of not more than 0.4 m/s^2 .
- N-3.2 The braking effect may commence with an initial braking force, which shall not be higher than 10 percent of the (sum of the) maximum stationary axle load(s) nor higher than 13 percent of the (sum of the) stationary axle load(s) of the unladen trailer.
- N-3.3 The braking forces may also be increased in steps. At higher levels of the braking forces than those referred to in N-3.2 these steps shall not be higher than 6 percent of the (sum of the) maximum stationary axle load(s) nor higher than 8 percent of the (sum of the) stationary axle load(s) of the unladen trailer. However, in the case of single-axle trailers having a maximum mass not exceeding 1.5 t, the first step shall not exceed 7 percent of the (sum of the) maximum stationary axle load(s) of the trailer. An increase of 1 percent of this value is permitted for the subsequent steps (example: first step 7 percent, second step 8 percent, third step 9 percent, etc; any further step should not exceed 10 percent). For the purpose of these provisions a two-axle trailer having a wheelbase shorter than 1 m will be considered as a single axle trailer.
- N-3.4 The prescribed braking force of the trailer of at least 50 percent of the maximum total axle load shall be attained - with maximum mass - in the case of a mean fully developed deceleration of the tractor/trailer combination of not more than 5.9 m/s^2 with single-axle trailers and of not more than 5.6 m/s^2 with multi-axle trailers. Trailers with close-coupled axles where the axle spread is less than 1 m are also considered as single-axle trailers within the meaning of this provision. Moreover, the limits as defined in N-3.4.1 shall be observed. If the braking force is regulated in steps, they shall lie within the range shown in N-3.4.1.
- N-3.4.1 Compatibility of the braking rate of the trailer and the mean fully developed deceleration of the tractor/trailer combination (trailer laden and unladen) (see Fig. 22).

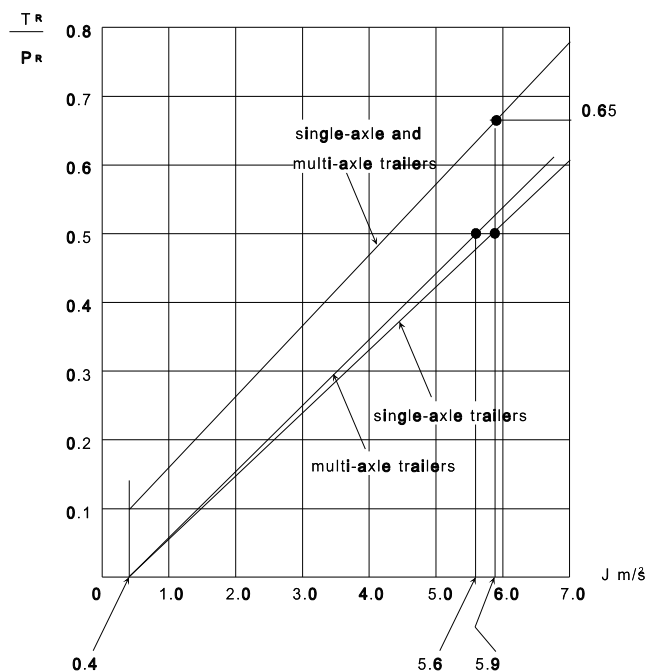


FIG. 22

**COMPATIBILITY OF THE BRAKING RATE OF THE TRAILER
AND THE MEAN FULLY DEVELOPED DECELERATION OF
THE TRACTOR/TRAILER COMBINATION (TRAILER LADEN
AND UNLADEN)**

NOTES

- 1 Limits indicated in the diagram refer to laden and unladen trailers. When the trailer unladen mass exceeds 75 percent of its maximum mass, limits shall be applied only to laden conditions.
- 2 Limits indicated in the diagram do not affect the provisions of this Annex regarding the minimum braking performances required. However, if braking performances obtained during test in accordance with provisions indicated in N-3.4 are greater than those required, said performances shall not exceed the limits indicated in the above diagram.

T_R = sum of braking forces at periphery of all wheels of trailer.

P_R = total normal static reaction of road surface on wheels of trailer.

J = mean fully developed deceleration of tractor/trailer combination.

N-3.5 The test shall be carried out with an initial speed of 60 km/h.

N-3.6 Automatic braking of the trailer shall be provided in accordance with the conditions of 5.2.2.9. If this automatic braking action requires electrical energy, a trailer braking force of at least 25 percent of the maximum total axle load shall be achieved for at least 15 min to satisfy the above-mentioned conditions.

ANNEX P

(Clause 6)

INERTIA DYNAMOMETER TEST METHOD FOR BRAKE LININGS**P-1.0 GENERAL**

- P-1.1 The procedure described in this Annex may be applied in the event of a modification of vehicle type resulting from the fitting of brake linings of another type to vehicles which have been approved in accordance with this standard.
- P-1.2 The alternative types of brake linings shall be checked by comparing their performance with that obtained from the brake linings with which the vehicle was equipped at the time of approval and conforming to the components identified in the relevant information document, a model of which is given in Annex A.
- P-1.3 The Test Agency responsible for conducting approval tests may at its discretion require comparison of the performance of the brake linings to be carried out in accordance with the relevant provisions contained in Annex C.
- P-1.4 Application for approval by comparison shall be made by the vehicle manufacturer or by his duly accredited representative.
- P-1.5 In the context of this Annex, vehicle shall mean the vehicle-type approved according to this standard and for which it is requested that the comparison shall be considered satisfactory.

P-2.0 TEST EQUIPMENT

- P-2.1 A dynamometer having the following characteristics shall be used for the tests:
- P-2.1.1 It shall be capable of generating the inertia required by P-3.1, and have the capacity to meet the requirements prescribed by C-1.5, C-1.6 and C-1.7 with respect to Type I, Type II and Type III tests.
- P-2.1.2 The brakes fitted shall be identical with those of the original vehicle-type concerned.
- P-2.1.3 Air cooling, if provided, shall be in accordance with P-3.4.
- P-2.1.4 The instrumentation for the test shall be capable of providing at least the following data:
- P-2.1.4.1 A continuous recording of disc or drum rotational speed.
- P-2.1.4.2 Number of revolutions completed during a stop, to a resolution not greater than one eighth of a revolution/
- P-2.1.4.3 Stop time.

- P-2.1.4.4 A continuous recording of the temperature measured in the centre of the path swept by the lining or at mid-thickness of the disc or drum or lining.
- P-2.1.4.5 A continuous recording of brake application control line pressure or force.
- P-2.1.4.6 A continuous recording of brake output torque.

P-3 TEST CONDITIONS

- P-3.1 The dynamometer shall be set as close as possible, with ± 5 percent tolerance, to the rotary inertia equivalent to that part of the total inertia of the vehicle braked by the appropriate wheel(s) according to the following formula:

$$I = MR^2$$

where

I = rotational inertia, in kgm^2 ;

R = dynamic tyre rolling radius, m; and

M = that part of the maximum mass of the vehicle braked by the appropriate wheel(s). In the case of a single-ended dynamometer, this part shall be calculated from the design braking distribution in the case of vehicles of categories M2, M3 and N when deceleration corresponds to the appropriate value given in C-2.1; in the case of vehicles of category T (trailers) the value of M will correspond to the load on the ground for the appropriate wheel when the vehicle is stationary and loaded to its maximum mass.

- P-3.2 The initial rotational speed of the inertia dynamometer shall correspond to the linear speed of the vehicle as prescribed in Annex C and shall be based on the dynamic rolling radius of the tyre.
- P-3.3 Brake linings shall be at least 80 percent bedded and shall not have exceeded a temperature of 180°C during the bedding procedure, or alternatively, at the vehicle manufacturer's request, be bedded in accordance with his recommendations.
- P-3.4 Cooling air may be used, flowing over the brake in a direction perpendicular to its axis of rotation. The velocity of the cooling air over the brake shall be:

$$v_{\text{air}} = 0.33 v$$

where

v = vehicle test speed at initiation of braking.

The temperature of the cooling air shall be the ambient temperature.

P-4 TEST PROCEDURE

- P-4.1 Five sample sets of the brake lining shall be subjected to the comparison test; they shall be compared with five sets of linings conforming to the original components identified in the information document concerning the first approval of the vehicle type concerned.
- P-4.2 Brake lining equivalence shall be based on a comparison of the results achieved using the test procedures prescribed in this Annex and in accordance with the following requirements.
- P-4.3 **Type 0 Cold Performance Test**
- P-4.3.1 Three brake applications shall be made when the initial temperature is below 100 °C. The temperature shall be measured in accordance with the provisions of P-2.1.4.4.
- P-4.3.2 In the case of brake linings intended for use on vehicles of categories M2, M3 and N, brake applications shall be made from an initial rotational speed equivalent to that given in C- 2.1, and the brake shall be applied to achieve a mean torque equivalent to the deceleration prescribed in that paragraph. In addition, tests shall also be carried out at several rotational speeds, the lowest being equivalent to 30 percent of the maximum speed of the vehicle and the highest being equivalent to 80 percent of that speed.
- P-4.3.3 In the case of brake linings intended for use on vehicles of category T, brake applications shall be made from an initial rotational speed equivalent to 60 km/h, and the brake shall be applied to achieve a mean torque equivalent to that prescribed in C-3.1. A supplementary cold performance test from an initial rotational speed equivalent to 40 km/h shall be carried out for comparison with the Type I test results as described in C-3.1.2.2.
- P-4.3.4 The mean braking torque recorded during the above cold performance tests on the linings being tested for the purpose of comparison shall, for the same input measurement, be within the test limits ± 15 percent of the mean braking torque recorded with the brake linings conforming to the component identified in the relevant application for vehicle type approval.
- P-4.4 **Type I Test (Fade Test)**
- P-4.4.1 With Repeated Braking
- P-4.4.1.1 Brake linings for vehicles of categories M2, M3 and N shall be tested according to the procedure given in C-1.5.1.
- P-4.4.2 With Continuous Braking
- P-4.4.2.1 Brake linings for trailers (category T) shall be tested in accordance with C-1.5.2 of Annex C.
- P-4.4.3 **Hot Performance**

- P-4.4.3.1 On completion of the tests required under P-4.4.1 and P-4.4.2, the hot braking performance test specified in C-1.5.3 shall be carried out.
- P-4.4.3.2 The mean braking torque recorded during the above hot performance tests on the linings being tested for the purpose of comparison shall, for the same input measurement, be within the test limits ± 15 percent of the mean braking torque recorded with the brake linings conforming to the component identified in the relevant application for vehicle type approval.
- P-4.5 **Type II Test (Downhill Behaviour Test)**
- P-4.5.1 This test is required only if, on the vehicle-type in question, the friction brakes are used for the Type II test.
- P-4.5.2 Brake linings for power-driven vehicles of category M3 (except for those vehicles required to undergo a Type IIA test according to C-1.6.4) and category N3, and trailers of category T4 shall be tested according to the procedure set out in C- 1.6.1.
- P-4.5.3 **Hot Performance**
- P-4.5.3.1 On completion of the test required under P-4.5.1, the hot performance test specified in C-1.6.3 shall be carried out.
- P-4.5.3.2 The mean braking torque recorded during the above hot performance tests on the linings being tested for the purpose of comparison shall, for the same input measurement, be within the test limits ± 15 percent of the mean braking torque recorded with the brake linings conforming to the component identified in the relevant application for vehicle type approval.
- P-4.6 **Type III Test (Fade Test)**
- P-4.6.1 **Test with Repeated Braking**
- P-4.6.1.1 Brake linings for trailers of category T4, shall be tested according to the procedure given in C-1.7.1 and C-1.7.2.
- P-4.6.2 **Hot Performance**
- P-4.6.2.1 On completion of the tests required under P-4.6.1 and P-4.6.2, the hot braking performance test specified in C-1.7.2 shall be carried out.
- P-4.6.2.2 The mean braking torque during the above hot performance tests on the linings being tested for the purpose of comparison shall, for the same input measurement, be within the test limits 15 percent of the mean braking torque recorded with the brake linings conforming to the component identified in the relevant application for vehicle type approval.
- P-5.0 INSPECTION OF BRAKE LININGS**
- P-5.1 Brake linings shall be visually inspected on completion of the above tests to check that they are in satisfactory condition for continued use in normal service.

ANNEX Q

Compatibility between towing vehicles and trailers with respect to ISO 11992 data communications

Q-1.0 General

- Q-1.1. The requirements of this annex shall only apply to towing vehicles and trailers equipped with an electric control line as defined in paragraph 2.24. of the Standard.
- Q-1.2. The ISO 7638 connector provides a power supply for the braking system or anti-lock braking system of the trailer. In the case of vehicles equipped with an electric control line as defined in paragraph 2.24. of the Standard this connector also provides a data communication interface via pins 6 and 7 – see paragraph 5.1.3.6. of the Standard.
- Q-1.3. This annex defines requirements applicable to the towing vehicle and trailer with respect to the support of messages defined within ISO 11992-2:2003 including Amd.1:2007.

Q-2.0 The parameters defined within ISO 11992-2:2003 including Amd.1:2007 that are transmitted by the electric control line shall be supported as follows:

- Q-2.1. The following functions and associated messages are those specified within this Standard that shall be supported by the towing vehicle or trailer as appropriate:

Q-2.1.1. Messages transmitted from the towing vehicle to the trailer:

Function / Parameter	ISO 11992-2:2003 Reference	AIS Reference
Service/secondary brake demand value	EBS11 Byte 3-4	J-3.1.3.2.
Two electrical circuits brake demand value	EBS12 Byte 3 Bit 1-2	Paragraph 5.1.3.2.
Pneumatic control line	EBS12 Byte 3 Bit 5-6	Paragraph 5.1.3.2.

Q-2.1.2. Messages transmitted from the trailer to the towing vehicle:

Function / Parameter

Function / Parameter	ISO 11992-2:2003 Reference	AIS Reference
VDC Active / passive ⁽¹⁾	EBS21 Byte 2 Bit 1-2	V-2.1.6.
Vehicle electrical supply sufficient / insufficient	EBS22 Byte 2 Bit 1-2	5.2.2.20.
Red warning signal request	EBS22 Byte 2 Bit 3-4	5.2.2.15.2.1., 5.2.2.16. & 5.2.2.20.
Supply line braking request	EBS22 Byte 4 Bit 3-4	5.2.2.15.2.
Stop lamps request	EBS22 Byte 4 Bit 5-6	5.2.2.22.1.
Vehicle pneumatic supply sufficient / insufficient	EBS23 Byte 1 Bit 7-8	5.2.2.16.

- Q-2.2. When the trailer transmits the following messages, the towing vehicle shall provide a warning to the driver:

Function / Parameter	ISO 11992-2:2003 Reference	Driver Warning Required
VDC Active / Passive ¹	EBS21 Byte 2 Bit 1-2	V-2.1.6.
Red warning signal request	EBS22 Byte 2 Bit 3-4	5.2.1.29.2.1.

Q-2.3. The following messages defined in ISO 11992-2:2003 including Amd.1:2007 shall be supported by the towing vehicle or trailer:

Q-2.3.1. Messages transmitted from the towing vehicle to the trailer:

No messages currently defined.

Q-2.3.2. Messages transmitted from the trailer to the towing vehicle:

Function / Parameter	ISO 11992-2:2003 Reference
Vehicle service brake active / passive	EBS22 Byte 1, Bit 5-6
Braking via electric control line supported	EBS22 Byte 4, Bit 7-8
Geometric data index	EBS24 Byte 1
Geometric data index content	EBS24 Byte 2

⁽¹⁾ VDC (Vehicle Dynamic Control) as defined within ISO 11992-2:2003 including Amd.1:2007 is defined within this Standard as Vehicle Stability Function - see paragraph 2.34 of the Standard.

Q-2.4. The following messages shall be supported by the towing vehicle or trailer as appropriate when the vehicle is installed with a function associated with that parameter:

Q-2.4.1. Messages transmitted from the towing vehicle to the trailer:

Function / Parameter	ISO 11992-2:2003 Reference
Vehicle type	EBS11 Byte 2, Bit 3-4
VDC (Vehicle Dynamic Control) Active / passive ⁽¹⁾	EBS11 Byte 2, Bit 5-6
Brake demand value for front or left side of vehicle	EBS11 Byte 7
Brake demand value for rear or right side of vehicle	EBS11 Byte 8
ROP (Roll Over Protection) system enabled/disabled ⁽²⁾	EBS12 Byte 1, Bit 3-4
YC (Yaw Control) system enabled/disabled ⁽³⁾	EBS12 Byte 1, Bit 5-6
Enable/disable trailer ROP (Roll Over Protection) system ⁽³⁾	EBS12 Byte 2, Bit 1-2
Enable/disable trailer YC (Yaw Control) system ⁽³⁾	EBS12 Byte 2, Bit 3-4
Traction help request	RGE11 Byte 1, Bit 7-8
Lift axle 1 - position request	RGE11 Byte 2, Bit 1-2
Lift axle 2 - position request	RGE11 Byte 2, Bit 3-4
Steering axle locking request	RGE11 Byte 2, Bit 5-6
Seconds	TD11 Byte 1
Minutes	TD11 Byte 2
Hours	TD11 Byte 3
Months	TD11 Byte 4
Day	TD11 Byte 5
Year	TD11 Byte 6
Local minute offset	TD11 Byte 7
Local hour offset	TD11 Byte 8

⁽¹⁾ VDC (Vehicle Dynamic Control) as defined within ISO 11992-2:2003 including Amd.1:2007 is defined within this Standard as Vehicle Stability Function - see paragraph 2.34 of the Standard.

⁽²⁾ ROP (Roll Over Protection) as defined within ISO 11992-2:2003 including Amd.1:2007 is defined within this Standard as Roll-Over Control – see paragraph 2.34.2.2. of the Standard.

⁽³⁾ YC (Yaw Control) as defined within ISO 11992-2:2003 including Amd.1:2007 is defined within this Standard as Directional Control – see paragraph 2.34.2.1. of the Standard.

Q-2.4.2. Messages transmitted from the trailer to the towing vehicle:

Function / Parameter	ISO 11992-2:2003 Reference
Support of side or axle wise brake force distribution	EBS21 Byte 2, Bit 3-4
Wheel based vehicle speed	EBS21 Byte 3-4
Lateral acceleration	EBS21 Byte 8
Vehicle ABS active / passive	EBS22 Byte 1, Bit 1-2
Amber warning signal request	EBS22 Byte 2, Bit 5-6
Vehicle type	EBS22 Byte 3, Bit 5-6
Loading ramp approach assistance	EBS22 Byte 4, Bit 1-2
Axle load sum	EBS22 Byte 5-6
Tyre pressure sufficient / insufficient	EBS23 Byte 1, Bit 1-2
Brake lining sufficient / insufficient	EBS23 Byte 1, Bit 3-4
Brake temperature status	EBS23 Byte 1, Bit 5-6
Tyre / wheel identification (pressure)	EBS23 Byte 2
Tyre / wheel identification (lining)	EBS23 Byte 3
Tyre / wheel identification (temperature)	EBS23 Byte 4
Tyre pressure (actual tyre pressure)	EBS23 Byte 5
Brake lining	EBS23 Byte 6
Brake temperature	EBS23 Byte 7
Brake cylinder pressure first axle left wheel	EBS25 Byte 1
Brake cylinder pressure first axle right wheel	EBS25 Byte 2
Brake cylinder pressure second axle left wheel	EBS25 Byte 3
Brake cylinder pressure second axle right wheel	EBS25 Byte 4
Brake cylinder pressure third axle left wheel	EBS25 Byte 5
Brake cylinder pressure third axle right wheel	EBS25 Byte 6
ROP (Roll Over Protection) system	EBS25 Byte 7, Bit 1-2

enabled/disabled ²	
YC (Yaw Control) system enabled/disabled ³	EBS25 Byte 7, Bit 3-4
Traction help	RGE21 Byte 1, Bit 5-6
Lift axle 1 position	RGE21 Byte 2, Bit 1-2
Lift axle 2 position	RGE21 Byte 2, Bit 3-4
Steering axle locking	RGE21 Byte 2, Bit 5-6
Tyre wheel identification	RGE23 Byte 1
Tyre temperature	RGE23 Byte 2-3
Air leakage detection (Tyre)	RGE23 Byte 4-5
Tyre pressure threshold detection	RGE23 Byte 6, Bit 1-3

Q-2.5 The support of all other messages defined within ISO 11992-2:2003 including Amd.1:2007 is optional for the towing vehicle and trailer.

(Clauses 5.1.3 and 6)

**TEST PROCEDURE TO ASSESS THE FUNCTIONAL COMPATIBILITY
OF VEHICLES EQUIPPED WITH ELECTRIC CONTROL LINES**

R-1.0 GENERAL

- R-1.1 This Annex defines a procedure that may be used to check towing and towed vehicles equipped with an electric control line against the functional and performance requirements referred to in 5.1.3.6.1. Alternative procedures may be used at the discretion of the certifying agency, if an equivalent level of checking integrity can be established.
- R-1.2 The references to ISO 7638 within this Annex apply to ISO 7638-1:2003 for 24V applications and ISO 7638-2:2003 for 12V applications.

R-2.0 INFORMATION DOCUMENT

- R-2.1 The vehicle manufacturer/system supplier shall supply to the certifying agency an information document that contains at least the following:
- a) A schematic of the vehicle braking system;
 - b) Evidence that the interface, including the physical layer, data link layer and the application layer and the respective position of supported messages and parameters, complies with ISO 11992;
 - c) A list of supported messages and parameters; and
 - d) The specification of the motor vehicle with respect to the number of control circuits that signal the pneumatic and/or electric control lines.

R-3.0 TOWING VEHICLES

R-3.1 ISO 11992 Trailer Simulator

The simulator shall,

- a) Have a connector meeting ISO 7638:2003 (7 pin) to connect to the vehicle under test. Pins 6 and 7 of the connector shall be used to transmit and receive messages complying with ISO 11992:2003 and its Amd. 1: 2007;
- b) Be capable of receiving all of the messages transmitted by the motor vehicle to be type approved and be capable of transmitting all trailer messages defined within ISO 11992-2:2003 and its Amd. 1: 2007;
- c) Provide a direct or indirect readout of messages, with the parameters in the data field shown in the correct order relative to time; and
- d) Include a facility to measure coupling head response time in accordance with E- 2.6.

R-3.2 Checking Procedure

R-3.2.1 Confirm that the manufacturer's/supplier's information document demonstrates compliance with the provisions of ISO 11992 with respect to the physical layer, data link layer and application layer.

R-3.2.2 Check the following, with the simulator connected to the motor vehicle via the ISO 7638 interface and whilst all trailer messages relevant to the interface are being transmitted:

R-3.2.2.1 **Control line signalling**

R-3.2.2.1.1 The parameters defined in EBS 12 byte 3 of ISO 11992-2:2003 shall be checked against the specification of the vehicle as follows:

Control Line Signalling	EBS 12 Byte 3	
	Bits 1 - 2	Bits 5 - 6
Service braking demand generated from one electrical circuit	00b	—
Service braking demand generated from two electrical circuits	01b	—
Vehicle is not equipped with a pneumatic control line (see Note)	—	00b
Vehicle is equipped with a pneumatic control line	—	01b

NOTE – This specification of vehicle is prohibited by footnote to 5.1.3.1.3 of this Standard.

R-3.2.2.2 Service/Secondary brake demand

R-3.2.2.2.1 The parameters defined in EBS 11 of ISO 11992-2:2003 shall be checked as follows:

Test condition	Byte reference	Electrical control line signal value
Service brake pedal and secondary brake control released	3 - 4	0
Service brake pedal fully applied	3 - 4	33280d to 43520d (650 to 850 kPa)
Secondary brake fully applied (see Note)	3 - 4	33280d to 43520d (650 to 850 kPa)

NOTE – Optional on towing vehicles with electric and pneumatic control lines when the pneumatic control line fulfills the relevant requirements for secondary braking.

R-3.2.2.3 **Failure warning**

- R-3.2.2.3.1 Simulate a permanent failure in the communication line to pin 6 of the ISO 7638 connector and check that the yellow warning signal specified in 5.2.1.29.1.2 is displayed.
- R-3.2.2.3.2 Simulate a permanent failure in the communication line to pin 7 of the ISO 7638 connector and check that the yellow warning signal specified in 5.2.1.29.1.2 is displayed.
- R-3.2.2.3.3 Simulate message EBS 22, byte 2 with bits 3 - 4 set to 01b and check that the red warning signal specified in 5.2.1.29.1.1 is displayed.
- R-3.2.2.4 **Supply line braking request**
- For power-driven vehicles which can be operated with trailers connected via an electric control line only.
- Only the electric control line shall be connected.
- Simulate message EBS 22, byte 4 with bits 3 - 4 set to 01b and check that when the service brake, secondary brake or parking brake is fully actuated the pressure in the supply line falls to 150 kPa (1.5 bar) within the following two seconds.
- Simulate a continuous absence of data communication and check that when the service brake, secondary brake or parking brake is fully actuated the pressure in the supply line falls to 150 kPa (1.5 bar) within the following two seconds.
- R-3.2.2.5 **Response time**
- Check that, with no faults present, the control line response requirements defined in E-2.6 are met.
- R-3.2.2.6. **Illumination of stop lamps**
- Simulate message EBS 22 byte 4 bits 5 to 6 set to 00 and check that the stop lamps are not illuminated.
- Simulate message EBS 22 byte 4 bits 5 to 6 set to 01 and check that the stop lamps are illuminated.
- R-3.2.2.7. **Intervention of trailer stability function**
- Simulate message EBS 21 byte 2 bits 1 to 2 set to 00 and check that the driver warning defined in V-2.1.6. is not illuminated.
- Simulate message EBS 21 byte 2 bits 1 to 2 set to 01 and check that the driver warning defined in V-2.1.6. is illuminated.
- R-3.2.3 **Additional Checks**
- R- 3.2.3.1 At the discretion of the testing agency the checking procedures defined above may be repeated with the non-braking functions relevant to the

interface in different states or switched off.

- R-3.2.3.2. Q-2.4.1. defines additional messages that shall under specific circumstances be supported by the towing vehicle. Additional checks may be carried out to verify the status of supported messages to ensure the requirements of 5.1.3.6.2. are fulfilled.

R-4.0 TRAILERS

R-4.1 ISO 11992 Towing Vehicle Simulator

The simulator shall,

- a) Have a connector meeting ISO 7638:2003 (7 pin) to connect to the vehicle under test. Pins 6 and 7 of the connector shall be used to transmit and receive messages complying with ISO 11992:2003 and Amd.1:2007;
- b) Have a failure warning display and an electrical power supply for the trailer;
- c) Shall be capable of receiving all of the messages transmitted by the trailer to be type approved and be capable of transmitting all motor vehicle messages defined within ISO 11992-2:2003 and Amd.1:2007;
- d) Provide a direct or indirect readout of messages with the parameters in the data field shown in the correct order relative to time; and
- e) Include a facility to measure brake system response time in accordance with E-3.5.2.

R-4.2 Checking Procedure

- R-4.2.1 Confirm that the manufacturer's/supplier's Information Document demonstrates compliance with the provisions of ISO 11992 2003 and Amd.1:2007 with respect to the physical layer, data link layer and application layer.

- R-4.2.2 Check the following, with the simulator connected to the trailer via the ISO 7638 interface and whilst all towing vehicle messages relevant to the interface are being transmitted.

R-4.2.2.1 Service Brake System Function

- R-4.2.2.1.1 The trailer response to the parameters defined in EBS 11 of ISO 11992-2:2003 and Amd.1:2007 shall be checked as follows.

The pressure in the supply line at the start of each test shall be ≥ 700 kPa (7 bar) and the vehicle shall be laden (the loading condition may be simulated for the purpose of this check).

- R-4.2.2.1.1.1 For trailers equipped with pneumatic and electric control lines:

Both control lines shall be connected.

Both control lines shall be signalled simultaneously.

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The simulator shall transmit message byte 3, bits 5 – 6 of EBS 12 set to 01_b to indicate to the trailer that a pneumatic control line should be connected.

Parameters to be checked:

Message transmitted by the simulator		Pressure at the brake chambers
Byte reference	Digital demand value	
3 - 4	0	0 kPa
3 - 4	33280d (650 kPa)	As defined in the vehicle manufacturer's brake calculation

R-4.2.2.1.1.2 Trailers equipped with pneumatic and electric control lines or an electric control line only.

Only the electric control line shall be connected

The simulator shall transmit the following messages:

Byte 3, bits 5 - 6 of EBS 12 set to 00_b to indicate to the trailer that a pneumatic control line is not available, and byte 3, bits 1 - 2 of EBS 12 set to 01_b to indicate to the trailer that the electric control line signal is generated from two electric circuits.

Parameters to be checked:

Message transmitted by the simulator		Pressure at the brake chambers
Byte reference	Digital demand value	
3 - 4	0	0 kPa
3 - 4	33280d (650 kPa)	As defined in the vehicle manufacturer's brake calculation

R-4.2.2.1.2 For trailers equipped with only an electric control line, the response to messages defined in EBS 12 of ISO 11992-2:2003 shall be checked as follows:

The pneumatic supply line at the start of each test shall be ≥ 700 kPa (7 bar).

The electric control line shall be connected to the simulator.

The simulator shall transmit the following messages:

Byte 3, bits 5 - 6 of EBS 12 set to 01_b to indicate to the trailer that a pneumatic control line is available.

Byte 3-4 of EBS 11 shall be set to 0 (no service brake demand)

The response to the following messages shall be checked:

EBS 12, Byte 3, Bit 1-2	Pressure in the Brake Chambers or Reaction of the Trailer
01 _b	0 kPa (service brake released)

00b	The trailer is automatically braked to demonstrate that the combination is not compatible. A signal should also be transmitted via Pin 5 of the ISO 7638:2003 connector (yellow warning).
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R-4.2.2.1.3 For trailers connected with only an electrical control line, the response of the trailer to a failure in the electric control transmission of the trailer which results in a reduction in braking performance to at least 30 percent of the prescribed value shall be checked by the following procedure.

R-4.2.2.1.3 For trailers connected with only an electrical control line, the response of the trailer to a failure in the electric control transmission of the trailer which results in a reduction in braking performance to at least 30 percent of the prescribed value shall be checked by the following procedure.

The pneumatic supply line at the start of each test shall be ≥ 700 kPa.

The electric control line shall be connected to the simulator.

Byte 3, bits 5-6 of EBS 12 set to 00b to indicate to the trailer that a pneumatic control line is not available.

Byte 3, bits 1-2 of EBS 12 set to 01b to indicate to the trailer that the electric control line signal is generated from two independent circuits.

The following shall be checked:

Test Condition	Braking System Response
With no faults present in the trailer braking system	Check that the braking system is communicating with the simulator and that Byte 4, bits 3-4 of EBS 22 is set to 00b.
Introduce a failure in the electric control transmission of the trailer braking system that prevents at least 30 percent of the prescribed braking performance from being maintained	Check that Byte 4, bits 3-4 of EBS 22 is set to 01b Or The data communications to the simulator has been terminated

R-4.2.2.2 **Failure warning**

R-4.2.2.2.1 Check that the appropriate warning message or signal is transmitted under the following conditions:

R-4.2.2.2.1.1 Where a permanent failure within the electric control transmission of the trailer braking system precludes the service braking performance being met, simulate such a failure and check that byte 2, bits 3 - 4 of EBS 22 transmitted by the trailer is set to 01b. A signal should also be transmitted via pin 5 of the ISO 7638 (Part 1 or 2) connector (yellow warning).

R-4.2.2.2.1.2 Reduce the voltage on pins 1 and 2 of the ISO 7638 (Part 1 or 2) connector to below a value nominated by the manufacturer which precludes the service braking system performance from being fulfilled and check that byte 2, bits 3 - 4 of EBS 22 transmitted by the trailer are set to 01b. A signal should also be transmitted via pin 5 of the

ISO 7638 (Part 1 or 2) connector (yellow warning).

R-4.2.2.2.1.3 Check compliance with the provisions of 5.2.2.16 by isolating the supply line. Reduce the pressure in the trailer pressure storage system to the value nominated by the manufacturer. Check that byte 2, bits 3 - 4 of EBS 22 transmitted by the trailer is set to 01b and that byte 1, bits 7 - 8 of EBS 23 is set to 00. A signal should also be transmitted via pin 5 of the ISO 7638 (Part 1 or 2) connector (yellow warning).

R-4.2.2.2.1.4 When the electrical part of the braking equipment is first energised check that byte 2, bits 3 - 4 of EBS 22 transmitted by the trailer is set to 01b. After the braking system has checked that no defects that require identification by the red warning signal are present the above message should be set to 00b.

R-4.2.2.3 **Response time checking**

R-4.2.2.3.1 Check that, with no faults present, the braking system response time requirements defined in E-3.5.2 are met.

R-4.2.2.4. **Automatically commanded braking**

In case the trailer includes a function where its operation results in an automatically commanded braking intervention, the following shall be checked:

If no automatically commanded braking intervention is generated, check that message EBS 22 byte 4 bits 5 to 6 are set to 00.

Simulate an automatically commanded braking intervention, when the resulting deceleration is $\geq 0.7 \text{ m/sec}^2$, check that message EBS 22 byte 4 bits 5 to 6 are set to 01.

R-4.2.2.5. **Vehicle stability function**

In the case of a trailer equipped with a vehicle stability function, the following checks shall be carried out:

When the vehicle stability function is inactive, check that message EBS 21 byte 2 bits 1 to 2 are set to 00.

Simulate an intervention of the vehicle stability control function as specified in V-2.2.4. and check that message EBS 21 byte 2 bits 1 to 2 are set to 01.

R- 4.2.2.6 Support of the electric control line

Where the trailer braking system does not support braking via the electric control line check that message EBS 22 byte 4 bits 7 to 8 are set to 00.

Where the trailer braking system supports the electric control line, check that message EBS 22 byte 4 bits 7 to 8 are set to 01.

R-4.2.3 **Additional Checks**

R-4.2.3.1 At the discretion of the certifying agency the checking procedures defined above may be repeated with the non-braking messages

relevant to the interface in different states or switched off.

Where repeat measurements of the brake system response time are carried out, variations in the value recorded may occur due to the reaction of the vehicle pneumatics. In all cases the prescribed response time requirements shall be met.

R-4.2.3.2. and P-2.4.2. defines additional messages that shall under specific circumstances be supported by the trailer. Additional checks may be carried out to verify the status of supported messages to ensure the requirements of 5.1.3.6.2. are fulfilled.

ANNEX S
(Clauses 5.1.5 and M-4.1)
**SPECIAL REQUIREMENTS TO BE APPLIED TO THE
SAFETY ASPECTS OF COMPLEX
ELECTRONIC VEHICLE CONTROL SYSTEMS**

S-1.0 GENERAL

This Annex defines the special requirements for documentation, fault strategy and verification with respect to the safety aspects of complex electronic vehicle control systems (see S-2.3) as far as this standard is concerned.

This Annex may also be called, by special paragraphs in this standard, for safety related functions which are controlled by electronic system(s).

This Annex does not specify the performance criteria for 'The System' but covers the methodology applied to the design process and the information which shall be disclosed to the certifying agency, for type approval purposes.

This information shall show that 'The System' respects, under normal and fault conditions, all the appropriate performance requirements specified elsewhere in this standard.

S-2.0 DEFINITIONS

For the purpose of this Annex, following definitions shall apply.

S-2.1 Safety Concept – A description of the measures designed into the system, for example within the electronic units, so as to address system integrity and thereby ensure safe operation even in the event of an electrical failure.

The possibility of a fall-back to partial operation or even to a back-up system for vital vehicle functions may be a part of the safety concept.

S-2.2 Electronic Control System – A combination of units, designed to co-operate in the production of the stated vehicle control function by electronic data processing.

Such systems, often controlled by software, are built from discrete functional components such as sensors, electronic control units and actuators and connected by transmission links. They may include mechanical, electro-pneumatic or electro-hydraulic elements.

The system referred to herein, is the one for which type approval is being sought.

S-2.3 Complex Electronic Vehicle Control Systems – Those electronic control systems which are subject to a hierarchy of control in which a controlled function may be over-ridden by a higher level electronic control system/function.

A function which is over-ridden becomes part of the complex system.

S-2.4 Higher-Level Control – Systems/functions those which employ additional processing and/or sensing provisions to modify vehicle behaviour by commanding variations in the normal function(s) of the vehicle control system.

This allows complex systems to automatically change their objectives with a priority which depends on the sensed circumstances.

S-2.5 Units – The smallest divisions of system components which will be considered in this Annex, since these combinations of components will be treated as single entities for purposes of identification, analysis or replacement.

S-2.6 Transmission Links – The means used for interconnecting distributed units for the purpose of conveying signals, operating data or an energy supply.

This equipment is generally electrical but may, in some part, be mechanical, pneumatic, hydraulic or optical.

S-2.7 **Range of Control** – Refers to an output variable and defines the range over which the system is likely to exercise control.

S-2.8 **Boundary of Functional Operation** – The boundaries of the external physical limits within which the system is able to maintain control.

S-3.0 **DOCUMENTATION**

S-3.1 **Requirements**

The manufacturer shall provide a documentation package which gives access to the basic design of ‘The System’ and the means by which it is linked to other vehicle systems or by which it directly controls output variables.

The function(s) of ‘The System’ and the safety concept, as laid down by the manufacturer, shall be explained.

Documentation shall be brief, yet provide evidence that the design and development has had the benefit of expertise from all the system fields which are involved.

For periodic technical inspections, the documentation shall describe how the current operational status of ‘The System’ is to be checked.

S-3.1.1 **Documentation shall be made available in following two parts:**

- a) The formal documentation package for the approval, containing the material listed in S-3 (with the exception of that of S-3.4.4) which shall be supplied to the Test Agency at the time of submission of the type approval application. This will be taken as the basic reference for the verification process set out in S-4.
- b) Additional material and analysis data of S-3.4.4, which shall be retained by the manufacturer, but made open for inspection at the time of type approval.

S-3.2 **Description of the Functions of ‘The System’**

A description shall be provided which gives a simple explanation of all the control functions of ‘The System’ and the methods employed to achieve the objectives, including a statement of the mechanism(s) by which control is exercised.

S-3.2.1 A list of all input and sensed variables shall be provided and the working range of these defined.

S-3.2.2 A list of all output variables which are controlled by ‘The System’ shall be provided and an indication given, in each case, of whether the control is direct or via another vehicle system. The range of control (see S-2.7) exercised on each such variable shall be defined.

S-3.2.3 Limits defining the boundaries of functional operation (see S-2.8) shall be stated where appropriate to system performance.

S-3.3 System Layout and Schematics

S-3.3.1 Inventory of Components

A list shall be provided; collating all the units of ‘The System’ and mentioning the other vehicle systems which are needed to achieve the control function in question.

An outline schematic showing these units in combination shall be provided with both the equipment distribution and the interconnections made clear.

S-3.3.2 Functions of the Units

The function of each unit of ‘The System’ shall be outlined and the signals linking it with other units or with other vehicle systems shall be shown. This may be provided by a labelled block diagram or other schematic, or by a description aided by such a diagram.

S-3.3.3 Interconnections

Interconnections within ‘The System’ shall be shown by a circuit diagram for the electrical transmission links, by an optical-fiber diagram for optical links, by a piping diagram for pneumatic or hydraulic transmission equipment and by a simplified diagrammatic layout for mechanical linkages.

S-3.3.4 Signal Flow and Priorities

There shall be a clear correspondence between these transmission links and the signals carried between units.

Priorities of signals on multiplexed data paths shall be stated, wherever priority may be an issue affecting performance or safety as far as this standard is concerned.

S-3.3.5 Identification of Units

Each unit shall be clearly and unambiguously identifiable (for example by marking for hardware and marking or software output for software content) to provide corresponding hardware and documentation association.

Where functions are combined within a single unit or indeed within a single computer, but shown in multiple blocks in the block diagram for clarity and ease of explanation, only a single hardware identification marking shall be used.

The manufacturer shall, by the use of this identification, affirm that the equipment supplied conforms to the corresponding document.

S-3.3.5.1 The identification defines the hardware and software version and, where the latter changes such as to alter the function of the unit as far as this standard is concerned, this identification shall also be changed.

S-3.4 Safety Concept of the Manufacturer

S-3.4.1 The manufacturer shall provide a statement which affirms that the strategy chosen to achieve ‘The System’ objectives will not, under non-fault conditions, prejudice the safe operation of systems which are subject to the prescriptions of this standard.

S-3.4.2 In respect of software employed in ‘The System’, the outline architecture shall be explained and the design methods and tools used shall be identified.

The manufacturer shall be prepared, if required, to show some evidence of the means by which they determined the realisation of the system logic, during the design and development process.

- S-3.4.3 The manufacturer shall provide the technical authorities with an explanation of the design provisions built into ‘The System’ so as to generate safe operation under fault conditions. Possible design provisions for failure in ‘The System’ are for example:

- a) Fall-back to operation using a partial system.
- b) Change-over to a separate back-up system.
- c) Removal of the high level function.

In case of a failure, the driver shall be warned, for example by warning signal or message display. When the system is not de-activated by the driver, for example; by turning the ignition (run) switch to ‘off’, or by switching off that particular function if a special switch is provided for that purpose, the warning shall be present as long as the fault condition persists.

- S-3.4.3.1 If the chosen provision selects a partial performance mode of operation under certain fault conditions, then these conditions shall be stated and the resulting limits of effectiveness defined.
- S-3.4.3.2 If the chosen provision selects a second (back-up) means to realise the vehicle control system objective, the principles of the change-over mechanism, the logic and level of redundancy and any built in back-up checking features shall be explained and the resulting limits of back-up effectiveness defined.
- S-3.4.3.3 If the chosen provision selects the removal of the higher level function, all the corresponding output control signals associated with this function shall be inhibited, and in such a manner as to limit the transition disturbance.
- S-3.4.4 The documentation shall be supported, by an analysis which shows, in overall terms, how the system will behave on the occurrence of any one of those specified faults which will have a bearing on vehicle control performance or safety.

This may be based on a failure mode and effect analysis (FMEA), a fault tree analysis (FTA) or any similar process appropriate to system safety considerations.

The chosen analytical approach (es) shall be established and maintained by the manufacturer and shall be made open for inspection by the Test Agency at the time of the type approval.

- S-3.4.4.1 This documentation shall itemise the parameters being monitored and shall set out, for each fault condition of the type defined in S-3.4.4 the warning signal to be given to the driver and/or to service/technical inspection personnel.

S-4.0 VERIFICATION AND TEST

- S-4.1 The functional operation of ‘The System’, as laid out in the documents required in S-3 shall be tested as follows:

S-4.1.1 Verification of the Function of ‘The System’

As the means of establishing the normal operational levels, verification of the performance of the vehicle system under non-fault conditions shall be conducted against the manufacturer's basic benchmark specification unless this is subject to a specified performance test as part of the approval procedure of this or another standard.

S-4.1.2 Verification of the Safety Concept of S-3.4

The reaction of ‘The System’ shall, at the discretion of the type approval authority, be checked under the influence of a failure in any individual unit by applying corresponding output signals to electrical units or mechanical elements in order to simulate the effects of internal faults within the unit.

S-4.1.2.1 The verification results shall correspond with the documented summary of the failure analysis, to a level of overall effect such that the safety concept and execution are confirmed as being adequate.

**ANNEX T
(Clause 6)**

**PERFORMANCE TESTING OF BRAKING SYSTEM COMPONENTS
PART-1 PERFORMANCE TESTING OF TRAILER BRAKING
COMPONENTS**

T-1.0 General

T-1.1 This Annex defines the test procedures applicable in defining the

performance of the following:

- a) Diaphragm brake chambers (see T-2);
- b) Spring brake actuators (see T-3);
- c) Trailer brakes - cold performance characteristics (see T-4); and
- d) Anti-lock braking systems (see T-5).
- e) Vehicle stability y function (see T-6.).

NOTE – Procedures for determining the fade test performance for trailer brakes and automatic brake wear adjustment devices are defined in Annex K.

T-1.2 The above test reports may be used in conjunction with the procedures defined in Annex U or at the time of evaluating a trailer which is being subject to actual performance requirements defined for the respective trailer.

T-2.0 PERFORMANCE CHARACTERISTICS FOR DIAPHRAGM BRAKE CHAMBERS

T-2.1 General

T-2.1.1 This section defines the procedure by which the thrust/stroke/pressure characteristics are determined for diaphragm brake chambers which are used in compressed air braking systems (see Note) to generate forces required in brakes with mechanical actuation.

For the purpose of this verification procedure the service brake section of a combined spring brake actuator is considered to be a diaphragm brake chamber.

NOTE – Other brake chamber designs may be approved upon presentation of equivalent information.

T-2.1.2 The verified performance characteristics declared by the manufacturer shall be used in all calculations relating to the brake compatibility requirements of Annex J, the Type 0 cold service braking performance requirements of Annex U and the determination of the available actuator stroke with respect to the verification of the hot performance of Annex K.

T-2.2 Test Procedure

T-2.2.1 Zero datum position of the brake chamber is to be taken as the non-pressurized position.

T-2.2.2 In nominal pressure increments of ≤ 100 kPa (1 bar), through a pressure range of 100 to ≥ 800 kPa, the corresponding thrust generated is to be monitored over the full stroke range available for a stroke displacement of ≤ 10 mm/s or a stroke increment of ≤ 10 mm and whilst not permitting the applied pressure to deviate ± 5 kPa.

T-2.2.3 For each pressure increment the corresponding average thrust (Th_A) and the effective stroke (s_p) shall be determined as per T-12.

T-2.3 Verification

T-2.3.1 With reference to item 3.1, 3.2 and 3.3 of T-6, a minimum of 6 samples are to be tested, with a verification report being issued, providing no one result deviates by more than 4 percent of the declared characteristics. However, with respect to the verification of the s_p characteristics the pressure range over which the 4 percent tolerance is fulfilled shall be recorded and specified in item 3.3.1 of T-6. Outside of this pressure range the tolerance may exceed 4 percent.

[With reference to T-6 (item 3.1, 3.2, 3.3 and 3.4), a minimum of 6 samples are to be tested, with a verification report being issued providing that the requirements of T-2.3.2, T-2.3.3 and T-2.3.4 are satisfied.]

T-2.3.2 The test results recorded shall be reported on a form, a model of which is shown in T-7 and shall be included with the verification report detailed within T-2.4.

With respect to the verification of average thrust (Th_A) - $f(p)$, a graph defining the acceptable performance variation shall be constructed following the model shown in Fig. 23, which is based on the manufacturers declared thrust to pressure relationship. The manufacturer shall also define the category of trailer for which the brake chamber may be used and the corresponding tolerance band applied.

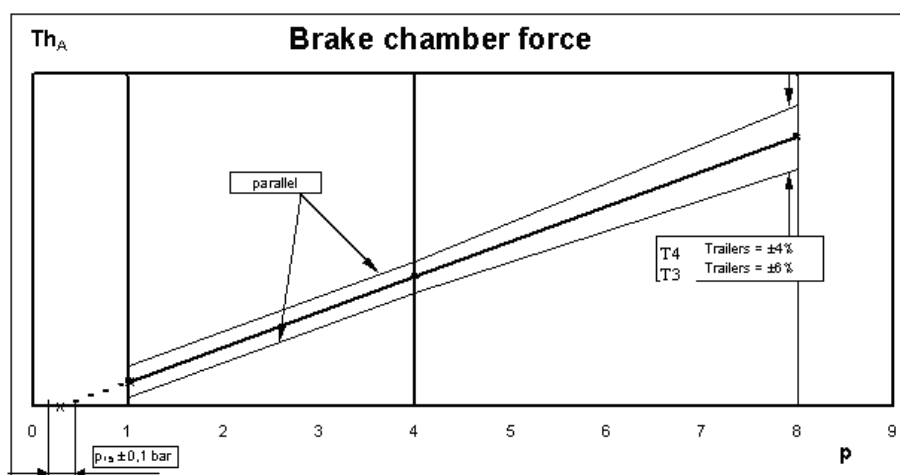


FIG. 23

VERIFICATION OF AVERAGE THRUST (Th_A) - $F(P)$

- T-2.3.3 It shall be verified that the pressure (p_{15}) required to produce a pushrod stroke of 15 mm from the zero datum position with a tolerance of ± 10 kPa by following one of the following test procedures:
- T-2.3.3.1 Utilizing the declared function of thrust (Th_A) - $f(p)$ the brake chamber threshold pressure (p_{15}) shall be calculated when $Th_A = 0$. It shall then be verified that when this threshold pressure is applied a pushrod stroke as defined in T-2.3.3 is produced.
- T-2.3.3.2 The manufacturer shall declare the brake chamber threshold pressure (p_{15}) and it shall be verified that when this pressure is applied the pushrod stroke defined in T-2.3.3 is produced.
- T-2.3.4 With respect to the verification of effective stroke (s_p) - $f(p)$, the measured value shall not be less than -4% of the s_p characteristics at the manufacturer's declared pressure range. This value shall be recorded and specified in item 3.3.1 of T-6. Outside of this pressure range the tolerance may exceed -4 percent.
- T-2.3.5 The test results recorded shall be reported on a form, a model of which is shown in T-7 and shall be included with the verification report, detailed in T-2.4.
- T-2.4 **Verification Report**
- T-2.4.1 The manufacturer's declared performance characteristics, verified by the test results recorded in accordance with T-2.3.2 shall be reported on a form, a model of which is shown in T-6.
- T-3.0 PERFORMANCE CHARACTERISTICS FOR SPRING BRAKES**
- T-3.1 **General**
- T-3.1.1 This section defines the procedure by which the thrust/stroke/pressure characteristics are determined for spring brakes (see Note) that are used in compressed air braking systems to generate forces required in brakes with mechanical actuation. For the purpose of this verification procedure the spring brake section of a combined spring brake actuator is considered to be a spring break.
- NOTE — Other spring brake designs may be approved upon presentation of equivalent information.
- T-3.1.2 The performance characteristics declared by the manufacturer shall be used in all calculations relating to the parking braking performance requirements of Annex U.
- T-3.2 **Test Procedure**
- T-3.2.1 Zero datum position of the spring brake chamber is to be taken as the fully pressurized position.
- T-3.2.2 In nominal stroke increments of ≤ 10 mm, the corresponding thrust generated is to be monitored over the full stroke range available at zero pressure.
- T-3.2.3 The pressure shall then be gradually increased until the stroke is 10 mm from the zero datum position, and this pressure, defined as the release pressure, shall be recorded.

T-3.2.4 The pressure shall then be increased to 850 kPa (8.5 bar), or the maximum working pressure declared by the manufacturer, whichever is lower.

T-3.3 Verification

T-3.3.1 With reference to T-8 (items 2.1, 3.1, 3.2 and 3.3), a minimum of 6 samples shall be tested; with a verification report being issued providing the following conditions are met:

T-3.3.1.1 Over a range of stroke from 10 mm to 2/3 of the maximum stroke, no one result, measured in accordance with T-3.2.2, deviates by more than 6 percent from the declared characteristics.

T-3.3.1.2 No one result, measured in accordance with T-3.2.3, exceeds the declared value.

T-3.3.1.3 Each spring brake continues to function correctly after completion of the test in accordance with T-3.2.4.

T-3.3.2 The test results recorded shall be reported on a form, a model of which is shown in T-9 and shall be included with the verification report detailed within T-3.4.

T-3.4 Verification Report

T-3.4.1 The manufacturer's declared performance characteristics, verified by the test results recorded in accordance with T-3.3.2 shall be reported on a form, a model of which is shown in T-8.

T-4.0 COLD PERFORMANCE CHARACTERISTICS FOR TRAILER BRAKES

T-4.1 General

T-4.1.1 This procedure covers the testing of the 'cold' performance characteristics of air operated S cam and disc brakes (see Note) fitted to trailers.

NOTE — Other brake designs may be approved upon presentation of equivalent information.

T-4.1.2 The performance characteristics declared by the manufacturer shall be used for all calculations relating to the braking compatibility requirements of Annex J and to the Type 0 cold service braking and parking braking performance requirements of Annex U.

T-4.2 Brake Factor and Brake Threshold Torque

T-4.2.1 The preparation of the brake shall be in accordance with T-4.4.2.

T-4.2.2 The brake factor is determined by using the following formula:

$$B_F = \frac{\Delta \text{Output torque}}{\Delta \text{Input torque}}$$

and shall be verified for each of the lining or pad materials specified in paragraph 4.3.1.3. below.

T-4.2.3 The brake threshold torque shall be expressed in a manner that remains

valid for variations of brake actuation and is denoted by the symbol C_o .

- T-4.2.4 The values of B_F shall remain valid for variations of the following parameters:
 - T-4.2.4.1 Mass per brake up to that defined in T-4.3.1.5.
 - T-4.2.4.2 Dimensions and characteristics of external components used to actuate the brake.
 - T-4.2.4.3 Wheel size/tyre dimensions.
- T-4.3 **Information Document**
 - T-4.3.1 The brake manufacturer shall provide the Test Agency with at least the following information:
 - T-4.3.1.1 A description of the brake type, model, size etc.
 - T-4.3.1.2 Details of the brake geometry
 - T-4.3.1.3 The make and type of brake lining(s) or brake pad(s)
 - T-4.3.1.4 The brake drum or brake disc material
 - T-4.3.1.5 The maximum technically permitted mass for the brake
 - T-4.3.2 Additional Information
 - T-4.3.2.1 Wheel and tyre sizes to be used for the test
 - T-4.3.2.2 The declared brake factor B_F
 - T-4.3.2.3 The declared threshold torque $C_{o,dec}$
- T-4.4 **Test Procedure**
 - T-4.4.1 Preparation
 - T-4.4.1.1 A graph defining the acceptable performance variation shall be constructed, following the model shown in Fig. 24, using the manufacturers declared brake factor.

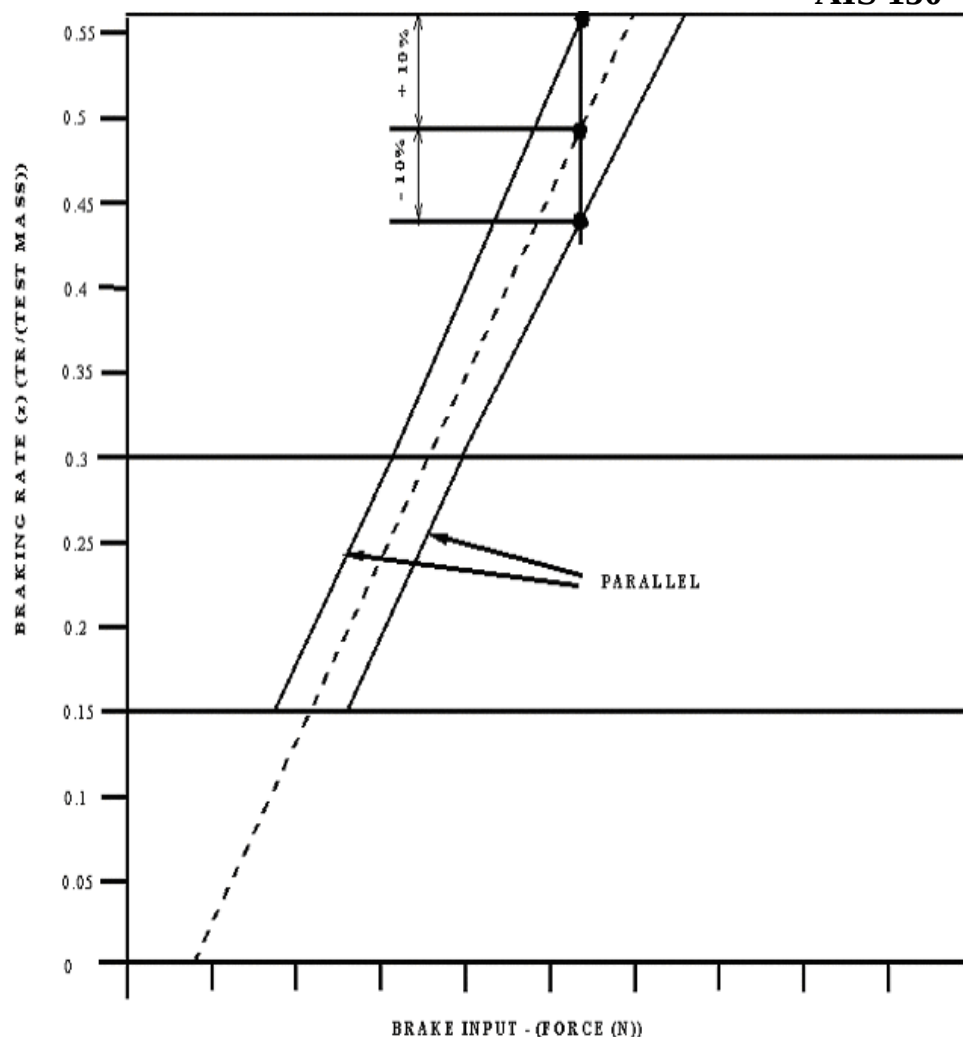


FIG. 24
Acceptable Performance Variation

- T-4.4.1.2 The performance of the device used to actuate the brake shall be calibrated within an accuracy of 1 percent.
- T-4.4.1.3 The dynamic tyre radius at the test loading shall be determined as prescribed for the test method.
- T-4.4.2 **Bedding in (Burnishing) Procedure**
- T-4.4.2.1 In the case of drum brakes the tests shall start with new brake linings and new drum(s), the brake linings shall be machined to achieve the best possible initial contact between the linings and drum(s).
- T-4.4.2.2 In the case of disc brakes the tests shall start with new brake pads and new disc(s), machining of the pad material shall be at the discretion of the brake manufacturer.
- T-4.4.2.3 Make 20 brake applications from an initial speed of 60 km/h with an input to the brake theoretically equal to 0.3 TR/Test Mass. The initial temperature at the lining/drum or pad/disc interface shall not exceed 100 °C before each brake application.

- T-4.4.2.4 Carry out 30 brake applications from 60 km/h to 30 km/h with an input to the brake equal to 0.3 TR/Test Mass and with a time interval between applications of 60 s (see Note). The initial temperature at the lining/drum or pad/disc interface on the first brake application shall not exceed 100 °C.
- NOTE** — If the track test method or the rolling road test methods are to be utilised, energy inputs equivalent to those specified shall be used.
- T-4.4.2.5 On completion of the 30 brake applications defined in T-4.4.2.4 and after an interval of 120 s carry out 5 brake applications from 60 km/h to 30 km/h with an input to the brake equal to 0.3 TR/Test Mass and with an interval of 120 s between applications (see Note under T-4.4.2.4).
- T-4.4.2.6 Make 20 brake applications from an initial speed of 60 km/h with an input to the brake equal to 0.3 TR/Test Mass. The initial temperature at the lining/drum or pad/disc interface shall not exceed 150 °C before each brake application.
- T-4.4.2.7 **Carry out a performance check as follows:**
- T-4.4.2.7.1 Determine the input torque to produce theoretical performance values equivalent to 0.2, 0.35 and 0.5 ± 0.05 TR/Test Mass.
- T-4.4.2.7.2 Once the input torque value has been determined for each braking rate, this value shall remain constant throughout each and subsequent brake applications (for example constant pressure).
- T-4.4.2.7.3 Make a brake application with each of the input torques determined in T-4.4.2.7.1 from an initial speed of 60 km/h. The initial temperature at the lining/drum or pad/disc interfaces shall not exceed 100 °C before each application.
- T-4.4.2.8 Repeat the procedures defined in T-4.4.2.6 and T-4.4.2.7.3 above until the performance of five consecutive non-monotonic measurements at each constant input value has stabilized within a tolerance of -10 percent of the maximum value.
- T-4.4.2.9. If the manufacturer can demonstrate by field test results, that the brake factor after this bedding in state is different from the brake factor which has developed on the road, additional conditioning is permissible.
- The maximum brake temperature, measured at the lining/drum or pad/disc interface, during this additional bedding in procedure shall not exceed 500 °C in the case of drum brakes and 700 °C in the case of disc brakes.
- This field test shall be an endurance run with the same type and model of brake as that to be recorded in K-4, test report. The results of at least 3 tests in accordance with T-4.4.3.4. conducted under the conditions of the laden Type-0 test, during the field test, shall be the basis for determining whether further conditioning is permissible. The brake tests shall be documented as prescribed in T-15.

The details of any additional conditioning shall be recorded and appended to the brake factor B_F in K-4.2.3.1., by specifying for instance the following test parameters:

- (a) Brake actuator pressure, the brake input torque or the brake output torque of the brake application;
- (b) Speed at the beginning and the end of the brake application;
- (c) Time in the case of a constant speed;
- (d) Temperature at the beginning and the end of the brake application or the duration of the brake cycle.

T-4.4.2.10 In the case of this procedure being carried out on an inertia dynamometer or rolling road, unlimited use of cooling air is permitted.

T-4.4.3 Verification Test

T-4.4.3.1 The temperature measured at the lining/drum or pad/disc interface shall not exceed 100 °C, at the start of each brake application.

T-4.4.3.2 The brake threshold torque shall be determined from the measured value of brake input by reference to a calibrated input device.

T-4.4.3.3 The initial speed for all brake applications is 60 ± 2 km/h.

T-4.4.3.4 A minimum of six consecutive brake applications shall be made from 0.15 to 0.55 TR/Test Mass at ascending increments of application pressure, following by six brake applications made with the same application pressures in descending increments.

T-4.4.3.5 For each of the brake applications in T-4.4.3.4 the braking rate is calculated, corrected to take account of rolling resistance, and plotted on the graph specified in T-4.4.1.1.

T-4.5 **Test Methods**

T-4.5.1 Track Test

T-4.5.1.1 The brake performance test shall be carried out on a single axle only.

T-4.5.1.2 The tests shall be carried out on a straight level track, with a surface affording good adhesion, and performed when there is no wind liable to affect the results.

T-4.5.1.3 The trailer shall be loaded (as closely as possible) to the maximum technically permitted mass for each brake, however, additional mass may be added if required to ensure that sufficient mass is over the axle under test to achieve a braking rate of 0.55 TR/(maximum technically permitted mass per brake) without wheel lock.

T-4.5.1.4 The dynamic rolling radius of the tyre may be verified at low speed, < 10 km/h, by measuring the distance travelled as a function wheel revolutions, the minimum number of revolutions required to determine the dynamic rolling radius is 10.

- T-4.5.1.5 The rolling resistance of the vehicle combination is to be determined by measuring the time taken for the vehicle speed to reduce from 55 to 45 km/h and the distance covered, when tested in the same direction in which the verification test will be carried out and with the engine disconnected and any endurance brake system disengaged.
- T-4.5.1.6 Only the brakes of the axle under test shall be actuated and reach an input pressure at the brake input device of 90 ± 3 percent (after maximum build up time of 0.7 s) of its asymptotic value. The test shall be carried out with the engine disconnected and any endurance braking system disengaged.
- T-4.5.1.7 The brakes shall be closely adjusted at the start of the test.
- T-4.5.1.8 The brake input for the purpose of calculating the brake threshold torque shall be determined by lifting the wheel and gradually applying the brake whilst the wheel is rotated by hand until resistance is detected.
- T-4.5.1.9 The final speed v_2 shall be calculated by the following formula:

$$v_2 = v_1 [(P_0 + P_1) / (P_0 + P_1 + P_2)]^{1/2}$$
 where
 v_1 = initial speed, km/h;
 v_2 = final speed, km/h;
 P_0 = mass of the towing vehicle under test conditions, in kg;
 P_1 = part of the mass of the trailer borne by the unbraked axle(s), in kg;
 and
 P_2 = part of the mass of the trailer borne by the braked axle(s), in kg.
- T-4.5.1.10 The braking performance of the axle under test shall be determined by calculating the deceleration determined from a direct measurement of velocity and distance between $0.8 v_1$ and v_2 , where v_2 shall not be less than $0.1 v_1$. This shall be deemed to be equivalent to the mean fully developed deceleration (MFDD) as defined in Annex C.
- T-4.5.2 **Inertia Dynamometer Test**
- T-4.5.2.1 The test shall be carried out on a single brake assembly.
 The test shall be carried out on a single brake assembly.
- T-4.5.2.2 The test machine shall be capable of generating the inertia required by T-4.5.2.5.
- T-4.5.2.3 The test machine shall be calibrated for speed and brake output torque within an accuracy of 2 percent.
- T-4.5.2.4 The instrumentation for the test shall be capable of providing at least the following data:
 a) A continuous recording of brake application pressure or force;
 b) A continuous recording of brake output torque;
 c) A continuous recording of the temperature measured at the lining/drum or pad/disc interface; and
 d) Speed during the test.

- T-4.5.2.5 The inertia (I_T) of the dynamometer shall be set as close as possible, with ± 5 percent tolerance, including the internal friction of the dynamometer, to that part of the linear inertia of the vehicle acting upon one wheel necessary for a performance of $0.55TR/\text{maximum}$ technically permitted mass according to the following formula:

$$I_T = Pd R^2$$

where

I_T =actual rotary inertia, in kgm^2 ;

R =tyre rolling radius defined by the formula $0.485 D$;

$D = d + 2H$ (see Note);

d =rim diameter conventional number, in mm;

H =nominal section height, in mm = $S_1 \times 0.01 Ra$;

S_1 =section width, in mm;

Ra =nominal aspect ratio; and

Pd =maximum technically permitted mass/brake as defined in T-4.3.1.5.

NOTE — Outer diameter of tyre, as defined in IS 15633 and IS 15636

- T-4.5.2.6 Cooling air at ambient temperature may be used, flowing at a velocity not exceeding $0.33 v$ over the brake in a direction perpendicular to its axis of rotation.
- T-4.5.2.7 The brake shall be closely adjusted at the start of the test.
- T-4.5.2.8 The brake input for the purpose of calculating the brake threshold torque shall be determined by gradually applying the brake until the onset of brake torque generation is observed.
- T-4.5.2.9 The brake performance shall be determined by applying the following formula to the measured brake output torque

$$\text{Braking rate} = \frac{M_t R}{I g}$$

where

M_t = average brake output torque (Nm) - based on distance

$I g$ = deceleration due to gravity (m/s^2)

R = tyre rolling radius defined by the formula $0.485 D$

The average brake output torque (M_t) shall be calculated from the deceleration determined from a direct measurement of velocity and distance between $0.8 v_1$ and $0.1 v_1$. This shall be deemed to be equivalent to the mean fully developed deceleration (MFDD) as defined in Annex C.

T-4.5.3 Rolling Road Test

T-4.5.3.1 The test shall be carried out on single axle with one or two brakes.

T-4.5.3.2 The test machine shall have a calibrated means of imposing load to simulate the required mass for the brake(s) to be tested.

T-4.5.3.3 The test machine shall be calibrated for speed and brake torque within an accuracy of 2 percent taking into account the internal friction characteristics. The dynamic rolling radius of the tyre (R) shall be determined by measuring the rotational speed of the rolling road and the unbraked wheels of the axle under test at a speed equivalent to 60 km/h, and calculated by the formula:

$$R = R_R \frac{n_D}{n_w}$$

where

R_R = radius of the rolling road,

n_D = rotational speed of rolling road, and

n_w = rotational speed of the unbraked wheels of the axle.

T-4.5.3.4 Cooling air at ambient temperature may be used, flowing at a velocity not exceeding 0.33 v over the brake(s).

T-4.5.3.5 The brake(s) shall be closely adjusted at the start of the test.

T-4.5.3.6 The brake input for the purpose of calculating the brake threshold torque shall be determined by gradually applying brake(s) until the onset of brake torque generation is observed.

T-4.5.3.7 The brake performance shall be determined by measuring the brake force at the periphery of the tyre calculated to braking rate, taking into account the rolling resistance. The rolling resistance of the loaded axle will be determined by measuring the force at the periphery of the tyre at a speed of 60 km/h.

The average brake output torque (M_t) shall be based on the measured values between the moment, the application pressure/force reaches its asymptotic value from the onset of pressure rise at the brake input device and when the energy input has reached the value W_{60} that is defined in T-4.5.3.8.

T-4.5.3.8 For determining the braking rate, an energy input W_{60} equivalent to the kinetic energy of the corresponding mass for the brake under test when braked from 60 km/h to standstill, shall be taken into account.

where

$$W_{60} = \int_0^{t(W_{60})} F_B \cdot v \cdot dt$$

T-4.5.3.8.1 If the test speed v cannot be maintained at 60 ± 2 km/h during the measurement of the braking rate according to T-4.5.3.8, the braking rate shall be determined from the direct measurement of brake force F_B and/or brake output torque M_t , so that the measurement of this/these parameter(s) are not affected by the dynamic forces of the inertia mass of the rolling road test machine.

T-4.6	Verification Report
T-4.6.1	The manufacturer's declared performance characteristics, verified by the test results recorded in accordance with T-4.4.3 shall be reported.
T-5	ANTI-LOCK BRAKING SYSTEMS (ABS)
T-5.1	General
T-5.1.1	This paragraph defines the procedure to determine the performance of a trailer anti-lock braking system.
T-5.1.2	Tests carried out on trailers of category T4 will be deemed to cover the requirements for trailers of T3 category.
T-5.2	Information Document
	The manufacturer of the ABS shall supply to the Test Agency an information document of the system(s) requiring performance verification. This document shall contain at least the information defined in T12.
T-5.3	Definition of Test Vehicles
T-5.3.1	Based on the information supplied in the information document, in particular the trailer applications defined in item 2.1 of T-10, the Test Agency shall carry out tests on representative trailers having up to three axles and equipped with the respective anti-lock braking system/configuration. Additionally, when selecting trailers for evaluation consideration shall also be given to the parameters defined in the following paragraphs.
T-5.3.1.1	Suspension type
	The method of evaluating the performance of the anti-lock braking system relative to the suspension type will be selected in the following way:
	a) Semi-trailers — for each suspension group for example balanced mechanical etc. a representative trailer shall be evaluated; and
	b) Full trailers — Evaluation to be carried out on a representative trailer equipped with any one suspension type.
T-5.3.1.2	Wheelbase
	For semi-trailers the wheelbase shall not be a limiting factor, but for full trailers the shortest wheelbase shall be evaluated.
T-5.3.1.3	Brake type
	Approval shall be limited to S cam or disc brakes but should other types become available then comparative testing may be required.
T-5.3.1.4	Load sensing device
	The utilization of adhesion shall be determined with the load sensing valve set to laden and unladen conditions. In all cases the requirements of M-2.7 shall apply.

- T-5.3.1.5 Brake actuation**
- Differentials in the level of actuation shall be recorded for evaluation during the tests to determine the utilization of adhesion. Results obtained from tests for one trailer may be applied to other trailers of the same type.
- T-5.3.2** For each type of trailer under test, documentation showing brake compatibility as defined in Annex J (Figs. 7 and 9) shall be made available to demonstrate conformity.
- T-5.3.3** For the purpose of the approval, semi-trailers and centre axle trailers shall be deemed to be of the same vehicle type.
- T-5.4 Test Schedule**
- T-5.4.1** The following tests shall be conducted by the Test Agency on the vehicle(s) defined in T-5.3 for each ABS configuration taking into consideration the application list defined in item 2.1 of T-10. However, worst case cross referencing may eliminate certain tests. If worst case testing is actually used, this should be stated in the test report.
- T-5.4.1.1 Utilization of Adhesion**
- Tests shall be carried out according to the procedure defined in M-6.2 or each ABS configuration and trailer type, as defined in the manufacturer's information document (see item 2.1. of T-10).
- T-5.4.1.2 Energy consumption**
- T-5.4.1.2.1 Axle loading**
- The trailer(s) to be tested shall be loaded so that the axle load is 2500 ± 200 kg or 35 percent ± 200 kg of the permissible static axle load whichever is the lower.
- T-5.4.1.2.2** It shall be ensured that 'full cycling' of the anti-lock braking system can be achieved throughout the dynamic tests defined in M-6.1.3.
- T-5.4.1.2.3 Energy Consumption Test**
- The test shall be carried out according to the procedure defined in M-6.1 for each ABS configuration.
- T-5.4.1.2.4** To enable trailers submitted for approval to be checked for conformity to the anti-lock energy consumption requirements (see M-6.1), the following checks shall be carried out:
- T-5.4.1.2.4.1** Prior to the commencement of the energy consumption test (T-5.4.1.2.3) in the case of brakes with nonintegrated brake wear adjustment the brakes shall be set to a condition where the relationship (R_i) of brake chamber push rod travel (s_T) against lever length (l_T) is 0.2. This relationship shall be determined for a brake chamber pressure of 650 kPa (6.5 bar).

Example $l_T=130$ mm,
 s_T at 650 kPa brake chamber pressure = 26 mm

$$R_l = s_T / l_T = 26/130 = 0.2$$

In the case of brakes with integrated automatic brake wear adjustment the brakes shall be set to the normal running clearance specified by the manufacturer.

Setting of the brakes as defined above shall be carried out when the brakes are cold (< 100 °C).

- T-5.4.1.2.4.2 With the load sensing valve set to the laden condition and the initial energy level set according to M-6.1.2 the energy storage device(s) shall be isolated from further supply of air. The brakes shall be applied with a control pressure of 650 kPa (6.5 bar) at the coupling head and then released. Further applications shall be made until the pressure in the brake chambers is the same as that obtained after following the test procedure defined in T-5.4.1.2.1 and T-5.4.1.2.2. The number of equivalent brake applications (n_{er}) shall be noted.

The equivalent number of static brake applications (n_e) is to be recorded in the test report.

where

$n_e = 1.2 \cdot n_{er}$ and is to be rounded up to the nearest whole integer

T-5.4.1.3 **Split friction test**

Where an anti-lock braking system is to be defined as category A system then all such ABS configurations shall be subject to the performance requirements of M-6.3.2.

T-5.4.1.4 **Low and high speed performance**

- T-5.4.1.4.1 With the trailer set as for utilization of adhesion evaluation, verification of the low and high speed performance shall be carried out according to M-6.3.1.

- T-5.4.1.4.2 Where a tolerance exists between the number of exciter teeth and tyre circumference, functional checks shall be carried out at the extremes of tolerance in accordance with M-6.3. This may be achieved by utilizing different tyre sizes or by producing special exciters to simulate frequency extremes.

T-5.4.1.5 **Additional checks**

The following additional checks shall be carried out with the towing vehicle unbraked and the trailer unladen.

- T-5.4.1.5.1 When the axle bogie passes from a high adhesion surface (k_H) to a low adhesion surface (k_L) where $k_H \geq 0.5$ and $k_H / k_L \geq 2$, with a control pressure at the coupling head of 650 kPa (6.5 bar), the directly controlled wheels shall not lock. The running speed and the instant of applying the trailer brakes is so calculated that with the anti-lock braking system full cycling on the high adhesion surface, the passage from one surface to the other being made at approximately 80 km/h and at 40 km/h.
- T-5.4.1.5.2 When the trailer passes from a low adhesion surface (k_L) to a high adhesion surface (k_H) where $k_H \geq 0.5$ and $k_H / k_L \geq 2$, with a control pressure at the coupling head of 650 kPa (6.5 bar), the pressure at the brake chambers shall rise to an appropriate high value within a reasonable time and the trailer shall not deviate from its initial course. The running speed and the instant of applying the brakes is so calculated that, with the anti-lock braking system full cycling on the low adhesion surface, the passage from one surface to the other occurs at approximately 50 km/h.
- T-5.4.1.6 Documentation relating to the controller(s) shall be made available as required by 5.1.5 and M-4.1.
- T-5.5 **Approval Report**
- T-5.5.1 An approval report shall be produced, the content of which is defined in T-11.
- T-6.0 Vehicle stability function**
- T-6.1. **General**
- T-6.1.1. This section defines a test procedure to determine the dynamic characteristics of a vehicle equipped with a vehicle stability function consisting of at least one of the following functions:
- (a) Directional control;
 - (b) Roll-over control.
- T-6.2. **Information document**
- T-6.2.1. The system/vehicle manufacturer shall supply to the Test Agency an information document of the control function(s) for which performance verification is required. This document shall contain at least the information defined in T-14.
- T-6.3. **Definition of test vehicle(s)**
- T-6.3.1. Based on the stability control function(s) and their application(s) defined in the manufacturer's information document the Test Agency shall carry out a performance verification. This may include one or more dynamic manoeuvres as defined in **V-2.2.3.** to this Standard on a trailer(s) having up to three axles which is representative of the application(s) defined in T-14.2.1

- T-6.3.1.1. When selecting the trailer(s) for evaluation, consideration shall also be given to the following:
- (a) Suspension type: for each suspension group, e.g. balanced pneumatic, a trailer of that specification shall be evaluated;
 - (b) Wheel base: wheel base shall not be a limiting factor;
 - (c) Brake type: approval shall be limited to trailers with S-cam or disc brakes but should other types become available, then comparative testing may be required;
 - (d) Braking system: the braking system of the trailer(s) to be evaluated shall comply with all of the relevant requirements of this Standard.
- T-6.4. **Test schedule**
- T-6.4.1. To evaluate the vehicle stability control function the tests used shall be agreed between the system/vehicle manufacturer and the Test Agency and shall include conditions, appropriate to the function being evaluated, that would without the intervention of the stability control function result in loss of directional control or roll-over. The dynamic manoeuvres, test conditions and results shall be included in the test report.
- T-6.5. **Towing vehicle**
- T-6.5.1. The towing vehicle used for evaluating the performance of the vehicle (trailer) stability function shall have the necessary pneumatic and electrical connections and if the towing vehicle is equipped with a vehicle stability function as defined in 3.34 that function shall be disabled.
- T-6.6. **Test report**
- T-6.6.1. A test report shall be produced, the content of which shall be at least that defined in T-15

PART 2 - PERFORMANCE TESTING OF MOTOR VEHICLE BRAKING COMPONENTS

T-7.1. General

Part 2 defines the procedures applicable in defining the performance of the following:

T-7.1.1. A vehicle stability function.

T-7.1.1.1. General

T-7.1.1.1.1. This section defines the procedure of determining the dynamic characteristics of a vehicle equipped with a vehicle stability function as specified in 5.2.1.32.

T-7.1.1.2. Information document

T-7.1.1.2.1. The system manufacturer shall supply the Test Agency with an information document on the vehicle stability control function(s) for which performance verification is required. This document shall contain at least the information defined in T-18 and shall be attached as an appendix to the test report.

T-7.1.1.3. Definition of test vehicle(s)

T-7.1.1.3.1. Based on the stability control function(s) and their application(s) defined in the system manufacturer's information document, the Test Agency shall carry out a vehicle based performance verification. This shall include one or more dynamic manoeuvres as defined in **V-2.1.3.** on a motor vehicle(s) which is representative of the application(s) defined in paragraph T-14.2.1. of the system manufacturer information document.

T-7.1.1.3.2. When selecting the motor vehicles(s) for evaluation, consideration shall also be given to the following:

- (a) Braking system: the braking system of the test vehicle(s) to be evaluated shall comply with all of the relevant requirements of this Standard;
- (b) Vehicle category – M₂, M₃, N₂, N₃;
- (c) Character of the vehicle;
- (d) Vehicle configuration(s) (e.g. 4x2, 6x2, etc.): each configuration to be evaluated;
- (e) Drive orientation (left or right hand drive): not a limiting factor – evaluation not required;
- (f) Single front axle steering: not a limiting factor – evaluation not required (see subparagraphs (g) and (h));
- (g) Additional steering axles (e.g. forced steering, self-steering): to be evaluated;
- (h) Steering ratio: to be evaluated – end-of-line programming or self-learning systems not a limiting factor;
- (i) Drive axles: to be taken into consideration with regard to the use

(loss) of wheel speed sensing in the determination of vehicle speed;

- (j) Lift axles: lift axle detection / control and lifted condition to be evaluated;
- (k) Engine management: communication compatibility to be evaluated;
- (l) Gearbox type (e.g. manual, automated manual, semi-automatic, automatic): to be evaluated;
- (m) Drive train options (e.g. retarder): to be evaluated;
- (n) Differential type (e.g. standard or self-locking): to be evaluated;
- (o) Differential lock(s) (driver selected): to be evaluated;
- (p) Brake system type (e.g. air over hydraulic, full air): to be evaluated;
- (q) Brake type (disc, drum (single wedge, twin wedge, S-cam)): not a limiting factor, however, should other types become available, then comparative testing may be required;
- (r) Anti-lock braking configurations: to be evaluated;
- (s) Wheelbase: to be evaluated

In the case where vehicles conforming to the minimum and maximum wheelbases as specified in the information document are not available at the time of testing, minimum and maximum wheelbase verification may be carried-out using system manufacturer test data for real vehicles with a wheelbase within 20 per cent of the actual minimum and maximum wheelbase vehicles being tested by the Test Agency;

- (t) Wheel type (single or twin): to be covered in the system manufacturer's information document;
- (u) Tyre type (e.g. structure, category of use, size): to be covered in the system manufacturer's information document;
- (v) Track width: not a limiting factor – covered by variations in the centre of gravity evaluation;
- (w) Suspension type (e.g. air, mechanical, rubber): to be evaluated;
- (x) Centre of gravity height: to be evaluated

In the case where vehicles conforming to the maximum centre of gravity height as specified in the information document are not available at the time of testing, maximum centre of gravity height verification may be carried-out utilising system manufacturer's test data for real vehicles with a centre of gravity height within +20 per cent of the actual maximum centre of gravity height of the vehicles being tested by the Test Agency;

- (y) Lateral acceleration sensor position: installation envelop as

specified by the system manufacturer to be evaluated;

- (z) Yaw rate sensor position: installation envelop as specified by the system manufacturer to be evaluated.

T-7.1.1.4. Test schedule

T-7.1.1.4.1. To evaluate the vehicle stability control function, the tests used shall be agreed upon between the system manufacturer and the Test Agency and shall include conditions, appropriate to the function being evaluated, that would without the intervention of the stability control function result in loss of directional control or roll-over control. The dynamic manoeuvres, test conditions and results shall be included in the test report.

The evaluation shall include the following, as appropriate:

T-7.1.1.4.1.1. Additional steering axles:

Evaluate the influence by a comparison of results with the axle in its normal steering mode and with the steering disabled so that it becomes a fixed axle, unless it is an end-of-line programming parameter.

T-7.1.1.4.1.2. Steering ratio:

Tests to be carried-out to determine the effectiveness of any end-of-line programming or self-learning using a number of vehicles with different steering ratios, or the approval is restricted to the steering ratios actually tested.

T-7.1.1.4.1.3. Lift axle:

Tests to be carried-out with the lift axle in the raised and lowered conditions, with position detection and signal transfer being evaluated to establish that the change in wheelbase has been recognized.

T-7.1.1.4.1.4. Engine management:

Control of the engine, or any other source(s) of motive power, to be shown to be independent from driver demand.

T-7.1.1.4.1.5. Drive train options:

The effect of any options to be shown, e.g. retarder management to be independent of the driver in the case of a retarder.

T-7.1.1.4.1.6. Differential type/differential lock(s):

Effect of self-locking or driver selected locking to be shown, e.g. function maintained, reduced or switched-off.

T-7.1.1.4.1.7. **Anti-lock braking configurations:**

Each anti-lock braking configuration shall be tested on at least one vehicle.

If the vehicle stability function is hosted on different systems (e.g. ABS, EBS), tests shall be carried-out on vehicles having the different hosting systems.

T-7.1.1.4.1.8. **Suspension type:**

Vehicles shall be selected on the basis of the suspension type (e.g. air, mechanical, rubber) of each axle or axle group.

T-7.1.1.4.1.9. **Centre of gravity height:**

Tests shall be carried-out on vehicles where it is possible to adjust the centre of gravity height so as to demonstrate that the roll-over control is able to adapt to changes in the centre of gravity height.

T-7.1.1.4.1.10. **Lateral acceleration sensor position:**

The effect of the lateral acceleration sensor being installed in different positions on the same vehicle shall be evaluated to confirm the installation envelop specified by the system manufacturer.

T-7.1.1.4.1.11. **Yaw rate sensor position:**

The effect of the yaw rate sensor being installed in different positions on the same vehicle shall be evaluated to confirm the installation envelop specified by the system manufacturer.

T-7.1.1.4.1.12. **Loading:**

Vehicles shall be tested in both the laden and unladen/part laden conditions to demonstrate that the vehicle stability function is able to adapt to differing conditions of load.

In the case of a semi-trailer tractor, tests shall be carried-out as follows:

- a) With a coupled semi-trailer, in the laden and unladen/part laden conditions, in which the roll-over control, if fitted, has been disabled.
- b) In the solo condition (without a coupled semi-trailer or imposed load);
- c) With a load simulating the laden condition (without a coupled semi-trailer).

T-7.1.1.4.2. **Evaluation of buses**

As an alternative, in the case of buses, trucks having the same braking system type may be used in the evaluation. However, at least one bus shall be included in the testing and the subsequent report.

T-7.1.1.5. **Test report**

T-7.1.1.5.1. A test report shall be produced, the content of which shall be at least that defined in T-19.

T 8 MODEL VERIFICATION REPORT FORM FOR DIAPHRAGM BRAKE CHAMBERS

Report No.....

- 1.0 Identification
- 1.1 Manufacturer: (Name and address)
- 1.2 Make(see Note 1)
- 1.3 Type:(see Note 1)
- 1.4 Part number:(see Note 1)
- 2.0 Operating conditions:
- 2.1 Maximum working pressure:
- 3.0 Performance characteristics declared by the manufacturer:
- 3.1 Maximum stroke (s_{max}) at 650 kPa(6.5 bar)(see Note 2)
- 3.2 Average thrust (Th_A) - f (p) (see Note 2)
- 3.3 Effective stroke (s_p) - f (p) (see Note 2)
- 3.3.1 Pressure range over which the above effective stroke is valid: (compare T-2.3.1) (see T-2.3.4).
- 3.4 Pressure required to produce a push rod stroke of 15 mm (p_{15}) based on $Th_A - f(p)$ or declared value. (see Notes 2, 3 and 4)
- 4.0 Scope of application
 - a) The brake chamber may be used on trailers of category T3 and T4
yes/no
 - b) The brake chamber may be used on trailers of category T3 only
yes/no
- 5.0 Name of Test Agency conducting the test:
- 6.0 Date of test
- 7.0 This test has been carried out and the results reported in accordance with Annex T.
Test Agency (see Note 4) conducting the test

Signed: Date:
- 8.0 Test documents:

NOTES

- 1 To be marked on brake chamber, however for inclusion within the test report only the parent part number is required, model variants need not be indicated.
- 2 Identification shall be amended when changes are made which have an influence on the performance characteristics, paragraphs T- 3.1, T-3.2 and T- 3.3.
- 3 Delete as appropriate.
- 4 For the purposes of the application of the characteristics defined in this report with respect to Annex J it shall be assumed that the relationship from p_{15} to the declared $Th_A - f(p)$ at a pressure of 100 kPa is linear.

T9 MODEL REFERENCE RECORD OF TEST RESULTS FOR DIAPHRAGM BRAKE CHAMBERS

Report No.

 Record of test results¹⁾ for part number.....

Pressure ²⁾ (p) kPa	Average Thrust (Th _A) N	Effective Stroke (S _p) mm

NOTES

1 To be created for each of the six samples tested.

2 Pressure "p" shall be actual pressure values used in the test as defined in T-2.2.2.

T 10 MODEL VERIFICATION REPORT FORM FOR SPRING BRAKES

Report No.....

1.0 Identification

1.1 Manufacturer: (Name and address)

1.2 Make:(see Note 1)

1.2 Type:(see Note 1)

1.3 Part number: (see Note 1)

2.0 Operating conditions:

2.1. Maximum working pressure:

3.0 Performance characteristics declared by the manufacturer:

3.1. Maximum stroke (s_{max}) (see Note 2)

3.2. Spring thrust (Th_s) - f (s) (see Note 2)

3.3. Release pressure (at 10 mm stroke) (see Note 2)

4.0 Date of test:

5.0 This test has been carried out and the results reported in accordance with Annex T.

Test Agency conducting the test

Signed:Date:

6.0 Test documents:

.....

NOTES

- 1** To be marked on the spring break, however for inclusion within the test report only the parent part number is required, model variants need not be indicated.
- 2** Identification shall be amended when changes are made which have an influence on the performance characteristics (see T-3.1, T-3.2 and T-3.3).

BRAKES	
Report No.....	
Record of test results ¹⁾ for part number:	
Stroke ²⁾ s (mm)	Thrust Th _s (N)
Release pressure (at 10 mm stroke) kPa	
NOTES 1 To be created for each of the six samples tested. 2 Stroke "s" will be the actual stroke values used in the test as defined in T-3.2.2 .	

- 1.0 General
 - 1.1 Name of manufacturer
 - 1.2 System name
 - 1.3 System variations
 - 1.4 System configurations (for example 2S/1M, 2S/2M etc.)
 - 1.5 Explanation of the basic function and/or philosophy of the system.
- 2.0 Applications
 - 2.1 List of trailer types and ABS configurations for which approval is required.
 - 2.2 Schematic diagrams of the system configurations installed on the trailers defined in item 2.1 above with consideration given to the following parameters:
 - Sensor locations
 - Modulator locations
 - Lift axles
 - Steering axles
 - Tube: type - bore size(s) and lengths
 - 2.3 Relationship of tyre circumference to the resolution of the exciter, including tolerances.
 - 2.4 Tolerance of tyre circumference between one axle and another fitted with the same exciter.
 - 2.5 Scope of application with respect to suspension type:
 - Air suspension: Any type of balanced 'trailing arm' air suspension
 - Other suspensions: to be defined by manufacturer, model and type (balanced/unbalanced).
 - 2.6 Recommendations on differential brake input torque (if any) in relation to the ABS configuration and trailer bogie.
 - 2.7 Additional information (if applicable) to the application of the anti-lock braking system.
- 3.0 Component description
 - 3.1 Sensor(s)
 - Function
 - Identification (for example part number(s))
 - 3.2 Controller(s)
 - General description and function
 - Identification (for example part number(s))
 - Safety aspects of controller(s)
 - Additional features (for example retarder control, automatic configuration, variable parameters, diagnostics)
 - 3.3 Modulator(s)
 - General description and function

Identification (for example part number(s))

Limitations (for example maximum delivery volumes to be controlled)

3.4 Electrical Equipment

Circuit diagram(s)

Powering methods

Warning lamp sequence(s)

3.5 Pneumatic Circuits

Braking schematics covering the ABS configurations as applied to the trailer types defined in 5.2.1.2.1.

Limitations on pipe, tube sizes and associated lengths that have an effect on system performance (for example between modulator and brake chamber)

3.6 Electromagnetic compatibility

3.6.1 Documentation demonstrating compliance with the provisions of M-4.4.

T13 TRAILER ANTI-LOCK BRAKING SYSTEM TEST REPORT

Test Report No:

1.0 Identification

- 1.1 Manufacturer of the anti-lock braking system (name and address)
- 1.2 System name/model
- 2.0 System(s) and Installation(s) approved
- 2.1 ABS configuration(s) approved (for example 2S/1M, 2S/2M etc.)
- 2.2 Range of application (type of trailer and number of axles)
- 2.3 Methods of powering:ISO 7638 (Part 1 or 2), ISO 1185 etc.
- 2.4 Identification of approved sensor(s), controller(s) and modulator(s)
- 2.5 Energy consumption - equivalent number of static brake applications.
- 2.6 Additional features for example retarder control, lift axle configuration, etc.
- 3.0 Test data and results
- 3.1 Test vehicle data
- 3.2 Test surface information
- 3.3 Test results:
 - 3.3.1 Utilisation of adhesion
 - 3.3.2 Energy consumption
 - 3.3.3 Split friction test
 - 3.3.4 Low speed performance
 - 3.3.5 High-speed performance
 - 3.3.6 Additional checks
 - 3.3.6.1 Transition from high to low adhesion surfaces
 - 3.3.6.2 Transition from low to high adhesion surfaces
 - 3.3.7 Failure mode simulation
 - 3.3.8 Functional checks of optional power connections
 - 3.3.9 Electro-magnetic compatibility
- 4.0 Limits of installation
- 4.1 Relationship of tyre circumference to the resolution of the exciter
- 4.2 Tolerance on tyre circumference between one axle and another fitted with the same exciter
- 4.3 Suspension type
- 4.4 Differential(s) in brake input torque within the trailer bogie
- 4.5 Wheel base of full trailer
- 4.6 Brake type
- 4.7 Tube sizes and lengths
- 4.8 Load sensing device application
- 4.9 Warning lamp sequence
- 4.10 System configurations and applications that comply with the category A requirements

4.11 Other recommendations / limitations (for example location of sensors, modulator(s), lift axle(s), steering axle(s))

5.0 Date of test:

This test has been carried out and the results reported in accordance with Annex T.

Test Agency (see Note) conducting the test

Signed:Date:

Attachment: Manufacturer's information document

T-14 VEHICLE (TRAILER) STABILITY FUNCTION INFORMATION DOCUMENT

1.0 General

1.1. Name of manufacturer

1.2. System name

1.3. System variations

- 1.4. Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control
- 1.5. System configurations (where appropriate)
- 1.6. System identification
- 2.0 Applications
 - 2.1. List of trailer types and configurations for which approval is required
 - 2.2. Schematic diagrams of the respective configurations installed on the trailers defined in item 2.1. above with consideration given to the following:
 - (a) Lift axles;
 - (b) Steering axles;
 - (c) Anti-lock braking configurations.
 - 2.3. Scope of application with respect to suspension type:
 - (a) Air suspension: any type of balanced "trailing arm" air suspension;
 - (b) Other suspensions: individually identified by manufacturer, model and type (balanced/unbalanced).
 - 2.4. Additional information (if applicable) to the application of the directional control and/or the roll-over control function(s)
- 3.0 Component description
 - 3.1. Sensors external to the controller
 - (a) Function;
 - (b) Limitations on the location of the sensors;
 - (c) Identification, e.g. part numbers.
 - 3.2. Controller(s)
 - (a) General description and function;
 - (b) Identification e.g. part numbers;
 - (c) Limitations on the location of the controller(s);
 - (d) Additional features.
 - 3.3. Modulators
 - (a) General description and function;
 - (b) Identification;
 - (c) Limitations.

- 3.4. Electrical equipment
 - (a) Circuit diagrams;
 - (b) Powering methods.
- 3.5. Pneumatic circuits

System schematics including anti-lock braking configurations associated with the trailer types defined in T-6.2.1.
- 3.6 Safety aspects of the electronic system in accordance with Annex R to this Standard
- 3.7. Electro-magnetic compatibility
- 3.7.1 Documentation demonstrating compliance with AIS-004 (Part 3) as required by 5.1.1.4.

T-15 VEHICLE (TRAILER) STABILITY FUNCTION TEST REPORT

Test Report No:

1.0 Identification

1.1. Manufacturer of the vehicle stability function (name and address)

1.2. System name / model

1.3. Control function

2.0 System(s) and installations approved:

2.1. Anti-lock braking configurations (where appropriate)

2.2. Range of application (trailer type(s) and number of axles)

2.3. System identification

2.4. Additional features

3.0 Test data and results

3.1. Test vehicle data (including the specification and functionality of the towing vehicle)

3.2. Test surface information

3.3. Additional information

3.4. Demonstrative tests/simulations used for the purpose of evaluating the directional control and the roll-over control as appropriate.

3.5. Test results

3.6. Assessment in accordance with Annex R to this Standard

4.0 Limits of installation

4.1. Suspension type

4.2. Brake type

4.3. Location of components on the trailer

4.4. Anti-lock braking configurations

4.5. Other recommendations/limitations (e.g. lifting axles, steering axles, etc.)

5.0 Attachments

6.0 Date of test:

7.0 This test has been carried out and the results reported in accordance with Annex T.

T16 SYMBOLS AND DEFINITIONS	
Symbol	Definition
B_F	Brake factor (input torque to output torque amplification ratio)
C_O	Threshold input torque (minimum torque necessary to produce a measurable brake torque)
D	Outer tyre diameter (overall diameter of an inflated new tyre)
d	A conventional number denoting the nominal rim diameter and corresponding to the diameter of the rim expressed either in inches or mm
F_B	Brake force
H	Nominal tyre section height (the distance equal to half the difference between the outer diameter of the tyre and the nominal rim diameter)
I	Rotary inertia
l_T	Brake lever length of reference test trailer
M_t	Average brake output torque
n_e	Equivalent number of static brake applications for the purpose of type approval
n_{er}	Equivalent number of static applications obtained during testing
n_D	Rotational speed of the rolling road
n_W	Rotational speed of the unbraked wheels of the axle
P_d	Maximum technically permitted mass for the brake
p	Pressure
p_{15}	Pressure in the brake chamber required to produce a pushrod stroke of 15mm from the zero datum position.
R	Dynamic tyre rolling radius (calculated using $0.485D$)
R_a	Nominal aspect ratio of the tyre (one hundred times the number obtained by dividing the number expressing the nominal section height of the tyre in mm by the number expressing the nominal section width in mm).
R_l	Ratio of S_T / l_T
R_R	Radius of the rolling road
S_1	Section width of the tyre (linear distance between the outsides of the sidewalls of an inflated tyre, excluding elevations due to labelling (marking), decorations or protective bands or ribs).
S	Actuator stroke (working stroke plus free stroke)
S_{Max}	Total actuator stroke
S_p	Effective stroke (the stroke at which the output thrust is 90 percent of the average thrust Th_A)
S_T	Brake chamber push rod travel of reference test trailer in mm
Th_A	Average thrust (the average thrust is determined by integrating the values between 1/3 and 2/3 of the total stroke S_{Max}) (see Fig. 25)

Th_s	Spring thrust of the spring brake
TR	Sum of braking forces at periphery of all wheels of trailer or semi-trailer
V	Rolling road test speed
V_1	Initial speed, when braking starts
V_2	Speed at end of braking
W_{60}	Energy input, equivalent to the kinetic energy of the corresponding mass for the brake under test when braked from 60 km/h to standstill
Z	Braking rate of vehicle

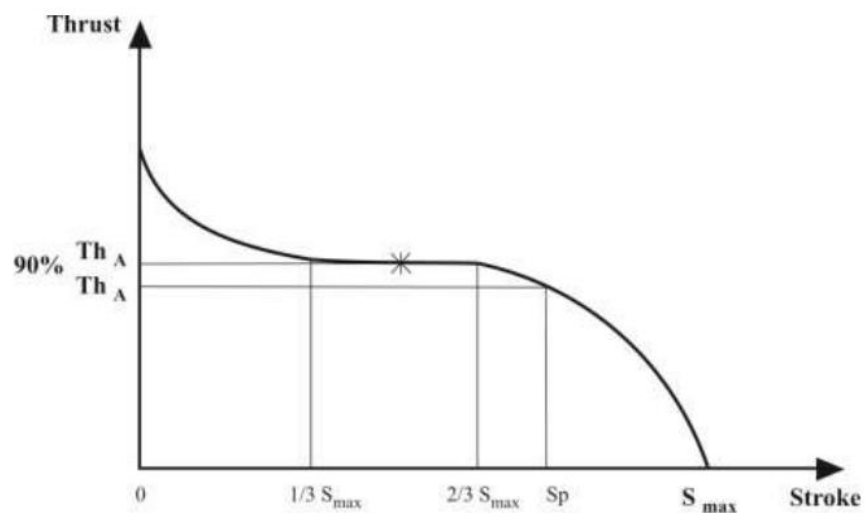


FIG. 25
DIAPHRAGM BRAKE CHAMBERS - DETERMINATION OF
EFFECTIVE STROKE

T-17 **Field test documentation form as prescribed in paragraph T-4.4.2.9.**

1. Identification

1.1. Brake:

Manufacturer

Make

Type

Model

Drum brake or disc brake⁽¹⁾

Data to identify the tested item

Technically permissible brake input torque $C_{\max}$

Automatic brake adjustment device: integrated/non-integrated⁽¹⁾

Brake drum or brake disc:

1.2. Internal diameter of drum or outside diameter of disc

Effective radius⁽²⁾

Thickness

Mass

Material

Data to identify the tested item

1.3. Brake lining or pad:

Manufacturer

Type

Identification

Width

Thickness

Surface area

Method of attachment

Data to identify the tested item

1.4. Actuator:

Manufacturer

Make

Size

Type

Data to identify the tested item

1.5. Automatic brake adjustment device (see note)

Note: Not applicable in the case of integrated automatic brake adjustment device.

Manufacturer

Make

Type

Version

Data to identify the tested item

1.6. Test vehicle data

Towing vehicle:

Identification – No.

Load on each axle

Trailer:

Identification – No.

Category: T_2 / T_3 / $T_4^{(1)}$

Full trailer / semi-trailer / central axle trailer⁽¹⁾

Number of axles

Tyres/rims:

Twin / single¹

Dynamic rolling radius R laden

Load on each axle

2.0 Test Data and results

2.1. Field test:

General description covering: distance travelled, time duration and location

2.2. Braking test:

2.2.1. Test track information

2.2.2. Test procedure

2.3. Test results:

Brake factor

Test 1

Date of test 1

Test 2

Date of test 2

Test 3

Date of test 3

Diagrams

⁽¹⁾ Strike out what does not apply

⁽²⁾ Applies only to disc brakes.

- 1.1. Name of manufacturer
- 1.2. System
- 1.3. System variants
- 1.4. System options
- 1.4.1. Control function (directional/roll-over/both) including an explanation of the basic function and/or philosophy of the control
- 1.5. System configurations (where appropriate)
- 1.6. System identification including software level identifier
- 2.0 Applications
- 2.1. List of motor vehicles by description and configuration that are covered by the information document
- 2.2. Schematic diagrams of the respective configurations installed on the motor vehicles defined in item 2.1. above with consideration given to the following:
 - (a) Lift axles
 - (b) Steering axles
 - (c) Anti-lock braking configurations
- 2.3. Scope of application with respect to suspension:
 - (a) Air
 - (b) Mechanical
 - (c) Rubber
 - (d) Mixed
 - (e) Anti-roll bar
- 2.4. Additional information (if applicable) to the application of the directional control and roll-over control functions, for example:
 - (a) Wheelbase, track, centre of gravity height
 - (b) Wheel type (single or twin) and tyre type (e.g. structure, category of use, size)
 - (c) Gearbox type (e.g. manual, automated manual, semi-automatic, automatic)
 - (d) Drive train options (e.g. retarder)
 - (e) Differential type/differential lock(s) (e.g. standard or self-locking, automatic or driver selected)
 - (f) Management of the engine or any other source(s) of motive power
 - (g) Brake type
- 3.0 Component description:
- 3.1. Sensors external to the controller
 - (a) Function
 - (b) Limitations on the location of the sensors

- (c) Identification (e.g. part numbers)
- 3.2. Controller(s)
 - (a) General description and function
 - (b) Functionality of internal sensors (if applicable)
 - (c) Hardware identification (e.g. part numbers)
 - (d) Software identification
 - (e) Limitations on the location of the controller(s)
 - (f) Additional features
- 3.3. Modulators
 - (a) General description and function
 - (b) Hardware identification (e.g. part numbers)
 - (c) Software identification (if applicable)
 - (d) Limitations
- 3.4. Electrical Equipment
 - (a) Circuit diagrams
 - (b) Powering methods
- 3.5. Pneumatic circuits

System schematics including anti-lock braking configurations associated with the motor vehicle types defined in item 2.1. of this T-18
- 3.6. Safety aspects of the electronic system in accordance with Annex R to this Standard
- 3.7. Electro-magnetic compatibility
 - 3.7.1. Documentation demonstrating compliance with AIS-004 (Part 3) as required by 5.1.1.4.

T 19 Vehicle (motor vehicle) stability function test report
 Test Report No.:

- 1. Identification:

- 1.1. Manufacturer of the vehicle stability function (name and address)
- 1.2. Applicant (if different from the manufacturer)
- 1.3. Systems
 - 1.3.1. System variants
 - 1.3.2. System options
 - 1.3.2.1. Control functions
- 2.0 System(s) and installations:
 - 2.1. Anti-lock braking configurations
 - 2.2. Vehicle applications
 - 2.2.1. Vehicle category (e.g. N₂, N₃, etc.)
 - 2.2.2. Character of the vehicle
 - 2.2.3. Vehicle configuration(s) (e.g. 4x2, 6x2, etc.)
 - 2.2.4. End of line programming
 - 2.3. System identification
 - 2.4. Functional description
 - 2.4.1. Directional control
 - 2.4.2. Roll-over control
 - 2.4.3. Low speed operation
 - 2.4.4. Off-road mode
 - 2.4.5. Drive train options
 - 2.5. Components
 - 2.6. Trailer detection and functionality
 - 2.7. Intervention warning
 - 2.8. Failure warning
 - 2.9. Stop lamp illumination
- 3.0 Assessed vehicle variables:
 - 3.1. General
 - 3.2. Brake system type
 - 3.3. Brake type
 - 3.4. Centre of gravity
 - 3.5. Management of the engine or other source(s) of motive power
 - 3.6. Gearbox type
 - 3.7. Installation configurations
 - 3.8. Lift axles
 - 3.9. Effect of load variations
 - 3.9.1. Roll-over control

- 3.9.2. Directional control
- 3.10. Steering ratio
- 3.11. Additional steering or steered axles
- 3.12. Suspension
- 3.13. Track width
- 3.14. Yaw rate and lateral acceleration sensor(s)
- 3.15. Wheelbase
- 3.16. Wheel type, tyre type, tyre size
- 4.0 Limits of Installation:
- 4.1. Suspension type
- 4.2. Brake type
- 4.3. Location of components
- 4.3.1. Yaw rate and lateral acceleration sensor(s) position
- 4.4. Anti-lock braking configuration(s)
- 4.5. Additional steered axle
- 4.6. Additional recommendations and limitations
- 4.6.1. Brake system type
- 4.6.2. Management of the engine or other source(s) of motive power
- 4.6.3. Lift axles
- 5.0 Test data and results:
- 5.1. Test vehicle data (including the specification and functionality of any trailer(s) used during the test(s))
- 5.2. Test surface information
- 5.2.1. High adhesion surface
- 5.2.2. Low adhesion surface
- 5.3. Measurement and data acquisition
- 5.4. Test conditions and procedures
- 5.4.1. Vehicle tests
- 5.4.1.1. Directional control
- 5.4.1.2. Roll-over control
- 5.5. Additional Information
- 5.6. Test results
- 5.6.1. Vehicle tests
- 5.6.1.1. Directional control

5.6.1.2. Roll-over control

5.7. Assessment in accordance with Annex R to this Standard

5.8. Compliance with AIS-004 (Part 3)

6.0 Attachments: ⁽¹⁾

7.0 Date of test:

8.0 This test has been carried out and the results reported in accordance with T-7

9.0 Test Agency conducting the test

10.0 Signature: Date:

⁽¹⁾ System supplier test data in support of the tolerance allowance as specified in paragraphs T-7.1.1.3.2.(s) and T-7.1.1.3.2.(x) shall be attached.

(2)

ANNEX U
(Clauses 6, T-1.2 and T-3.1.2)
ALTERNATIVE PROCEDURE FOR THE TYPE APPROVAL OF TRAILERS

U-1.0 GENERAL

- U-1.1 This Annex defines an alternative procedure for type approving trailers, utilising information from test reports issued in accordance with Annex K and Annex T.
- U-1.2 On completion of the verification procedures described in U-3 to U-8, the Testing Agency shall issue a type approval certificate conforming to the model specified in A-2.
- U-1.3 For the purposes of the calculations defined within this Annex the centre of gravity height shall be determined in accordance with the method defined in U-9.

U-2.0 APPLICATION FOR TYPE APPROVAL

- U-2.1 The application for type approval of a trailer type with regard to the braking equipment shall be submitted by the trailer manufacturer. In support of the approval the trailer manufacturer shall supply to the Testing Agency at least the following
- U-2.1.1 A copy of the ECE or EU type approval certificate and an information document of a trailer hereafter referred to as the "reference trailer" on which the service braking performance comparison is to be based. This trailer will have been subject to the actual tests defined in Annex C to this Standard for the appropriate trailer or equivalent EU Directive. A trailer that has been approved to the alternative procedure defined in this Annex shall not be used as a reference trailer.
- U-2.1.2 Copies of the Annex K and Annex T test reports.
- U-2.1.3 A documentation package that contains the relevant verification information including the relevant calculations for the following:

Performance Requirements	Reference to Clause
Cold service braking performance	U-3.0
Parking brake performance	U-4.0
Emergency brake performance	U-5.0
Failure of brake distribution system	U-6.0
Anti-lock braking	U-7.0
Vehicle stability function	U-8.0
Functional check	U-9.0

- U-2.1.4 A trailer, representative of the trailer type to be approved hereafter referred to as the ‘subject trailer’.
- U-2.2 The manufacturer of the ‘reference trailer’ and ‘subject trailer’ shall be the same.
- U-3.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING THE TYPE 0 TEST SERVICE BRAKING COLD PERFORMANCE**
- U-3.1 To demonstrate compliance with the Type 0 test service braking cold performance it shall be verified, by calculation, that the ‘subject trailer’ has sufficient brake force (TR) available to achieve the prescribed service braking performance and that there is sufficient adhesion available on a dry road surface (assumed to have a coefficient of adhesion of 0.8) to utilize this brake force.
- U-3.2 **Verification**
- U-3.2.1 The requirements of C-1.2.7 and C-3.1.2 (cold performance requirement and achievement without wheel lock, deviation or abnormal vibration) are considered to be satisfied by the subject trailer if it meets the verification criteria described in the following paragraphs, in both the laden and unladen conditions:
 - U-3.2.1.1 The wheelbase of the subject trailer shall not be less than 0.8 times the reference trailer wheelbase.
 - U-3.2.1.2 Any difference in the brake input torque between one axle and another within a bogie of the ‘subject trailer’ shall not differ from that of the ‘reference trailer’.
 - U-3.2.1.3 The number and arrangement of axles that is; lifting, steering, etc. of the ‘subject trailer’ shall not differ from that of the reference trailer.
 - U-3.2.1.4 The percentage distribution of the laden static axle load of the subject trailer shall not differ from that of the reference trailer by more than 10 percent.
 - U-3.2.1.5 For semi-trailers, a graph, in accordance with U-10, shall be constructed and from this graph, it shall be verified that,
 - a) $TR_{Max} \geq TR_{pr}$ [that is line (1) shall not be below line (3)]; and
 - b) $TR_L \geq TR_{pr}$ [that is line (2) shall not be below line (3)].
 - U-3.2.1.6 For centre axle trailers, a graph, in accordance with U-11, shall be constructed and, from this graph, it shall be verified that,
 - a) $TR_{Max} \geq TR_{pr}$ [that is line (1) shall not be below line (3)], and
 - b) $TR_L \geq TR_{pr}$ [that is line (2) shall not be below line (3)].
 - U-3.2.1.7 For full trailers, a graph, in accordance with U-12, shall be constructed and, from this graph, it shall be verified that,
 - a) $TR_{Max} \geq TR_{pr}$ [that is line (1) shall not be below line (2)], and
 - b) $TR_{Lf} \geq TR_{prf}$ [that is line (4) shall not be below line (3)], and
 - c) $TR_{Lr} \geq TR_{pr}$ [that is line (6) shall not be below line (5)].

U-4.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING THE

PARKING BRAKE PERFORMANCE**U-4.1 General**

U-4.1.1 This procedure provides an alternative to physically testing trailers on a gradient and ensures that trailers fitted with spring brake actuated parking mechanisms can meet the prescribed parking brake performance. This procedure shall not be applied to trailers fitted with parking mechanisms operated by means other than spring brakes. Such trailers shall be subjected to the physical test prescribed in Annex C.

U-4.1.2 The prescribed parking braking performance shall be demonstrated by calculation, using the formulae contained in U-4.2 and U-4.3.

U-4.2 Parking Performance

U-4.2.1 The parking brake force at the periphery of the tyres of axle(s) braked by the spring brake actuated parking mechanism shall be calculated using the following formula:

$$T_{pi} = (T_{hs} A_l - C_o) A_n A_{BF} / R_s$$

U-4.2.2 The normal reaction of the road surface on the axles of a stationary trailer facing uphill and downhill on an 18 percent gradient shall be calculated using the following formulae:

U-4.2.2.1 In the case of full trailers:

a) Facing uphill

$$N_{FU} = \left(PR_F - \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{FUi} = \frac{N_{FU}}{i_F}$$

$$N_{RU} = \left(PR_R + \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{RUi} = \frac{N_{RU}}{i_R}$$

b) Facing downhill

$$N_{FD} = \left(PR_F + \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{FDi} = \frac{N_{FD}}{i_F}$$

$$N_{RD} = \left(PR_R - \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{RDi} = \frac{N_{RD}}{i_R}$$

U-4.2.2.2 In the case of centre axle trailers:

a) Facing uphill

$$N_{RU} = \left(P + \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{RUi} = \frac{N_{RU}}{i_R}$$

b) Facing downhill

$$N_{RDi} = \frac{N_{RD}}{i_R}$$

U-4.2.2.3 In the case of semi-trailers:

a) Facing uphill

$$N_{RU} = \left(P - \frac{P_s \times E_R}{E_L} + \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{RUi} = \frac{N_{RU}}{i_R}$$

b) Facing downhill

$$N_{RD} = \left(P - \frac{P_s \times E_R}{E_L} - \frac{\tan P \times h \times P}{E_L} \right) \cos P$$

$$N_{RDi} = \frac{N_{RD}}{i_R}$$

U-4.3 **Verification**

The parking brake performance of the trailer shall be verified using the following formulae:

$$\left(\frac{\sum A_{Di} + \sum B_{Di}}{P} + 0.01 \right) \times 100 \geq 18\%$$

and

$$\left(\frac{\sum A_{Ui} + \sum B_{Ui}}{P} + 0.01 \right) \times 100 \geq 18\%$$

U-5.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING THE EMERGENCY/ AUTOMATIC BRAKING PERFORMANCE

U-5.1 General

To demonstrate compliance with the automatic braking performance requirements, either a comparison is made between the chamber pressure required to achieve the specified performance and the asymptotic chamber pressure after disconnection of the supply line, as defined in U-5.2.1, or it shall be verified that the brake force provided by the axle(s) fitted with spring brakes is sufficient to achieve the specified performance, as defined in U-5.2.2.

U-5.2 **Verification**

U-5.2.1 The requirements of C-3.3 are considered to be satisfied by a subject trailer, if the asymptotic chamber pressure (p_c) after disconnection of the supply line is greater than the chamber pressure (p_c) to achieve a performance of 13.5 percent of the maximum stationary wheel load. The supply line pressure being stabilized at 700 kPa prior to disconnection.

U-5.2.2 The requirements of C-3.3 are considered to be satisfied by a spring brake equipped subject trailer if:

$$\Sigma T_{pi} \geq 0.135 (PR)(g)$$

where T_{pi} is calculated in accordance with U-4.2.1.

U-6.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING BRAKING PERFORMANCE IN THE CASE OF FAILURE OF THE BRAKING DISTRIBUTION SYSTEM

U-6.1 **General**

U-6.1.1 To demonstrate compliance with the braking performance requirements in the case of a failure of the braking distribution system, a comparison is made between the chamber pressure required to achieve the specified performance and the chamber pressure available when a failure in the braking distribution system exists.

U-6.2 **Verification**

U-6.2.1 The requirements of J-6, are considered to be satisfied by the subject trailer if the pressure defined in U-6.2.1.1 is greater than or equal to the pressure defined in U- 6.2.1.2., in both laden and unladen conditions.

U-6.2.1.1 The chamber pressure (p_c) of the subject trailer, when $p_m = 650$ kPa, the supply line pressure = 700 kPa and a failure in the braking distribution system exists.

U-6.2.1.2 The chamber pressure (p_c) to achieve a braking rate of 30 percent of the service braking performance prescribed for the subject trailer.

U-7.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING THE ANTI-LOCK BRAKING PERFORMANCE

U-7.1 **General**

Testing of a trailer in accordance with Annex M may be waived at the time of type approval of the trailer provided that the anti-lock braking system (ABS) complies with the requirements of Annex U.

U-7.2 **Verification**

U-7.2.1 Verification of Components and Installation

The specification of the ABS installed on the trailer to be type approved shall be verified by satisfying each of the following criteria:

U-7.2.1.1	Paragraph		Criteria
	a)	Sensor(s)	No change allowed
	b)	Controller(s)	No change allowed
	c)	Modulator(s)	No change allowed

U-7.2.1.2 Tube size(s) and lengths

	a)	Reservoir supply to modulator(s)	
		Minimum inside diameter	May be increased
		Maximum overall length	May be reduced
	b)	Modulator delivery to brake chambers	
		Inside diameter	No change allowed
		Maximum overall length	May be reduced

U-7.2.1.3	Warning signal sequence	No change allowed
U-7.2.1.4	Differentials in brake input torque within a bogie	Only approved differentials (if any) permitted
U-7.2.1.5	For other limitations refer to item 4 of the test report as described in T-11.	Installation to be within the scope of the limitations defined - No deviations allowed

U-7.3 Verification of Reservoir Capacity

U-7.3.1 As the range of braking systems and auxiliary equipment used on trailers is diverse it is not possible to have a table of recommended reservoir capacities. To verify that adequate storage capacity is installed, testing may be conducted according to M-6.1 or by the procedure defined below:

U-7.3.1.1 In the case of brakes with nonintegrated brake wear adjustment the brakes on the subject trailer shall be set to a condition where the relationship (R_1) of brake chamber push rod travel (s_T) against lever length (l_T) is 0.2.

Example:

$$l_t = 130 \text{ mm}$$

$$R_e = s_T/l_T = s_T/130 = 0.2$$

s_T = Push rod travel at 650 kPa brake chamber pressure

$$= 130 \times 0.2 = 26 \text{ mm}$$

- U-7.3.1.2 In the case of brakes with integrated automatic brake wear adjustment the brakes shall be set to a normal running clearance.
- U-7.3.1.3 Setting of the brakes as defined above shall be carried out when the brakes are cold ($\leq 100\text{ }^{\circ}\text{C}$).
- U-7.3.1.4 With the brakes adjusted according to relevant procedure defined above and the load sensing device(s) set to the laden condition and the initial energy level set according to M-6.1.2, the energy storage device(s) shall be isolated from further supply. The brakes shall be applied with a control pressure of 650 kPa at the coupling head and then fully released. Further brake applications shall be made up to the number n_e determined from the test conducted in accordance with T-5.4.1.2.4.2 and defined in paragraph 2.5 of the anti-lock braking system approval report. During this application, the pressure in the operating circuit shall be sufficient to provide a total braking force at the periphery of the wheels equal to not less than 22.5 percent of the maximum stationary wheel load and without causing automatic application of any braking system not under the control of the anti-lock braking system.
- U-7.4. Trailers with more than 3 axles may utilize the T-13 ABS test report provided the following conditions are fulfilled:
 - U-7.4.1. Irrespective of trailer type at least one third of the axles in an axle group shall have all wheels directly controlled, with the wheels on the remaining axles being indirectly controlled ⁽¹⁾.
 - U-7.4.2. Utilization of Adhesion: The minimum utilization of adhesion specified within M-6.2. is deemed to be achieved when following conditions are fulfilled:
 - U-7.4.2.1. The relationship of the number of wheels directly or indirectly controlled by one or more pressure modulators and the location of the directly controlled wheels within the axle group shall correspond to those defined within paragraph 2.2. of the Information Document referred to in T-5.2;
 - U-7.4.2.2. The utilization of adhesion of the installed configuration is shown in the test report as meeting the requirements of M-6.2.
 - U-7.4.3. Energy Consumption: The number of equivalent static brake applications defined within 2.5. of the test report may be used in conjunction with the verification procedure of U-7.3.. Alternatively the test procedure specified within M-6.1. may be used;
 - U-7.4.4. Low speed performance: Additional verification is not required;
 - U-7.4.5. High speed performance: Additional verification is not required;
 - U-7.4.6. Category A performance: The split friction requirements specified within M-6.3.2. are deemed to be fulfilled when the number of wheels which are subject to independent left/right control is equal to or greater than the number of wheels controlled using "select low" axle control;
 - U-7.4.7. Surface transition performance: Additional verification not required;
 - U-7.4.8. Installation limitations: in all cases the following limitations shall apply:

⁽¹⁾When the number of axle(s) in an axle group is divided by 3 and the resulting number is less than 1, at least one axle shall be directly controlled. When the number of axles in the axle group is divided by 3 and the resulting number is not a whole number, a further axle shall be directly controlled in addition to the number of axles indicated by the whole number value.

- U-7.4.8.1. Any installation limitations defined within items 2.1. to 2.7. of the Information Document referred to in paragraph T-5.2 shall apply;
- U-7.4.8.2. Only products identified and referenced in the information document and test report may be installed;
- U-7.4.8.3. The maximum delivery volume controlled by each pressure modulator shall not exceed the volume specified within T-14.3.3.;
- U-7.4.8.4. An axle with directly controlled wheels may only be lifted when any axle which is indirectly controlled from this direct controlled axle is lifted in parallel;
- U-7.4.8.5. All other installation limitations specified within paragraph 4. of the test report shall apply.

U-8.0 ALTERNATIVE PROCEDURE FOR DEMONSTRATING THE PERFORMANCE OF A TRAILER EQUIPPED WITH A VEHICLE STABILITY FUNCTION.

U-8.1. Evaluation of a trailer in accordance with V-2. may be waived at the time of trailer type approval provided that the vehicle stability function complies with the relevant requirements of Annex T to this Standard.

U-8.2. Verification

U-8.2.1. Verification of components and installation

The specification of the braking system, in which the stability control function is integrated and installed on the trailer to be type approved shall be verified by satisfying each of the following criteria:

	Condition	Criteria
U-8.2.1.1.	(a) Sensor(s)	No change allowed
	(b) Controller(s)	No change allowed
	(c) Modulator(s)	No change allowed
U-8.2.1.2.	Trailer types as defined in the test report	No change allowed
U-8.2.1.3.	Installation configurations as defined in the test report	No change allowed
U-8.2.1.4.	For other limitations refer to paragraph 4. of the test report as described in T-15.	No change allowed

U 9.0 FUNCTIONAL AND INSTALLATION CHECKS

U 9.1 The Certifying agency / Approval authority shall carry out functional and

installation checks covering the following paragraphs:

U 9.1.1 Anti-lock Function

This shall be limited to a dynamic check of the anti-lock braking system. To ensure full cycling it may be necessary to adjust the load sensing device or utilize a surface having a low tyre to road adhesion. If the anti-lock system does not have an Annex T approval, the trailer shall be tested in accordance with Annex M and comply with the relevant requirements contained in that Annex.

U 9.1.2 Response Time Measurement

The Certifying agency shall verify that the subject trailer conforms to the requirements of Annex E.

U 9.1.3 Static Energy Consumption

The Certifying agency shall verify that the subject trailer conforms to the requirements of Annex F and Annex G as appropriate.

U 9.1.4 Service Brake Function

The Certifying agency shall verify that there are no abnormal vibrations during braking.

U 9.1.5 Parking Brake Function

The Certifying agency shall apply and release the park brake to ensure correct function.

U 9.1.6 Emergency/Automatic Braking Function

The Certifying agency shall verify that the subject trailer complies with the requirements of 5.2.1.18.4.2.

U 9.1.7 Vehicle and Component Identification Verification

The Certifying agency shall check the subject trailer against the details contained in the type approval certificate.

U-9.1.8. Vehicle stability function

For practical reasons verification of the vehicle stability function shall be limited to an installation check as defined in U-8.2. and observation of the correct warning signal sequence to ensure no faults are present.

U 9.1.9 Additional Checks

The Certifying agency may request additional checks to be carried out, if necessary.

U 10.0 METHOD OF CALCULATING THE CENTRE OF GRAVITY HEIGHT

The centre of gravity height for the complete vehicle (laden and unladen) can be calculated as follows (see Fig. 26):

h_1 = centre of gravity height of axle(s) assembly (inc. tyres, springs, etc) = R . 1.1

h_2 = centre of gravity height of frame (laden) = $(h_6 + h_8) \cdot 0.5$

h_3 = centre of gravity height of payload and bodywork (laden)

$$= (h_7 \cdot 0.3) + h_6$$

h_4 = centre of gravity height of frame (unladen) = $h_2 + s$

h_5 = centre of gravity height of bodywork (unladen) = $(h_7 \cdot 0.5) + h_6 + s$

where

h_6 = frame height, top

h_7 = body dimensions, inside

h_8 = frame height, bottom

P = total mass of the trailer

PR = total mass on all wheels of a semi-trailer or centre axle trailer

R = radius of the tyre

s = spring deflection between laden and unladen

W_1 = mass of axle(s) assembly (inc. tyres, springs, etc) = $P \cdot 0.1$

W_2 = mass of the frame = $(P_{unl} - W_1) \cdot 0.8$

W_3 = mass of payload and bodywork

W_4 = mass of bodywork = $(P_{unl} - W_1) \cdot 0.2$

LADEN:

$$h_{Rlad} = \frac{h_1.W_1 + h_2.W_2 + h_3.W_3}{P_{lad}}$$

UNLADEN:

$$h_{Runl} = \frac{h_1.W_1 + h_4.W_2 + h_5.W_4}{P_{unl}}$$

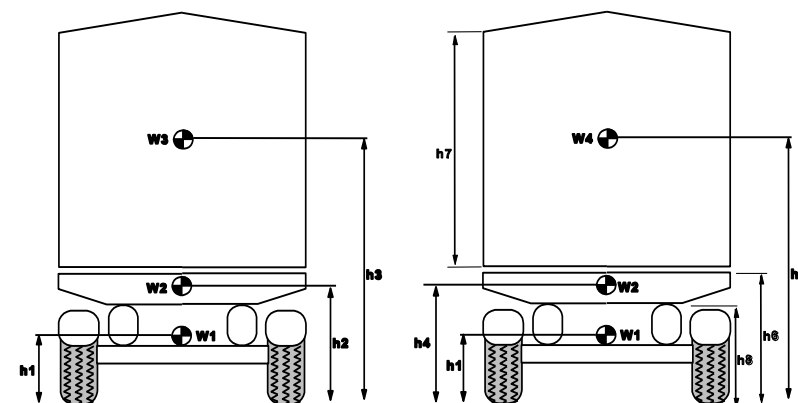
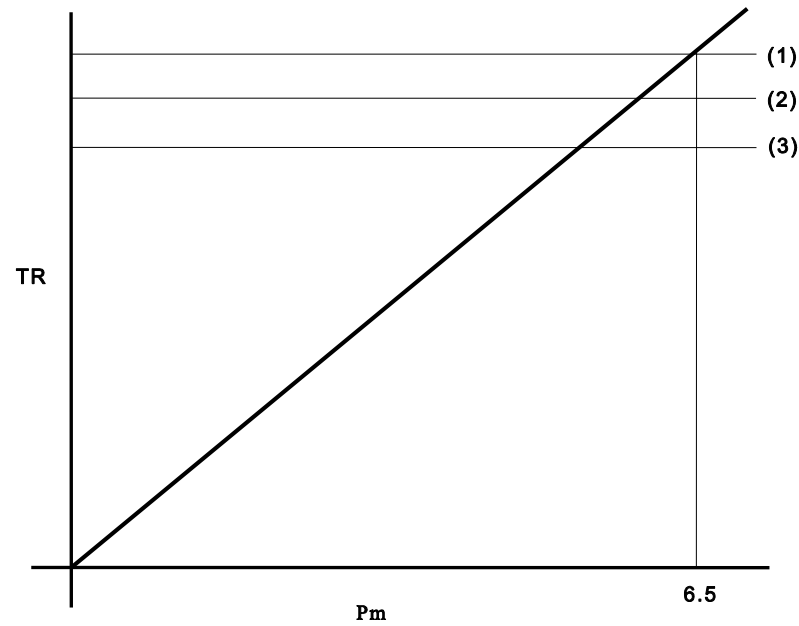


FIG. 26

CENTRE OF GRAVITY HEIGHT NOTES

- 1 For flat bed type trailers, a maximum height of 4 m shall be used.
- 2 For trailers where the exact centre of gravity height of the payload is not known, it shall be taken as 0.3 times the inside dimensions of the body.
- 3 For trailers with air suspension, the value of s shall be taken as zero.
- 4 For semi-trailers and centre axle trailers, replace P with PR , wherever it occurs.

U 11.0 VERIFICATION GRAPH FOR U-3.2.1.5 SEMI-TRAILERS



(1) = TR_{Max} , when $p_m = 650$ kPa and supply line = 700 kPa

(2) = $F_{Rdyn} \cdot 0.8 = TR_L$

(3) = $0.45 \cdot F_R = TR_{pr}$

where

the value of z_c being calculated using the following formula:

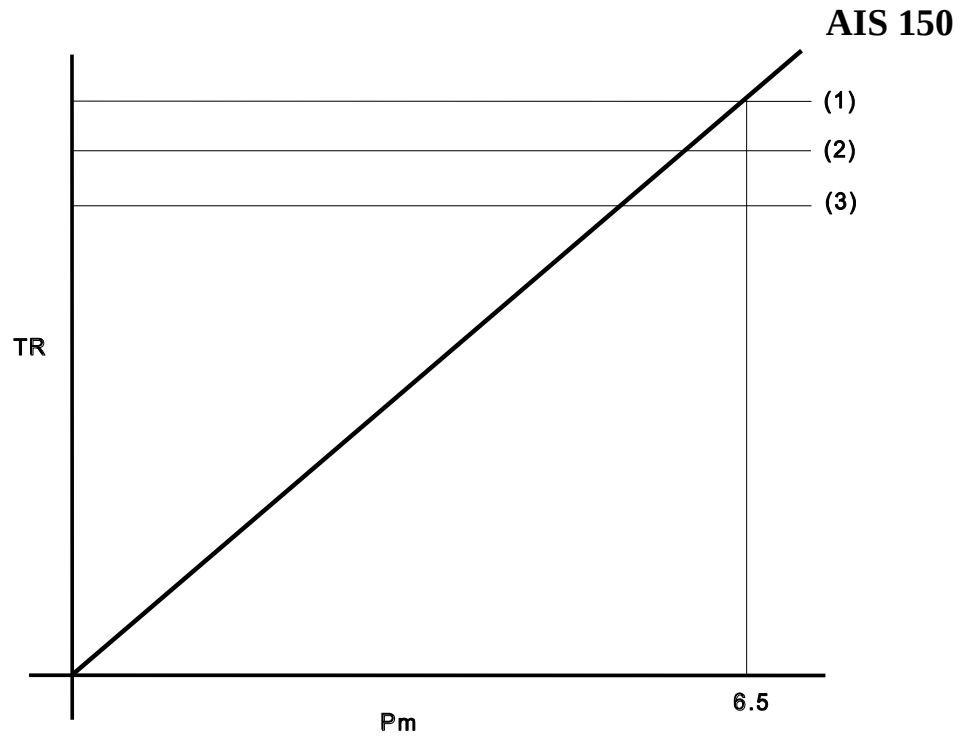
$$F_{Rdyn} = F_R - \frac{(TR_{pr} \times h_k) + (p \times g \times Z_c (h_R - h_k))}{E_R}$$

$$z_c = 0.45 - 0.01 \left(\frac{F_R}{(P + 7000)g} \right) + 0.01$$

NOTES

- 1 The value of 7000 above represents the mass of a towing vehicle with no trailer attached.
- 2 For the purpose of these calculations, closely spaced axles (having an axle spread of less than 2 m) may be treated as one axle.

U 12.0 Verification graph for paragraph U-3.2.1.6. - Centre axle trailers



(1) = TR_{Max} , when $p_m = 650$ kPa and supply line = 700 kPa

(2) = $F_{Rdyn} \cdot 0.8 = TR_L$

(3) = $0.5 \cdot F_R = TR_{pr}$

where

$$F_{Rdyn} = F_R - \frac{(TR_{pr} \times h_k) + (p \times g \times Z_c(h_R - h_k))}{E_R}$$

the value of z_c being calculated using the following formula:

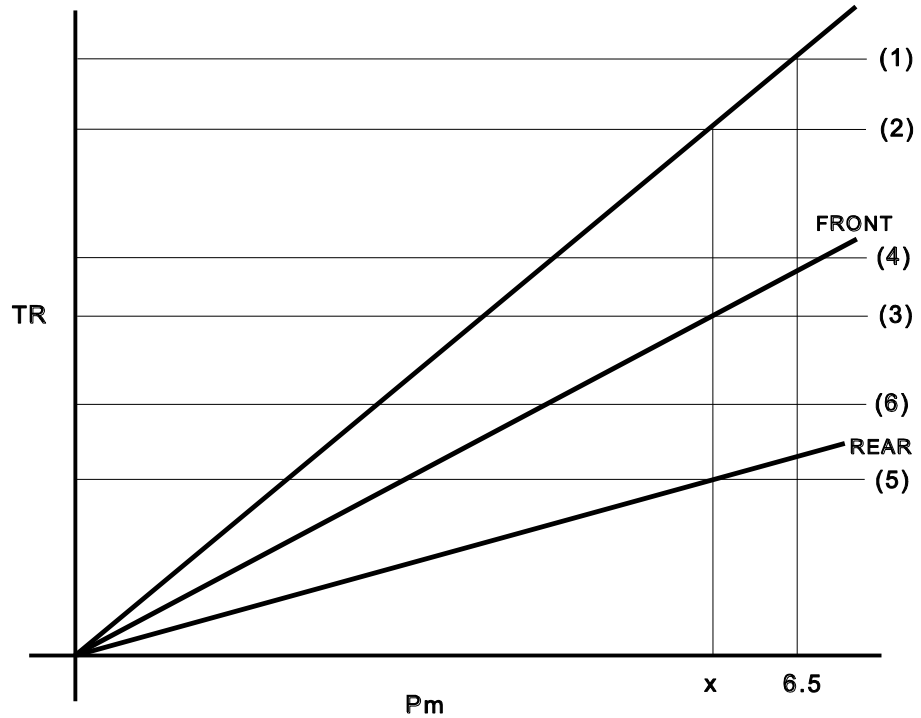
$$z_c = 0.5 - 0.01 \left(\frac{F_R}{(P + 7000)g} \right) + 0.01$$

NOTES

- 1 The value of 7000 above represents the mass of a towing vehicle with no trailer attached.
- 2 For the purpose of these calculations, closely spaced axles (having an axle spread of less than 2 m) may be treated as one axle.

U 13.0

VERIFICATION GRAPH FOR U-3.2.1.7 - FULL TRAILERS



(1) = TR_{Max} , when $p_m = 650$ kPa and supply line = 700 kPa

(2) = $0.5 \cdot F_R = TR_{pr}$

(3) = $TR_{prf} = TR_f$, when $p_m = x$

(4) = $F_{fdyn} \cdot 0.8 = TR_{Lf}$

(5) = $TR_{prf} = TR_r$, when $p_m = x$

(6) = $F_{rdyn} \cdot 0.8 = TR_{Lr}$

where

$$F_{fgdyn} = F_f + \frac{P \times g \times Z_c \times h_R}{E}$$

and

$$F_{rdyn} = F_r - \frac{P \times g \times Z_c \times h_R}{E}$$

the value of z_c being calculated using the following formula:

$$z_c = 0.5 - 0.01 \left(\frac{F_R}{(p + 7000)g} \right) + 0.01$$

NOTES

1 The value of 7 000 above represents the mass of a towing vehicle with no trailer attached.

2 For the purpose of these calculations, closely spaced axles (having an axle spread of less than 2 m) may be treated as one axle.

U 14.0

SYMBOLS AND DEFINITIONS

Symbol	Definition
A_{Di}	T_{pi} when $T_{pi} \leq 0.8 N_{FDi}$ for front axles, or $0.8 N_{FDi}$ when $T_{pi} > 0.8 N_{FDi}$ for front axles
B_{Di}	T_{pi} when $T_{pi} \leq 0.8 N_{RDi}$ for rear axles, or $0.8 N_{RDi}$ when $T_{pi} > 0.8 N_{RDi}$ for rear axles
A_{Ui}	T_{pi} when $T_{pi} \leq 0.8 N_{FUi}$ for front axles, or $0.8 N_{FUi}$ when $T_{pi} > 0.8 N_{FUi}$ for front axles
B_{Ui}	T_{pi} when $T_{pi} \leq 0.8 N_{RUi}$ for rear axles, or $0.8 N_{RUi}$ when $T_{pi} > 0.8 N_{RUi}$ for rear axles
B_F	brake factor
C_o	threshold camshaft input torque (minimum camshaft torque necessary to produce a measurable brake torque)
E	wheelbase
E_L	distance between the coupling support leg or landing legs to the centre of the axle(s) of a centre axle trailer or semi-trailer
E_R	distance between king-pin and centre of axle or axles of semi-trailer
F	force (N)
F_f	total normal static reaction of road surface on front axle(s)
F_{fdyn}	total normal dynamic reaction of road surface on front axles(s)
F_r	total normal static reaction of road surface on rear axle(s)
F_{rdyn}	total normal dynamic reaction of road surface on rear axle(s)
F_R	total normal static reaction of road surface on all wheels of the trailer or semi-trailer
F_{Rdyn}	total normal dynamic reaction of road surface on all wheels of the trailer or semi-trailer
g	acceleration due to gravity (9.81 m/s^2)
h	height above ground of centre of gravity
h_K	height of fifth wheel coupling (king pin)
h_r	height of centre of gravity of trailer
i	axle index
i_F	number of front axles
i_R	number of rear axles
l	lever length
n	number of spring brake actuators per axle
N_{FD}	total normal reaction of road surface on front axle(s) when facing downhill on an 18 percent gradient

N_{FDi}	normal reaction of road surface on front axle i when facing downhill on an 18 percent gradient
N_{FU}	total normal reaction of road surface on front axle(s) when facing uphill on an 18 percent gradient
N_{FUi}	normal reaction of road surface on front axle i when facing uphill on an 18 percent gradient
N_{RD}	total normal reaction of road surface on rear axle(s) when facing downhill on an 18 percent gradient
N_{RDi}	normal reaction of road surface on rear axle i when facing downhill on an 18 percent gradient
N_{RU}	total normal reaction of road surface on rear axle(s) when facing uphill on an 18 percent gradient
N_{RUi}	normal reaction of road surface on rear axle i when facing uphill on an 18 percent gradient
p_m	pressure at coupling head of control line
p_c	pressure in the brake chamber
P	mass of individual vehicle
P_s	static mass at fifth wheel coupling at trailer mass P
PR	total normal static reaction of road surface on wheels of trailer or semi-trailer
PR_F	total normal static reaction of road surface on the front axles on level ground
PR_R	total normal static reaction of road surface on the rear axles on level ground
R_s	static laden radius of the tyre, calculated using the following formula: $R_s = 1/2 \, dr + F_R \cdot H$ where dr = nominal rim diameter H = design section height = $1/2 (d - dr)$ d = rim diameter convention number F_R = factor, as defined by ETRTO (Engineering Design Information 1994, page CV.11)
T_{pi}	brake force at periphery of all wheels of axle i provided by the spring brake(s)
Th_s	spring thrust of the spring brake

TR	sum of braking forces at periphery of all wheels of the trailer or semi-trailer
TR _f	sum of braking forces at periphery of all wheels of the front axle(s)
TR _r	sum of braking forces at periphery of all wheels of the rear axle(s)
TR _{max}	sum of maximum available braking forces at periphery of all wheels of the trailer or semi-trailer
TR _L	sum of braking forces at periphery of all wheels of the trailer or semi-trailer at which the limit of adhesion is reached
TR _{Lf}	sum of braking forces at periphery of all wheels of the front axle(s) at which the limit of adhesion is reached
TR _{Lr}	sum of braking forces at periphery of all wheels of the rear axle(s) at which the limit of adhesion is reached.
TR _{pr}	sum of braking forces at periphery of all wheels of the trailer or semi-trailer required to achieve the prescribed performance
TR _{prf}	sum of braking forces at periphery of all wheels of the front axle(s) required to achieve the prescribed performance
TR _{prr}	sum of braking forces at periphery of all wheels of the rear axle(s) required to achieve the prescribed performance
Z _c	braking rate of the vehicle combination, with the trailer only braked
cos P	cosine of angle subtended by slope of 18 percent and horizontal plane = 0.984 18
tan P	tangent of angle subtended by slop of 18 percent and horizontal plane = 0.18

ANNEX V

Special requirements for vehicles equipped with a vehicle stability function**V-1.0 General**

- V-1.1. This annex defines the special requirements for vehicles equipped with a vehicle stability function, pursuant to 5.2.1.32., 5.2.1.33 and 5.2.2.23..
- V-1.2. In meeting the requirements of this annex the "other vehicles" as mentioned in V-2.1.3. and V-2.2.3. below shall not differ in at least the following essential respects:
 - V-1.2.1. The character of the vehicle;
 - V-1.2.2. In the case of power-driven vehicles the axle configuration (e.g. 4x2, 6x2, 6x4);
 - V-1.2.3. In the case of trailers the number and arrangement of axles;
 - V-1.2.4. The front axle steering ratio in the case of power-driven vehicles when the vehicle stability function does not include it as a end-of-line programmable feature or as a self-learning feature;
 - V-1.2.5. Additional steered axles in the case of power-driven, and steered axles in the case of trailers;
 - V-1.2.6. Lift axles;

V-2.0 Requirements

- V-2.1. Power-driven vehicles
 - V-2.1.1. Where a vehicle is equipped with a vehicle stability function as defined in 3.34, the following shall apply:

In the case of directional control the function shall have the ability to automatically control individually the speed of the left and right wheels on each axle or an axle of each axle group by selective braking based on the evaluation of actual vehicle behaviour in comparison with a determination of vehicle behaviour demanded by the driver. ⁽¹⁾

In the case of roll-over control the function shall have the ability to automatically control the wheel speeds on at least two wheels of each axle or axle group by selective braking or automatically commanded braking based on the evaluation of actual vehicle behaviour that may lead to vehicle roll-over.⁽¹⁾

⁽¹⁾Additional interaction with other vehicle systems or components is allowed. Where these systems or components are subject to special Standards, such interaction shall comply with the requirements of those Standards, e.g. interaction with the steering system shall comply with the requirements set out in IS 11948:2010 for corrective steering.

In both cases, the function is not required:

- (a) When the vehicle speed is below 20 km/h;
- (b) Until the initial start-up self-test and plausibility checks have been completed;
- (c) When the vehicle is being driven in reverse;
- (d) When it has been automatically or manually disabled. In this case, the following conditions shall apply as appropriate:
 - i) When a vehicle is equipped with a means to automatically disable the vehicle stability function to provide increased traction by modifying the functionality of the drive train, the disablement and its re-instatement shall be automatically linked to the operation which changes the functionality of the drive train;
 - ii) When a vehicle is equipped with a means to manually disable the vehicle stability function, the vehicle stability function shall be automatically reinstated at the initiation of each new ignition cycle;
 - iii) A constant optical warning signal shall inform the driver that the vehicle stability function has been disabled. The yellow warning signal specified in V-2.1.5. may be used for this purpose. The warning signals specified in 5.2.1.29. shall not be used.

V-2.1.2. To realise the functionality defined above a vehicle stability function shall include, in addition to selective braking and/or automatically commanded braking, at least the following:

- (a) The ability to control engine power output;
- (b) In the case of directional control: The determination of actual vehicle behaviour from values of yaw rate, lateral acceleration, wheel speeds, and from the driver's control inputs to the braking and steering systems and to the engine. Only on-board generated information shall be used. If these values are not directly measured, the evidence of the appropriate correlation with directly measured values under all driving conditions (e.g. including driving in a tunnel) shall be shown to the Test Agency at the time of type approval;
- (c) In the case of roll-over control: The determination of actual vehicle behaviour from values of the vertical force on the tyre(s) (or at least lateral acceleration and wheel speeds) and from the driver's control inputs to the braking system and to the engine. Only on-board generated information shall be used. If these values are not directly measured, the evidence of the appropriate correlation with directly measured values under all driving conditions (e.g. including driving in a tunnel) shall be shown to the Test Agency at the time of type approval;
- (d) In the case of a towing vehicle equipped according to 5.1.3.1.: The ability to apply the service brakes of the trailer via the respective control line(s) independently of the driver.

V-2.1.3. The vehicle stability function shall be demonstrated to the Test Agency by dynamic manoeuvres on one vehicle which has the same vehicle stability function as the vehicle type to be approved. This may be realized by a comparison of results obtained with the vehicle stability function enabled and disabled for a given load condition. As an alternative to carrying-out dynamic manoeuvres for other vehicles and other load conditions, fitted with the same vehicle stability system, the results from actual vehicle tests or computer simulations may be submitted.

As an alternative to the above, a test report conforming to T-7.1.1 may be used.

The use of the simulator is defined in V-3.

The specification and validation of the simulator is defined in V-4.

Until unified test procedures are agreed, the method by which this demonstration is carried out shall be agreed between the vehicle manufacturer and the Test Agency and shall include the critical conditions of directional control and roll-over control as appropriate to the vehicle stability function installed on the vehicle with the method of demonstration and results being appended to the type approval report. This may be carried-out other than at the time of type approval.

As a means of demonstrating the vehicle stability function any of the following dynamic manoeuvres shall be used ⁽¹⁾

Directional control	Roll-over control
Reducing radius test	Steady state circular test
Step steer input test	J-turn
Sine with dwell	
J-turn	
μ -split single lane change	
Double lane change	
Reversed steering test or "fish hook" test	
Asymmetrical one period sine steer or pulse steer input test	

To demonstrate repeatability the vehicle will be subject to a second demonstration using the selected manoeuvre(s).

(1)Should the use of any of the above defined manoeuvres not result in loss of directional control or roll-over as appropriate an alternative manoeuvre may be used in agreement with the Test Agency.

- V-2.1.4. Interventions of the vehicle stability function shall be indicated to the driver by a flashing optical warning signal fulfilling the relevant technical requirements of AIS-071. The indication shall be present as long as the vehicle stability function is in an intervention mode. The warning signal specified in paragraph 5.2.1.29.1.2. of this standard shall not be used for this purpose.

Additionally, interventions by systems related to the vehicle stability function (including traction control, trailer stability assist, corner brake control, and other similar functions that use throttle and/or individual torque control to operate and share common components with vehicle stability function) may also be indicated to the driver by this flashing optical warning signal.

Interventions of the vehicle stability function used in any learning process to determine the vehicle operational characteristics shall not generate the above signal.

The signal shall be visible to the driver, even in daylight, such that the driver can easily verify the satisfactory condition of the signal without leaving the driver's seat.

- V-2.1.5. A vehicle stability function failure or defect shall be detected and indicated to the driver by an optical warning signal fulfilling the relevant technical requirements of AIS-071. The warning signal specified in paragraph 5.2.1.29.1.2. of this Standard shall not be used for this purpose. The warning signal shall be constant and remain displayed as long as the failure or defect persists and the ignition (start) switch is in the 'on' (run) position.

- V-2.1.6. In the case of a power-driven vehicle equipped with an electric control line and electrically connected to a trailer with an electric control line the driver shall be warned by a specific optical warning signal fulfilling the relevant technical requirements of AIS-071 whenever the trailer provides the information "VDC Active" via the data communications part of the electric control line. The optical signal defined in paragraph 2.1.4. above may be used for this purpose.

V-2.2. Trailers

- V-2.2.1. Where a trailer is equipped with a vehicle stability function as defined in paragraph 2.34. of this Standard, the following shall apply:

In the case of directional control the function shall have the ability to automatically control individually the speed of the left and right wheels on each axle or an axle of each axle group by selective braking based on the evaluation of actual trailer behaviour in comparison with a determination of the relative behaviour of the towing vehicle.¹

In the case of roll-over control the function shall have the ability to automatically control the wheel speeds on at least two wheels of each axle or axle group by selective braking or automatically commanded braking based on the evaluation of actual trailer behaviour that may lead to roll-over.¹

V-2.2.2. To realise the functionality defined above a vehicle stability function shall include, in addition to automatically commanded braking and where appropriate selective braking, at least the following:

(a) The determination of actual trailer behaviour from values of the vertical force on the tyre(s), or at least lateral acceleration and wheel speeds. Only on-board generated information shall be used. If these values are not directly measured, the evidence of the appropriate correlation with directly measured values under all driving conditions (e.g. including driving in a tunnel) shall be shown to the Test Agency at the time of type approval.

V-2.2.3. The vehicle stability function shall be demonstrated to the Test Agency by dynamic manoeuvres on one vehicle which has the same vehicle stability function as the vehicle type to be approved. This may be done by a comparison of results obtained with the vehicle stability function enabled and disabled for a given load condition. As an alternative to carrying-out dynamic manoeuvres for other vehicles and other load conditions, fitted with the same vehicle stability system, the results from actual vehicle tests or computer simulations may be submitted.

As an alternative to the above, a test report conforming to T-6 may be used.

The use of the simulator is defined in T-8.

The specification and validation of the simulator is defined in T-9.

Until unified test procedures are agreed, the method by which this demonstration is carried out shall be agreed between the trailer manufacturer and the Test Agency and shall include the critical conditions of roll-over control and directional control as appropriate to the vehicle stability function installed on the trailer with the method of demonstration and results being appended to the type approval report. This may be carried-out other than at the time of type approval.

As a means of demonstrating the vehicle stability function any of the following dynamic manoeuvres shall be used:²

Directional control	Roll-over control
Reducing radius test	Steady state circular test
Step steer input test	J-turn
Sine with dwell	
J-turn	
μ -split single lane change	
Double lane change	
Reversed steering test or "fish hook" test	
Asymmetrical one period sine steer or pulse steer input test	

- V-2.2.4. To demonstrate repeatability the vehicle will be subject to a second demonstration using the selected manoeuvre(s).

Trailers equipped with an electric control line, when electrically connected to a towing vehicle with an electric control line, shall provide the information "VDC active" via the data communications part of the electric control line when the vehicle stability function is in an intervention mode. Interventions of the vehicle stability function used in any learning process to determine the trailer operational characteristics shall not generate the above information.

- V-2.2.5. To maximise the performance of trailers that utilise "select-low" such trailers are permitted to change control mode to "select-high" during an intervention of the "Vehicle Stability Function".

V-3.0 Use of the dynamic stability simulation

The effectiveness of the directional and/or roll-over stability control function of power driven vehicles and trailers of categories M, N and O, may be determined by computer simulation.

- V-3.1. Use of the simulation

- V-3.1.1. The vehicle stability function shall be demonstrated by the vehicle manufacturer to the Type Approval Authority or Test Agency with the same dynamic manoeuvre(s) as for the practical demonstration in V-2.1.3. or V-2.2.3.

- V-3.1.2. The simulation shall be a means whereby the vehicle stability performance may be demonstrated with the vehicle stability function enabled or disabled, and in the laden and unladen conditions.

- V-3.1.3. The simulations shall be carried out with a validated modelling and simulation tool. The simulation tool shall only be used when each relevant parameter of the vehicle to be type-approved, as listed in V-4.1.1. is included in the simulation tool and when the value of each parameter falls within its respective validated range. The verification shall be carried out using the same manoeuvre(s) as defined in V-3.1.1.

The method by which the simulation tool is validated is given in V-4.

- V-3.1.3.1. A vehicle manufacturer using a validated simulation tool that was not directly validated by themselves for a vehicle type-approval shall carry-out at least one confirmation test.

This confirmation test shall be conducted in conjunction with a Test Agency and shall be a comparison between an actual vehicle test and a simulation using one of the manoeuvres as defined in V-3.1.1. The confirmation test shall be repeated in the event of any change to the simulation tool.⁽¹⁾

The results of the confirmation test shall be attached to the type-approval documentation.

- V-3.1.4. The availability of the simulation tool software, to the software version used, shall be maintained for a period of not less than 10 years following the date of the approval of the vehicle.

⁽¹⁾The necessity of a confirmation test shall be subject to a discussion between the vehicle manufacturer, the Test Agency

V-4.0 Dynamic stability simulation tool and its validation

V-4.1. Specification of the simulation tool

V-4.1.1. The simulation tool shall take into account the main factors which influence the directional and roll motion of the vehicle.

V-4.1.1.1. The simulation tool shall take into account the following vehicle parameters as applicable: ⁽¹⁾

- (a) Vehicle category;
- (b) Character of the vehicle;
- (c) Gearbox type (e.g. manual, automated manual, semi-automatic, automatic);
- (d) Differential type (e.g. standard or self-locking);
- (e) Differential lock(s) (driver selected);
- (f) Brake system type (e.g. air over hydraulic, full air);
- (g) Brake type (e.g. disc, drum (single wedge, twin wedge, S-cam));
- (h) Tyre type (e.g. structure, category of use, size);
- (i) Suspension type (e.g. air, mechanical, rubber).

V-4.1.1.2. The simulation model shall include at least the following parameters as applicable ⁽¹⁾

- (a) Vehicle configuration(s) (e.g. 4x2, 6x2, etc., identifying axle functionality (e.g. free running, driven, lifted, steered) and position);
- (b) Steering axles (working principle);
- (c) Steering ratio;
- (d) Drive axle(s) (effect on wheel speed sensing and vehicle speed);
- (e) Lift axle(s) (detection/control and wheelbase change effect when lifted);
- (f) Engine management (communication, control and response);
- (g) Gearbox characteristic(s);
- (h) Drive train option(s) (e.g. retarder, regenerative braking, auxiliary propulsion system);
- (i) Brake characteristic(s);
- (j) Anti-lock braking configuration;
- (k) Wheelbase;
- (l) Track width;
- (m) Centre of gravity height;
- (n) Lateral acceleration sensor position;
- (o) Yaw rate sensor position;
- (p) Loading.

V-4.1.1.3. The Test Agency conducting the validation shall be provided with an information document covering at least the points in V-4.1.1.1. and V-4.1.1.2.

V-4.1.2. The vehicle stability function shall be added to the simulation model by means of:

- a) A subsystem (software model) of the simulation tool as software-in-the-loop,
or
- b) An actual electronic control unit in a hardware-in-the-loop configuration.

V-4.1.3. In the case of a trailer, the simulation shall be carried out with the trailer coupled to a representative towing vehicle.

⁽¹⁾ Parameters not included shall limit the use of the simulation tool.

- V-4.1.4. Vehicle loading condition
- V-4.1.4.1. The simulation tool shall be able to take into account the laden and unladen conditions.
- V-4.1.4.2. The simulation tool shall, as a minimum, meet the following criteria:
- (a) A fixed load;
 - (b) A given mass;
 - (c) A given mass distribution; and
 - (d) A given height of the centre of gravity.
- V-4.2. Validation of the simulation tool
- V-4.2.1. The validity of the applied modelling and simulation tool shall be verified by means of comparisons with a practical vehicle test(s). The test(s) utilised for the validation shall be those which, without control action, would result in loss of directional control (under-steer and over-steer) and/or roll-over control as appropriate to the functionality of the stability control function installed on a vehicle.
During the test(s) the following motion variables, as appropriate, shall be recorded or calculated in accordance with ISO 15037 Part 1:2006 or Part 2:2002 as relevant:
- (a) Yaw velocity;
 - (b) Lateral acceleration;
 - (c) Wheel load or wheel lift;
 - (d) Forward velocity;
 - (e) Driver input.
- V-4.2.2. The objective is to show that the simulated vehicle behaviour and operation of the vehicle stability function is comparable with that seen in practical vehicle tests.
The ability of the simulation tool to be used with parameters that have not been validated by a practical vehicle test shall be shown by conducting simulations with varied parameter values. The results of these simulations shall be checked to be logical and similar in comparison to the results of known practical vehicle tests.
- V-4.2.3. The simulation tool shall be deemed to be validated when its output is comparable to the practical test results produced by the same vehicle(s) during the manoeuvre(s) selected from those defined with V-2.1.3. or V-2.2.3., as appropriate.
The simulation tool shall only be used with regard to features for which a comparison has been made between real vehicle tests and simulation tool results. The comparisons shall be carried-out in the laden and unladen condition to show the different conditions of load can be adapted to and to confirm the extreme parameters to be simulated, e.g.:
- (a) Vehicle with shortest wheelbase and highest centre of gravity;
 - (b) Vehicle with longest wheelbase and highest centre of gravity.
- In the case of the steady state circular test the under-steer gradient shall be the means of making the comparison.
In the case of a dynamic manoeuvre, the relationship of activation and sequence of the vehicle stability function in the simulation and in the practical vehicle test shall be the means of making the comparison.
- V-4.2.4. The physical parameters that are different between the reference vehicle and simulated vehicle configurations shall be modified accordingly in the simulation.
- V-4.2.5. A simulation tool test report shall be produced, a model of which is defined in V-5, and a copy attached to the vehicle approval report.

V-5.0 Vehicle stability function simulation tool test report

Test report number:

- V-5.1. Identification
 - V-5.1.1. Name and address of the simulation tool manufacturer
 - V-5.1.2. Simulation tool identification: name/model/number (hardware and software)
- V-5.2. Simulation tool
 - V-5.2.1. Simulation method (general description, taking into account the requirements of V-4.1.1.)
 - V-5.2.2. Hardware/software in the loop (see V-4.1.2.)
 - V-5.2.3. Vehicle loading conditions (see V-4.1.4.)
 - V-5.2.4. Validation (see V-4.2.)
 - V-5.2.5. Motion variables (see V-4.2.1.)
- V-5.3. Scope of application:
 - V-5.3.1. Vehicle category:
 - V-5.3.2. Character of the vehicle:
 - V-5.3.3. Vehicle configuration:
 - V-5.3.4. Steering axles:
 - V-5.3.5. Steering ratio:
 - V-5.3.6. Drive axles:
 - V-5.3.7. Lift axles:
 - V-5.3.8. Engine management:
 - V-5.3.9. Gearbox type:
 - V-5.3.10. Drive train options:
 - V-5.3.11. Differential type:
 - V-5.3.12. Differential lock(s):
 - V-5.3.13. Brake system type:
 - V-5.3.14. Brake type:
 - V-5.3.15. Brake characteristics:
 - V-5.3.16. Anti-lock braking configuration:
 - V-5.3.17. Wheelbase:
 - V-5.3.18. Tyre type:
 - V-5.3.19. Track width:
 - V-5.3.20. Suspension type:
 - V-5.3.21. Centre of gravity height:
 - V-5.3.22. Lateral acceleration sensor position:
 - V-5.3.23. Yaw rate sensor position:
 - V-5.3.24. Loading:
 - V-5.3.25. Limiting factors:

- V-5.3.26. Manoeuvre(s) for which the simulation tool has been validated:
- V-5.4. Verifying vehicle test(s)
- V-5.4.1. Description of vehicle(s) including the towing vehicle in case of trailer testing:
 - V-5.4.1.1. Vehicle(s) identification: make/model/VIN
 - V-5.4.1.1.1. Non-standard fitments:
 - V-5.4.1.2. Vehicle description, including axle configuration/suspension/wheels, engine and drive line, braking system(s) and vehicle stability function content (directional control/rollover control), steering system, with name/model/number identification:
 - V-5.4.1.3. Vehicle data used in the simulation (explicit)
- V-5.4.2. Description of test(s) including location(s), road/test area surface conditions, temperature and date(s):
- V-5.4.3. Results laden and unladen with the vehicle stability function switched on and off, including the motion variables referred to in V-4.2.1. as appropriate:
- V-5.5. Simulation results
 - V-5.5.1. Vehicle parameters and the values used in the simulation that are not taken from the actual test vehicle (implicit):
 - V-5.5.2. Results laden and unladen with the vehicle stability function switched on and off for each test conducted under paragraph V-5.4.2., including the motion variables referred to in V-4.2.1., as appropriate:
- V-5.6. Concluding statement

The simulated vehicle behaviour and operation of the vehicle stability function is comparable with that of practical vehicle tests.

Yes/No
- V-5.7. Limiting factors
- V-5.8. This test has been carried out and the results reported in accordance with V-4.

Test Agency conducting the test ⁽¹⁾

Signed:

Date:

Type Approval Authority⁽¹⁾

⁽¹⁾To be signed by different persons if the Test Agency and the Type Approval Authority are the same organisation.

ANNEX W

Requirements for the brake electric/electronic interface of an automated connector

W-1.0 General

This annex defines the requirements applicable to installations where the connection and disconnection of the brake electric/electronic interface between the towing vehicle and the towed vehicle is achieved by an automated connector.

This annex also considers the case where a vehicle is equipped with both ISO 7638 connector and an automated connector.

W-2.0 Categories of automated connectors Automated connectors are classified in different categories: (See Note)

Note: New categories of couplings may be added later on for new/innovative technical solutions, when standard interfaces will be defined and agreed upon.

Category A: Automated connector for tractor/semi-trailer combinations shall meet the requirements of W-4. All automated connectors within this category are compatible together.

Category B: Automated connectors for tractor/semi-trailer combinations that do not meet all the requirements of W-4. They are not compatible with category A. Interfaces of category B are not necessarily compatible to all type of interfaces within this category.

Category C: Automated connectors for combination other than tractor/semitrailer shall meet the requirements of W-5 (See Note) . All automated connectors within this category are compatible together.

Note: Until a standard is defined and agreed upon, no automated connector shall be defined as being of category C.

Category D: Automated connectors for combinations other than tractor/semi-trailer that do not meet all the requirements of W-5. They are not compatible with category C. Interfaces of category D are not necessarily compatible to all type of interfaces within this category.

W-3. Requirements

The brake electric/electronic interface of the automated connector shall achieve the same functional requirements as specified for the ISO 7638 connector throughout this standard and its Annexes.

W-3.1. The contacts (pins and sockets) for the brake electric/electronic interface shall have the same electrical characteristics and functionality as the ISO 7638 contacts.

- W-3.1.1. The data contacts of the brake electric/electronic interface shall be used to transfer information exclusively for braking (including ABS) and running gear (steering, tyres and suspension) functions as specified in ISO 11992-2:2003 including its Amendment 1:2007. The braking functions have priority and shall be maintained in the normal and failed modes. The transmission of running gear information shall not delay braking functions.
- W-3.1.2. The power supply, provided by the brake electric/electronic interface, shall be used exclusively for braking and running gear functions and that required for the transfer of trailer related information not transmitted via the electric control line. However, in all cases the provisions of 5.2.2.18. shall apply. The power supply for all other functions shall use other measures.
- W-3.2. In the case of semi-trailer combinations equipped with an automated connector the maximum length of the cable for braking data communication shall be:
- (a) Tractor: 21 m;
 - (b) Semi-trailer: 19 m;
- in the running mode.
- In all other cases the conditions of 5.1.3.6. and 5.1.3.8. apply with respect to maximum cable lengths.
- W-3.3. Vehicles being equipped with both a connector conforming to ISO 7638 and an automated connector shall be built in such a way that only a single path is possible for the functioning of the electric control transmission or in the transmission of information in accordance with ISO 11992-2:2003 including Amendment 1:2007. See W-4 for examples.
- In the case of automatic path selection the priority shall be given to the automated connector.
- W-3.4. Any trailer equipped with an automated connector shall be equipped with a spring braking system according to Annex G.
- W-3.5. The manufacturer applying for type approval shall submit an information document describing the functionality and any limitations in the use of the automated connector and any associated equipment, including information about the category according to paragraph 2. of this annex.
- In the case of automated connectors of categories B and D, the means to identify the type of automated connector shall also be described to ensure identification of compatibility.
- W-3.6. The vehicle user's handbook provided by the manufacturer shall warn the driver of the consequences of not checking the compatibility of the automated connector between the towing vehicle and the trailer. Information about mixed mode operation shall also be provided if applicable.
- To enable the driver to check the compatibility, vehicles fitted with an automated connector shall have a marking specifying the category according to paragraph 2. of this annex. For categories B and D also the type of the installed automated connector shall be shown. This marking shall be indelible and visible to the driver when standing on the ground beside the vehicle.

W-4.0 Examples of the layout of an automated connection between vehicles

Automated connection and manual connection equipped vehicles: data bus requirements.

Diagrams for electrical connections show routing of signals of pins 6 and 7 according to ISO 7638.

Legend**Electrical**

E1 ISO 11992-2 node in tractor, e.g. ECU ABS/EBS

E2 Tractor ISO 7638 socket

E3 Tractor ISO 7638 plug for automated connector

E4 Tractor part of automated connector

E5 Trailer ISO 7638 plug for automated connector

E6 Trailer ISO 7638 socket

E7 Trailer part of automated connector

E8 ISO 7638 coiled cable

E9 ISO 7638 park socket

E10 ISO 11992-2 node in trailer, e.g. ECU ABS/EBS

I Cable from E1 to E2

II Cable from E10 to E6

III Cable from E5 to E7

IV Cable from E3 to E4

Pneumatic

P1 Trailer control valve mounted on tractor

P2 T-piece

P3 Pneumatic coupling head on tractor (control and supply)

P4 Tractor part of automated connector

P5 Pneumatic coupling head on trailer (control and supply)

P6 Pneumatic valve to seal the unused terminal (double check valve) (control and supply)

P7 Trailer part of automated connector

P8 Pneumatic coiled tube (control and supply)

P9 Pneumatic park socket (control and supply)

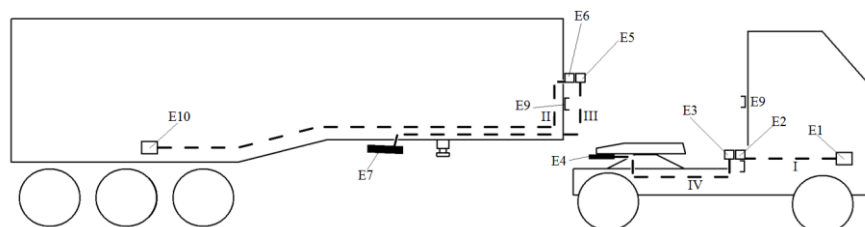
Tractor and semi-trailer examples

I. Automated connection and manual connection equipped vehicles

Automated connection mode

Figure A

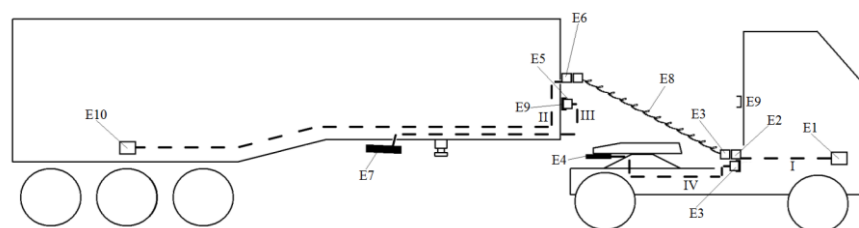
Point-to-point connection ECU Tractor (E1) and ECU Trailer (E10) via ACV. Automated connection mode: No coiled cables connected, Connection between E1 and E10 when E4 and E7 are connected (i.e. when the fifth wheel is coupled)



Manual connection mode

Figure B

Point-to-point connection ECU Tractor (E1) and ECU Trailer (E10) via coiled cable. Manual mode: Coiled cables connected, Connections between E3 and E4 as E5 and E7 are not in use

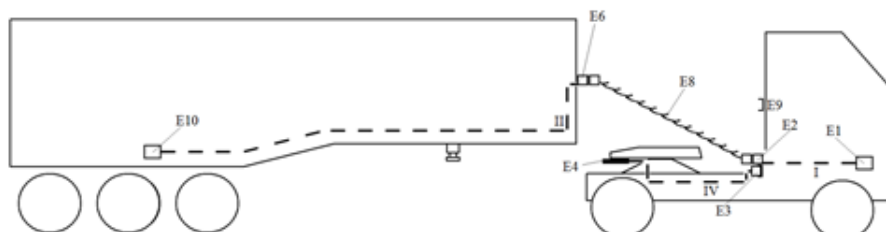


II. Only one part of the vehicle combination is equipped with an automated connection

Manual mode A (only the tractor equipped with automated connection)

Figure C

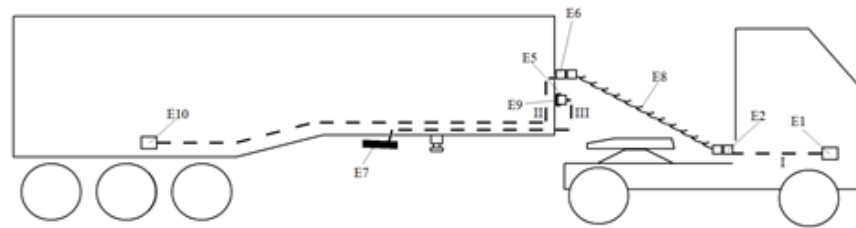
Point-to-point connection ECU Tractor (E1) and ECU Trailer (E10) when the fifth wheel is closed. Coiled cables connected, Line E3 to E4 is not in use



Manual mode B (only the semi-trailer equipped with automated connection)

Figure D

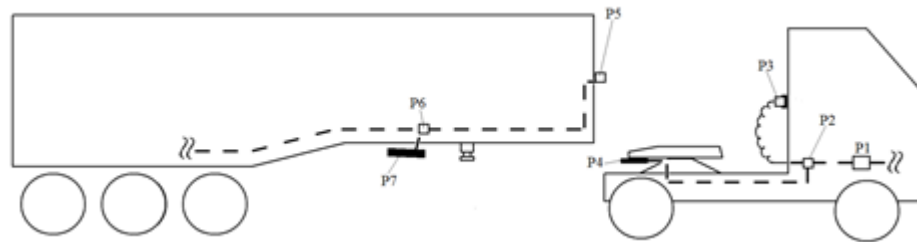
Point-to-point connection ECU Tractor (E1) and ECU Trailer (E10). Coiled cables connected, Line E5 to E7 is not in use



Automated connection mode

Figure E

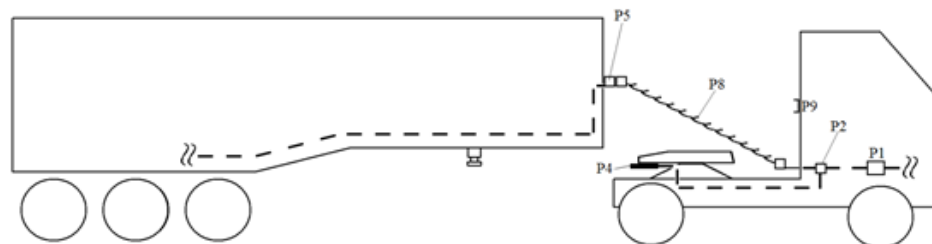
Pneumatic connection Tractor and Trailer via ACV. Automated connection mode: No coiled cables connected, Connection between tractor and trailer when P4 and P7 are connected (i.e. when the fifth wheel coupled)



Manual mode A (only the tractor equipped with automated connection)

Figure F

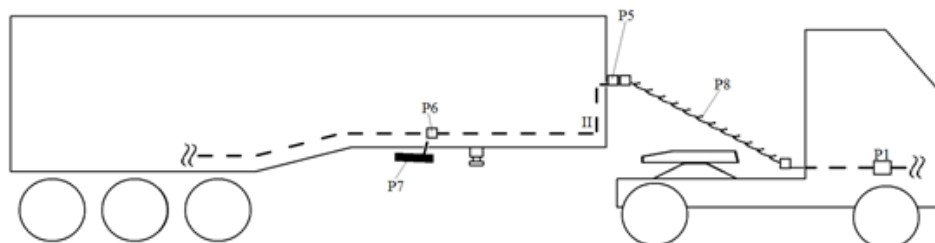
Pneumatic connection Tractor and Trailer via coiled tube. Coiled tubes connected, Line P2 to P5



Manual mode B (only the semi-trailer equipped with automated connection)

Figure G

Pneumatic connection Tractor and Trailer via coiled tube. Coiled tubes connected, Line P1 to P5



W-5.0 Couplings of category A shall conform with the relevant provisions of ISO 13044-2:2013 to ensure compatibility of braking systems of tractor and semi-trailer

W-6.0 (Reserved) To be defined at a later date."

ANNEX X
(Foreword)
CLAUSE WISE REFERENCES OF ECE AND THIS STANDARD

Clause Ref of this standard	Title	Corresponding Clause of UN R
1.	Scope	1
2	References	NA
3	Definitions	2
4	Application for approval	3
5	Specification	5
6	Tests	6
7	Modification of vehicle type of braking system and extension of approval	7
NA	Conformity of Production	8
NA	Braking equipment, devices, methods and conditions not covered by this standard	Annex 1
Annex A	Communication concerning the approval or extension or refusal or withdrawal of approval or production definitely discontinued of a vehicle type with regard to braking,	Annex 2
	List of vehicle data for the purpose of Standard No. 90 approvals	Annex 2, Appendix 1
Annex B	Test Procedures	Annex 3
Annex C	Braking tests and performance of braking systems	Annexure 4
	Procedure for monitoring the state of battery charge	Annexure 4, Appendix 1
Annex D	Additional provisions applicable to certain vehicles meant for carriage of dangerous and hazardous goods	Annex 5
Annex E	Method of measuring the response time on vehicles equipped with compressed-air braking systems	Annex 6
	Example of a simulator	Annex 6, - Appendix
Annex F	Provisions relating to energy sources and energy storage devices (Energy accumulators)	Annex 7
Annex G	Provisions relating to specific conditions for spring braking systems	Annex 8

Annex H	Provisions relating to parking braking systems equipped with a mechanical brake-cylinder locking device (Lock actuators)	Annex 9
Annex J	Distribution of braking among the axles of vehicles and requirements for compatibility between towing vehicles and trailers	Annex 10
Annex K	Cases in which Type I and/or Type II (or Type II A) tests do not have to be carried out	Annex11
Annex L	Conditions governing the testing of vehicles equipped with inertia (overrun) braking systems	Annex12
	Figures 1-8	Annex 12 Appendix 1
	Test Report on inertia-braking system control device	Annex 12 Appendix 2
	Brake Test Report	Annex 12 Appendix 3
	Test report on the compatibility of the inertia brake control device the transmission and the brakes on the trailer	Annex 12 Appendix 4
Annex M	Test requirements for vehicles fitted with anti-lock systems	Annex 13
	Symbols and definitions	Annex 13 Appendix 1
	Utilization of adhesion	Annex13 Appendix 2
	Performance on differing adhesion surfaces	Annex13 Appendix 3
	Method of selection of the low-adhesion surfaces	Annex 13 Appendix 4
Annex N	Test conditions for trailers with electrical braking systems	Annex 14
	Compatibility of the braking rate of the trailer and the mean fully developed deceleration of the tractor/trailer combination (trailer laden and unladen)	Annex 14 Appendix
Annex P	Inertia dynamometer test method for brake linings	Annex15

Annex Q	Interpretation of ISO 11992-2 and ISO 11992-3 for the purpose of clause 5.1.3.6	Annex 16
Annex R	Test procedure to assess the functional compatibility of vehicles equipped with electric control lines	Annex 17
Annex S	Special requirements to be applied to the safety aspects of complex electronic vehicle control systems	Annex 18
Annex T	Performance testing of braking components	Annex 19
	Model verification report form for diaphragm brake chambers	Annex19 Appendix 1
	Model reference record of test results for diaphragm brake chambers	Annex19 Appendix 2
	Model verification report form for spring brakes	Annex19 Appendix 3
	Model reference record of test results for spring brakes	Annex 19 Appendix 4
	Trailer anti-lock braking system information document	Annex 19 Appendix 5
	Trailer anti-lock braking system test report	Annex 19 Appendix 6
	Symbols and definitions	Annex19 Appendix 7
Annex U	Alternative procedure for the type approval of trailers	Annex 20
	Method of calculating the centre of gravity height	Annex 20 Appendix 1
	Verification graph for 3.2.1.5 - semi-trailers	Annex 20 Appendix 2
	Verification graph for 3.2.1.6 - central axle trailers	Annex 20 Appendix 3
	Verification graph for 3.2.1.7 - full trailers	Annex 20 Appendix 4
	Symbols and definitions	Annex 20 Appendix 5
Annex V	Special requirements for vehicles equipped with a vehicle stability function	Annex 21
Annex W	Requirements for the brake electric/electronic interface of an automated connector	Annex 22

ANNEX Y
COMMITTEE COMPOSITION – AISC Panel

Name	Organization
Convener	
Mr. A. A. Badusha	The Automotive Research Association of India (ARAI)
Members	Representing
Mr. K. B. Patil	The Automotive Research Association of India (ARAI)
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Mr. Suchindran M	Toyota Kirloskar Motor Pvt. Ltd. (SIAM)
Mr. Gururaj Ravi / Mr. Rajesh Vyas / Mr. Raj Kumar Diwedi	Maruti Suzuki India Ltd.(SIAM)
Mr. Rajendra Khile/ Mr. Karuppasamy	Renault Nissan Technology and Business Centre (SIAM)
Mr. Mohan Kumar Muthusamy	VE Commercial Vehicles Ltd. (SIAM)
Mr. Makarand Brahme	Volkswagen India Pvt. Ltd. (SIAM)
Mr. Vikrant Lokhande	Volvo Trucks – VECV (SIAM)
Mr. Vishal P. Jain	Isuzu Motors India
Mr. P. Venugopal / Mr. A. Vijayan	Brakes India Ltd.
Mr. S. Balachandran / Mr. D. Prabhakaran / Mr. Sachin Deshmukh	WABCO
Mr. Arun Bisht	Knorr- Bremse India

ANNEX Z
COMMITTEE COMPOSITION – AISC

Chairperson	
Mrs. Rashmi Urdhwareshe	Director The Automotive Research Association of India, Pune
Members	Representing
Representative from	Ministry of Road Transport and Highways (Dept. of Road Transport and Highways), New Delhi
Representative from	Ministry of Heavy Industries and Public Enterprises (Department of Heavy Industry), New Delhi
Shri S. M. Ahuja	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises, New Delhi
Shri Shrikant R. Marathe	Former Chairman, AISC
Shri R.R. Singh	Bureau of Indian Standards, New Delhi
Director	Central Institute of Road Transport, Pune
Director	Global Automotive Research Centre, Chennai
Director	International Centre for Automotive Technology, Manesar
Director	Indian Institute of Petroleum, Dehra Dun
Director	Vehicles Research and Development Establishment, Ahmednagar
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Representatives from	Society of Indian Automobile Manufacturers
Shri R. P. Vasudevan	Tractor Manufacturers Association, New Delhi
Shri Uday Harite	Automotive Components Manufacturers Association of India, New Delhi

Member Secretary
Shri Vikram Tandon
Dy. General Manager
The Automotive Research Association of India, Pune

* At the time of approval of this Automotive Industry Standard (AIS)